An adaptive mixing framework for a personalised listening experience (AMPLE)

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Declaration

I hereby declare that the work presented in this thesis has not been submitted for any other degree or professional qualification and that it is the result of my own independent work.

Date

Abstract

Film sound delivery remains largely reliant on static, one-size-fits-all mixes that often fail to translate effectively across contemporary listening conditions, despite advances in audio technology. Viewers frequently report problems relating to dialogue intelligibility, loudness variation, and imbalance between soundtrack elements, particularly when consuming film content on mobile devices, televisions, or within noisy environments.

Adaptive mixing approaches are investigated in this research as a means of supporting more context-responsive film sound reproduction across mobile and stationary playback systems. A mixed-method research design combining literature analysis, audience surveys, professional industry surveys, and controlled listening experiments was employed. Audience data were collected from approximately 500 participants to examine playback hardware preferences, listening behaviour, subtitle usage, listening environments, and sound element preferences across cinema, mobile, and stationary contexts. Additional data were gathered from professional re-recording mixers to investigate current workflows, monitoring practices, and delivery constraints associated with television and streaming productions.

Results demonstrated substantial variability in audience listening preferences, playback hardware, and environmental conditions, highlighting limitations in fixed delivery approaches. Notably, over 60% of surveyed audiences reported frequent volume adjustments to maintain intelligibility. Professional responses similarly indicated that, although practitioners recognise the need for greater flexibility in soundtrack delivery, existing workflows remain constrained by established standards, production practices, budget limitations, and time pressures, often prioritising dialogue clarity above other soundtrack elements.

Drawing upon these findings, the adaptive mixing for a personalised listening experience (AMPLE) framework is proposed. Object-based audio principles alongside narrative-priority metadata are used to support context-responsive adaptation of soundtrack elements during playback. Two listening studies (N=25 each) evaluated user interaction with adaptive soundtrack control under both reference listening conditions and simulated environmental masking conditions. Findings indicated consistent prioritisation of dialogue clarity across listening scenarios, while preferences relating to music, ambience, and sound effects varied according to playback environment and contextual noise conditions.

Collectively, the research contributes an empirically grounded adaptive mixing framework that integrates audience behaviour, playback context, and professional production constraints within a unified model for film sound delivery. Findings suggest that adaptive and object-based approaches may support improved soundtrack translation across diverse consumer listening conditions while maintaining narrative priorities within the mix.

Publications associated with this research.

Shalabi, Ahmed; McGregor, Iain; 2024; Capturing Film Mixers' Workflows; Audinate, Cambridge, UK; Edinburgh Napier University, Edinburgh, UK; Paper

Gelby, Ahmed; McGregor, Iain; 2023; Capturing audience film sound preferences; Audinate, 4380 S Macadam Avenue, Suite 255, Portland, OR 97239, USA; Edinburgh Napier University, 10 Colinton Rd, Edinburgh EH10 5DT UK; Paper 179

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Chapter 1: Introduction

In this section, the research area, and the background are introduced, followed by the research aims, and objectives declaration. The thesis's structure is outlined and the contributions to the broader professional community field are identified followed by a summary of the chapter and overview of the next one.

1.1 Background

Film sound delivery is increasingly experienced across a wide range of playback devices and listening environments, yet professional mixing practices remain oriented toward fixed output formats and standardised conditions (Alexander, 2020; Basil, 2001; Bump, 2020; Campbell & Mathurin, 2021; Costa & Duarte, 2022; Dziabkin, 2019; Falkowski-Gilski & Uhl, 2020; Hagen, 2022; Jarvey, 2023; Lawler, 2020; Looms, 2018; Market.us, 2020; Rich Media, 2020; Spangler, 2019; Statista, 2023; Walton, 2018).

Viewers frequently report problems with dialogue intelligibility, excessive loudness variation, and imbalance between sound elements when consuming content on television and streaming platforms, particularly in domestic or mobile contexts (BBC, 2017; BBC Academy, 2020; Brown, 2016; Dalziel, 2022; Johnson, 2017; Kean et al., 2015a; Plunkett, 2016; Skovenborg & Lund, 2009; Sontag, 1996; L. Ward, 2017).

Although industry bodies such as the ITU, AES, and SMPTE have introduced standards intended to regulate loudness and delivery formats, these approaches depend on static mixes and predetermined downmixing strategies that assume stable playback conditions (AES TD1004.1.15-10, 2015; EBU-TECH 3344, 2016; ITU-R BS.1770-4, 2015; SMPTE ST 2036-1:2014, 2014; ST 2110-30:2017, 2017; BBC & DPP, 2020; Netflix, 2021). This structural reliance on fixed rendering limits their capacity to respond to differences in playback hardware, ambient noise, and individual listening preferences.

Cinema sound benefits from tightly controlled playback conditions and extensive calibration standards, yet research strongly suggests that recommended monitoring levels are not always followed in practice (Bosun 2001; Guru, Martens, and Lee 2011a; Hill, Hawskford, and Newell 2016a; Rumsey 2018a; Toole 2015a; Völker 1996a). This divergence contributes to progressive increases in loudness and reductions in dynamic range (Ghouti, 2017; P. Newell, 2014b; P. Newell & Newell, 2016; Polychromatic, 2019; Thornton, 2017a, 2017b, 2018a).

These factors may contribute to reduced consistency in the listening experience and can affect the transmission of creative intent across playback contexts. For television and streaming audiences, complaints about poor audibility and intrusive background music remain persistent. These complaints often prompt viewers to enable subtitles, adjust volume repeatedly, or disengage from content. Such responses interrupt narrative engagement and may encourage prolonged exposure to elevated listening levels (Aazh et al., 2015; BBC, 2017; BBC Academy, 2020; Brown, 2016; Dalziel, 2022; DESTATIS, 2009; Johnson, 2017; Löhler et al., 2019; Madans et al., 2021; Plunkett, 2016; Shirley & Kendrick, 2006; Shirley & Oldfield, 2015; Sontag, 1996; L. Ward, 2017).

Existing research on automatic and adaptive mixing has focused primarily on accessibility features or music production workflows (Ellis-Geiger, 2016; Fenton, 2018; Nam et al., 2021; Oldfield & Shirley, 2011; Pestana et al., 2015; J. Reiss et al., 2017; J. D. Reiss, 2011). Limited attention has been given to film sound or to the combined influence of user behaviour, playback devices, and listening environments.

Channel based audio, which remains dominant in film production, is intrinsically tied to specific speaker layouts and offers little flexibility for user interaction or cross device consistency (Füg et al., 2016; Lee et al., 2014; Rumsey, 2015b). Object based audio represents sound elements as discrete objects accompanied by metadata that describes their narrative role and spatial attributes. Rendering can therefore be adapted at playback for different speaker configurations, devices, and listening contexts. This technical capability provides a basis for personalisation and contextual adaptation, yet no established framework applies it systematically to the everyday variability of film sound consumption (Jackson et al. 2017; Rumsey 2012, 2016b, 2018b).

This research is motivated by the gap between professional mixing practices and the conditions under which audiences listen to film sound. It proposes that an adaptive mixing framework informed by user behaviour, playback hardware, and listening environment can support a more consistent and intelligible listening experience while preserving narrative priorities within the mix.

The contribution of this work lies in integrating these three dimensions within a single framework for film sound rather than treating them as separate concerns. The focus of the proposed framework is adaptive mixing logic rather than interface design. Personalisation is implemented through the treatment of sound elements as objects ranked by narrative importance and rendered dynamically in response to contextual information available at playback. The issues identified in this section directly motivate the aims and research questions set out in the following section.

1.2 Research Questions

This research follows a sequential logic to address the systemic gap between professional mixing standards and the reality of consumer listening. First, it is necessary to define the limitations of the current delivery models (RQ1). Second, to propose a solution, the specific mixing variables that audiences seek to control must be quantified (RQ2). Finally, these insights must be synthesised into a technical framework that automates this personalisation (RQ3).

RQ1: How does the current delivered listening experience impact audiences across different platforms, and where do existing static delivery models fail to meet their needs?

RQ2: What are the psychoacoustic preferences and listening behaviours of audiences when watching films on mobile or stationary hardware under varying environmental conditions?

RQ3: How can the key components of an adaptive mixing framework be architected and presented to provide customizable, context-aware functionality that resolves the identified translation gap?

To answer these questions, the thesis structure functions as a cumulative investigation:

RQ1 (The Problem Definition): Addressed in Chapters 3 and 4. Chapter 3 utilises a large-scale survey (N=500) to identify where the current one-size-fits-all model fails the audience, while Chapter 4 contrasts this with industry constraints. This defines the specific technical and demographic barriers that the new framework must overcome.

RQ2 (The Empirical Requirement): Investigated in Chapters 3, 6, and 7. While Chapter 3 gathers self-reported habits, Chapters 6 and 7 rigorously test these claims through active mixing experiments. These chapters isolate the specific mixing parameters (for example, dialogue level, dynamic Range) that users adjust to compensate for hardware limitations and environmental noise, providing the empirical basis for the adaptive framework.

RQ3 (The Technical Solution):
Addressed through the definition of the AMPLE framework in Chapter 5 and its empirical evaluation in Chapters 6 and 7. Drawing directly from the limitations identified in RQ1 and the user data captured in RQ2, this question drives the development of the prototype. It evaluates whether an object-based approach can support closer alignment between professional mixing intentions and consumer listening experiences.

Contribution to Knowledge and Industry Application:

The answers to these questions provide the foundation for an adaptive mixing framework. By moving beyond simple loudness compliance and investigating the specific psychoacoustic preferences of the audience (RQ2), this research provides empirical evidence that may inform future developments in post-production workflows.

The findings may provide practical relevance for industry by:

Operationalising Metadata: Identifying which soundtrack stems (for example, dialogue and ambience) may benefit from independent control during transmission.

Supporting Improved Translation: Providing mixers with empirical data on how their content is experienced across non-cinema listening environments.

Supporting Listening Practices: Addressing factors associated with repeated volume adjustment, which may arise from intelligibility difficulties, by allowing users to adjust mix balance rather than relying solely on increased global gain.
Accordingly, the adaptive mixing framework for a personalised listening experience (AMPLE) is not assumed at the outset but instead emerges progressively through the synthesis of audience evidence, professional

constraints, and empirical findings presented throughout the subsequent chapters.

1.3 Aims

1. To conduct a literature review on film sound mixing, delivery, accessibility, and the audience's sonic experience.

2. To gather requirements on audience behaviour, preferences, and feedback related to the listening experience when watching films in the cinema, and on both mobile and stationary devices.

3. To design and conduct experiments where audience members mix their own film soundtracks on both mobile and stationary devices.

4. To use insights from these experiments to inform the development of an adaptive mixing framework, supporting personalised listening experiences for film on mobile and stationary devices.

1.4 Objectives

To answer the research questions and address the identified gap, the objectives of this research are to establish the limitations of current film sound delivery practices, derive design requirements from empirical evidence, and synthesise these into a technically grounded framework.

The specific objectives are as follows:

Objective 1: To analyse current limitations in production and delivery: To conduct a critical review of literature on film sound production, professional workflows, and quality control (QC). This identifies the structural constraints in existing practices and situates the research problem within contemporary scholarship.

Objective 2: To characterise audience listening behaviour: To obtain quantitative and qualitative data from film audiences regarding playback hardware, listening environments, and loudness preferences. This characterises the specific patterns of listening behaviour that must be accounted for in an adaptive design.

Objective 3: To identify professional constraints and opportunities: To gather data from professional dubbing mixers regarding current workflows, monitoring practices, and delivery strategies. This is so that the proposed solution remains compatible with the realities of professional production.

Objective 4: To derive explicit design requirements: To analyse the audience and industry datasets using statistical and interpretive methods. The objective is

to translate raw data into specific design requirements that link audience behaviour with professional practice.

Objective 5: To define the Adaptive Mixing Framework (AMPLE): To use these requirements to define a framework that operationalises playback context and user behaviour within a single technical workflow.

Objective 6: To validate the framework through empirical testing: To evaluate this framework through controlled listening experiments (prototype 1 and prototype 2). This assesses whether the framework supports clearer and more consistent listening experiences across different playback conditions,
providing empirical evidence in relation to the research questions.

1.5 Thesis structure

This thesis is organised to address the research questions in a progressive sequence, moving from problem definition through framework development to empirical evaluation.

Chapter 1: Introduction. Introduces the research problem by situating film sound practice within contemporary listening contexts. It defines the aims, research questions (RQ1–RQ3), and the scope of the study, outlining the disconnect between professional standards and consumer listening conditions.

Chapter 2: Literature Review. Reviews relevant literature on film sound production, mixing practices, delivery standards, and the psychology of the listening experience. It identifies the structural limitations in existing approaches that motivate the investigation of adaptive mixing.

Chapter 3: Audience Requirements Gathering. Addresses the first two research questions by reporting findings from a large-scale audience study (N=500). It examines listening environments, playback hardware, and perceived sound quality, deriving initial design considerations and user requirements for the proposed framework.

Chapter 4: Industry Constraints and Workflows. Complements the audience analysis through examination of professional mixing workflows. It extends the requirements for adaptation from the perspective of current production practice, identifying technical constraints that the framework must navigate.

Chapter 5: Framework Definition. Synthesises findings of Chapters 3 and 4 to address RQ3. It defines the object-based adaptive framework (AMPLE), detailing its design principles and the translation of user requirements into technical specifications.

Chapter 6: Prototype 1 – Evaluation of Listener Preference. Evaluates the framework through the first listening experiment. It examines object-based control under reference listening conditions, establishing a baseline for audience preferences and identifying the dialogue anchor behaviour observed throughout the thesis.

Chapter 7: Prototype 2 – Evaluation in Context. Extends the evaluation to simulated real-world listening conditions, through simulated commuting environments. It examines how the framework responds to environmental noise and playback constraints under changing listening conditions.

Chapter 8: General Discussion and Conclusion. Draws together findings from the empirical studies and prototype evaluations. It addresses the research questions, discusses contributions to knowledge, identifies limitations, and outlines directions for future research in adaptive audio.

1.6 Summary

This chapter has established the gap between professional film sound delivery practices and the listening conditions experienced by contemporary audiences. It has presented that existing standards and workflows provide limited support for variation in playback devices, environments, and listener behaviour, and that this gap motivates the need for adaptive approaches to film sound mixing. The research questions and objectives have been defined in response to this problem, framing the study as an investigation of audience experience, professional constraints, and their synthesis within an adaptive mixing framework. The structure of the thesis reflects this progression, moving from problem definition and requirements gathering through framework development to empirical evaluation.

Chapter 2: Literature review

This chapter contains a review and summary of film sound mixing technologies and practices, as well as listening experiences and user design approaches. The review begins with an overview of film sound and production stages, and film mixing inclusive of delivery, as well as feedback channels available to filmmakers and key stakeholders in the industry. The review continues with listeners' perspectives and how the listening experience is assessed, inclusive of the environment.

2.1 Film Sound: structural constraint and narrative priority

This section analyses the standard film sound workflow to identify the structural origins of the intelligibility gap. Rather than providing a historical overview of production techniques, this review focuses specifically on the hierarchical separation of sound elements, referred to as stems, and the environmental assumptions made during the re-recording process.

2.1.1 The Hierarchical Construction of the Soundtrack

Contemporary film sound is not a faithful recording of reality, but a constructed super-reality designed to support narrative immersion (Chion & Gorbman, 2009). This construction relies on the strict segregation of audio into three distinct narrative categories (dialogue, sound effects, and music), often referred to as the food groups of the mix (Purcell, 2007; D. Yewdall, 2012).

2.1.1.1 Dialogue

Dialogue is universally prioritised as the primary carrier of narrative information. The production mixer's sole objective is to capture clean speech (Holman, 2012), often at the expense of environmental context. If the production track is compromised by noise, it is replaced via automated dialogue replacement (ADR) to ensure pristine intelligibility.

Relevance to Research: This prioritization confirms that dialogue intelligibility is the industry's primary goal. The failure of this goal in consumer environments – as identified in Chapter 1– represents a failure of delivery, not intent.

2.1.1.2 Foley and Sound Effects

Because production microphones are typically directional – designed to reject background noise – the physical acoustic world is often synthesized artificially in post-production (B. Wright, 2014). This includes, Foley, human interactions like footsteps, and hard effects, like gunshots, or door slams.

Relevance to Research: This artificial reconstruction means that the background noise in a film is a discrete, controllable audio layer – a stem. It is not baked into the dialogue. This supports the technical feasibility of the adaptive framework (AMPLE) proposed in Chapter 5, which relies on the ability to independently attenuate these effects.

2.1.1.3 Music

Music functions as the emotional guide, often filling the spectral space surrounding the dialogue (Cohen, 2001).

Relevance to Research: Music frequently occupies the same spectral bandwidth as dialogue. In consumer environments where listeners cannot adjust this internal balance, heavy music scoring becomes a primary source of frequency masking, reinforcing the need for object-based separation at the playback endpoint.

2.1.2 The Re-Recording Stage: Mixing in a Vacuum

The discrete elements described above are combined during the re-recording stage. This process is guided by the spotting session, during which the director and supervising sound editor determine the narrative purpose of each sound cue (Beauchamp, 2005; D. Yewdall, 2007).

However, a critical limitation in this workflow is the listening environment itself. Mixing decisions are made in reference listening environments – acoustically treated stages with low noise floors (Kerins, 2011), typically at or below noise criterion 20 (NC-20) and high dynamic range capabilities (85dB SPL) (Prodec Group, 2026; SMPTE, 2010c).

The mixer balances the whisper of a character against a loud explosion based on the assumption that the audience can hear the full dynamic range. However, as noted in Chapter 3, modern audiences frequently consume content in high-noise environments, for example, transport where this dynamic range is physically impossible to perceive.

2.1.3 The Delivery Bottleneck

The standard workflow culminates in the creation of dialogue, music, and effects, referred to as DME stems, saved as separate tracks. However, for consumer delivery, these stems are typically summed into a single print master (LoBrutto, 1994).

This workflow highlights a paradox in current practice. The industry spends significant resources separating and controlling audio elements, often referred to as pre-dubs or stems, to create a perfect mix for the cinema. Yet, by baking these elements together for final delivery, the industry removes the flexibility required for the mix to survive on a mobile device.

Therefore, the solution to the Research Questions (RQ2, RQ3) does not require inventing a new production method. Instead, it requires extending the existing stem-based workflow to the consumer endpoint, allowing the re-balancing that currently happens on the dub stage to occur dynamically at the point of playback.

2.2 Mixing for film

Mixing sound for films - whether it is for cinema, television, or streaming - is an iterative process that involves several creative and technical decisions. In which re-recording mixers – synonymous with dubbing mixers – serve the director's creative intent by constructing a hyper-realistic and immersive sound experience for the audience. The term re-recording self-defines the process, i.e., taking something already recorded and distilling it down by the mixing processes to a more convenient representation from many audio tracks (Holman, 2012).

To create the film's final soundtrack, mixers combine the relevant stems – dialogue, music, and sound effects – for processing. These stems are pre-mixed from individual tracks. See Figure *2-1*.

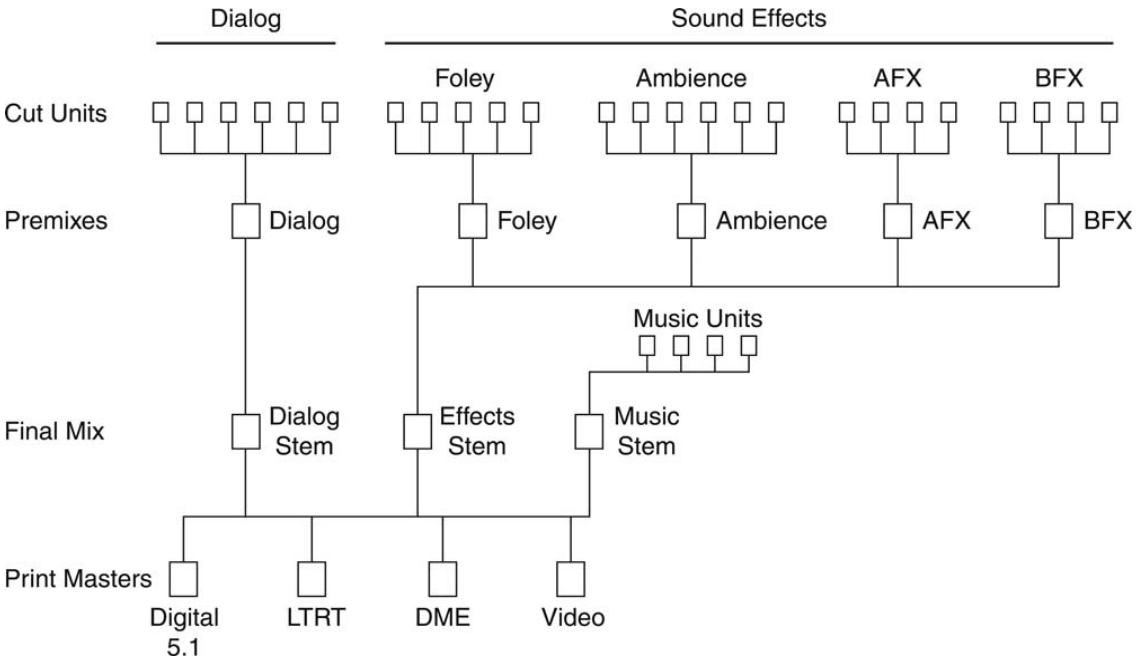

Figure 2-1 Typical signal flow and routing of premix stems in a film sound re-recording workflow (Holman, 2018a, 2018b).

Dubbing mixers continuously manipulate those stems to achieve the best sound quality and creative intent to support the narrative. This exercise is done on an acoustically treated space often called – a dub stage, that is arranged like a film theatre with one to three mixers, the lead mixer is often responsible for dialogue and possibly for music.

The stage is arranged like a film theatre with one to three mixers, where the lead mixer is often responsible for dialogue and possibly for music. The stage is arranged with the mixers having the best audience seats and working simultaneously on the program material, which enables them to listen to the film soundtrack under the same conditions as a member of the audience would do in a cinema (Enhanced Media, 2019; Holman, 2012). Figure *2-2* shows Warner's dubbing stage in London, UK.

Figure 2-2 Stage one - Warner Bros De Lane Lea – London (Warner Bros, 2020)

Film sound has evolved from its early inception in 1923 by DeForest (Adams, 2012; DeForest, 1923). Cinema sound is intended to be a fully calibrated medium which offers end-to-end control, which is provided by controlled acoustics, known speakers' positions inclusive of pre/post-production standards and common workflows (DCI, 2020c; SMPTE, 2010b, 2012). This is further driven by the rigorous quality assurance process that offers an ample level of consistency in a theatrical setting (Brock et al., 2009a, 2009b; Gedemer, 2013a, 2013b, 2015, 2016a, 2016b; Hill et al., 2015; Rumsey, 2015a).

Mixing for television and streaming is typically done in an acoustically treated studio which is almost always smaller than a film dubbing stage, much like cinema, various standards like the Advanced Television Systems Committee (ATSC) A/85 or European Broadcast Union (EBU) Tech 3344 have different recommended sound pressure levels for different sized rooms (ATSC, 2014; EBU-TECH 3344, 2016). The topic of which room we should be mixing television in is out of the scope of this study, nevertheless, the physical size of the mixing space has a significant impact on television and cinema mixes such as, imaging distortion especially in the low frequency range (Cox et al., 2004; Csadi et al., 2019; R. J. Hughes et al., 2016; Olive & Martens, 2007). Figure *2-3* shows Jumpbuck Film & TV studio in London, UK.

Figure 2-3 Jumpbuck Film & TV Studio- London *(Jumbuck Film & TV, 2020)*.

Unlike cinema, television and streaming do not offer the same level of end-to-end control, nor does it stand as a fully calibrated system. For television consumers, playback hardware varies from smartphone built-in speakers to customised home theatres in varying acoustic environments, for example, a quiet living room or a noisy train commute. Background sounds in a listening space and user-preferred playback levels also vary, such as watching the news relatively quietly compared to a football match, that could also vary depending on how many people are watching together.

To address the variance in consumer playback devices, content creators frequently utilize downmixing or fold-down techniques. These methods encode multichannel content to ensure backward compatibility across different native mixing formats, as a significant portion of the audience listens in stereo or mono, see Figure *2-4*.

This downmixing process occurs either at the transmission centre or is rendered dynamically at the consumer's endpoint by the playback hardware. Downmixing multichannel signals inadvertently causes an increase in loudness during specific program segments (S. Norcross & Lavoie, 2011), resulting in an output consistently higher than the original discrete mix. Comparable results have also been reported in similar studies (Riedmiller et al., 2003a).

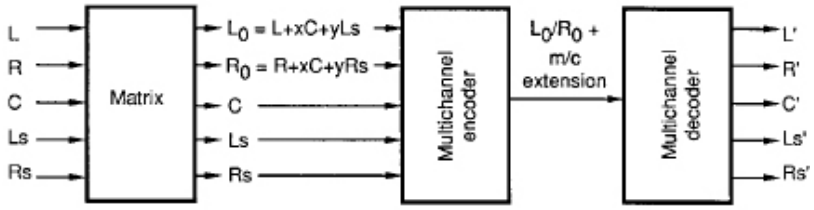

Figure 2-4 Compatibility matrixing of surround information for MPEG-2-BC (Rumsey, 2012).

To ensure some level translation for different playback systems and backward compatibility, some television mixers tend to monitor different speaker configurations. Mixers often switch between different monitoring modes, a few also sometimes print a stereo mix during the mastering process to listen to how the mix will translate on different reproduction configurations, and whether any adjustment is needed prior to delivery.

Compared to cinema, the time constraints for television and streaming mixing are significantly tighter. Mixing a single episode of a television show can take anywhere from a single day to a full week. Given these constraints, the majority of the schedule is dedicated to balancing production dialogue and automated dialogue replacement (ADR). Consequently, this accelerated post-production schedule often restricts mixers from thoroughly monitoring their work across different consumer speaker configurations (Enhanced Media, 2019).

Quality assurance is the final part of the delivery stage of television and streaming programmes. All programmes delivered to broadcasters must be fully editorially and technically checked and ready for transmission prior to delivery whether it is traditional television or streaming, the practice ensures consistent quality for sound and picture throughout with problems. Most quality control checks have a particular focus on dialogue to ensure that it is clear and

understandable at first-time viewing. Streaming services like Netflix publish specifications and best practices, an example of which can be seen in (Netflix, 2020b). Figure *2-5* shows broadcast loudness targets by region. Some industry professionals argue that the delivery specification for television or streaming could be restricting creativity, nevertheless, it is to preserve the best possible experience for television and streaming consumers, while protecting the creative freedom of the creators (Thornton, 2021b).

Delivery Spec	Territory	Target Loudness (LUFS)	Maximum True Peak	Gate Used
EBU R128	Europe	-23	-1	Yes
ATSC A/85	US & Canada	-24	-2	Yes (initially No)
TR-B32	Japan	-24	-1	Yes
Op-59	Australia	-24	-2	Yes

Figure 2-5 Delivery specifications by region (Thornton, 2021b).

Whether it is for cinema, television, or streaming, mixing sound for film is an exercise that requires technical and creative decisions to realise the director's artistic intent. The process is extensive and lengthy; some mixing processes and techniques are noted below.

2.2.1 Levels and Panning

Balancing the relative levels of individual elements is arguably the most critical task during the re-recording process. Utilizing a digital audio workstation (DAW) or mixing console, dubbing mixers adjust stems to ensure that narrative priorities – such as dialogue – remain prominent against music and sound effects (Holman, 2012; D. Yewdall, 2007). Panning is subsequently employed to place these sound sources across a set of output channels. In multi-channel cinema mixing, a dedicated centre channel is predominantly reserved for dialogue, physically anchoring speech to the screen.

However, in consumer stereo systems where no physical centre channel exists, a phantom centre is often reproduced. Studies have highlighted that phantom centres exhibit timbral differences compared to discrete centre channels, indicating that spatial downmixing inherently alters the perceived spectral balance of the dialogue (Rumsey, 2012).

2.2.2 Dynamic range and Gain structure

Dynamic range is the ratio between the loudest undistorted sound and the quietest discernible sound; it must be strictly managed to prevent unintentional masking between combined sound elements.

Mixers employ compressors, expanders, and limiters to manage the dynamic range, ensuring that transient peaks do not cause clipping and quieter details remain audible (Holman, 2012; D. Yewdall, 2007). However, these dynamic relationships are established and optimized exclusively for a calibrated dub stage with known speaker placements and a controlled noise floor (Rumsey, 2012;

Toole, 2015). When this dynamic range is reproduced in a completely different acoustic reality outside the studio – such as a noisy commute or an uncalibrated domestic space – quieter details are frequently masked by environmental noise, rendering the original dynamic intent ineffective for the listener (Lund, 2006a; Pearsons et al., 1977).

2.2.3 Frequency response processing

Alongside level control, manipulating the frequency response is essential for achieving sonic consistency across a film's soundtrack. Equalisation (EQ) and filtering are routinely utilized to carve out spectral space for competing elements, remove unwanted low-frequency rumble, and seamlessly match automated dialogue replacement (ADR) to original production recordings (Holman, 2010; Izhaki, 2012).

This process relies on the assumption of a standardized playback environment, specifically utilizing the X-curve frequency response calibration, typical of cinema auditoria. The X-curve dictates an electro-acoustic target response that remains relatively flat up to 2 kHz before applying a steady high-frequency roll-off – typically 3 dB per octave – to compensate for large room acoustics, reverberation buildup, and high-frequency screen attenuation (P. Newell, 2014b; SMPTE, 2010c).

Consequently, the delicate spectral balance achieved on the dub stage often translates poorly when played back through consumer devices, which typically lack this specific high-frequency roll-off and possess restricted overall frequency responses, or when masked by the low-frequency bias typical of transport noise.

2.2.4 Time-based effects

Time based effects like reverbs, delays and shifters have different utilisations during re-recording, while reverbs, like Altiverb are often used in recorded music, they are used to match ADR with production sound or trigger localisation cues (Audio Ease, 2025). Brief time delays, like EchoBoy, are used for timbral thickening or layering and longer ones result in progressively separate echoes (Sound Toys, 2025). Shifters are used for voices that are meant to be in a time range different from an actor’s original take, they are units that take the waveform of one take and shift extend or shrink it, examples of time based effects in film include: The Reluctant Dragon 1941 (Disney, 2020; IMDb, 2020) or EVE in the film WALL.E (Burt, 2016; IMDb, 2008; Pixar, 2008)

2.2.5 Delivery

Once re-recording is done, transferring, and preparing deliverables is next, this stage satisfies the business needs of distribution, promotion, localization, and archiving, subsequent processes are manufacturing ones, the task is often accomplished with the support and assistance of mix technicians and broadcast engineers. Films are progressively released into cinemas, then on Blu-ray and streaming services, on cable television services, and finally on Freeview TV, while this stage might be a technical stage primarily to ensure the mix meets the current standards, mixers often take it as a one last opportunity for a sanity check and do any intentional tweaks needed (Freeview, 2025).

Deliverables are the terms frequently used by cinema and television industries, note, that executives and content producers consider deliverables to be the assets that represent the value of the product; for this reason, formats, and backups to store said assets is arguably one of the most important tasks in the postproduction pipeline. Mixers are the ones that prepare separate deliverables that conform to the limits of every individual release format and since different playback hardware and mediums vary, each deliverable must be rendered with respect to the capability of the specific target medium.

For a multiplatform release, there are different technical standards and deliverables for each distribution channel, i.e., theatrical, cinema, consumer physical media such as Blu-ray, streaming or broadcast like Netflix and BBC. Nevertheless, all deliverables are focused on three main aspects: technical specifications such as, loudness, peaks, or file types, naming convention and file content, plus any other requirements that are needed for the purpose of distribution, marketing, localisation, or archiving, this can also differ per region or jurisdictions.

For cinema, standards and recommendations published by organisations like Digital Cinema Initiative, SMPTE and ITU, provide detailed guidance on, measurement, calibration, and acoustics inclusive of speaker layout and room volume, such standards ensure consistency of playback quality and drive the concept of end-to-end control embedded in the cinema medium. The standards apply for playback auditoria and dubbing theatres, in which the monitoring environment at the listening position is typically set to 85 dB/A, 82 dB/A in the surrounds for 5.1 and 7.1 formats (DCI, 2018a, 2020b, 2020a; ISO, 2016; ITU, 2012b; SMPTE, 2010a, 2010b, 2012).

A typical set of deliverables for a cinema mix will be a full theatrical full mix deliverable, theatrical mix minus narration, theatrical music and effects (M&E), associated stems or sub mix of music, dialogue, and sound effects, if the budget and time constraints allow then a nearfield of those will be also created, this list will need to be created for all required formats, i.e., Dolby Atmos, 5.1, 2.0.

For both film and television shows that have multilingual distribution, common deliverables are M&E; music and effects, with the dialogue supplied as a separate track only for translation and synchronization, due to the faster postproduction schedules in television productions, deliverables are sometimes

simplified, a show will have a primary master, original language comp and one secondary master, and an M&E for multilingual distribution (Holman, 2012).

Unlike cinema, loudness specifications and peak limitations are unique to streaming and broadcast platforms.

Material	Recommended Maximum Peaks
Uncompressed Music	**-3 dBTP**
Compressed Music (depending on degree of compression)	**-10 dBTP**
Heavy M & E (gunshots, warfare, aircraft, loud traffic, etc.)	**-3 dBTP**
Background M & E (office/street noise, light mood music etc.)	**-18 dBTP**

Figure 2-6 BBC true peak recommendations *(BBC & DPP, 2020)*

Figure *2-6* shows the recommended maximum true peaks for BBC asset delivery, the document specifies a target loudness of –23 LUFS with a tolerance level of ±1.0LU, Audio is to be measured over the full program according to European Broadcast Union (EBU) Tech 3343. The document is based on AS-11, the technical document published by the Digital Production Partnership (DPP), its members are the UK's biggest broadcast networks such as ITV, BBC, Sky and Channel 4 (EBU & Camerer, 2016; Phill Tudor et al., 2021).

Another example is Netflix which requires the programme to be at a target loudness of -27 LKFS with a tolerance level of +/- 2 LU measured over the full program according the (International Telecommunication Union (ITU) 1770-4, where as Apple TV+ requests that the average loudness of dialogue/speech of the master must be within the range of -31 LKFS to -10 LKFS, with a true peak of -6 dB and a target loudness of –24db, also measured according to ITU 1770-4. in addition, some deliverable documents call for volume reductions on the music stems relative to dialogue. Table *2-1* shows a breakdown of requirements for 3 different platforms (Apple Inc, 2021; ITU, 2015; Netflix, 2021).

Table 2-1 Target loudness by platform

	Target Loudness	Peak (True peak)	Standard measurement
Apple TV+	-24 LKFS	-6 dB	ITU 1770-4
Netflix	-27 LKFS	-2 dB	ITU 1770-4
BBC	–23 LUFS	-3 dB	EBU Tech 3343

A typical streaming/broadcast might require the following as deliverables: a full mix print, M&E mix with Foley and sound effects, associated stems or sub mixes of music, dialogue and sound effects and as mentioned previously, all the deliverables will be compliant with the loudness and peak specification for the playback platform, for example, a project will have a separate set of the previously mentioned deliverables for each mix format; Atmos, 5.1 and 2.0, plus an Audio Definition Model (ADM) or Dolby Atmos Master Format (DAMF) and occasionally a theatrical mix for award seasons or festivals. For cinema and TV, much like the creative aspect of sound mixing, there is an immense amount of technical detail to create a mix that will reach the audience successfully (Holman, 2012; Urban & Foglia, 2021).

2.2.6 Channels, objects, and Metadata

During the production of Fantasia, Disney was influenced by Korsakov's Flight of the Bumblebee and wanted to have it featured in the musical while playing in all parts of the theatre (Algar et al., 1940; Furmanek, 2021; Holman, 2012; Korsakov, 1900; Stough, 2015). At the time, single or monaural sources were used in theatres, such systems were ineffective, limited, and impaired by noise and amplitude distortion. This was especially apparent in symphonic music, and while dialogue and action benefited from the mono point source sound, music, and sound effects suffered from phase distortion (Holman, 1990, 2007, 2012).

To address this, RCA and Disney engineers developed a multichannel reproduction system dubbed, Fantasound for the animated feature, the sound was diffused throughout the cinema. Fantasound was excluded in later screenings, only to reappear with 'This is Cinerama', along with discrete 7 channel sound and the development of other surround formats took off at the time. Other than Fantasia, cinema often incorporated stereo reproduction, with dialogue panned to match the visuals, this was a lengthy process; this practice gradually faded in favour of mono centred dialogue plus stereo music and sound effects (Holman, 1990, 2007, 2012).

Development of the multichannel formats is attributed to the need for clarity, increased localisation, and enhanced involvement of the audience in cinema auditoria. Early recordings systems from the 1800s to early 1900s were single channel only or monophonic until the Paris exhibition of 1881, where Clement Ader's demonstrated early experiment of stereo reproduction, many years after; stereo reproduction became a commercial reality (Hertz, 1981; Rumsey, 2012).

To create superior spatialisation compared to 2.0 stereophony, ideas from the likes of Bell labs, Blumlein, Steinberg and Snow and others have driven surround sound layout and production techniques since the 1930s (Klipsch, 1964; O'Brien, 2018). More loudspeakers were added later, and professionals attempted to extend such spatialisation techniques to work for more speakers; however, they only considered pairs of channels or three at a time (Blumlein, 1931; Shankleman, 2008; Steinberg & Engineering, 1934; "Symposium on Auditory Perspective," 1964). For cinema applications with its wide screen, the need to cover a broad listening area and the large distance between left and right speakers dictated the use of three front channels; left, centre, right (Holman, 2007; Rumsey, 2012).

The three-channel configuration allowed for a wider front than (2.0) stereo with the centre channel focused the central image, anchoring the dialogue clearly mid screen. Unlike two-channel stereo, the centre channel does not suffer timbral modifications since it is a real source. To enable superior viewing or listening experience in cinema applications, and to provide enveloping effects, a four-channel system was introduced adding an additional effects channel or surround channel to the three front channels, Warner and 20th Century are credited with introducing the format in the 1950s (Gerzon, 1974; Rumsey, 2012).

The four-channel system covered a wide audience area by utilizing multiple surround loudspeakers in cinemas to create a distributed reproduction. However,

because these speakers were fed from a single surround channel, they offered the exact same signal, resulting in poor envelopment and spaciousness. Most psychoacoustic research suggests that human ears require decorrelated signals to perceive genuine spatial envelopment.

The successor 5.1 channel configuration addressed this limitation by allowing stereo effects and room ambience to accompany a primarily front-orientated sound stage similar to conventional 3-channel stereo with the rear channels for ambience and effects, In a 5.1 configuration; the '.1' element is a dedicated low frequency effects (LFE) channel or sub-bass channel with a limited bandwidth (D. Griesinger, 1997, 1999; D. H. Griesinger, 1998; Kendall, 1995; Rao, 2017).

Layout and configuration for the 5.1 surround system is specified in ITU-R BS.775, the document has recommendations on screen size and angles in which speakers are to be setup, shown in Figure *2-7*. The front ±30° is for 2.0 compatibility which results in a narrow front sound stage, to increase envelopment the surrounds are situated at ±110, multiple speakers are also allowed to be distributed evenly between ±60° and ±150° (ITU, 2012a).

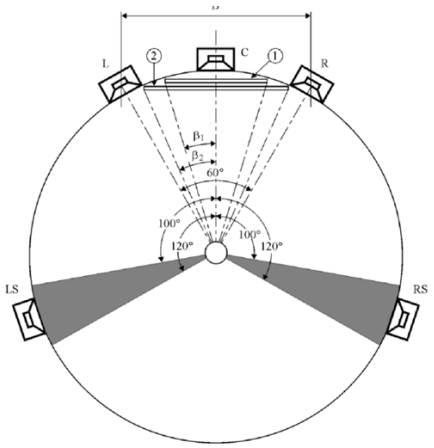

Figure 2-7 Reference loudspeaker arrangement (ITU, 2012a).

The '.1' in the format represents the low frequency channel with a sub that extends to a maximum of 120 Hz, its primary function is in sound-for- picture applications where sound effects and other noises are used often. Note that the LFE speaker for consumer systems is optional, thus the recommendation is that mixes should always be done with that in mind, and a programme should sound good even if the LFE channel is not reproduced (Dolby Labs, 2000; SMPTE, 1999; Woszczyk et al., 2005).

Although today the 5.1 surround sound format is widely used in cinema auditoria, note that the layout has the following limitations to consider; the front sound wall is considerably narrow due to a compromise for backwards compatibility with 2.0 – two-channel applications, similarly pan laws for music applications and multi microphone techniques are not optimised for three loudspeakers, consequently, the centre channel can be a challenge for music playback, in addition, the left and right surrounds` position or angle, results in a large hole in the sound image behind the listening position, the format also lacks 360 imaging capability (Robjohns, 2001a, 2001d, 2001c, 2001b, 2001e; Roginska & Geluso, 2017a; Rumsey, 2018b, 2012).

It was due to such limitations that other formats emerged, such as the 7.1-channel configuration; the format adds two further loudspeakers to the 5.1-channel configuration, the additional speakers are located at centre-left (CL) and centre-right (CR) to cover screen width in larger sized cinemas. Another variation of the format adds two loudspeakers used for back surround left, and back surround right, another notable format is Holman's 10.2 surround, it includes seven channels: left and right wide, left height, left, centre, right, left and right height, three surround channels: left, back, right surrounds, and two LFE channels: Left and Right. Proprietary systems like Lexicon's Logic 7 and Dolby EX have also been promoted, the latter included a centre rear channel to the standard 5.1-channel setup (Robjohns, 2001a, 2001d, 2001c, 2001b, 2001e; Roginska & Geluso, 2017a; Rumsey, 2018b, 2012).

2.2.6.1 Digital Delivery Overview

Digital delivery for surround in cinemas and at the consumers' end emerged in the 1990s with AC-3 (Bosi & Goldberg, 2003). This format introduced critical features including dialogue normalisation and dynamic range control, with the flexibility for a 2.0 channel version that could be rebuilt in the decoder. As the industry evolved, lossless formats with higher data rates became available, such as Dolby TrueHD supporting up to 16 channels. Other notable delivery formats include Digital Theatre Systems (DTS), Dolby Atmos, and MPEG with its most recent iteration MPEG-H (Bleidt et al., 2017; Dolby Atmos, 2015; DTS, 2022).

These formats can handle a broad number of surround channels, formats and fully immersive layouts, but most importantly, they introduced objects, a mechanism that fundamentally shifts audio production away from fixed loudspeaker layouts towards content that renders to whatever speakers are available at the reproduction end (Bosi et al., 1997, 1999; Bosi & Goldberg, 2003; Herre et al., 2015b, 2015a; Liu et al., 1998; Mesarovic et al., 1999; Murtaza et al., 2015; Roginska & Geluso, 2017b; Rumsey, 2012; M. Smyth, 1999; S. Smyth et al., 2011; S. M. F. Smyth et al., 1996; You et al., 1999).

The MPEG standard specifies the reference model of spatial audio object coding (SAOC), the technology can reconstruct, adjust, and render a multitude of audio objects based on a lesser number of transmitted channels and additional data. Unlike channels, audio objects represent individual signals that can be controlled independently of one another by metadata or users, objects can be rendered and placed at the chosen location at different levels (Herre et al., 2014, 2015b; Herre & Disch, 2007).

2.2.6.2 Object-Based Audio

Unlike channels, audio objects represent individual signals that can be controlled independently through metadata or user interaction. Objects can be rendered and placed at chosen locations at different levels, providing first-time flexibility in audio production and delivery. Audio objects are formally defined as audio elements with associated metadata that represent the creator's intent by specifying the translation from audio elements to loudspeaker signals (Roginska & Geluso, 2017b).

The MPEG standard specifies the reference model of spatial audio object coding (SAOC), a technology that can reconstruct, adjust, and render multiple audio

objects based on fewer transmitted channels plus additional data. SAOC achieves loudspeaker format agnosticism without discrete audio signal encoding. Instead, it encodes object level differences (OLD), inter-object cross coherences (IOC), and downmix channel level differences (DCLD) into a parameter bitstream, while supporting backwards compatibility (Bosi et al., 1997; ISO, 2018; Murtaza et al., 2015). A variation offering much lower bit rate coding is parametric audio coding, which encodes audio into core signals and a stream of parameters that define features needed to rebuild an approximation of the original (Breebaart et al., 2007).

2.2.6.3 Spatial Rendering - Perceptual Considerations

A fundamental consideration in object audio is the distinction between environmental and observer reference frames (Proulx et al., 2016; Rabini et al., 2019). In post-production, mixers adopt an environmental or allocentric approach, with panning tools laid out accordingly -Figure *2-8*. This allocentric referencing in 3-dimensional space allows audio objects to maintain the same relative position across any screen size, enabling optimal reproduction in various room sizes and shapes (Roginska & Geluso, 2017b). In cinema auditoria, loudspeaker positions relative to the screen, audience, and room provide reference points for rendering algorithms and authoring tools to map audio signals to loudspeakers. Consequently, auditory events are perceived at intended locations in 3D space, with coordinate metadata generated using the same reference (Begault, 2004; Rumsey, 2006, 2012).

Figure 2-8 panning tool - Pro Tools (Komar, 2017)

Current rendering algorithms use higher-level perceptual cues, such as interaural time/level differences, to render objects through amplitude panning. The core idea is to represent the desired source direction vector as a linear combination of loudspeaker direction vectors and solve for the loudspeaker gain vector,

applying these gains produces a virtual sound source between speakers (Dickins et al., 1999; Lossius et al., 2009; Pulkki, 1997).

Most renderers compute wide-band gain values, though research suggests additional benefits of frequency-dependent panning gains, which reduce timbral coloration and improve localization stability by adapting to the frequency-specific diffraction caused by the listener's head (Laitinen et al., 2014; Pulkki, 1997; Pulkki et al., 1999).

Audience perception of object position is affected by timbral fidelity, spatial accuracy, and smoothness - factors that represent design trade-offs in rendering algorithms. Studies suggest that a listener's distance from room centre dictates optimal rendering strategies: dual-balance panning performs well at centre, while direction-based panning yields better results at far distances (Kostadinov et al., 2010a, 2010b; Tsingos et al., 2014).

2.2.6.4 Production Metadata

Re-recording mixers face significant challenges in maintaining definition and spatial isolation among key mix elements. Current mixing formats have introduced additional object controls including speaker exclusion zones and snap-to-speaker mode (Robinson, Mehta, and Tsingos 2012; Robinson and Tsingos 2014) .

Speaker exclusion allows engineers to mask certain zones, ensuring specific objects won't render in those areas. The metadata generated during mixing dynamically reconfigures how the object is handled by the renderer. Snap-to-speaker mode renders objects wholly from the single nearest loudspeaker without phantom panning.

In television and streaming productions, final mix elements are often sent as audio objects to be rendered at the receiver end, conforming to the final speaker configuration (Mann et al. 2013) with metadata enabling personalisation and optimum sound (Riedmiller et al., 2015). Content creators can define different configurations and sound experiences, for example, in sports where audiences can select commentator-free mixes.

Metadata also defines object placements and gains for variable speaker layouts. Categorisation tags (music, dialogue) enable playback hardware to process gain changes relative to ambient noise in the playback environment. Metadata can assign unique programme IDs for further personalisation, allowing recall of settings used on similar programmes or genres. While audiences may desire unrestricted control, content creators can define strict rules for object rendering, including overall mix rules or inter-object mutual exclusion, preventing problematic combinations like simultaneous bilingual dialogue objects (Roginska & Geluso, 2017b).

2.2.6.5 Managing Complexity and Delivery Efficiency

For complete end-to-end object-based adoption in the film industry, managing complexity for efficient delivery, rendering, and post-processing is crucial. Some current strategies prioritise primary objects and discard lesser ones and/or reduce spatial complexity – like MPEG I Layer 3 (MP3) coding strategies (Kelly and Tew 2002; Painter and Spanias 2000; Roginska and Geluso 2017b; Tsingos

and Lemaitre 2005). This approach requires assessing the relative importance of each source while dynamically adapting content, using metrics like those proposed by (Kayser et al., 2005).

Note that object audio mono coding lacks direct backwards compatibility, requiring greater capacity than 5.1 or 7.1 delivery because full object decoding and re-rendering is needed for legacy channel-based layouts. Workflows like independent and core object coding overcome these drawbacks, while approaches like parametric and joint coding address artefacts from dynamic object prioritising (Herre & Disch, 2007; Riedmiller et al., 2015; Tsingos et al., 2004).

2.2.6.6 Implementation Benefits and Trade-offs

Object audio provides flexible and scalable content delivery mechanisms compared to traditional channel-based playback. For Cinema, it offers spatial coherence and adaptability to different target formats, streamlining high-resolution height-inclusive audio delivery for producers and creating efficient, less time-consuming mastering processes for engineers (Robinson & Tsingos, 2014; Roginska & Geluso, 2017a; Rumsey, 2015b; Tsingos et al., 2014).

At the delivery or playout stage, object-based audio retains both spatial high resolution and artistic intent. However, its effectiveness may be constrained in non-deterministic or overly fixed playback environments, particularly during reproduction. It offers enhanced immersion through added height perception, flexibility across speaker layouts and acoustic settings, as well as personalisation, interoperability, and adaptability to diverse playback hardware (Robinson & Tsingos, 2014; Roginska & Geluso, 2017a; Rumsey, 2015b; Tsingos et al., 2014).

For home audiences, studies suggest object-based presentation can be preferred to channel-based virtualisation, driven by a perceptual preference for the improved spatial stability and envelopment it offers, even if the listener is unaware of the underlying technology. Additionally, object audio enables efficient transmission, requiring fewer audio tracks for playback with relatively small file sizes compared to traditional formats (Robinson & Tsingos, 2014; Roginska & Geluso, 2017b).

2.3 Workflow and Creative Intent

Because directors and producers often need to showcase unfinished films to studio executives, potential distributors, or test audiences for feedback before release, they are generally reluctant to present these works using only the unedited production sound. As a result, it is common for the producer to contact the supervising sound editor to request temporary sound elements -such as punches, body falls, or furniture breaking -to be added for preview purposes, commonly known as temp dubs.

While temp dubs can be helpful for the sound team to begin exploring conceptual ideas, they are also a source of tension. On one hand, they offer an early opportunity to experiment with sound design; on the other, directors and producers often become overly attached to these temporary tracks after hearing them repeatedly in the editing room over several weeks. This overfamiliarity can

ultimately stifle the creative process for the sound team, whose final work may be constrained by these placeholders; Compounding this issue, test screenings are becoming more common, and the associated production costs are steadily rising (Sadoff, 2006; Szaniawski, 2020; D. Yewdall, 2012). This reliance on temporary mixes highlights a critical rigidity in the current workflow: the locked nature of the mix often prioritises the producer's familiarity over the end-user's acoustic reality, a constraint that adaptive mixing seeks to address.

2.3.1 Predubs to final Mix

The predub process is where the session is distilled from hundreds of prepared sound elements down into more manageable groupings for the dubbing mixer and while budgets and schedule vary, the average is often four weeks excluding preparing deliverables with a conservative number of fixes.

Producers or directors often discuss schedule with the supervising sound editor (SSE) after the film postproduction has been planned and budgeted for. The SSE's approach then becomes how to best utilise the schedule to meet a certain price point or budget, they allocate the larger portion of the available schedule for predubs to go through the raw material, which will result in smoother and quicker a final mix session (Johnsen, 2018; Wells, 2019; D. L. Yewdall, 2012).

A low budget motion picture can be a challenge, especially at the mixing stage; the collaboration and interdependency of the dubbing mixer and the sound editors for; dialogue, Foley, sound effects is magnified when there is no room to waste budget. A common approach for sound designers/editors to become mixers on such projects, which is also attributed to the continuity of the creative vision, as well as offering a more fluid and seamless transition from the spotting notes to the final mix stems. On completion of the final mix, the three basic elements of the mix – dialog, music, and sound effects – are on their own separate stems or files. When these three files are played together in sync and are level matched, the combined playback produces a single continuous soundtrack. For this research, this standard industry practice of segregating stems (DME) confirms that the technical architecture required for object-based delivery is already inherent in the production workflow, even if currently discarded at the final delivery stage. At this point, the supervising sound editor runs a continuity playback of the entire film with the final mix to take notes and make changes if necessary (Holman, 2012; D. L. Yewdall, 2012).

There is a considerable high correlation between temporariness and creativity, a film organisation, being a temporary creative one, exists to develop an idea and once the task is done; the organisation members disband (Morley, 1977). For mixers, producers, and directors every project requires some level of experimentation and creativity. While location shooting is often the first element that comes to mind when one mentions film production, the reality is that the majority of feature films are constructed in discrete post-production stages

Every stage of film production as previously discussed in 2.1 involves a group of professionals with a specific set of skills that work collaboratively to perform a specific function, such stages are momentary organisations the cooperate and disperse. The director's job in this organisation is to manage and inspire those

groups and their creative work within time and budget, essentially being the creative process project manager (Bordwell, 1991; Morley, 1977).

At the mixing stage, the workflow will typically involve coordinating with the supervising sound editor to discuss and decide the particular needs and an appropriate number of tracks for the session and often times left for the mixers to decide. Note that supervising sound editors often come onto the stage with a precise design and vision they already have been developing, from the Foley artist and sound designer on up through their sound editorial staff (D. Yewdall, 2012).

Tasks in a session will be split between different mixers, one for dialogue and music whilst another mixes sound effects. Most mixers will prefer to work in reels, dividing the film mixing session into twenty-minute chunks, to make the mix more manageable especially in difficult to meet deadlines. This in turn makes it easier to send reels to the director for review and feedback in case they are not present in the dubbing stage, whether the director is present or not, mixers' job is ultimately to apply their skills and knowledge to present the director's creative intent in the best way it could possibly be (Baker, 2019; Kiser, 2020; Riehle, 2007, 2009; Screen Skills, 2020; Siebum, 2018a, 2018b).

Genres in film mixing are often approached differently, comedies typically have a focus on dialogue with subtle effects occasionally applied on jokes, and sometimes hyper-realistic sound effects are utilised for key moments. A similar approach could be observed in television sitcoms, where some episodes could have no music except for the titles (Evans & Hayward, 2016; Lambert, 2012). This is due to the audience focus on jokes which means mostly that the anchor point is dialogue and Foley to a large extent. Conversely, Horror or Sci-fi, sound design, music and providing contrast takes priority. For example: big scares preceded by created silence, is a common approach used by mixers to compliment the narrative. In such genres, Foley is often used to set the tone of a scene, such as: keys jingling or heavy footsteps to provide contrast (Crowley & Tibbo, 2015).

Mixers will often approach a dubbing session from the ground up and view it as building something out of nothing. In pre-dubbing, mixers distil the hundreds of prepared sound elements into more manageable groupings. For example, elements, such as dialog, ADR, Walla Group or loop group – a team of voice

actors in post-production for film, TV, or games who create background crowd murmur, are put together. Figure *2-9* shows an example of a mixing flowchart.

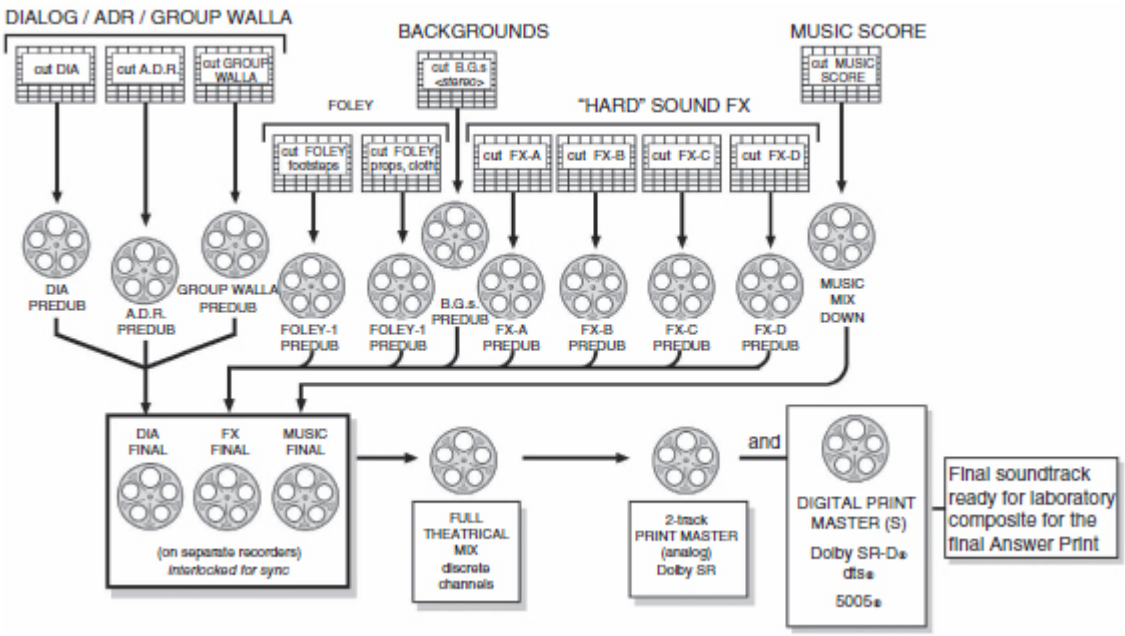

Figure 2-9 -Re-recording mixing flow chart (D. Yewdall, 2012).

In some films, music soundtrack is intense; it is a challenge to get the dialogue stem to sit well in the mix where it is audible and intelligible and does not bury everything else. As mentioned in 2.2, several tools could be creatively applied to position the music score and modify depth such as dynamic EQ, sometimes mixers will use the surround speakers to provide contrast and extend dimension and often change spatial positions of the ambience tracks or music score to guide the audience's focus on-screen (Nair, 2015a, 2015b). Note that creativity in the studio is welcome from mixers, however, directors and producers have a vision that dubbing mixers are there to fulfil (Pereira 2018; Siebum 2018a, 2018b).

While the specific value for an audience might vary from case to case, the cinema experience is primarily driven by the film makers' creative intent. As a story telling medium, film could be treated as an exercise of the creative process by both the filmmaker, inclusive of their entire crew, and an audience, since it provokes and enhances parts of the mind that make sense of the raw input of their senses (Conrad, 2016).

2.3.2 Dialogue

Dialogue editors take ownership of everything that has been recorded during production or on set, they take a draft finished film from the video editor, arrange edited sound, and organise them. This could be anything going through additional outtakes, committing quieter passages, compelling breaths, and less obvious lip smacks; to completely replacing washed out takes with clean close-ups. This practice enhances scenes, making them effectively deeper and boosts clarity for the narrative.

Note that the mechanisms of filmmaking, from camera movements, rustling crew, set noise, or otherwise removed by the dialogue editor. They assist in presenting

the story and its characters in the best possible light free of what is behind the camera, similarly they are also involved in the automated dialogue replacement (ADR) process, replacing problematic on set takes and strengthen performances (Kempton, 2004; Purcell, 2007) .

Dialogue editing is done on a digital audio workstation (DAW), with a monitoring environment sensibly resembling the mixing room, thus evading surprises during mixing, some editors like to work on headphones to get more detail before checking their work on loudspeakers (Purcell, 2013a). As technology improved and got cheaper over the years, budgets degenerated, it is argued that this triggered low quality production sound as dialogue editors are increasingly able to save unusable takes.

Editors now can efficiently arrange a scene, eliminate transient noises, to clean and modify unsuitable room tones, and to easily fix camera angle transitions, however, technology has not resolved all production problems, this could either enhance creativity with the director or evidence to be troublesome it with an indecisive one (Purcell, 2013b). A dialogue editor could spend an average of 250 hours editing, inclusive of ADR, it is common that the film sequence changes long after it's supposedly been locked, picture editors often produce a report from their picture workstation, dialogue editors use a conformation application which compares lists from two sessions and presents a new document with the changes (Jaeckle, 2013; Kempton, 2004; Purcell, 2007, 2013c).

To prepare for the mix, dialogue editors typically provide only those tracks that are useful for the re-recording mixer, they often coordinate with the mixer beforehand to arrange and prepare their session. At this point, they check for sync and anything that might have been missed. At the dialogue premix phase, mixers and dialogue editors work together to accomplish the following: A smooth flow of room tone from one scene to another, control any bad sounds such as sibilance; unless left deliberately as a character choice, eliminate excess ambient noise and add focus and perspective to scenes, as well as matching ADR scenes.

Note that mixers run a complicated room, and dialogue editors spend several days together with the mixer to realize how to get the best dialogue mix possible, they apply their knowledge and skill as well as their creativity to get the most out of the dialogue premix phase. While directors might claim they want to attend the dialogue premix, most of them end up staying away, they only come onto the stage only when a reel – twenty minutes chunk is finished.

Directors will take notes with the dialogue editors during playback and often attach timecode – a standardized sequence of numbers (HH:MM:SS:FF) added to audio and video to uniquely identify each frame – to their notes, once the notes are reviewed, editors will work with the mixer to address criticisms in the final mix, unless something requires going back to the premix (Jaeckle, 2013; Purcell, 2007, 2013c; Waves, 2010).

Dialogue is arguably the foundation of film soundtrack – at the final mix phase, the dialogue track is put together with music and effects. Ultimately, the final mix establishes the overarching acoustic balance, dynamic hierarchy, and spatial

imaging of the soundtrack (Purcell, 2007, 2013c; Waves, 2010). However, because this balance is calibrated within a pristine acoustic predictable environment, the – dialogue is king – philosophy often fails to translate to consumer devices. This disconnect between the studio's intent, and the playback reality is the primary motivator for the intelligibility adjustments proposed in the adaptive framework.

2.3.3 Foley

Films are often considered to be mainly a visual medium, however, without sound effects films would fall short. Foley adds a layer of depth and immersion to audiences in a film among other things; from the sound of footsteps creeping closer, to the rustling of clothing as characters move. Foley is performed in sync by artists during postproduction, often replacing sounds recorded on set, most films use a mix of on-set recordings and Foley effects. Foley artists recreate fully immersive sounds often using unexpected and creative ways, it is best practice for them to break down scenes into individual sound events, then perform them and record them in real-time with the film. Some sounds are conventional, like front door keys, for which, the original object is accessible. Some are complicated and are created in an unusual manner like searing meat in a pan to make rain sound (B. Hughes & Wakefield, 2015; MasterClass, 2021a).

Both the artist and Foley mixer partner to create outstanding Foley track for a film, often they have a similar philosophy about the work, although this is not always the case. It is the artist and performer of the Foley, that determines where it will be done and who the mixer will be. The Foley mixer records the sound the artist makes, they mostly prefer a Foley stage with very little reflection, often using a more classical approach by mixing the Foley flat which makes matching the sound easier for dubbing mixers at the final stage. Experienced Foley mixers clearly distinguish between an inadequate recording because of the equipment versus poor artists performance, communicating such reasons is essential in the session. Foley mixers are ultimately responsible for the quality of the recording which means working with the room's sonic character and addressing issues during a Foley recording session (Ament, 2014). In the context of mobile consumption (RQ2), these subtle Foley textures could be the first elements to be masked by environmental noise. This suggests that an adaptive framework requires specific logic to either boost these cues or deprioritize them in favour of dialogue clarity.

Like picture editors, Foley mixers must not overcut the Foley and risk sterilising an artists' character performance, since Foley is often considered a type of aural extension of the actors' intent. Once the Foley has been cued, performed, and edited, it goes to the dubbing stage for the predubs to final mix. Traditionally re-recording mixers arrange Foley into six tracks of footsteps and six tracks of props, often done after ADR, and sound effects, for tighter schedules, it has become common for one stage to be predubbing dialogue while another might be predubbing Foley with independent films having one person dub an entire project to cope with budget and time constraints.

During Foley predubs, mixers often work with Foley sounds against the dialogue tracks, or it might be mixed with sound effects, due to time constraints sometimes the dubbing mixer does not always get as much as they would like, however some directors will negotiate to get the financing necessary for postproduction sound, so they have the time to get the quality they feel the professionals need to do the best sound possible for the film (Ament, 2014; B. Hughes & Wakefield, 2015; Klyce & Fincher, 2021; Pond, 2012).

When predubs are complete, re-recording starts, Foley then becomes less of a focus for the mixer as an independent element but rather how it complements the dialogue and the actors' performance. Nevertheless, ample room is still available to make changes in the Foley if needed, the director might decide that they want to hear something different in a scene once music is added. Dubbing mixers approach Foley often as they approach dialogue, they dub Foley using the same setting that they use for the dialogue to recreate an identical sonic sense of space that the actors are in, for mixers, Foley is about perspective and adding live to dialogue as it extends the actors' aural performance (Ament, 2014).

2.3.4 Sound effects

Often, the sound design team has an extensive workload, they layer effects from their personal or commercial libraries to create an immersive experience in accordance with the director's creative intent. Sound effects come in different categories, spot or hard effects cover auditory events such as doors, vehicles, and punches, and could be used to replace or augment effects recorded on set. Background effects or backgrounds are often described as enveloping effects that widen and surround audience, they are commonly extended, steady, and looping sounds that give audience a different perception of what is on screen or the setting in which a particular scene is happening. Designed effects are otherworldly sounds, or sounds that are difficult to record, or ones that must be created from scratch such as monster growls, meteor strikes, and spaceships (344 Audio, 2021).

Spotting sessions are essential for a well-produced final soundtrack, they are an opportunity for the director to communicate the desired style and scope of work in a precise scene-by-scene discussion with the sound effects department. At this point the director clearly articulates their intent and desire. Furthermore, the process is an opportunity for the sound effects team to share thoughts, ideas and collaborate. It is common practice for the condition of the production dialogue track to be determined prior to the spotting sessions, this enables the supervising sound editor to understand the scope of works, and budget. A spotting session often requires certain delicacy when discussing problematic topics, such as inadequate acting performance or picture editing, or poor direction, the session commands great observation skills from the sound effects team to accurately recall and describe what they see (Baxter, 2007; Camerer, 2001; Viers, 2008; D. L. Yewdall, 2012).

During the predubbing process, the effects mixer typically operates alongside the head dubbing mixer as a unified team. This collaborative approach allows them to systematically manage the vast number of auditory elements and stem routings before advancing to the final mix session (D. L. Yewdall, 2012).

2.3.5 Music

A film score or soundtrack is original music written specifically to accompany a motion picture; it usually comprises several pieces or cues which could be orchestral, choral, or other instrumental ensembles. Cues are often synchronised or inserted at specific points during a motion picture to enhance drama and emotional impact (Savage, 2008). Often, composers consistently collaborate with a director on many productions, for example, John Williams with Steven Spielberg and George Lucas, or Tim Burton with Danny Elfman (Connal, 2009; McBride, 2010; White, 2000).

At early cuts of a film, temporary scores could be used as a guide to what the filmmakers' creative intent is in terms of tone, tempo, and intensity for the final soundtrack, a prominent example of this approach is, *A Space Odyssey* (Kubrick, 1998), the director used pieces of classical music as they worked better than Alex North's original score pieces could (Henderson, 2009).

The use of temporary music – referred to as temp in the studio – requires licensing rights with sometimes excessive costs if it remains when the film gets released. More importantly, directors could acquire an attachment to the temporary music as they continuously listen to it with the motion picture, even some well-established composers could struggle to counter a filmmakers' love for a temporary score, this is termed temp love, the term is not exclusive for attachment to the music track; it occurs with sound effects too (MasterClass, 2021b).

During mixing, music editors often finish all the music editing and track preparation following the recording session with each cue grouped together and labelled as needed. Some music editors send a cue sheet for the dubbing mixer to follow so they understand which tracks have music in them and when they will play. It is common for the post-production producer to direct the dubbing session with the director.

Once dialogue, Foley, and effects are examined and levelled in a scene, music is placed across the digital audio workstation (DAW) session, to help incite emotion or the action where needed. When there are a lot of sound elements going on in a scene, and with all three elements (dialogue, music, sound effects) fighting for attention, mixers will repeatedly go over that scene until all the levels achieve that perfect, delicate balance where everything is in its right place in the sound field and all the sounds complement each other. Yet this delicate balance creates a rigid hierarchy that frequently collapses in non-ideal listening environments. The inability of the end-user to re-negotiate this balance – specifically the music-to-dialogue ratio – constitutes a central limitation in current delivery standards that this thesis aims to resolve.

A final playback and fixes session is scheduled, where the director, producer and often music, effects and dialogue editors are present taking notes regarding what works and what does not, such notes are subjective sometimes. The postproduction process could be a challenging environment for music composers; dialogue is the anchor element, and sometimes great instrumental solos could be partially or completely masked by a loud sound effect such as an

explosion or otherwise. For this reason, composers often send their apologies and appoint the music editor, to protect the music and yet make sure it is in service to the narrative (Ledesma, 2011b, 2011a).

The relationship nuances between a music editor and a composer, is like that of the director and film editor, where the director envisions how the story is going to be, and the editor assembles it, an arguably more important relationship is that of the composer and director. Discussions that happen during demo and spotting sessions, helps in accomplishing everything that the director wants in a successful soundtrack. It is common practice following the spotting session, for the music editor to go through the picture and make detailed notes for the composer to assign a specific cue number to each sequence (Zottmann, 2018).

The nature of this workflow, where sound elements and stems are maintained separately until the final stage of delivery, indicates that the underlying architecture to facilitate object-based adaptive audio is already present within standard practice. This suggests that implementing user-centric adaptation at the endpoint would not necessarily require major changes to the core mixing process, but rather a shift in how delivery standards package and transmit these existing stems.

2.4 Feedback channels and Quality Control

Audience feedback can provide powerful insight for content creators; there are ways that afford opportunities for audience feedback and participation that could influence the creation and development of a motion picture. Access to filmmakers permitted by new and old media, gives audience the power to shape the film industry.

2.4.1 Audience Feedback Tools

Reviews of films on dedicated sites like IMDb, Metacritic, and Rotten Tomatoes (IMDb, 2022; Rotten Tomatoes, 2022) in addition to social media, can act as an irreplaceable analytics tool for content creators as it gives them an almost immediate audience response feedback loop. Social media offers consumers the power to pass judgement following the release. On occasion audiences would pass judgment following the release trailers, or leaked teasers and clips as well as exclusive screenings before a film's full public release.

Similarly, user comments as well as posts from news sites, forums and social media platforms can be aggregated through marketing tools, such as, SEM software marketing, which combines keyword research, and online ranking data, among other analytics (SEM, 2023). This can provide valuable insights about the audience response towards the content (Harris Interactive, 2021; Meier, 2018; Recco, 2013; Seavers, 2019; Stream Circle, 2016).

Website reviews typically analyse plots and characters; they allow readers to know and understand the whole narrative of the film. Reviews often provide a plot description, with a focus on the film's artistic merits (Bordwell, 1991; Griff, 2014; Laique, 2022). Commercial studio producers and executives arrange test screenings or production screenings; Harold Lloyd and Hal Roach are often cited as the pioneers of this approach as noted in (T. Dirks, 1996).

Screenings are for journalists, publicists, and the film cast, as well as critics. During such screenings, the selected audience sample –who get invited to attend– provide feedback to the film producers prior to its completion using surveys, email and crowd response. This allows further refinements before the film is finalised and delivered for distribution, such test screenings are conducted in person at a cinema auditorium or virtually using dedicated secure services like iScreeningRoom or Frame.io (Frame.io, 2023; iScreeningRoom, 2023). Unlike legacy test screening, where feedback is collected using rating cards, email communications or observing audience applause and response; virtual screenings offer content producers a streamlined platform to collect data and response from viewing using survey and feedback tools as well as other analytics.

The screening process is sometimes negatively looked upon by filmmakers, as it is claimed to reduce their creative efforts to being a product that needs development in the market; such practice is increasingly widespread outside Hollywood (Griff, 2014; Harris, 2013; ICO, 2021; Meyncke, 2010). Independent productions do the same before the official release date, coordination for this is often done by public relations' (PR) staff, and their network of media outlets as well as industry support personnel. Feedback is often collected by PR and marketing staff and communicated with the creators to measure audience reaction and their response. Surveys and questionnaires are also a common way to collect audience feedback, as it allows viewers to suggest changes to the content (Griff, 2014; Harris, 2013; ICO, 2021; Meyncke, 2010).

Typically, a market research company like Nielsen's National Research Group (NRG) (Bond, 2015) recruit viewers from their database or Internet film sites. Following the screening, audience complete a questionnaire covering key elements of the film, for example, characters, plot, and whether they would recommend it. Consequently, a focus group is selected from the larger audience sample, to further explore their response, with key stakeholders; producers, director, and distributor, often observing the group's discussion.

This practice primes slight tension between creatives and executives in the film industry; Hollywood studios and other investors find test screening invaluable and offer an opening for corrections missed in production or in post, whereas creatives observe this exercise as dumbing down practice to appeal to mass market at the expense of a creative vision. This is discussed in (Miller et al., 2005); Miller argues that such distinctions are unproductive although preview screenings have proven invaluable to major studios; there remains a stigma associated with Hollywood-style test screenings, focus groups and other commercial tools (Barnwell, 2019; Darby, 2009; Griff, 2014; Guback, 1969; Gwynn, 2017; Miller et al., 2005).

At this stage of filmmaking, the production studio exercises great control over the film as it becomes a creative asset; investors and funders are able to view the production and screenings are then held to ultimately decide the fate of the film; this could be replacing the score, changing composers, adding special effects or a new ending (Armstrong, 2008; Broxton, 2004; Christopher, 1981; Clemmensen, 2004; de Winter, 2008; Ebert, 2003; Glen & Hearn, 2001;

Goldwasser, 2004; Jones & Jolliffe, 2006; Lawrence, 2007; Maher, 2010; Müller & Klemens, 2002; Pfeiffer & Lewis, 2002; Smith, 2006; West & Rovin, 1994).

Nevertheless, the director often retains the right to edit the film and submit his version i.e., the director's cut (DGA, 2017, 2020). Consequently, the producer has the right to further edit the film and create the final version or the final cut, this is often driven by feedback from early preview and test screenings, and market research data, which are often cited as key indicators or drivers that impact the film studios' decision-making process. They often determine the film final form and its distribution as well as exhibition channels and territories (Brook, 2014; Harris, 2013; Lambie, 2014).

Film festivals are another invaluable feedback and audience communication tool for filmmakers, they are utilised as means of promotion locally, and regionally too. Film festivals are often characterised by their rich and diverse content; they are a valuable platform for directors to introduce and demonstrate their work to a wide demographic and a direct communication tool with their audience. Festivals are often publicised by experienced directors as the ultimate test screening; veteran filmmakers actively attend and engage with audience to get feedback in the same way commercial film studios test screen their films with industry personnel and publicists (Grove, 2019; Powers, 2020). In countries with a film industry tradition, festivals have been present for the last six or seven decades or more; globally it has been estimated that more than 3500 film festivals exist, the International Federation of Film Producers Associations (FIAPF) classifies and accredits festivals globally (Pedersen & Mazza, 2011).

It is common for festivals to engage with film production and distribution, some festivals have created their own distribution labels, while others have teamed up with TV channels or with specific streaming platforms. This means that they are not just an exhibition medium, but a participant in the creative cycle, inclusive of production and networking. They also provide a forum space for interpersonal encounters, negotiations between companies and creatives and pitching sessions, as well as development fund awards; festivals have become a platform for almost immediate feedback for creators, offering a platform to introduce their work and discuss topics shown in the film as well as the filmmaking process (Bershen, 2010; Elsaesser, 2019; Iordanova, 2015; Iordanova & Rhyne, 2009; Pedersen & Mazza, 2011; Screen Daily, 2016; Teo, 2009).

The feedback and approval processes in film production are heavily influenced by marketing data, primarily due to the significant financial investment and tight production schedules involved. However, the sub-phases of post-production - such as sound mixing, editing, and visual effects -remain crucial. Changes during this stage are often heavily guided by studio executives. As business-driven professionals, they rely on test screenings, audience surveys, and publicity strategies to predict and maximize a film's financial success, particularly its box office performance both domestically and internationally. This type of data is publicly available through sources such as ComScore and Box Office Mojo by IMDb Pro (Comscore, 2023; Dean, 2005; IMDb Pro, 2023). However, a critical limitation of these feedback mechanisms is their focus on narrative satisfaction rather than technical intelligibility. While test screenings reveal if an audience liked the sound, they rarely capture data on how that sound translated to specific

devices (e.g., tablets or smartphones). Consequently, the industry lacks a feedback loop for the actual listening experience outside the cinema, leaving a blind spot that this research aims to illuminate.

2.4.2 Quality control

Quality control (QC) is arguably one of the most important parts of the post-production workflow whether it is a theatrical, television or a streaming release, quality control ensures that a film meets the specifications and requirements of broadcasters, distribution channels, or film festivals (DPP, 2015; Google, 2020; Netflix, 2020a; Ross, 2018). For cinema, the digital cinema initiative (DCI) has released documentation that defines technical specifications and requirements for the mastering, distribution, and theatrical playback of digital cinema content, note that technical acceptance is a material condition of nearly all acquisition and distribution agreements. This means a filmmaker/studio will almost certainly be required to guarantee that the film is free of QC issues. (DCI, 2020c, 2020a, 2020b).

Table 2-2 QC checks example

Video QC	Conversion or compression artifacts, production issues like equipment in shot, transfer issues, editorial issues, title/action safe issues, misspelled on-screen text, mixed aspect ratios, broadcast compliance, animation or render errors, data corruptions
Audio QC	Artifacts, production issues, compression issues, static, sync with video, dropouts, pops/ticks, broadcast compliance, etc.
Localisation QC	Presence of textless elements

The QC process includes a complete review of the delivered files, audio specification such as audio data rate, audio bit depth, sampling rate and codec. Once all the technical specifications are met; the production house will make a digital cinema distribution master (DCDM), the interim master before the digital cinema package (DCP) and then the DCP package (J. Smyth, 2016). Table *2-2* shows a brief example description of the type of checks QC is for, Figure *2-10* shows an example report from a QC vendor.

	AUDIO	Mismatched dialogue / sync on "Huh?" (CH3,7
	AUDIO	English word in M&E in ghost whispers? ("evi
01:52:00:00	VIDEO	Aliasing on top and other edges of matte / liq confirmed intentional
	AUDIO	Dialogue out of sync ("Get back in the house (CH3,7&8)
01:52:05:08	VIDEO	Aliasing on vfx puddle, vfx motion stutter (int
	AUDIO	Recorded-in hot audio peak/distortion (CH1-3
	AUDIO	Recorded-in hot audio peak/distortion (CH1-3
	VIDEO	Density shift / Blacks change - confirmed inte
	AUDIO	Awkward dialogue edit ("I // just wanna talk t (CH3,7&8) - best source
SCENE	AUDIO	Grunts/efforts in M&E (CH9&10)
01:54:36:05	AUDIO	Dialogue does not match girl's lips (CH3,7&8)
01:54:37:19	AUDIO	Dialogue does not match man's lips (CH3,7&8

Figure 2-10 Example part of QC report (Enhanced Media, 2021).

For streaming, services like Netflix, Amazon and Google, they have a well-defined quality control (QC) process that is available for all content creators, it involves catalogue QC, where the content is examined on internal web players provided by QC suppliers. Master QC for technical quality checks of the video and audio assets is done; source files are examined on QC vendors' systems as well. Localization QC, for subtitle originators, is done to ensure the best quality possible (Google, 2020; Netflix, 2020a). Similarly, for TV productions, broadcasters have released documentation to outline the technical quality and delivery requirements of programmes, such as (ATSC, 2014; BBC & DPP, 2020; DPP, 2015; Phill Tudor et al., 2021). Figure *2-11* shows an example of the cinema supply chain.

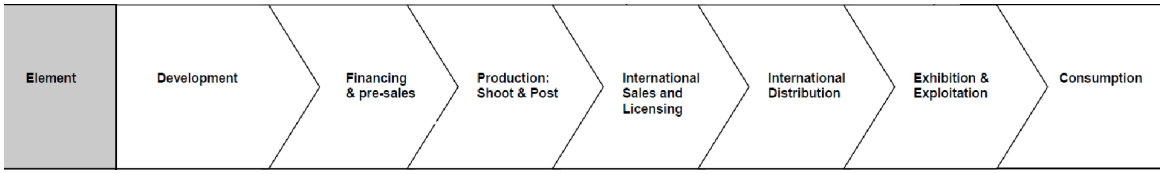

Figure 2-11 Film Supply Chain Example (Benghozi et al., 2015).

Note, that there is a different QC process to address every aspect, such as catalogue QC, which protects the audience experience, for example, typos in burned in subtitles, missing translation in localised release versions, or sync issues. During these process, transcoded material is examined on internal web players provided by QC suppliers. Master QC, ensure the optimum quality and compliance for distribution, this is inclusive of video and audio assets' technical quality; it is often handled by QC vendors (Enhanced Media, 2021; Netflix, 2020a; Ross, 2018).

For international releases or multi-lingual releases, localisation QC enforces a style guide and translation consistency. Note that none of these processes address the clarity of a sound mix on different playback hardware. This distinction is vital for this thesis: current QC protocols are designed to ensure transmission safety, for example, preventing clipping or sync errors rather than perceptual quality. A mix can pass every technical QC check for Netflix or the BBC and still be unintelligible on a mobile device in a noisy environment. This compliance vs. experience gap confirms that existing standards are insufficient to guarantee audio quality for the end-user, necessitating the adaptive approach proposed in Chapter 5.

While current feedback loops and quality control checks are highly effective for ensuring narrative satisfaction and technical compliance in ideal conditions, they present a notable limitation regarding non-ideal playback. They do not systematically capture how the mix translates to consumer hardware under environmental masking. Identifying this limitation establishes the academic rationale for the empirical audience testing undertaken in Chapter 3, which aims to investigate these real-world listening behaviours.

2.5 Accessibility

Advances in audio technology enable new approaches to accessibility. This section explores equity and inclusion for audio-visual media, examining the challenges for hearing and visually impaired audiences and the limitations of current technological solutions.

2.5.1 Equity & Inclusion

Estimates from World Health Organization (WHO), report that over 5% of the world suffer from disabling hearing loss which roughly totals to over 430 million people across the globe, and at least 2.2 billion people have a near or distance vision impairment (WHO, 2022). However, for the purpose of film sound delivery, the demographic of concern extends far beyond the clinically disabled.

All adults eventually experience Presbycusis – a symmetrical progressive loss of hearing over many years – and Presbyopia (vision loss). Crucially, Presbycusis specifically affects high-frequency sensitivity (3kHz–6kHz), which is the precise frequency range required for consonant articulation and speech intelligibility. This suggests that a significant portion of the general audience struggles with dialogue clarity yet falls outside the scope of traditional accessibility legislation (Tawfik et al., 2020; Wolffsohn & Davies, 2019). Figure *2-12* show examples of different levels of hearing loss.

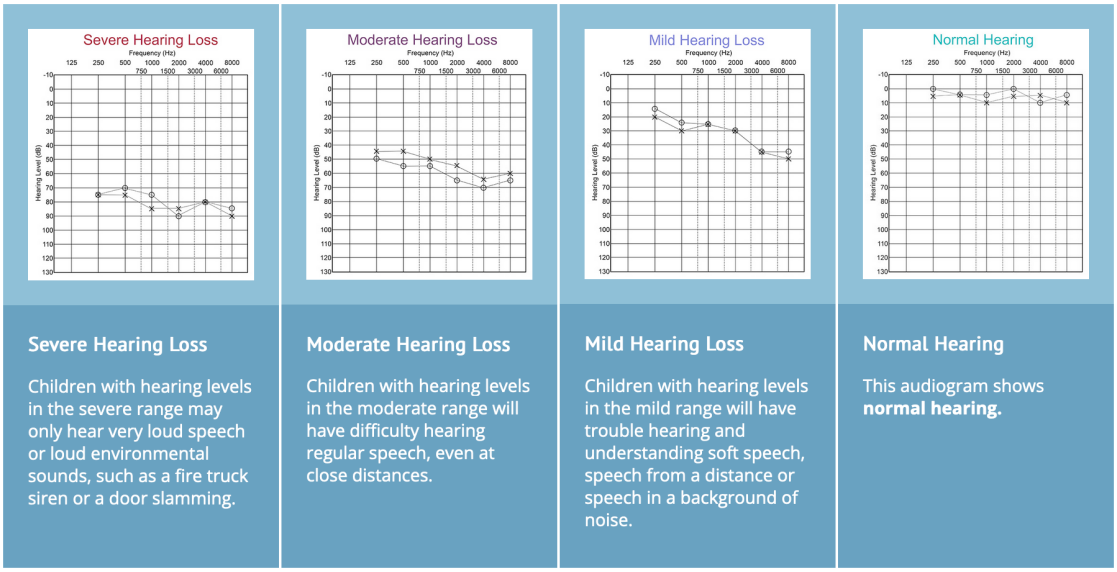

Figure 2-12 Examples showing different levels of hearing loss (Baby Hearing, 2023).

Furthermore, the listening environment itself often mimics the effects of impairment. While cinema sound levels (see Figure *2-13*) are calibrated for high dynamic range playback, these levels are unsustainable in domestic or mobile settings. Consequently, when a standard mix is played on a mobile device, the noise floor masks the lower-level details that a hearing-impaired listener relies on for context. This intersection of physiological loss and environmental masking creates a double disadvantage for the listener, which current static mixing standards fail to address.

Figure *2-13* shows measured sound levels in cinema which range from (58 dB/A to 140+ dB/A), A weighting is not an accurate representative of how human audio perception works; it is only accurate within the range of 20 to 55 dBs, and it does not provide indication of subjective loudness as C weighting, which is a more accurate representative of the response of the human ear from 85 dB to 140 dBs (Harrison, 2004; ITU-R BS.1770-4, 2015; Long, 2014; Martyr & Rogers, 2021; McMinn, 2013; Sinnott, 2018; Truax, 1999).

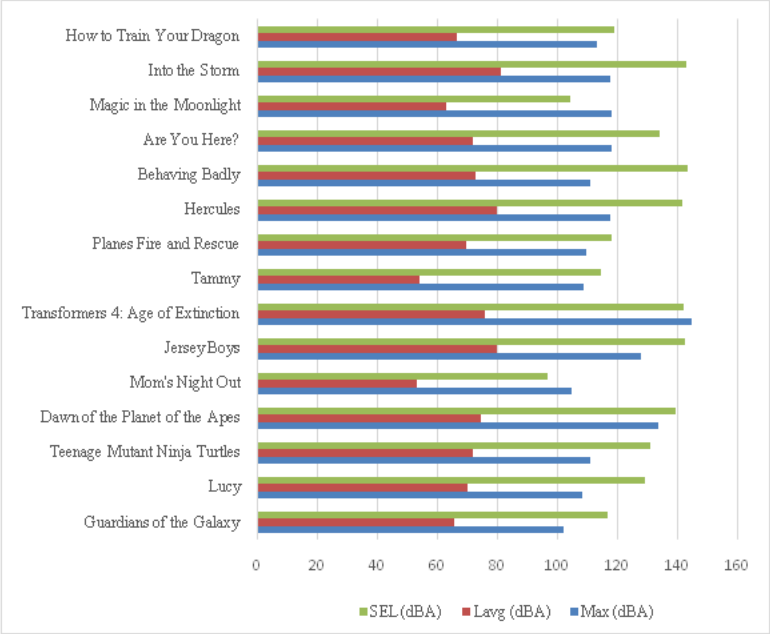

Figure 2-13 Cinema measured sound levels, SEL (Sound Exposure Level) and (Average Sound Level) (Pitts L, 2019).

Like their abled counterparts, those with visual and hearing impairments, have the right of access to information, communication, and mass media services (Article 21, 2008; Scope, 2022; United Nations, 2022). Beyond legal compliance, access to cinema and streaming provides vital social inclusion, combating the increased rates of depression and isolation found among adults with hearing disabilities (Australian Broadcasting Corporation Act 1983, 1983; Monzani et al., 2008; Shirley et al., 2019; L. A. Ward & Shirley, 2019a, 2019b).

To increase accessibility, legislation in many regions mandates services such as audio description (AD) and subtitles. However, while these technologies provide access to the narrative, they often conflict with the sonic experience (Code on Television Access Services, 2021; Hättich & Schweizer, 2020; Heinisch, 2015; Krejtz et al., 2012; Matamala & Orero, 2010, 2016; Noguer, 2013; Reviers, 2016; Romero-Fresco, 2015).

For cinema, television, and streaming, access to culture for persons with hearing or visual disabilities is achieved by use of accessibility services such as audio description (AD) or subtitles. AD represents commentary on what is happening on screen, whereas subtitles contain descriptions of important sound effects and speaker identification information as well as dialogue. Such approaches are also of use for the elderly, children, and non-native speakers (Fryer, 2016; Jankowska, 2008; Krejtz et al., 2012; Walczak, 2017a). Note that while these technologies provide access to the narrative, they often conflict with the sonic experience.

2.5.2 Technologies for the visually and hearing impaired.

For the visually impaired, AD offers a verbal commentary of visual action. However, because AD is typically mixed as an additional layer on top of the existing soundtrack, it introduces a masking problem. AD users report that the loudness of the film's music and effects often renders the description

unintelligible, or conversely, the description masks critical sound design cues (Udo & Fels, 2009; Whitfield & Fels, 2015). This highlights a mixing failure: simply adding more audio (AD) to a fixed stereo mix increases spectral clutter. An adaptive framework – as proposed in this thesis – would conversely allow for the attenuation of music and effects specifically when AD is active, resolving this masking conflict.

While researchers have proposed next-generation AD systems that utilize object-based audio to balance these elements (Lopez, 2015; Wang et al., 2021), these remain largely academic prototypes and have not yet influenced the standard delivery specifications for streaming or broadcast.

For the hearing impaired, subtitles remain the primary solution. However, relying on subtitles is arguably a workaround for poor audio mixing rather than a complete solution. Between 60–87% of viewers struggle with television dialogue, citing background music and background noise as the primary interference factors (Kochkin, 2009). This finding is critical to the research problem: a significant majority of users are turning to text not solely because of physiological hearing loss, but because the static mix balance prevents them from understanding the speech. This confirms a failure of the audio mix to adapt to the user's need for a higher signal-to-noise ratio (SNR).

2.5.3 Engagement experience

Engagement in film relies on narrative understanding – the ease with which a viewer comprehends the story (Busselle and Bilandzic 2008, 2009c), The literature identifies attentional focus as a key dimension of this engagement. Crucially, if a listener must exert excessive cognitive effort to decode speech from background noise, this attentional focus is broken, and immersion is lost.

Studies investigating immersion suggest that clarity is a prerequisite for presence. Improving the vocal delivery of audio description significantly increased levels of presence for visually impaired participants (Walczak, 2017a, 2017b; Walczak & Fryer, 2017). However, current accessibility standards treat access as a binary add-on (Subtitles: On/Off).

The literature strongly suggests that audibility is not binary, but a spectrum affected by age, environment, and hardware. Therefore, a static mix – even with subtitles – is insufficient. To restore true narrative engagement, the delivery system must transition from a fixed mix to an adaptive one that can mechanically compensate for the loss of intelligibility identified in this review.

2.6 Listening environments

Cinema sound differs from other forms of auditory reproduction, by and of itself; it is a collective social experience, and each member of the audience does not have a volume control. The experience is designed by the creators under a suite of calibrated and standardised conditions (Hill, Hawskford, and Newell 2016b; Newell 2014a; Toole 2015a). Unlike cinema, the experience for television and streaming is personal and individual, although could be shared at times.

Both mediums, television and streaming, are supported on a wide choice of mobile and stationary hardware with varying playback capabilities, most of which have either built-in loudspeakers or could be connected to a loudspeaker system (Alam et al., 2017; Android, 2022; Counterweight, 2020; Kotus et al., 2014; Qualcomm, 2021; C. Thomas, 2020).

The interoperability of the hardware allows consumers to use these devices in their living space or on the go, an audience member has the option to start a film on one device and continue on another. While television and streaming suite of standards like, ATSC A/85 or TR-B32, see Figure *2-14*, offer some level of loudness consistency for programmes, they fail to address, the extreme difference in reproduction capability and playback environment (Mcknight et al., 1996; Yang et al., 2020; Zhang et al., 2002). They also exclude a large demographic like the hearing and visually impaired, as well as the ageing population.

2.6.1 Cinema

The controlled reference Dolby laboratories introduced the industry standard where a reference pink noise signal is used to adjust the audio monitor level to 85 dB/C. This implies that every theatre playing back the film at the calibrated setting reproduces the volume level selected by the mixer at the dub stage (Allen, 1997; Gedemer, 2013b, 2015, 2016a, 2017; Murphy, 2015; J. Newell et al., 2015).

However, sound technology has evolved significantly since these standards were established. While loudspeaker headroom has increased, the loudness wars – an industry-wide trend of applying heavy dynamic range compression to maximize perceived overall volume at the expense of dynamic contrast – have led to shrinking dynamic range in trailers and commercials, forcing cinema operators to manually lower playback volumes to avoid complaints (TASA, 2015). This variability creates a fundamental disconnect: the mix is created in a stable, calibrated environment (85dB SPL), but is increasingly experienced in uncalibrated contexts where the reference level is arbitrary. This breaks the trust between the mixer's fader movements and the audience's perception

2.6.2 Domestic Environment

The reverberant field unlike the controlled acoustics of a cinema, a professional dubbing space uses low reverberation time to enhance localisation and intelligibility. In a typical domestic setting, the listening position is 1.8 times the critical distance (Olive & Toole, 1989; Toole, 1986a). This means a typical home setting is reverberant field dominated. Consequently, the domestic environment imparts different timbral characteristics to the mix, altering the frequency

response and direct-to-reverberant ratios compared to the studio (Beranek & Martin, 1996; Holman, 1991).

Unlike a domestic environment, cinema is a predominantly a controlled medium, where the sound field is made to be as consistent as possible across a number of venues (Allison, 1974; Winchell & Gourgon, 1982), this is achieved by building the mixing or dubbing stage like an auditorium to achieve reproduction consistency across a number of cinemas globally.

Such consistency is governed by standards that address, reverberation time and its variance in relation to room volume, electro-acoustic frequency response at listening positions, and direct to early reflections ratios listening positions as well as sound pressure level (SPL) calibration (DCI, 2018b, 2018a, 2020c; RP 2096-1:2017, 2017; SMPTE, 1998, 2010a, 2010c, 2012, 2014).

To scale down the cinema experience for a domestic setting, key differences need to be considered. Unlike a domestic setting, a professional dubbing space uses low reverberation time to enhance localisation and intelligibility, and listening positions are typically at half two – thirds of the critical distance to the loudspeakers located behind the perforated screen. In a typical domestic setting, the listening position is 1.8 times the critical distance.

This means that a typical home setting is reverberant field dominated compared to a professional setting, and as such, the domestic environment will have a comparably different timbral characteristics. Other factors to consider are, frequency response, frequency dependant direct to reverberant ratio, as well as reflections of walls, ceilings, and objects around the room plus the influence of those reflections along with increased sound ingression from external sources (Ando, 1985; Beranek, 1996; Holman, 1991; ISO 226:2003, 2003; Schroeder et al., 2005; Theile, 1986; Toole, 1986b, 1986a).

For audiences at home, television and streaming loudness inconsistency and volume leaps from one programme to another increasingly bother listeners. On TV, the transition from a film's dialogue to a heavily compressed commercial is perceived as particularly annoying (Kostek, Milarska, and Zakrzewski 2015; Lund 2006a; Sharma, Nicolau, and Mas 2022; Soulodre, Lavoie, and Norcross 2003; Weibel et al. 2019).

The issue of under-levelling and over-levelling are observed, resulting in level differences of more than 15 dB and results in an inconsistent disruptive experience. To facilitate better consistency between programmes, organisations like ITU, EBU and ARIB partnered with industry leaders to investigate the standardization of broadcast loudness, the result was a set of loudness measurements and delivery recommendations by organisations such as the international Telecommunication Union (ITU), and industry leaders (BBC, 2019; EBU & Camerer, 2016; Emmett & Girdwood, 1994; ITU, 2015; Klar & Spikofski, 2002; Spikofski & Klar, 2004).

There is a difference between the average loudness of the overall program and the dialogue component, and loudness standards vary as to which method is employed. Each broadcaster or streaming platform has their own delivery specification, Figure *2-14* and Figure *2-15* show the specification by region and platform.

Delivery Spec	Territory	Target Loudness (LUFS)	Maximum True Peak	Gate Used
EBU R128	Europe	-23	-1	Yes
ATSC A/85	US & Canada	-24	-2	Yes (initially No)
TR-B32	Japan	-24	-1	Yes
Op-59	Australia	-24	-2	Yes

Figure 2-14Delivery specification by region (Thornton, 2021a).

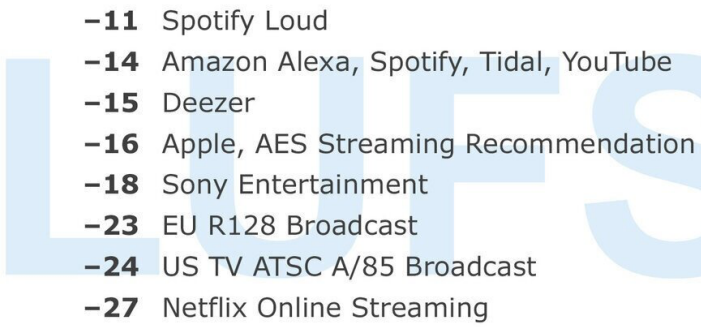

Figure 2-15 Loudness by platform (Thornton, 2021a).

For traditional broadcast or streaming, loudness measurement and control are now an essential and often mandatory requirement for broadcasting as well as for other forms of distribution and delivery, revisions of such standards address new audio formats such as object audio as well as legacy ones (ATSC, 2017; Bosi & Goldberg, 2003; Dolby Atmos, 2015; Herre et al., 2015b; Norcross et al., 2016; ST 2110-30:2017, 2017).

The technologies available today for broadcasters and streaming platforms, facilitates the delivery of ambitious, bandwidth heavy formats that have wider dynamic range and frequency response than their legacy counterparts. For content creators – especially filmmakers – it means fewer compromises on the picture as well as on the audio, however audio still needs optimization for a non-predictable presentation environment i.e., different than a cinema (Mapp, 2016; Thornton, 2018b, 2022).

While loudness normalization standards, for example, EBU R128, addresses the average volume between programs, it does not address the internal balance of the mix. A normalised film played in a reverberant living room may still suffer from unintelligible dialogue because the hard mix structure cannot adapt to the room's acoustic masking properties (EBU, 2016).

2.6.3 Shared environments

Research on preferred loudness levels for consumers at home and dynamic range tolerance is not abundant. Consumers have a well-defined dynamic range tolerance (DRT), and that speech is correctly identified when average levels are within a certain limit (Lund 2006b; Travaglini, Alemanno, and Lantini 2012; Ventry et al. 1971); even if sudden/momentary disturbances, or loud effects appear, conversely once the level fluctuates outside the tolerance area too often, the listener gets disturbed, and the listening experience is disrupted.

In this context, DRT is defined as the preferred level average window plus a peak level Headroom and will depend on the consumer's listening physical setting. Figure *2-16* shows tolerance for consumers under different settings, note the difference between cinema and in-flight, it could take potential audible dynamic range down from 40+ dB to only about 14 dB.

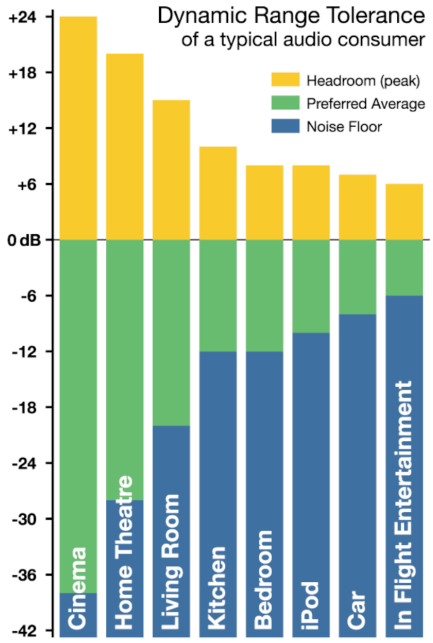

Figure 2-16-Dynamic Range Tolerance for consumers under different settings (Lund 2006a).

Typically, within an urban environment or transport settings, where there is significant background noise such as train, plane, or a vehicle, a wider dynamic range can be difficult to achieve, the trade-off is often reproduction distortion.

There is also potential damage to listeners' ears, driven by the tendency to increase listening levels to compensate for background noise. Studies on this topic suggest that consumers increase headphone levels by at least 5 dB above their preferred comfortable level when trying to overcome environmental masking (Airo et al., 1996; K. N. Dirks et al., 2021; Hadzi-Nikolova et al., 2013; Han et al., 2016; Holgers & Pettersson, 2005; Kathleen Pichora-Fuller et al., 1995; M Maassen, 2001; Naono et al., 1977; Pearsons, Bennett, 1977; Pearsons et al., 1977; Shimokura & Soeta, 2012; Walton et al., 2016; Weisser & Buchholz, 2019).

This arguably holds true in environments where mobile devices are often used. Note that exposure to noise typical of urban setting results in speech interference, such exposure could be in the excess of 70 dB (Fidell 1977; Lund

2006a). While Active Noise Control (ANC) might improve this to a certain extent, it introduces other aspects to consider that it introduces other aspects to consider that require further research, as noted in (Bitzer & Brandt, 2010; Darlington & Guiu, 2010; Kajikawa et al., 2012; Liebich et al., 2018; S. Olive et al., 2018; Schlieper et al., 2019; Struck, 2019).

Table *2-3* illustrate typical noise levels in transport settings, ranging from 65 dB (Cars) to over 90 dB (Subway).

Table 2-3 - Typical noise levels (Lund 2006a).

	SPL A weighted	SPL C weighted
Living Room, Urban	55 dB	70 dB
Inside Car	65 dB	85 dB
Inside Jet	75 dB	90 dB
Walk in Traffic	80 dB	92 dB
Subway	90 dB	100 dB

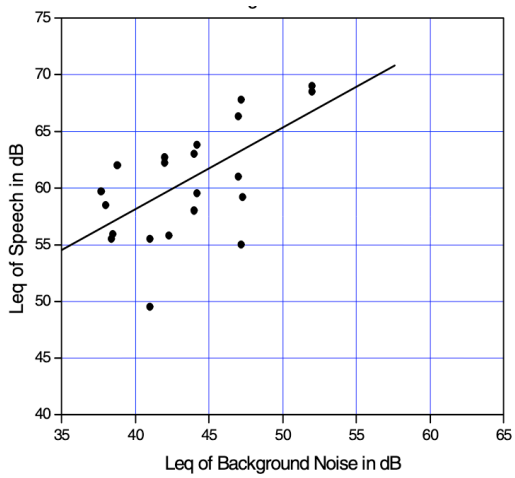

Figure 2-17 - Speech Levels Heard from TV as a Function (Pearsons et al., 1977).

Previous work by Pearsons et al. (1977) and Airo et al. (1996) indicates that listeners consistently increase playback volume to compensate for this noise, typically by 0.7 dB to 1 dB for every 1 dB increase in background noise. This behaviour highlights a critical user requirement for the adaptive framework: the listener is attempting to manually maintain a Signal-to-Noise Ratio (SNR) for dialogue. However, by raising the Global Gain of the entire mix, they inadvertently raise the noise floor of the content (SFX/music) as well, leading to unsafe listening levels.

The concept of most comfortable loudness (MCL) is common in otological – pathological anatomy – illness publications and less so in sound for visual media publications. Studies by (T. Letowski & Emanuel, 1995; Richards, 1975; Ventry et al., 1971) on normal hearing adults report an MCL average of 49.3 dB/A, whereas more recent studies such as (Benjamin, 2002, 2004; J. C. Riedmiller et al., 2003b; Soulodre et al., 2003) report an average of 57.7 dB/A within an 18 dB range for dialogue MCL.

Note, the increase could be explained; being stimulus dependant as the later studies focused on using dialogue and everyday content as opposed pure tones, broad- and narrow-band noise. Similarly, the average of MCL could have increased as our habits consuming content change and as sound technology improves.

In the study done by (Benjamin, 2004b), measurements were for TV and home theatre listening, the impact of background noise on listener behaviour was not considered in those studies, arguably a huge limitation. Figure 2-18 shows the data. The study suggests that a +2.91/-3.78 dB level change prompted listeners to describe the sound as noticeably louder or quieter, and +6.22/-9.22 dB elicits a reaction of too loud or too quiet. See Figure *2-18*.

Environment	Level	Range	SD	n
Television	57.7 dB	18 dB	4.4 dB	42
Home Theater	64.8 dB	23 dB	6.6 dB	17

Figure 2-18 - Preferred dialogue listening levels, dB/A (Benjamin, 2004b).

Transport environments, especially automotive, are noisy with negative influences on spectral, spatial, and temporal attributes of a reproduced sound field. Methods in use to research the subject typically utilise direct measurements of noise levels and recordings analysis in vehicles under conditions of typical use. The level and spectrum of noise in a vehicle is dictated by make/model, running surface, and speed, the noise spectrum is unique different from a domestic that is it is angled towards the low end; typical noise levels are about 0 dB SPL in the 20 kHz 1/3-octave band, 65 dB SPL in the 1k Hz 1/3-octave band, and 100 dB SPL in the 20Hz 1/3-octave band (Benjamin & Crockett, 2005a; House, 1989).

Previous work measuring noise in a vehicles such as (Brinkerhoff & Schwarz, 1958), reports a noise level of 73.2 dB/C SPL, and a noise level of 54.9 dB/A SPL, later studies by (Bozzoli et al., 2002; Geddes, 1990) reports comparably higher numbers as shown in Figure *2-19*; those levels are greater compared to a typical domestic environment, as a consequence, it prompts passengers to increase the level of the programme material they are listening to. In a study preferred listening levels by, data collected stationary vehicle – engine not running – shows an average listening level of 73.73 dB SPL, whereas the data collected in a running vehicle suggest a spread of 60 dB(A) to about 80 dB(A) depending on make/model, road surface, and speed.

Research indicates that vehicle passengers typically increase playback volume by an average of 0.83 dB(A) for every 1 dB increase in background noise (Benjamin & Crockett, 2005a).

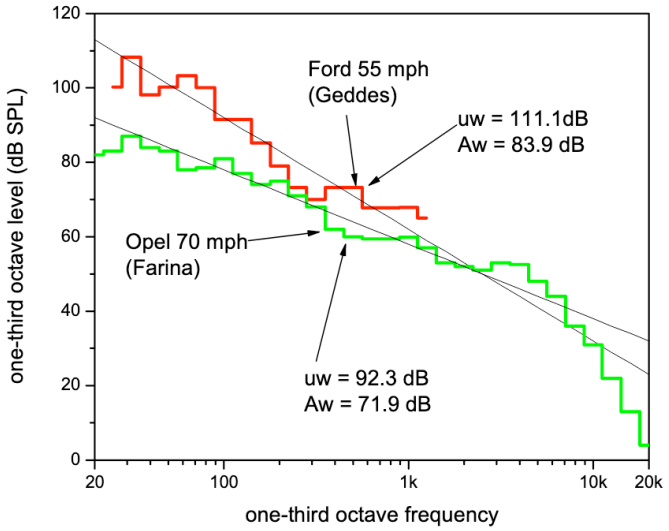

Figure 2-19 - Typical vehicle interior noise 1/3rd octave *(Benjamin & Crockett, 2005b; Bozzoli et al., 2002; Geddes, 1990)*.

Similar studies have been conducted to assess listening levels and noise exposure in a train environment. In a study published in 2008, the authors considered using music and speech-led programme material and considered background noise from trains, under and over ground, as well as busses and road traffic (Dance & Wash, 2008). The study reports an average of listening level 66.3 dB/A when the train is idle, and 88 dB/A when the train is moving for speech focused programme material with comparable results for music. This suggests that the listening level for any of the signals with train noise was above the safe noise exposure level. In another study, nine different subway lines inside Seoul were measured, music, gaming and dialogue-led programme material were used as stimuli (Han et al. 2016). The study reported that individuals in their twenties exhibited a higher average listening level compared to those aged fifty and above, despite the younger demographic generally possessing better hearing sensitivity. Figure *2-20* shows volume levels reported.

Another notable study was done on Sweden's central station, where passers-by persons were invited to measure the level on their portable music devices (Kähärï et al., 2011). The study reports an average listening level of 83 dB/A, a similar study was done in New Zealand to assess listening levels on the city commuter transport network, the authors used music and podcasts as stimuli, the study reports an average background noise level of 71.7 dBA, and an average listening level of 90 dB/A, the study suggests, that the listening level for podcasts was lower in the quiet condition (K. N. Dirks et al., 2021). However, it increased exponentially than the levels of the music, with an increased level of background noise. Although the study focused on buses as means of transport, the findings could provide insight in how commuters use their personal volume control to combat background noise generated by the method of transport; be it bus, train, or something else. Similar notable studies on the subject report comparable results (Himstedt, 2017; Nadri et al., 2012; Okokon et al., 2018; Silva & Correia, 2012).

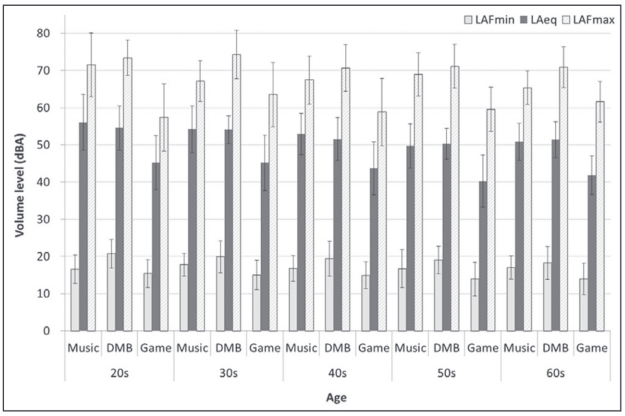

Figure 2-20 - Volume levels for LAFmin, LAeq, and LAFmax; 3 mobile apps and 5 age groups (Han et al., 2016).

Other notable work on environmental noise influences is illustrated in a listening study that was conducted in 2016, in which 22 participants were asked to adjust the balance of background and foreground audio objects to their preference whilst in the presence of reproduced environmental noise (Walton et al., 2016). The study demonstrated that environmental noise can significantly influence preferred background – foreground balance; consistent with other works such as (Bozzoli et al., 2002; Hodgetts et al., 2009; Kean et al., 2015b; Mason et al., 2015; Pardoe et al., 2020; Reis et al., 2009; Tan et al., 2014; Torcoli et al., 2019). However, cluster analysis of the data exposed that this should not be generalised to the whole population as one participant cluster preferred unchanged to increased foreground ratios in the presence of environmental noise.

While most of the previously studies and others like (Fligor & Cox, 2004; Keith et al., 2008; Keppler et al., 2010; Levey et al., 2011; Williams, 2009) focus on preferred listening levels of personal listening devices, they also provide insight on the risks of hearing loss due to exposure. The results might not be always comparable in a direct approach; this is highly attributed to different measurements and analysis being used, a notable paper by (Muchnik et al., 2012) highlights such concerns, among the findings - the paper reports that younger adults use higher volume levels, adopting a riskier behaviour, the same is reported in (Levey et al., 2011).

To mitigate hearing damage, manufacturers have implemented safety features such as Apple's headphone audio level notifications and Jabra's PeakStop technology (Apple Inc., 2021; GN Group, 2020). However, significant variability exists even within the same product generation. For instance, Apple's implementation varies by region – some territories enforce mandatory volume limiting while others allow user override.

This hardware inconsistency adds a layer of unpredictability for the content creator. Furthermore, these safety tools function as global limiters, attenuating the entire mix when thresholds are met. This creates a paradox for the commuter: they raise the volume to overcome environmental masking (as shown in Figure *2-20*), only to trigger a safety limiter that pushes the dialogue back down below the intelligibility threshold. This cycle confirms that global volume control is an insufficient tool for safe listening in noisy environments, necessitating the internal mix adaptation proposed in this research.

Some degree of external passive noise attenuation is also achieved by the occlusion of ears, as a by-product of a circumoral designed headphone. While active noise cancellation (ANC) is of utmost utility in so many scenarios (e.g., communicating inside a plane or close to a helicopter), it demands an aggressive noise elimination mechanism that is extremely complex and not suitable for entertainment; in such cases, noise-based automatic adjustments rather than elimination or cancellation is more suitable, note that ANC devices lack adaptability to users' preference.

Furthermore, ANC systems are typically built for maximal sound reduction irrespective of the incident direction of the sound. As such, when desired sound is present, the state-of-the-art methods add a separate system to reconstruct it. This often results in distortion and latency. In (Xiao et al., 2022), the authors propose a multichannel system that addresses such issues, the proposed system provide better noise reduction with less effort, however, it assumed that there was one desired source, in multiple desired sources, the system's performance is yet to be validated. In addition, the system was not examined in a non-ideal environment.

2.7 How is the listening experience evaluated?

Sound quality is a multi-faceted construct that extends beyond simple technical fidelity. To model how audiences form judgments in complex environments, this section examines the theoretical frameworks of sound quality and the limitations of current standardized testing methods when applied to non-ideal listening contexts.

2.7.1 Defining Sound Quality: From Fidelity to Experience

Sound quality varies significantly depending on the application. It can be broadly defined as the adequacy of a sound relative to the user's expectations (Blauert & Jekosch, 2003). However, for the purpose of mixing evaluation, multilevel model provides a more precise framework (T. Letowski, 1989; T. R. Letowski & Scharine, 2017). Letowski posits that quality is driven by two underlying factors:

1. Timbral Fidelity: The spectral characteristics of the sound for example, voice clarity.
2. Spatial Fidelity: The distribution of sound sources, for example, immersion.

Validation studies suggest that timbral quality accounts for approximately 70% of the overall quality rating (Rumsey et al., 2005). This weighting is critical for this research: it suggests that in bandwidth-limited environments, like mobile playback, preserving the timbral integrity of dialogue should take precedence over spatial immersion.

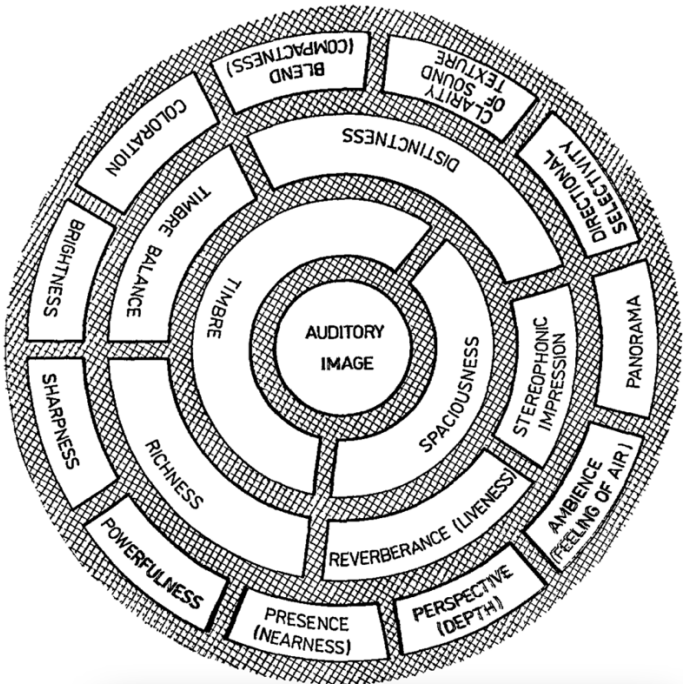

Figure 2-21 Multilevel auditory assessment language (T. Letowski, 1989).

Moving beyond simple fidelity, the concept of quality of experience (QoE), integrates the listener's internal reference and expectations (Raake, 2006). This distinction is vital: a high-quality cinema mix may result in a low-quality experience if played on a smartphone on a noisy train, because the user's expectation of intelligibility is not met. Therefore, this thesis adopts a QoE-based approach, evaluating the success of a mix based on user preference and narrative comprehension rather than technical transparency alone.

2.7.2 Perception Bandwidth and Listening Strategy

Human perception is a finite resource. The brain's effective data rate for conscious perception is severely limited, approximating 40 bits per sec (Lund & Mäkivirta, 2017). Consequently, listeners adopt different strategies depending on the cognitive load:

1. Slow Listening: Analytical, detailed evaluation, for example, a mixing engineer.
2. Easy Listening: Passive consumption, for example, a casual viewer.

Standard subjective tests often force participants into a slow listening mode, which does not reflect real-world consumption. To capture ecological validity, the evaluation methods in this research Chapter 6: Chapter 7: simulate easy listening conditions, where the participant reacts to the narrative flow rather than searching for microscopic artifacts.

2.7.3 Limitations of Standardized Test Methods (ITU)

The audio industry relies heavily on standardized listening tests such as ITU-R BS.1116 and BS.1534 (MUSHRA). These methods utilize a multi-stimulus comparison to grade the quality of a test signal against a pristine reference (ITU, 2015). While MUSHRA is the standard for evaluating codecs (e.g., detecting compression artifacts), it is ill-suited for evaluating mixing preference. MUSHRA requires a reference signal assumed to be perfect. In mixing, there is no single perfect mix; preference is subjective and context dependent. These tests focus on detecting impairments (what is wrong) rather than measuring satisfaction (what is right).

Method of Adjustment Given these limitations; this research moves away from static grading scales. Instead, it draws upon the Adjustment/Satisfaction Test (A/ST) proposed by (Torcoli et al., 2018). In this paradigm, the user actively manipulates parameters to find their optimal setting. This active mixing approach provides richer data than a simple 1-5 rating, as it reveals how users compensate for environmental constraints. Consequently, the experimental design in this thesis Chapter 6: Chapter 7: utilizes a Method of Adjustment, allowing participants to remix stems directly to quantify their ideal listening preference.

2.8 Design Approaches

The continuous advancement and maturation of mobile and stationary device technologies have transformed media consumption. They enable users to consume content at anytime and anywhere, this could be at home, in an airport or through a commute. This requires the sound or listening experience for the consumed content to be consistent across a multitude of devices with minimal to no disruption (Alexander, 2020; Bump, 2020; Dziabkin, 2019; Falkowski-Gilski & Uhl, 2020; Lawler, 2020; Market.us, 2020; Rich Media, 2020; Spangler, 2019).

To address this, an agile system that adapts the film sound mix for viewers could be developed. It could also allow for, different speaker types and layouts, playback hardware, listening environment and user behaviour and preferences with little to no user interaction. The next section looks at design approaches and how such a user focused system can be developed.

The following design approaches are not reviewed as a full human-computer interaction analysis. Rather, they are introduced selectively to establish the conceptual foundations required for an adaptive listening framework – where user context influences system behaviour. The emphasis therefore is not on interface design itself, but on how design principles inform the translation of perceptual needs into adaptive audio decision-making.

2.8.1 User centred design

User centred design (UCD) (Norman & Draper, 2017), describes design processes where users are considered and always involved during the various stages of the design process. The approach is broad with a considerable number of methods. It is perhaps simpler to view such methods as a spectrum of many ways in which users are involved in a design.

At one end, designers might consult with users at limited stages during the process about their needs during requirements gathering and usability testing. On the other hand, some methods involve users throughout; by treating them as equal partners in the design, which generally results in a deeper impact by users on a finished product.

Consequently, this research moves beyond treating the audience as passive receivers. Instead, it adopts a user-centred approach where the listener's perceptual requirements – specifically intelligibility in noise – dictate the system's behaviour. By integrating the end-user into the requirements gathering phase (Chapter 3:), the resulting framework ensures that the mix is defined by the user's context.

2.8.2 Usability

The term usability is typically used to replace user friendly, which could be vague and subjective. There are many different strategies to making a product usable. The definitions currently in use derive from a few views of what usability is and how it should be measured. (Bevan, Kirakowski, and Maissel 1991; Soegaard 2021), proposed that Usability relates to the user interaction of an individual with a product or system.

Usability can be measured by evaluating user performance, satisfaction rating and acceptability. The term focuses on utility; the ability to be utilized by individuals with ease and efficiency (Bevan, 1995; ISO, 2019; Shackel & Richardson, 2008). Note, most publications agree that usability is context dependant; driven by the interaction between tools, problems, users, and context (Naur, 1992; Taylor & Newman, 1999).

In the context of this thesis, usability is re-framed not as interface navigation, but as the 'listenability' of the content. If a mix requires excessive cognitive load to decode due to environmental masking, it fails the fundamental usability metric of efficiency. Therefore, the evaluation metrics used in this study focus on 'Ease of Listening' and 'Narrative Comprehension' as the primary indicators of system usability.

2.8.3 Inclusive Design

Inclusive Design is an approach in which developers ensure that products and services are accessible and usable by as many people as reasonably possible, regardless of age, ability, or circumstance (Design Council, 2006) .Its roots lie in the UK, where it is formally described in the British Standard on Managing Inclusive Design (BSI, 2005). Unlike universal design, which aspires to design for everyone without adaptation, inclusive design acknowledges practical constraints through the principle of reasonable inclusion, accepting that some extreme cases may require specialist solutions

The push for both inclusive and universal design has been driven by demographic change, particularly an ageing population, and by a global movement to integrate disabled persons into all areas of society. This has led to legislative measures guaranteeing disabled individuals' right to access the built

environment, public spaces, and digital services (Commission of European Communities 1999; DDA 1995; DOJ 2017; UK Government 2010).

The principle of inclusive design rests on two key aspects:

1. Human diversity – linking normal with able-bodied is inaccurate, as physical and mental capabilities vary widely across individuals and over the course of a lifetime.
2. The role of environment – disability arises in the interaction between a person and their environment; poor design can disable just as good design can enable.

This definition is crucial for the proposed framework. By acknowledging that 'disability' can be situational (e.g., a person with perfect hearing on a noisy train), the Adaptive Mixing Framework adopts an Inclusive Design philosophy. It aims to provide a 'reasonable' adaptation for the hearing-impaired and the environmentally impaired alike, rather than forcing a single 'Universal' mix that fails both groups.

2.8.4 User eXperience

While user experience (UX) encompasses a broad range of interactions, for the purpose of this research, the focus is on the role of empathy in the designer-user relationship. Empathy allows designers to identify with the feelings and needs of users, moving beyond viewing them merely as test subjects for functional practicality (P. Wright & McCarthy, 2008). In the context of film sound, this empathetic approach challenges the static – one-size-fits-all – delivery model by acknowledging the user's specific environmental context and limitations. Empathy is crucial for personal experiences, a principle that underpins the development of the adaptive framework in this thesis, which prioritises user agency and comfort over rigid technical adherence (Suri, 2010).

Similar discussions are reported in (Mattelmäki & Battarbee, 2002), the authors argue that while designing for practicality is needed, empathy is crucial for personal experience and private contexts; they proceed by stating that design could be a personal connection between designer and user that facilitates seeing and understanding users from their own perspective and as people with feelings rather than test subjects. This is expanded upon in (P. Wright & McCarthy, 2005), arguing that, in an empathic relationship, the designer doesn't give up their role to become the user because that would limit their ability to create something new. Instead, the designer tries to understand the user's world from their own perspective as a designer.

Empathy develops within a continuous relationship between designer and user, where the designer learns about the other user’s needs by responding with empathy, making mistakes at times, and then adjusting responses accordingly. If this relational aspect of empathy is absent, then designers can only rely on their own reactions and perspective to guide them. In (P. Wright & McCarthy, 2008), the authors identify three common characteristics of empathic methods:

1. The designer's direction to the other person, and whether they are determined to understand and help.
2. Their attention to their empathic response and how it is adjusted to the perspective and needs of the user.
3. The kind and quality of relationship between designer and user that they propose, most particularly whether it is likely to provide opportunities to attune to the needs and emotional responses of the user.

2.8.5 Adaptive Audio Systems

Adaptive systems are often directly or indirectly interactive, as such, they benefit greatly from a development lifecycle that ensures user involvement from the early design stages (Weibelzahl et al., 2020). In that regard, an adaptive system is a set of modular entities that, for an integrate whole, together can respond to contextual input, be it, environmental, hardware conditions, or user requirements to optimize performance. Example of such systems is noise cancellation, speech enhancement, and music production tools. Foundational work in this area is the development of adaptive noise cancellation techniques, pioneered by Bernard Widrow and John Kaunitz in the 1970 (Bitzer and Brandt 2010; Liebich et al. 2018; Widrow et al. 1975). This approach, relies on a secondary sensor to capture interference and adaptively filters it out, improving signal clarity, the technique largely laid the groundwork for modern adaptive audio processing applications.

Another significant contribution to the field is the exploration of cross-adaptive audio effects, as reviewed in (J. D. Reiss & Brandtsegg, 2018a). This paper reviews cross-adaptive processes where multiple audio signals influence each other's processing, within the context of music performance, production and automated mixing. It highlights applications in creative and performative contexts, with some intersection with user-centred design.

Adaptive systems within the context of sound mixing can be traced back to 1975 (Dugan, 1975, 1989), the work of Dugan describe means of automating microphone gain handling, particularly for speech applications. The field underwent significant transformation between 2007 and 2010, arguably through the work of (Hafezi and Reiss 2015; De Man and Reiss 2013; Pestana et al. 2015; Reiss 2011; Reiss et al. 2017; Tom, Reiss, and Depalle 2019). The contributions expanded the concept to encompass more signal processing, stereo panning, equalisation, delay, polarity correction, source enhancement and acoustic feedback prevention

The developments initiated a period sustained improvement in automatic mixing research (J. Reiss et al., 2017). In the last decade, the field has broadened considerably, and proliferated systems and methods to automating a range of production tasks. A notable trend is the increased adoption of machine learning techniques moving beyond traditional rule-based automation (Benito and Reiss 2017; Reiss et al. 2017; Tang, Fazenda, and Cox 2018; Woodcock et al. 2018).

While initial efforts in automatic mixing primarily focused on level balancing, the more recent work previously mentioned, has tackled the automation of composite signal processors. This expansion of scope has triggered parallel research into the perceptual aspects of mixing and the intricacies of the mix process itself.

Other available adaptive systems in audio dynamically adjust to user needs or environmental conditions, are particularly available in applications like hearing aids. Two key theoretical frameworks include adaptive filter theory and array signal processing, which enable real-time adaptation to optimize audio output (Aazh et al., 2015; Beracoechea et al., 2006; Lough et al., 2023; Moore, 1987).

One of the key approaches is adaptive beamforming and noise reduction techniques for hearing aids, the approach focuses on differential beamformers that account for microphone distance and head shading effects (Puder, 2006). The approach involves utilising adaptive filter algorithms, which adjust filter coefficients to minimize interference and enhance speech intelligibility. This approach also presents a combined noise reduction and Gain optimization procedure that adapts to the user's hearing loss and instantaneous signal-to-noise ratio. Other developments adjust to spatial and spectral differences between speech and noise, these enable hearing aids to adapt to complex acoustic environments, improving user experience, specifically for assistive listening (Bentler et al., 2004; Bunnell, 1990; Chung, 2004; Kuk et al., 2005; Leeuw & Dreschler, 1991).

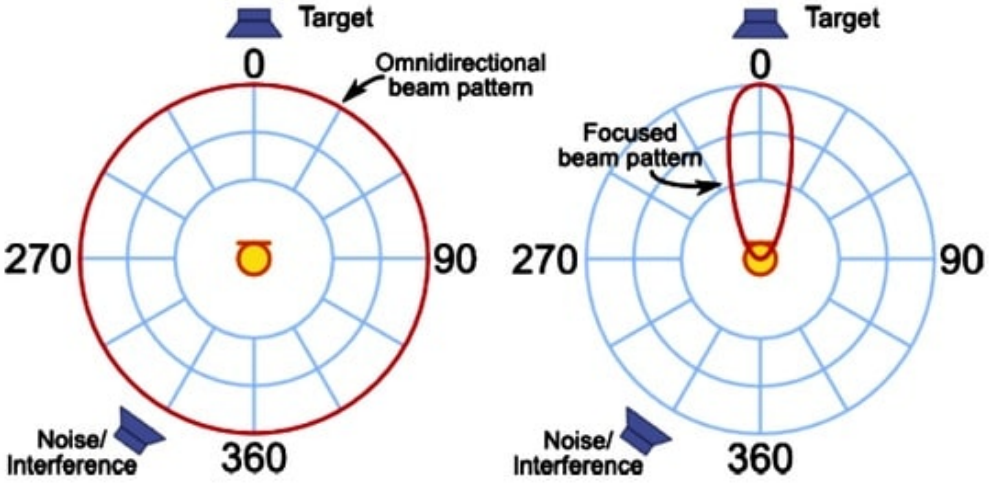

Figure 2-22 Beamforming in hearing aids (Staab 2015).

Despite these advances, current approaches to automatic and adaptive mixing remain largely constrained by:

- Music-Focus: Most of the literature (Hafezi & Reiss, 2015; J. D. Reiss, 2011; J. D. Reiss & Brandtsegg, 2018b; Tom et al., 2019b) focuses on music production, which prioritizes spectral balance over narrative intelligibility.

- Static Output: Most systems automate the mix creation but still render a static file, failing to adapt to the user's changing environment.

This identifies the primary gap addressed by this thesis: the need for a system that applies these automatic mixing principles dynamically at the playback endpoint, using environmental and listening context to drive the processing decision-making process.

Much of the existing work remains fragmented and incremental. It focuses on the micro-automation of established mixing practices – predominantly within music production workflows – rather than exploring new models of audio delivery and distribution at the playback stage. While there is considerable innovation in the field, there remains a large opportunity to operationalise models that address real-world audience needs through adaptive technologies.

Within this research, user-focused design concepts and empathy are adopted strictly to inform the technical audio architecture. In this context, empathy is operationalised to calculate and compensate for the listener's environmental and listening constraints – such as acoustic masking during a commute – to preserve dialogue intelligibility. Instead of forcing the listener to adapt to a static studio mix, the audience's acoustic reality becomes the primary context for system performance.

This requirement directs the data collection in Chapter 3 and the active mixing experiments in Chapters 6 and 7. Ultimately, what is delivered is the AMPLE framework – an architectural workflow to adjust the listening experience based on environmental metadata, acoustic interference, and hardware constraints, ensuring translation across diverse listening conditions.

2.9 Summary

The literature suggests a clear shift in film consumption from the controlled acoustic conditions of cinema to fragmented listening contexts shaped by streaming services and personal devices (Livingstone 2021; Navarro 2021).

While cinema sound is produced and reproduced within standardised environments, contemporary audiences increasingly experience film sound through highly variable hardware and uncontrolled acoustic settings. This transition has introduced substantial challenges for sound translation, intelligibility, and immersion (Fischetti et al. 1993; Hill, Hawksford, and Newell 2016c; Rumsey 2018a).

Although technical standards regulate overall programme loudness, they do not account for differences in playback hardware, listening level, or acoustic context. As a result, mixes that perform acceptably in professional environments often translate poorly to domestic and mobile listening situations (Toole, 2015). In addition, current professional workflows do not systematically consider the listener’s environment or personal preferences, and few adaptive systems exist that can modify sound presentation dynamically at the point of playback.

From the literature, three primary research gaps have been identified:

1. The Standardisation Gap: Existing loudness standards control signal level but do not address variation in playback conditions or device capability.
2. The Context Gap: Professional mixing practice does not incorporate contextual information about the listener's environment or behaviour.
3. The Agency Gap: Although accessibility features exist, there is no established framework that allows all users to adjust mix balance while preserving creative intent.

This thesis addresses these gaps by examining professional mixing workflows alongside empirical evidence of audience listening behaviour. On this basis, it proposes an adaptive framework for personalised sound reproduction on mobile and desktop devices, with the aim of improving intelligibility, consistency, and engagement across diverse listening contexts. The limitations identified across the literature therefore establish the necessity of an empirical requirements phase, in which audience listening behaviour must be examined directly to derive the design constraints carried forward into Chapter 3.

Chapter 3: Requirements Gathering-End users

The suite of standards controlling cinema as a medium offers audience a sonic experience that is consistent with the director's creative intent. For directors and re-recording mixers, this standardization provides reassurance that the final soundtrack will accurately translate across different cinema auditoria (RP 2096-1:2017, 2017; SMPTE, 2010b, 2012, 2017).

Much like cinema, television and streaming loudness standards were developed to provide a consistent sonic experience for viewers while maintaining the producers' creative intent (AES TD1004.1.15-10, 2015; EBU-TECH 3344, 2016; ITU-R BS.1770-4, 2015; SMPTE ST 2036-1:2014, 2014; ST 2110-30:2017, 2017; BBC & DPP, 2020; Netflix, 2021).

However, current standards and recommendations do not fully address the wide variety of modern playback hardware and reproduction environments. Furthermore, they fail to account for significant variances in user-preferred playback levels and the specific auditory needs of a wider demographic, particularly the ageing population, the visually impaired, and audiences with hearing loss (BBC, 2017; BBC Academy, 2020; Brown, 2016; Dalziel, 2022; Johnson, 2017; Plunkett, 2016; Sontag, 1996; L. Ward, 2017).

This chapter describes the approach developed to gather requirements, identify viewers' behaviour, preferred playback hardware and the associated levels as well as the listening environment. The chapter describes the background and a method to obtain the end users' point of view, the results are also reported followed by the discussion section.

This half of the requirements-gathering process, focusing on the audience is presented, while Chapter 4: follows, it applies a similar method to professionals, examining re-recording mixers' workflow, decision-making process, and preferred playback hardware for entertainment. This chapter directly addresses RQ1, concerning the adequacy of current listening experiences across platforms, and RQ2, examining audience preferences and listening behaviour in mobile and stationary contexts.

3.1 Background

Streaming and TV services are increasingly compatible and available on a variety of playback hardware, for example, Apple TV, Roku. This enables a wide and diverse demographic to watch content across different reproduction devices, for example, smartphones, tablets, portable gaming consoles and in different settings, for example, home, airports, shopping centres, during a commute.

Interoperability between the different hardware available offers the user a synchronisation option; enabling users to consume content on a laptop with built-in speakers, switch to a handheld device with earbuds, and continue at a different time on soundbar-equipped televisions (Alexander, 2020; Bump, 2020; Dziabkin, 2019; Falkowski-Gilski & Uhl, 2020; Lawler, 2020; Market.us, 2020; Rich Media, 2020; Spangler, 2019; Walton, 2018; Walton et al., 2018).

It is crucial for the content sound mix to translate well and remain consistent across different playback devices and environments. The variance in the reproduction hardware, and their associated sound pressure levels (SPL), inclusive of the environment alter the sonic experience for the listeners, especially the perceived loudness, spectral and spatial properties of the soundtrack. As a result, viewers end up with a disruptive and inconsistent listening experience that lacks immersion across multiple devices (Bosun 2001; Fischetti et al. 1993; Generale and King 2017; Guru et al. 2011b; Hill et al. 2016c; Olive and Welti 2012a; Olive et al. 2013; Rumsey 2018a; Y Suzuki et al. 2003; Toole 2015a; Völker 1996b).

The current suite of standards for TV and streaming does not fully address such issues, they lack comprehensive feedback from a wide audience demographic, especially the ageing population, the visually impaired and audience with hearing loss. As such, a questionnaire concerning usage, and sonic preferences inclusive of the playback environment can help gather information from audience and push towards a more inclusive and user-centred quantitative research, to help fill the gaps in the current workflows, standards and delivery specifications.

3.2 Method

To identify audience usage patterns, playback preferences, and contextual listening conditions, an online survey was distributed to members of the public. The survey addressed film consumption across cinema, stationary, and mobile devices, with particular attention to listening environments, playback hardware, and perceived limitations of current sound delivery.

3.2.1 Questionnaire Design

The questionnaire adopted a mixed-methods design, combining closed-ended questions for quantitative analysis with open-ended prompts to capture qualitative insight. Likert-type scales were used throughout, with both five-point and seven-point formats employed. Scale selection was informed by prior validated instruments in the literature to support comparability. Five-point scales were used primarily for frequency-of-use and behavioural questions to minimise respondent burden, while seven-point scales were applied where greater sensitivity was required, such as perceived loudness comfort and satisfaction. Loudness-related items were adapted from (Benjamin, 2004b), and device preference scales were informed by (Lopatka et al., 2014).

It should be noted that the survey comprised targeted questions, scoped and extracted from several previously utilised studies (Atkin & Litman, 1986; Basil, 2001; BFI, 2017; De Rosa & Burgess, 2018; Gedemer, 2013b, 2016b; Ghouti, 2017; Griff, 2014; Kelly, 1991; Webster, 1986). Rather than employing the complete validated instruments – which often measure variables outside the focus of this research – questions were selected strictly for their relevance to film sound consumption and audience behaviour. While adapting and utilising partial instruments could limit the ability to conduct rigorous psychometric reliability testing against the original scales, this study does not aim to establish a standardized clinical diagnostic tool. This targeted scoping ensures high ecological validity, directly capturing the specific, real-world frustrations users experience with their listening experience.

3.2.2 Participants and Sampling

Participants were recruited using a convenience sampling strategy via academic networks, professional contacts, social media platforms, and mailing lists. This approach enabled the recruitment of a broad cross-section of film viewers across a wide age range (18–70 years) and multiple geographic regions. While this sampling strategy limits statistical generalisation, it was appropriate for an exploratory requirements-gathering study intended to identify trends, preferences, and recurring issues rather than to estimate population parameters.

Participation was voluntary and unpaid. All responses were anonymised using participant identifiers (PS1, PS2, etc.). A total of 500 valid responses were collected. Of these, 36.2% identified as female, 62% male, and 1.8% non-binary/prefer not to say. The mean age was 38 years (range: 18–69). Regarding hearing health, 80% reported no hearing loss, while 18.2% reported mild loss, 1.2% moderate, and 0.4% severe. Responses were gathered from sixty countries. See Figure *3-1*.

Ethical approval was granted by the University's Research and Integrity Committee, and the survey was implemented using the University-hosted NOVI platform (Novi, 2022). The survey remained open for twelve weeks. Full documentation, including participant information sheets and consent materials, is provided in Appendix A.

3.2.3 Instrument Development

The survey instrument was developed through an iterative process informed by existing literature on audience listening behaviour, playback environments, and sonic preference.

- Demographics: Followed established guidance from (Brant et al., 2015; ITU P.919, 2020).
- Hardware & Environment: Items addressing listening levels and environment were adapted from (Gershon et al., 2006; Oldfield & Shirley, 2011; Worthington et al., 2009)
- Sonic Experience: Questions concerning satisfaction were formed after (Easley et al., 2009; Rane et al., 2022).
- Cinema Experience: Questions were informed by (Gedemer, 2015, 2016b; Holman, 1991)

The survey was piloted internally with supervisors and colleagues prior to launch to ensure clarity and relevance (Passmore et al., 2002).

3.2.4 Data Analysis Strategy

Quantitative data were analysed using descriptive statistics to identify frequency distributions of hardware usage and environmental habits. Open-ended qualitative responses were analysed using Inductive Content Analysis (Braun & Clarke, 2019; Elo & Kyngäs, 2008a; Gedemer, 2016b; Holcomb, 2016; Huberman et al., 2002; Thrane, 2022). Responses were reviewed iteratively to identify recurring themes related to listening experience, control, and frustration. These themes were subsequently coded into higher-level attributes, for example, control, immersion, convenience, allowing qualitative findings to be statistically compared against the quantitative results.

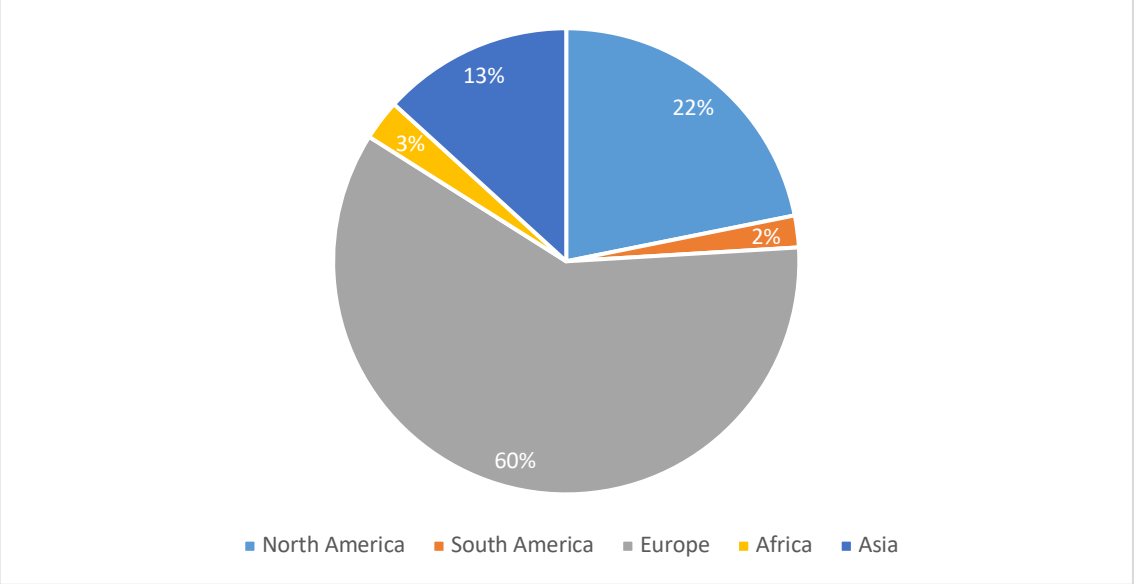

Figure 3-1 Geographic regions

3.2.5 Survey

Five hundred responses were collected. Upon clicking the survey link, participants were briefed with the information sheet, in which they read about the study and the survey questions. A mandatory consent form followed the information sheet, and by ticking 'I agree' on the form, participants were acknowledging and to participate in the study.

They were tasked to answer the questions to the best of their abilities, an option to correct their answers or continue with the survey later was available. The survey was made of up 55 questions total with the survey logic being designed in such a way as to keep participants from having to answer questions that were incongruous with answers from previous questions.

A further section was also included at the end of the survey. Apart from the consent form, age was the only other compulsory question, the raw data was imported to Microsoft Excel for sorting and further descriptive and Inferential analysis. Descriptive analysis was utilised to measure central tendencies, dispersion, and the frequency of events across the dataset. This descriptive approach allowed for a comparative evaluation of listening habits and volume adjustment behaviours between mobile, stationary, and cinematic environments.

3.3 Results

The following section presents the results from the online survey (N=500), compiled directly from the NOVI platform (Novi, 2022). The data indicates a notable divergence between the controlled environment of the cinema and the variable nature of mobile and stationary playback. Broadly, the results point to dissatisfaction with current mix levels on personal devices, the prevalence of environmental noise in non-cinema settings, and the adoption of compensatory behaviours such as volume adjustment and subtitle usage to manage intelligibility.

3.3.1 Listening Environments & Ambient Noise

To understand the acoustic challenges facing the modern audience, participants were asked to identify sources of noise intrusion. The results indicate that mobile consumption is rarely a solitary or quiet experience.

Transport & recreation: As shown in Figure *3-2*, the highest reported noise sources were public transport (69.0%) and recreation (63.9%). Specifically, within transport, noise was reported on buses (66.7%), trains (63.9%), and planes (45.9%). This high prevalence of background noise suggests that mobile listening is frequently characterised by environmental masking.

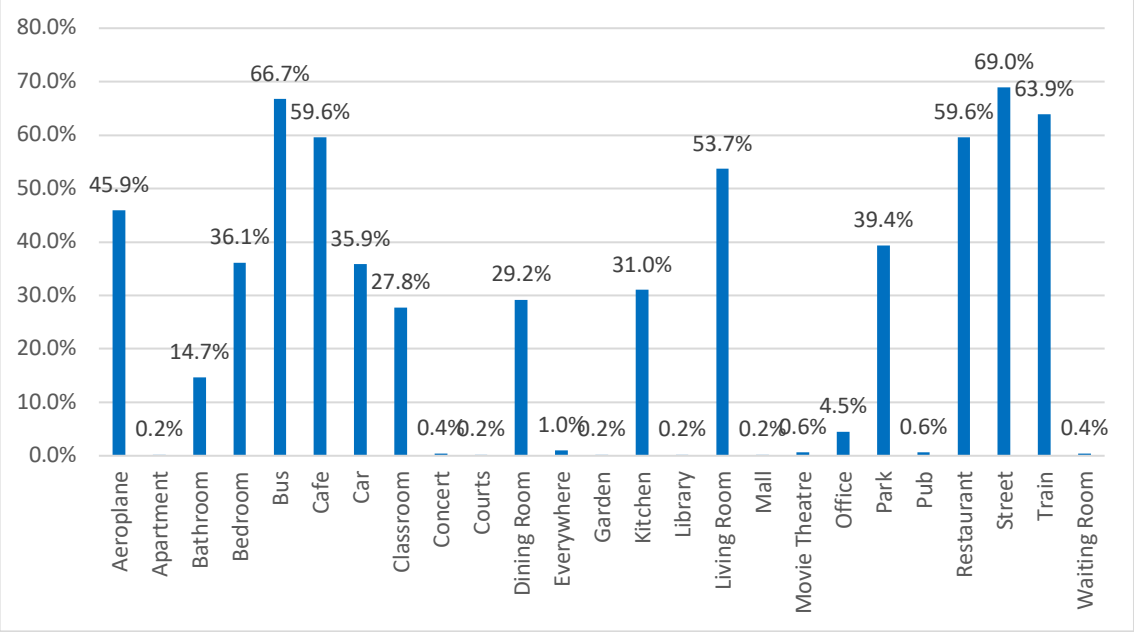

Figure 3-2 Noise pollution from other people's devices summary (n=490)

Domestic Interruption: In the domestic environment, 53% of participants reported noise intrusion, primarily in living rooms (53.7%) and kitchens (31%). Furthermore, Figure *3-3* highlights a social constraint: over 35% of participants reported being asked to lower their volume in domestic settings – living Room: 36.8%, bedroom: 31.9%. This data indicates a conflict for the user: the need for higher volume to overcome domestic noise often contrasts with social pressure to lower playback levels, potentially affecting the signal-to-noise ratio (SNR) for dialogue.

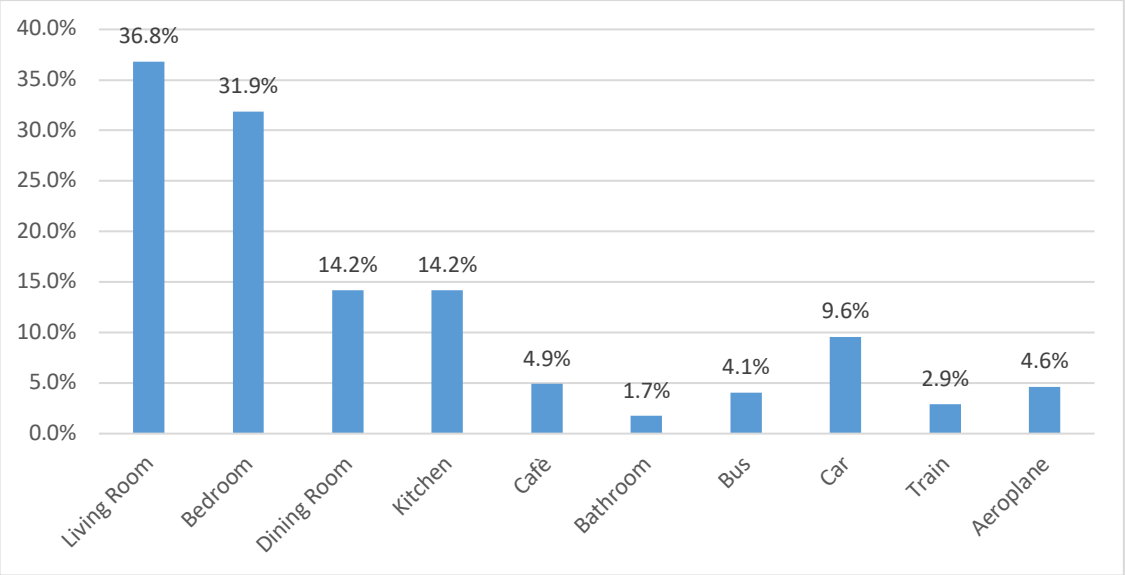

Figure 3-3 Induced noise (breakdown)(n=345)

3.3.2 Platform Choice & Hardware Diversity

Usage frequency data on consumption habits (Figure 3-4) reveals that while cinema attendance remains periodic (70% attend several times a year), stationary device usage is frequent (34% several times a month). Notably, 52% of participants indicated a preference for non-cinema viewing – stationary: 45%, mobile: 7%, suggesting that the reference experience for many participants is no longer the theatre.

Hardware Heterogeneity: The survey evidences the fragmentation of the playback ecosystem.

- Mobile: 47% use iOS devices, 24% Android. Crucially, regarding listening hardware (Figure *3-6*), there is a split between headphones (33.9%) and built-in device speakers (33.0%).
- Stationary: 83% use Android-based systems (e.g., Smart TVs). While 36.5% use built-in TV speakers, only 2.8% reported using a dedicated Dolby Atmos system (Figure *3-7*)

The reliance on built-in speakers (mobile and TV) highlights the prevalence of hardware with limited frequency response and dynamic range capabilities.

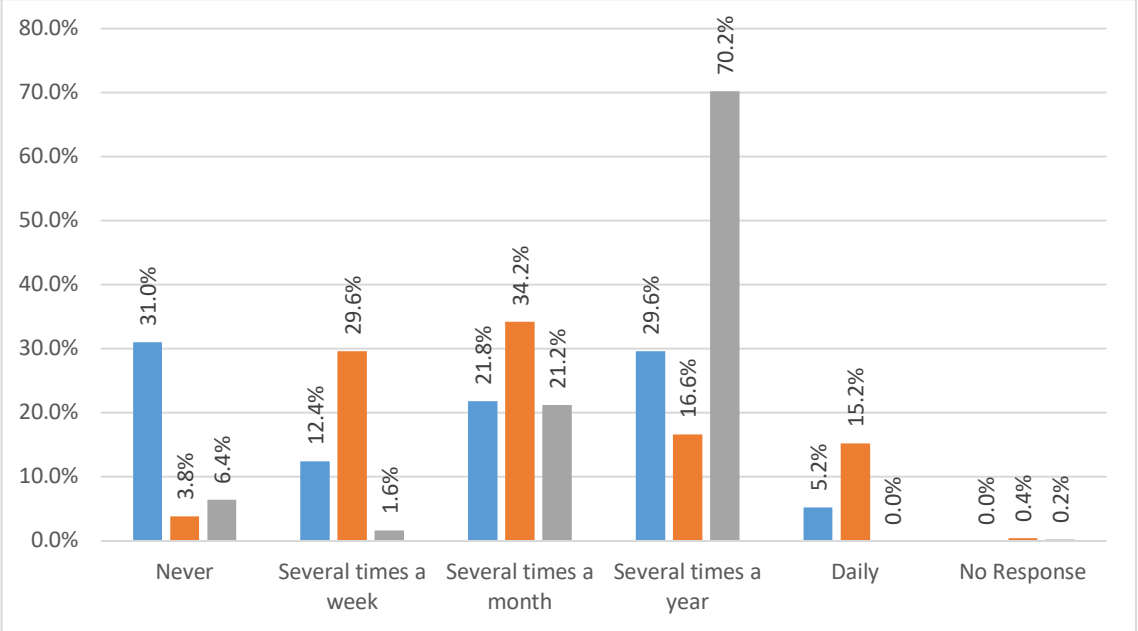

Figure 3-4 Percentage of users for mobile (n=345), stationary (n=498) and cinema (n=497)

To further contextualise these consumption habits, Figure *3-5* details the specific mobile hardware brands utilised by the respondents, highlighting the dominance of a few key manufacturers in the mobile playback ecosystem.

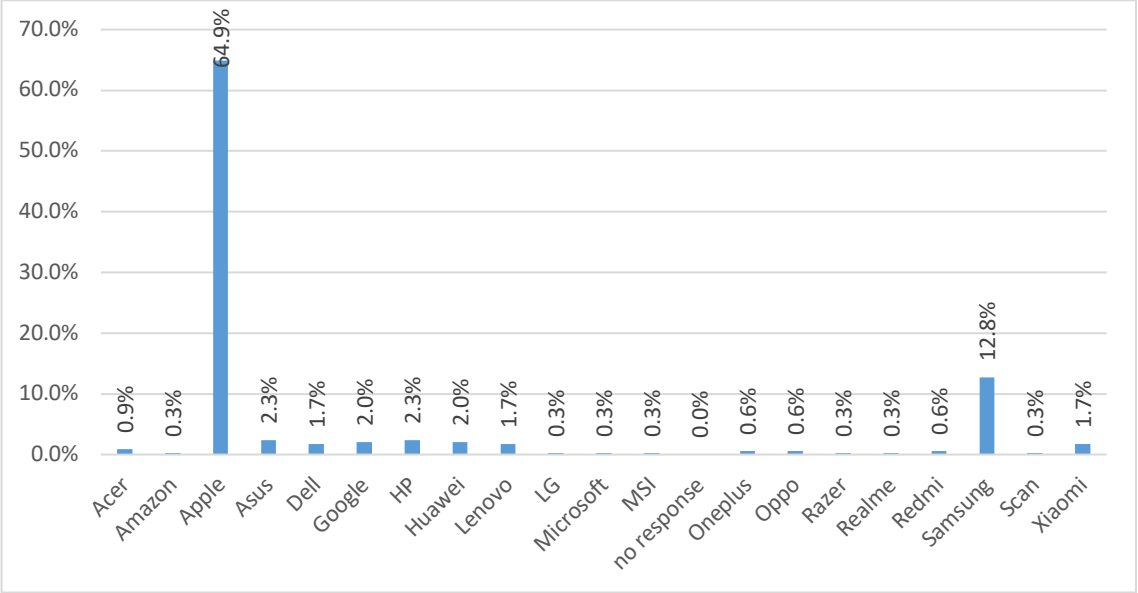

Figure 3-5 Mobile device make often used (n=345)

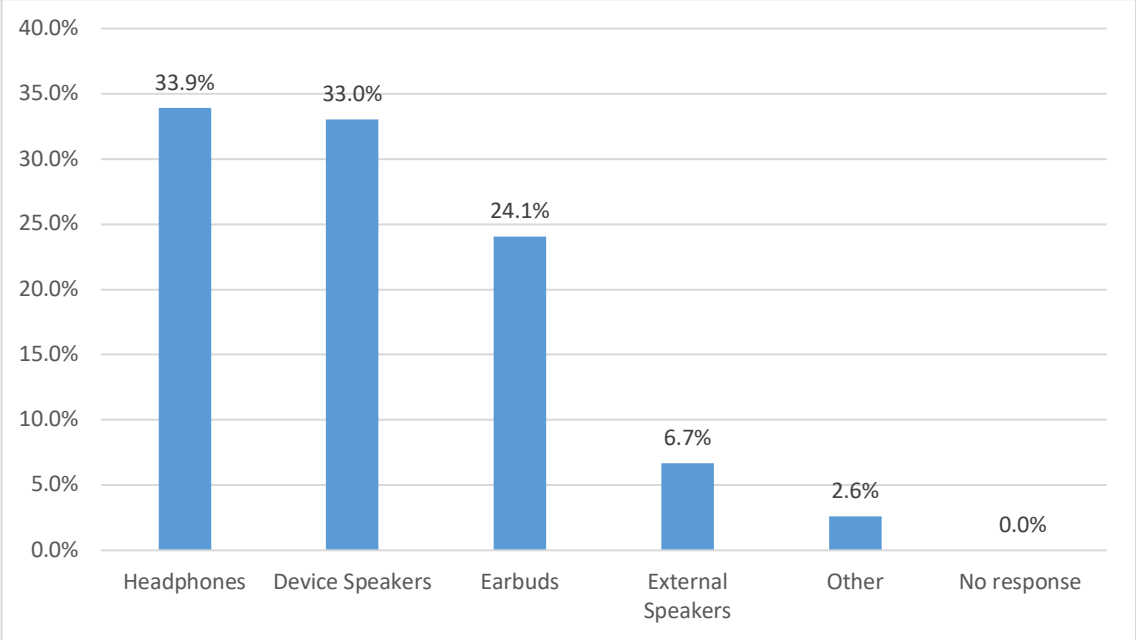

Figure 3-6 Listening device on mobile (n=345)

While mobile device users rely heavily on headphones or integrated micro-speakers, a similar reliance on built-in hardware is observed in domestic settings. Figure 3-7 illustrates the distribution of listening devices utilized for stationary playback, confirming that advanced surround systems remain a minority setup.

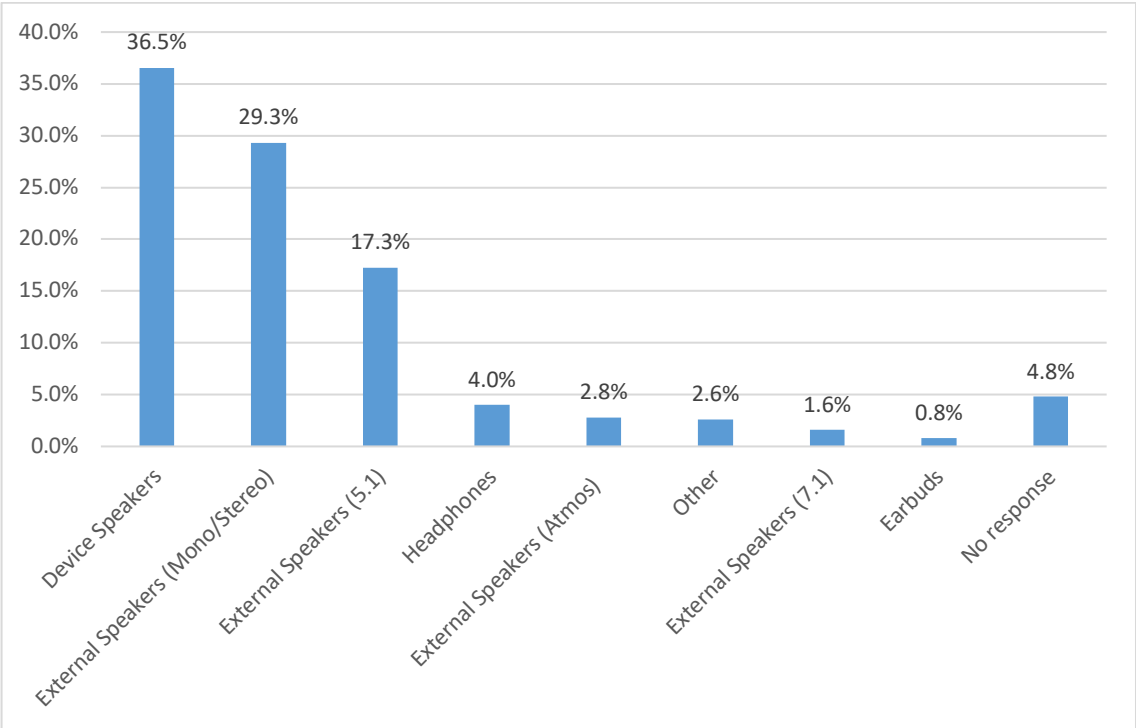

Figure 3-7 Listening device on stationary devices (n=498)

3.3.3 Volume Adjustment & Loudness Comfort

Participants were asked about their interaction with volume controls to gauge the adequacy of current dynamic range standards.

Frequency of Adjustment: Figure 3-8 shows that over 70% of participants alter the volume sometimes or often on mobile and stationary devices. In contrast, less than 2% reported never altering the volume.

- Cinema Loudness: 64.6% rated cinema levels as loud.
- Mobile/Stationary: Rated largely as comfortable (70%), but with a trend towards higher volume settings on mobile (40.9% set volume >75%) compared to stationary (20.7%).

The high frequency of volume adjustment on personal devices suggests that users are actively managing dynamic range to maintain a comfortable listening level across scenes.

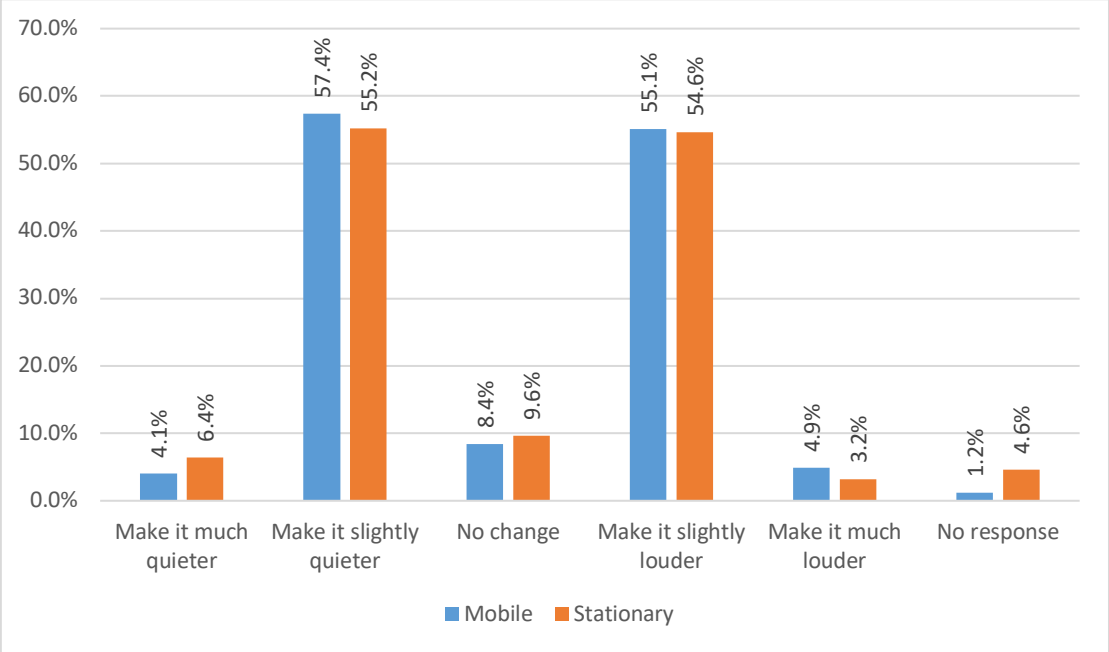

Figure 3-8 Volume alterations (how) mobile (n=345) – stationary (n=498)

Figure 3 9 shows typical volume set on each context as indicated by participants.

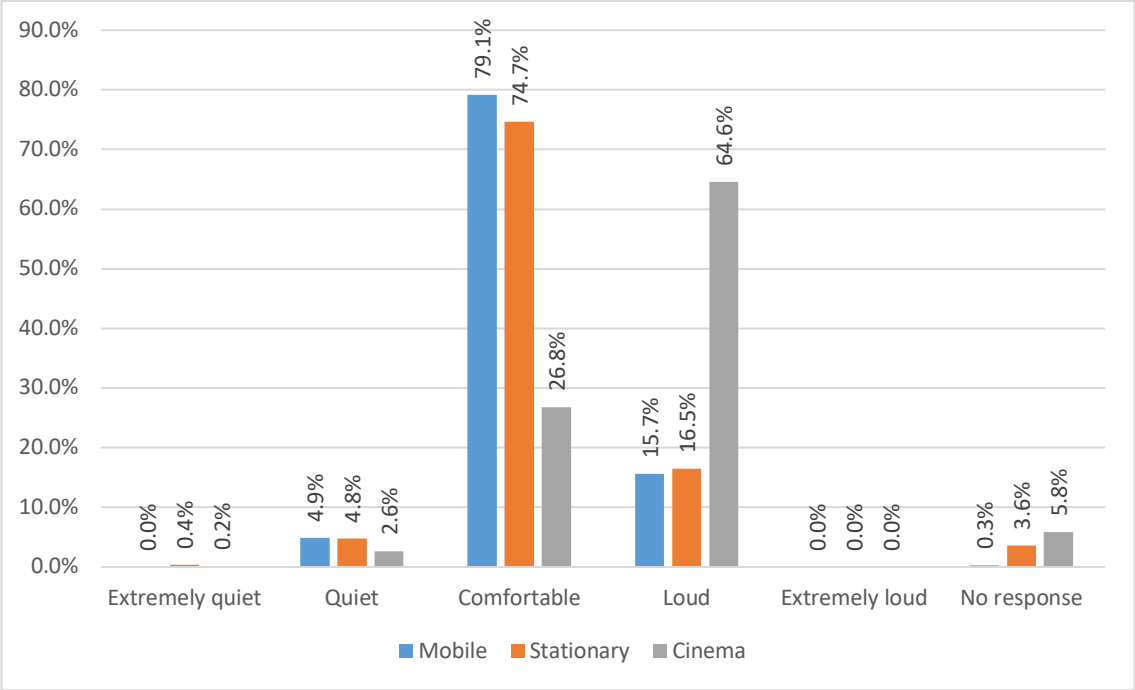

Figure 3-9 Typical volume, mobile(n=345), stationary(n=498), cinema(n=497)

3.3.4 The Intelligibility Gap: Dialogue, Music, and Subtitles

This section addresses the core research question regarding mix balance preferences.

Subtitle Usage: As shown in Figure 3-10 , there is a heavy reliance on subtitles outside the cinema. While 73.6% never use closed captions in cinemas, usage on mobile and stationary devices is significantly higher, with 10% always using them. This suggests that subtitles are being utilised to compensate for audio intelligibility issues rather than strictly for hearing impairment.

Stem Preference: Participants were asked how they would alter specific stems if given the choice (Figure *3-11*, Figure *3-12*, Figure *3-13*). The results suggest a consistent preference for dialogue dominance:

- Dialogue: >70% wanted dialogue to be louder on mobile/stationary devices.
- Music: >60% wanted music to be quieter.
- Effects: 50% wanted effects to be adjusted (typically quieter).

These preferences indicate a clear user requirement for a mix with boosted dialogue and attenuated music/effects, effectively narrowing the dynamic range to suit the device capabilities.

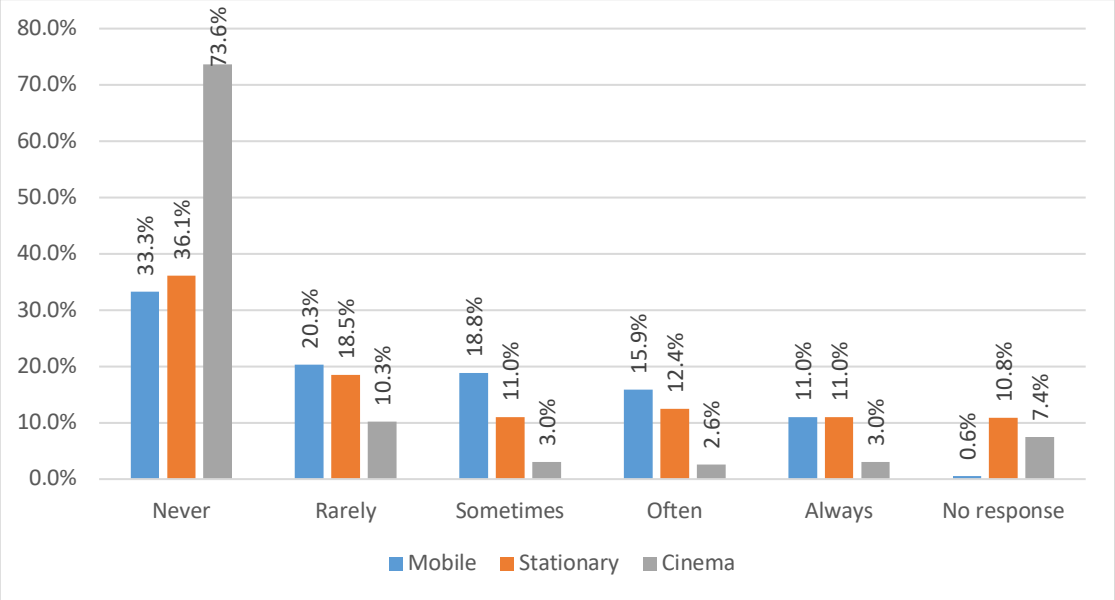

Figure 3-10 How often subttiles used mobile (n=345), stationary (n=498) and cinema (n=497)

Figure *3-11* illustrates the strong demand for dialogue enhancement.

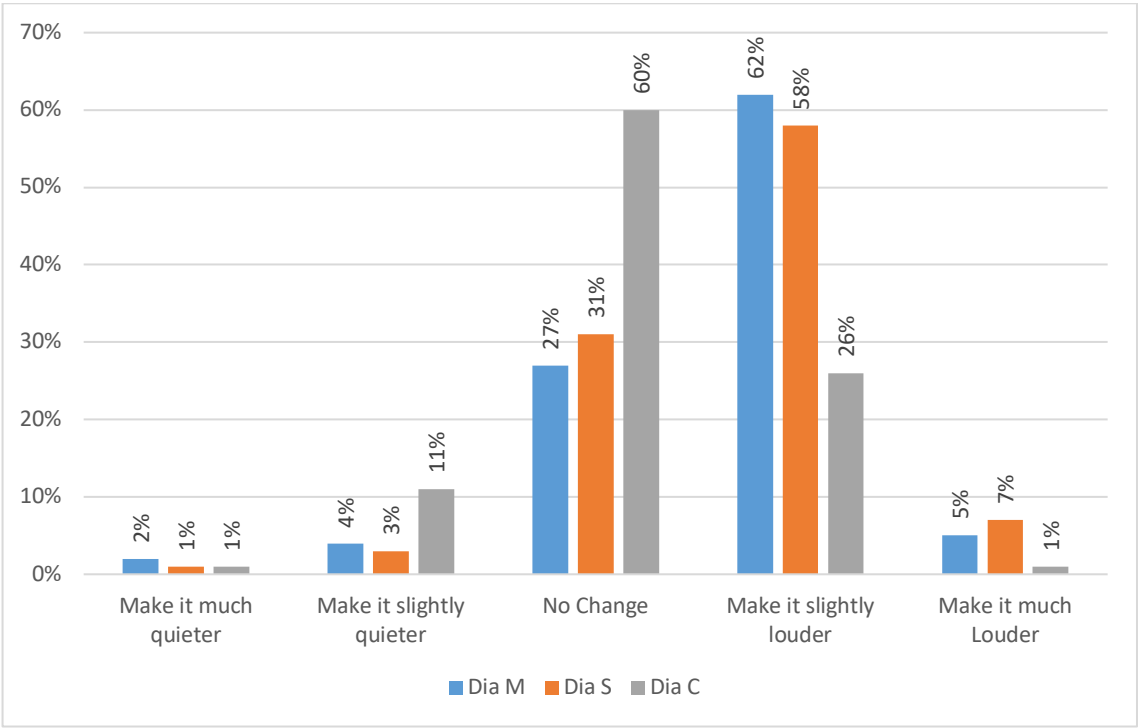

Figure 3-11 Dialogue levels

Figure 3-12 shows the requirement for the music stem, indicating how users attempt to manually balance the spectral competition between elements.

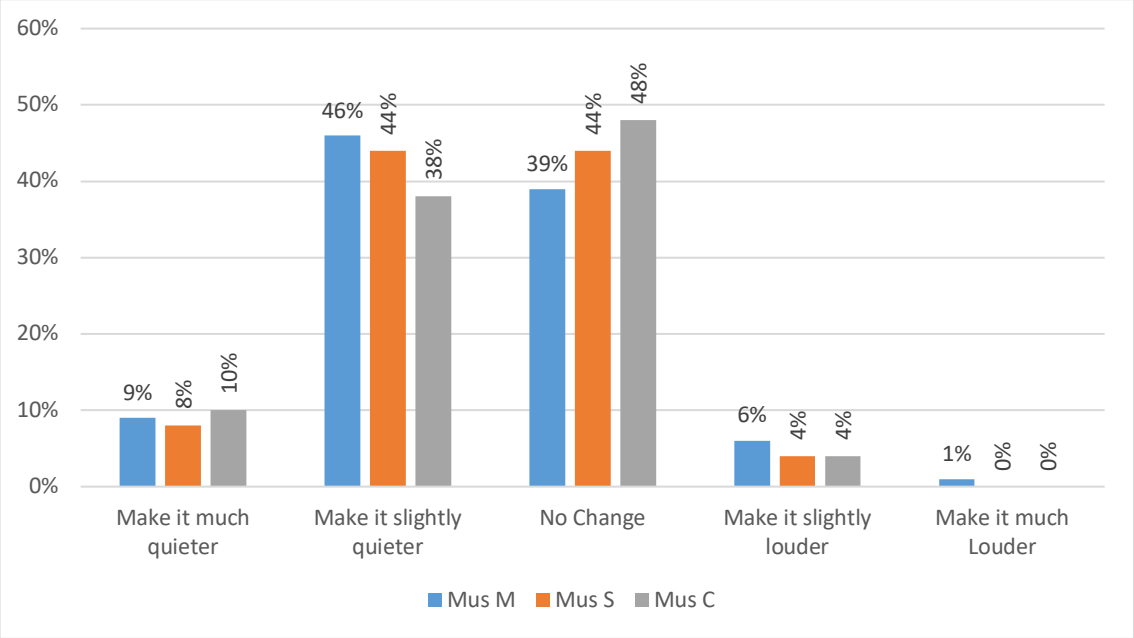

Figure 3-12 Music levels

Figure *3-13* details participant preferences regarding sound effects, confirming that non-narrative audio elements are frequently perceived as excessively loud in consumer environments.

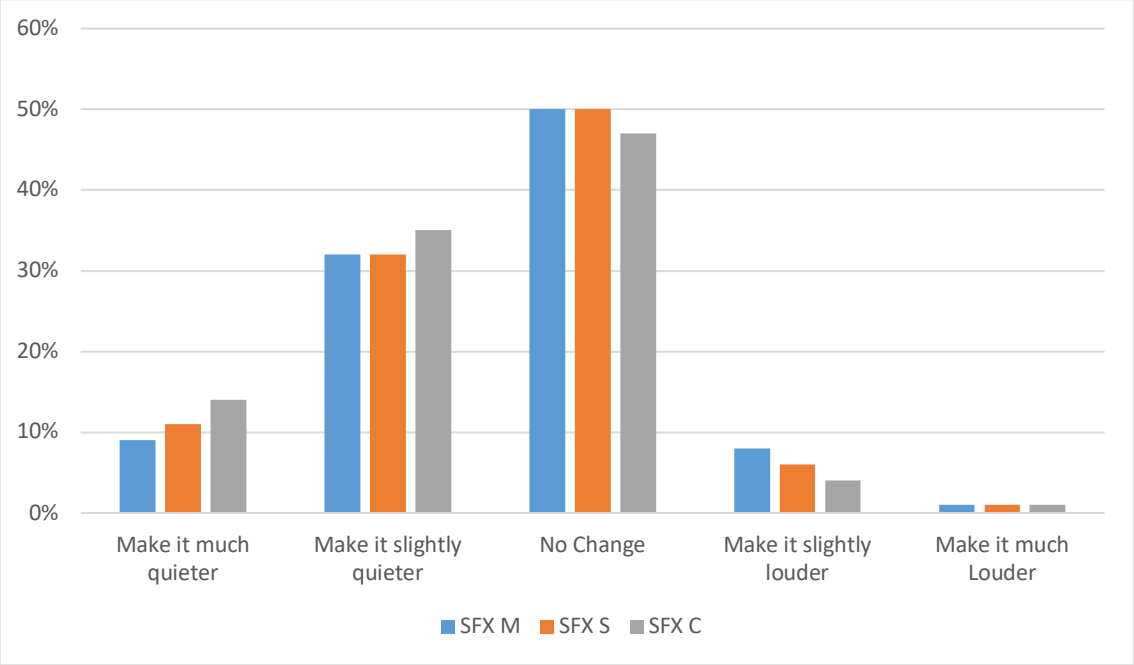

Figure 3-13 SFX Levels

3.3.5 Qualitative Drivers of Preference

To contextualise the quantitative data, open-ended responses were analysed using inductive content analysis. The recurring themes (Table *3-1*) illustrate the factors influencing user preference (Braun & Clarke, 2019; Elo & Kyngäs, 2008a; Gedemer, 2016b; Holcomb, 2016; Huberman et al., 2002; Thrane, 2022).

Key Themes Identified:

1. Control/Accessibility: Participants expressed frustration with the inability to hear dialogue without adjusting the volume during playback.
2. Convenience vs. Immersion: While immersion and sound quality were cited as reasons to prefer cinema, comfort and control were the primary drivers for mobile/stationary preference.
3. Creative Intent vs. Reality: Comments indicated that while users respect creative Intent, they prioritize intelligibility when watching in less-than-ideal environments.

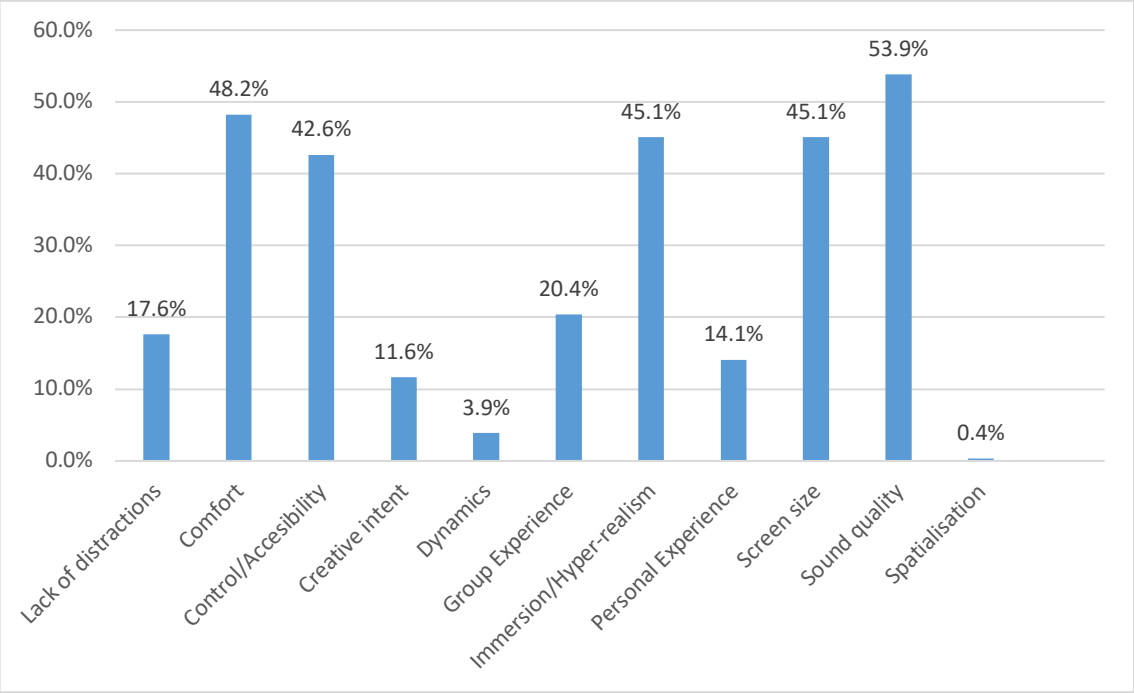

Figure 3-14 Comments

Table 3-1 Attributes - audience comments

Lack of distractions	Absence of anything that impedes audience from watching
Comfort	Convenience, a state of ease and freedom
Control/Accessibility	Ability to rewind, pause, turn on/off subtitles
Creative intent	original and purposeful vision of the filmmaker
Dynamics	Variation in volume, intensity, or loudness of a film soundtrack
Group experience	Collective/shared experience with a group while watching a film.
Personal experience	Individual/private experience in one's own environment
Immersion/Hyper-realism	Engagement/immersion in the film's world

Screen size	Visual quality, screen resolution and metric size
Sound quality	Type, size of speakers, and their layout
Spatialisation	Being in a 3-dimensional representation of a sound experience

3.4 Discussion

The survey results reveal a significant disconnect between the professional reference experience and the reality of modern consumption. While cinema remains a benchmark for immersion, the audience has largely migrated to mobile and stationary devices, prioritising convenience and control over absolute fidelity. This section interprets these findings through the lens of the research gaps identified in Chapter 2: the Standardisation Gap, the Context Gap, and the Agency Gap.

3.4.1 Hardware Variability and the Wild West of Playback

The results demonstrate a fragmented hardware landscape that defies the single-standard approach of cinema calibration.

Manufacturer Dominance: The clear split between Apple (Mobile) and Samsung (Stationary) preferences suggests that mixers are delivering content to two distinct sonic signatures. As shown in Figure 3-15, Figure 3-17, the frequency response of these devices varies, particularly in the low-frequency extension and presence regions.

The Implication: A mix balanced on a calibrated dubbing stage cannot possibly translate to both an iPhone EarPods (with its specific spectral roll-off) and a Samsung smart television (with its limited dynamic range). The standardisation gap is therefore not just a lack of rules, but a physical impossibility of optimizing a single delivery for such divergent hardware.

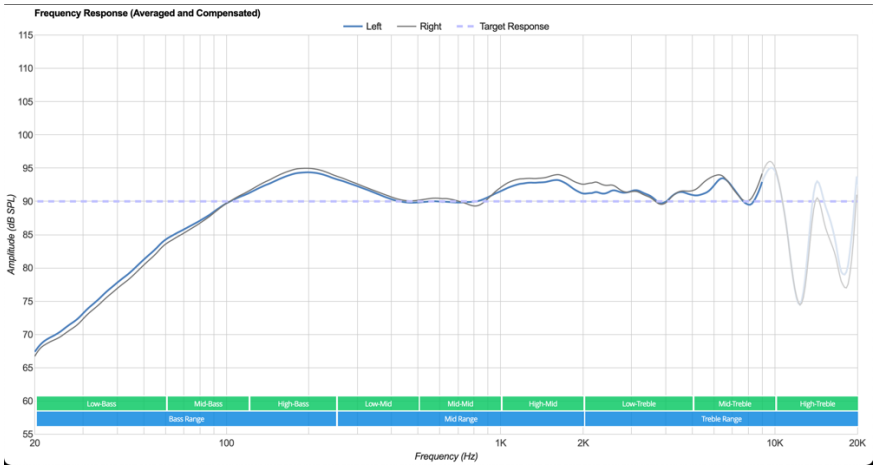

Figure 3-15 Apple Earpods frequency response (1st model) (Rtings, 2023).

As shown in Figure *3-16*, subsequent generations of mobile listening devices exhibit distinct frequency profiles, requiring the audio mix to translate across constantly shifting acoustic targets

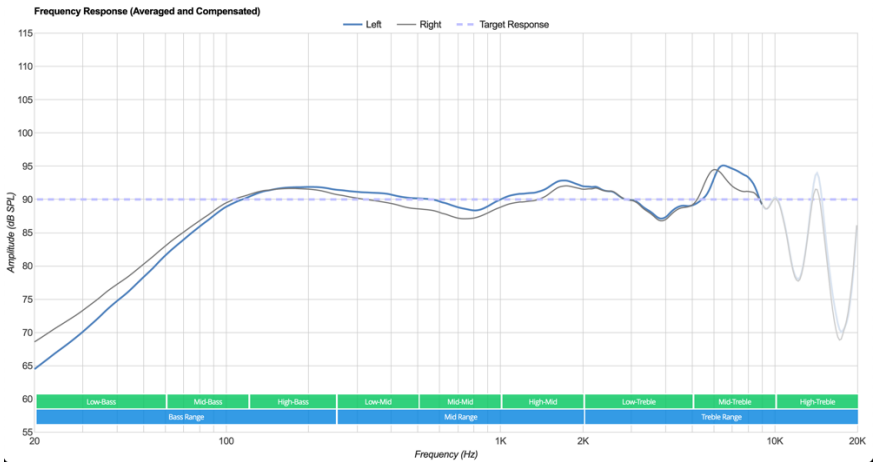

Figure 3-16 Apple AirPods frequency response (3rd generation) (Rtings, 2023).

Figure 3-17 provides an example of the frequency response differences between modern television sets, further confirming the impossibility of a single, static mix translating perfectly across all endpoints.

Product	SUPPORTED RESOLUTIONS Resolution	FREQUENCY RESPONSE Frequency Response	FREQUENCY RESPONSE Low-Frequency Extension	FREQUENCY RESPONSE Std. Dev. @ 70	FREQUENCY RESPONSE Std. Dev. @ 80	FREQUENCY RESPONSE Std. Dev. @ Max	FREQUENCY RESPONSE Max	FREQUENCY RESPONSE Dynamic Range Compression
TCL 5 Series/S546 2021 QLED	4k	graph	134.54 Hz	5.00 dB	4.81 dB	5.23 dB	88.2 dB SPL	3.67 dB
Hisense U6H	4k	graph	126.99 Hz	4.00 dB	3.99 dB	5.84 dB	87.9 dB SPL	4.45 dB
Hisense A65K	4k	graph	119.87 Hz	2.70 dB	3.25 dB	5.57 dB	86.0 dB SPL	4.96 dB
TCL Q6/Q650G QLED	4k	graph	142.54 Hz	5.75 dB	5.85 dB	6.30 dB	90.0 dB SPL	2.87 dB
Sony X90L/X90CL	4k	graph	95.14 Hz	2.66 dB	2.96 dB	4.81 dB	90.6 dB SPL	4.64 dB
Samsung CU7000/CU7000D	4k	graph	95.14 Hz	3.70 dB	4.09 dB	4.59 dB	81.1 dB SPL	3.35 dB
Sony X80K	4k	graph	106.79 Hz	1.98 dB	2.43 dB	5.47 dB	91.3 dB SPL	5.19 dB
Sony X75K	4k	graph	119.87 Hz	2.29 dB	2.26 dB	4.52 dB	91.7 dB SPL	4.91 dB
Sony X77L/X77CL	4k	graph	106.79 Hz	2.33 dB	2.52 dB	5.13 dB	86.0 dB SPL	5.84 dB
Samsung CU8000	4k	graph	119.87 Hz	2.68 dB	3.11 dB	4.70 dB	86.9 dB SPL	4.37 dB
LG UR8000	4k	graph	89.80 Hz	3.64 dB	3.78 dB	5.25 dB	87.7 dB SPL	3.62 dB
Hisense A6H	4k	graph	151.02 Hz	5.87 dB	6.03 dB	7.45 dB	90.0 dB SPL	3.07 dB
TCL QM8/QM850G QLED	4k	graph	89.80 Hz	4.74 dB	4.67 dB	5.09 dB	92.1 dB SPL	2.43 dB
Samsung S90C OLED	4k	graph	84.76 Hz	2.80 dB	3.63 dB	4.69 dB	86.1 dB SPL	4.78 dB
Sony X93L/X93CL	4k	graph	84.76 Hz	2.55 dB	2.45 dB	3.53 dB	90.8 dB SPL	3.57 dB
Samsung Q70/Q70A QLED	4k	graph	151.02 Hz	4.18 dB	6.09 dB	6.56 dB	83.4 dB SPL	3.96 dB
Sony X85K	4k	graph	100.79 Hz	1.90 dB	1.96 dB	4.84 dB	93.0 dB SPL	5.31 dB
Samsung Q60C [Q60, Q60CD] QLED	4k	graph	126.99 Hz	2.68 dB	3.08 dB	5.30 dB	88.1 dB SPL	4.32 dB
Samsung Q70C [Q70, Q70CD] QLED	4k	graph	142.54 Hz	2.94 dB	3.37 dB	5.23 dB	86.9 dB SPL	3.85 dB
Hisense U7H	4k	graph	95.14 Hz	2.73 dB	4.23 dB	6.69 dB	86.5 dB SPL	5.16 dB
TCL 5 Series/S555 2022 QLED	4k	graph	126.99 Hz	4.50 dB	4.45 dB	5.15 dB	84.1 dB SPL	2.25 dB
TCL Q7/Q750G QLED	4k	graph	100.79 Hz	3.22 dB	3.49 dB	5.15 dB	85.9 dB SPL	3.18 dB

Figure 3-17 Example of difference in response for TVs(Vafaei, 2018).

3.4.2 The Environmental Conflict: Immersion vs. Intelligibility

The data indicates that while sound quality is a driver for cinema attendance, comfort and lack of distractions drive mobile usage.

- The Noise Factor: The high prevalence of background noise in transport settings (69%) forces users into a compensatory behaviour: volume riding. As noted by Participant 13, the appeal of personal devices is the ability to "personalise it to what I prefer," contrasting with the fixed levels of the cinema.
- The Safety Paradox: The tendency to increase volume to overcome noise (masking) poses a hearing health risk. The survey confirms that users are manually fighting the dynamic range of the content to maintain intelligibility. This suggests that the current industry practice of wide dynamic range mixing for mobile is actively detrimental to user experience in noisy environments.

3.4.3 The Remix Demand: Evidence for Control

Perhaps the most critical finding is the audience's desire to alter the mix balance.

- Dialogue is the anchor element: The overwhelming preference (>60%) for louder dialogue and quieter music is not merely a preference; it is a functional requirement for intelligibility.
- Qualitative Validation: Comments from participants like P243 and P338 emphasize the desire for flexibility and control over volume. This directly challenges the director's intent argument often used to justify locking the mix. The results suggest that many participants prioritise intelligibility when consuming content outside ideal listening environments.

3.4.4 The Case for Adaptive Mixing

The convergence of these findings – hardware variability, environmental masking, and the demand for dialogue control – supports the need for the Adaptive Mixing Framework (AMPLE) proposed in this thesis. A static stereo mix, no matter how well produced, cannot adapt to the spectral limitations of an EarPod or the noise floor of a subway train. The solution must therefore shift from optimizing the file to optimizing the playback engine, granting the user the agency to adjust the narrative elements (dialogue) independently of the environmental elements (music/SFX).

3.5 Limitations

Certain limitations relating to sampling, representation, and self-reported behaviour should be considered when interpreting the findings presented in this chapter. Although the survey obtained responses from a relatively large participant pool across multiple geographic regions, the demographic distribution remained uneven, particularly in relation to gender representation. Consequently, the dataset may not fully reflect the diversity of listening behaviours, playback preferences, and media consumption practices present across the wider population of film audiences.

Geographic representation also remained weighted toward regions with greater access to streaming platforms, consumer playback technologies, and stable digital infrastructure. Listening practices within emerging economies, lower-income regions, or areas characterised by different patterns of media access may therefore be underrepresented within the dataset. Since playback environments, device availability, domestic noise conditions, and mobile media usage can vary considerably across economic and cultural contexts, broader international representation would strengthen the generalisability of the findings.

Interpretation of the results should additionally account for the limitations associated with questionnaire-based self-report methods. Responses concerning perceived dialogue clarity, immersion, subtitle usage, listening level, or environmental interference remain inherently subjective and may be influenced by individual listening habits, technical familiarity, memory recall, or personal interpretation of terminology used within the survey instrument. Accordingly, the findings are better understood as indicators of perceived audience experience

and behavioural tendency rather than objective measurements of auditory perception or acoustic performance.

Additional methodological limitations emerge from the adapted structure of the survey instrument itself. While many questions were informed by validated instruments and established listening research, the questionnaire was intentionally scoped and modified to address context-specific issues relating to contemporary film sound consumption. This adaptation improved relevance to real-world listening conditions and audience behaviours, although it also limits direct psychometric comparison with the original instruments from which individual measures were derived. Consequently, the survey should not be interpreted as a standardised psychoacoustic assessment tool, but rather as a context-sensitive instrument designed to identify recurring behavioural patterns and user requirements associated with contemporary film sound consumption.

Importantly, the purpose of this dataset was not to establish universal psychoacoustic models or statistically representative population-wide listening profiles. Instead, the survey was designed to identify recurring listening behaviours, contextual frustrations, audience requirements, and patterns of contemporary film sound consumption across varied listening conditions. Findings from this chapter should therefore be interpreted primarily as requirements-oriented evidence used to inform the development of the proposed adaptive mixing framework.

3.6 Design Implications for Framework Development

Based on the empirical requirements gathered in this chapter, this section distils the key findings into actionable design specifications. These implications do not introduce new data but rather translate the audience behaviours and frustrations identified in the survey into specific technical requirements for the Adaptive Mixing Framework (AMPLE) detailed in Chapter 5: .

1. Independent Dialogue Control
    a. Evidence: Users reported actively adjusting the global volume to fix intelligibility issues, often riding the remote during dynamic scenes.
    b. Design Implication: The findings suggest that independent dialogue adjustment may address a recurring audience need.
2. Environmental Context Awareness
    a. Evidence: Mobile users reported frequent frustration with ambient noise (transport/public spaces) masking the content, leading to unsafe listening levels.
    b. Design Implication: The system requires a mechanism to attenuate supportive elements (music/backgrounds) in response to noisy environments. This is to protect the anchor element (dialogue), so

it remains audible by improving the Signal-to-Noise Ratio (SNR) internally, rather than simply raising the global volume.

3.7 Summary

In this chapter, an online survey was described to understand audience listening habits and sonic preferences with regard to watching films. First, data were collected on participants' demographics such as age, gender, and region they live in. Next, they were asked about their preferred playback hardware and environment, this included questions on the types of devices they watch films on, levels and speakers they use. Participants were also asked to rate the sonic quality and their listening experience in different mediums, the next series of questions asked about any improvements they would like to do on different elements of a film soundtrack such as dialogue, music, or effects. Finally, a series of open-ended questions were asked during the survey to gain further insight into the answers given and to see how the data may be related.

A comparative analysis was carried out on various sections of data to get a complete understanding of audience demographics, playback environments and listening behaviour when watching films. The results show the variety of preferred playback hardware and listening devices by the audience for mobile and stationary settings. They also demonstrate the variety of preferred physical settings and their associated background noise as well as the diversity of the demographics and age groups. As such, one sized option could not address these, but rather a system that could be customised to individual needs.

The requirements identified in this section are derived directly from audience behaviours and listening experiences observed in the present study. Together, they define the user centred constraints that any adaptive solution must address in order to improve clarity and usability across varied listening contexts. The following chapter extends this analysis by examining professional production workflows and technical constraints, ensuring that audience needs are considered alongside industry practice. Taken together, these complementary perspectives provide the foundation from which the Adaptive Mixing Framework (AMPLE), presented in Chapter 5, is derived.

Chapter 4: Requirements Gathering-Professionals

Cinema has traditionally operated as a controlled delivery environment, providing directors and re-recording mixers with a high degree of confidence that the soundtrack will translate predictably across exhibition venues. Comparable assurances are sought within television and streaming production through delivery specifications and loudness standards (ATSC, 2013; BBC & DPP, 2020; Netflix, 2019), intended to support consistency for both audiences and content creators. However, these specifications do not fully account for the diversity of contemporary playback hardware, listening environments, or audience demographics, nor do they capture the practical constraints faced by professionals during the mixing process.

This chapter addresses that gap by examining the workflows, monitoring practices, and decision-making strategies of re-recording mixers working across cinema, television, and streaming contexts. Through direct engagement with professional practitioners, the chapter identifies the technical, creative, and operational factors that shape current mixing practice, providing a professional perspective that complements the audience-focused requirements established in Chapter 3. The findings presented here inform the design and evaluation of the adaptive mixing framework developed and tested in Chapters 5 and 6. In doing so, this chapter contributes to RQ1 by contextualising the translation challenges identified by audiences and directly informs RQ3 by examining how existing professional workflows may constrain or enable adaptive and personalised mixing approaches.

4.1 Background

Re-recording mixers, often referred to as dubbing mixers, play a central role in post-production sound by combining dialogue, music, and sound effects into a final soundtrack that reflects the director's artistic intent. This work is carried out in dubbing stages that are designed to replicate the geometry, acoustic behaviour, and calibrated playback conditions of commercial cinema auditoria.

Through adherence to established standards, both the dub stage and the cinema environment operate as controlled systems, offering a high degree of predictability in soundtrack reproduction. Within this context, mixers develop practices and decision-making strategies that are closely tied to the assumptions of calibrated loudness, stable acoustics, and fixed playback conditions. These conditions underpin professional confidence that a mix will translate consistently across cinema venues. Understanding this workflow is essential for examining how such practices may be challenged when content is later experienced through television and streaming platforms, where playback hardware, listening environments, and user control vary substantially.

4.2 Method

To identify current practices amongst dubbing mixers and gain contextual understanding of their decision-making process, an online survey was developed. This instrument investigated mixers' monitoring environments, speaker layouts,

and their specific workflow approaches for translating cinema content to streaming and TV standards.

4.2.1 Research Design and Sampling Strategy

Unlike the general population survey in Chapter 3, this study required a purposive sampling strategy to target a specific subset of expert practitioners. Recruitment focused on active industry professionals through specific guilds (e.g., CAS, AMPS), the Audio Engineering Society (AES), and verified professional social networks.

This targeted peer-to-peer approach successfully recruited established practitioners, including award-winning re-recording mixers and supervising sound editors. This ensured that respondents possessed the requisite technical knowledge to provide valid insights into commercial mixing workflows and delivery constraints. Consequently, the sample size (N=104) is considered appropriate for this specialized professional population.

4.2.2 Participants

A total of 104 professional mixers participated. The demographic data reflects the current gender disparity in the audio industry, with 94.2% identifying as male, 4.8% as female, and 1% undisclosed.

- Age: Mean age of 47 (Range: 18–70).
- Experience: Participants reported a mean of 22 years working in film sound, with a specific mean of 17 years as active re-recording mixers.
- Hearing Health: 73.1% reported no hearing loss, while 21.2% reported mild loss and 3.8% moderate loss.
- Geography: Participants represented 26 countries, ensuring a global perspective on delivery standards (see Figure 4-1)

Participants were anonymised using the convention MS1, MS2, etc., and no monetary compensation was offered.

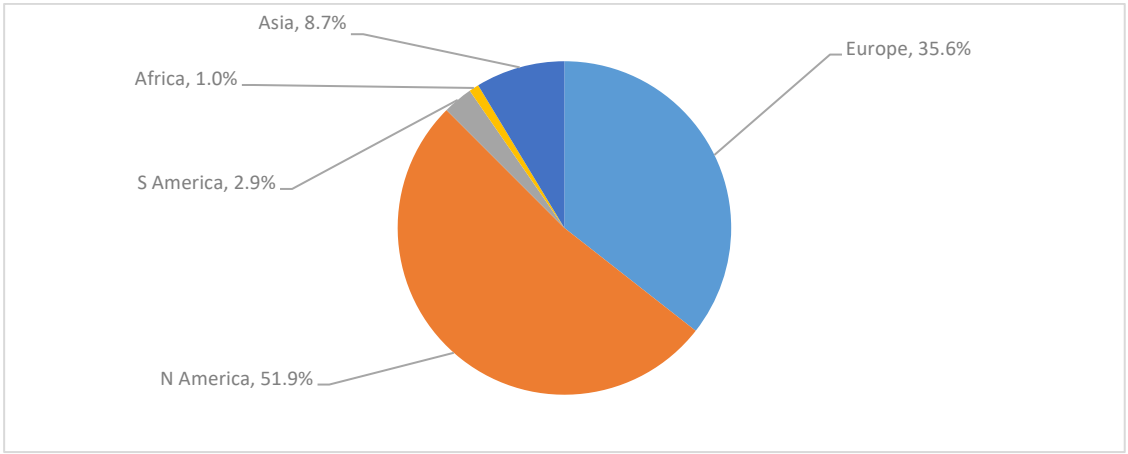

Figure 4-1 Geographic regions

4.2.3 Instrument Development

The survey instrument was modelled after previous literature concerning mixing approaches, monitoring layout, and content translation. Items were refined through iterative pilot testing with the supervision team and industry colleagues. The questionnaire addressed three primary constructs:

a. Monitoring Environment: Questions concerning room size, calibrated monitoring levels, and speaker distance were modelled after (Gedemer, 2015, 2016b; Ghouti, 2017).
b. Workflow & Translation: Items regarding the handling of stems (dialogue, music, effects) and dynamic range compression for television/streaming were based on workflows described by (Fonseca, 2015; Holman, 2012; B. Hughes & Wakefield, 2015).
c. Sonic Preference: Questions regarding the mixers' own preferences for entertainment playback were adapted from (Benjamin, 2002, 2004a; Oldfield & Shirley, 2011).

To minimise bias in self-reporting (Howard, 1980), questions used precise, technical language suitable for trained listeners.

4.2.4 Procedure and Data Analysis

Ethical approval was granted by the University's Research and Integrity Committee. The survey was hosted on NOVI and remained open for eight weeks (Novi, 2022). Full documentation is provided in Appendix B.

Data Analysis: Quantitative data, for example, room dimensions, monitoring levels, were analysed using descriptive statistics to identify standard industry practices. Open-ended responses regarding mixing strategies were analysed using Inductive Content Analysis, allowing for the identification of recurring themes in how professionals navigate the trade-off between creative intent and technical delivery specifications.

4.2.5 Survey

One hundred and four responses were collected. Upon clicking the survey link, participants were briefed with the information sheet, in which they read about the study and the survey questions. A mandatory consent form followed the information sheet, and by ticking 'I agree' on the form, participants were acknowledging and consenting to participate in the study. They were tasked to answer the questions to the best of their abilities, an option to correct their answers or continue with the survey later was available. The survey was made of up 55 questions total with the survey logic being designed in such a way as to keep participants from having to answer questions that were incongruous with answers from previous questions. A further comment section was also included

at the end of the survey, apart from the consent form, age was the only other compulsory question, the raw data was imported to Microsoft Excel for sorting and further descriptive and Inferential analysis. Descriptive analysis was utilised to measure mean, dispersion as well as frequency of events (Microsoft Corporation, 2026).

4.3 Results

The following section presents the results from the online survey (N=104). The findings detail professional workflows, including facility specifications, monitoring practices, and specific approaches to stem mixing for television and streaming.

4.3.1 Professional Experience and Workflow

Participants were asked to describe the primary facility where they conduct their work.

- Facility Size: As shown in Figure *4-2*, 31% of respondents (n=80) work in large dubbing stages (>15,000 cubic feet), while 17.5% reported working in rooms the size of a domestic living room or smaller.

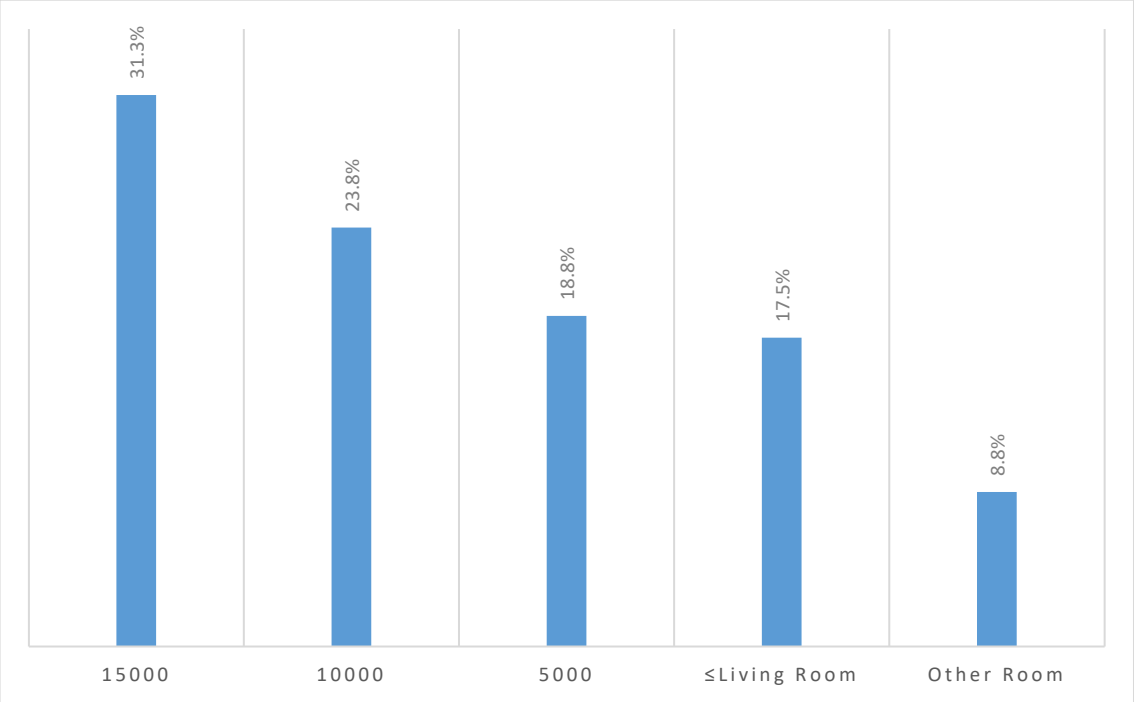

Figure 4-2 Work-in facility size

- Deliverables: When asked about creating dedicated mix passes for different formats (n=103), 68.9% reported yes. Figure 4-3 details the specific formats, with Streaming Platforms (61.2%) and Legacy TV (30.1%) being the most common secondary deliverables.

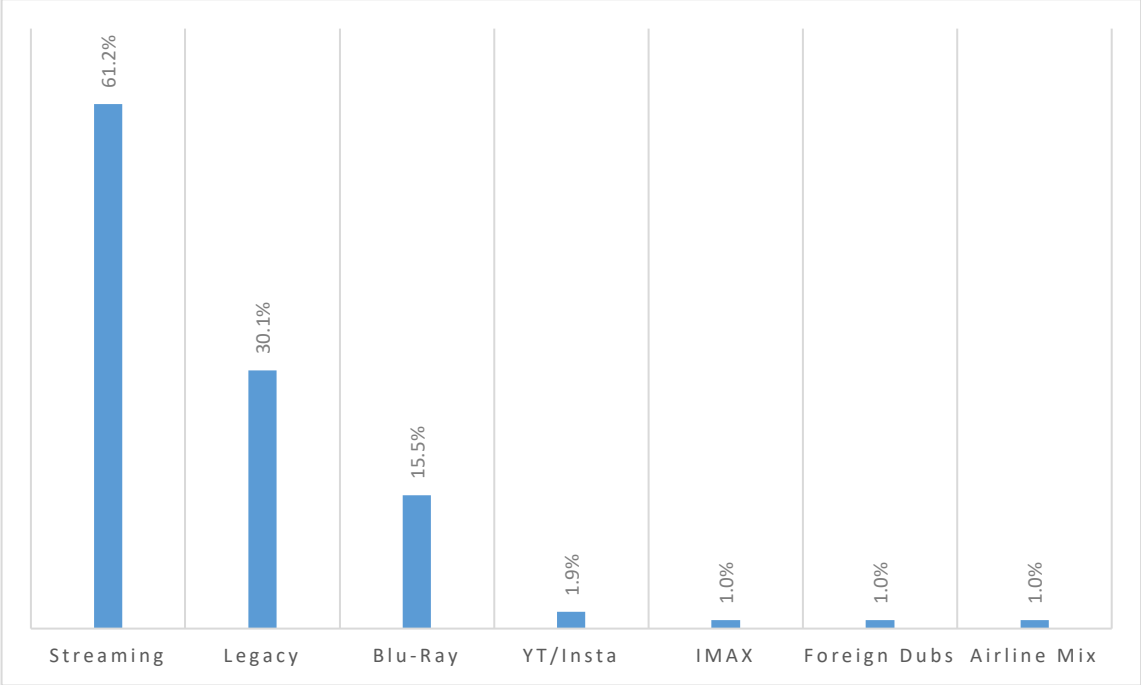

Figure 4-3 Different mix passes

Room Switching: Notably, 56% of respondents (n=67) reported moving to a smaller room (for example, a near-field suite) to mix non-theatrical passes, while 43.3% remain in the large dub stage. For streaming passes, the most common room size reported was equivalent to a domestic living room (<5,000 cubic feet), as illustrated in Figure *4-4*.

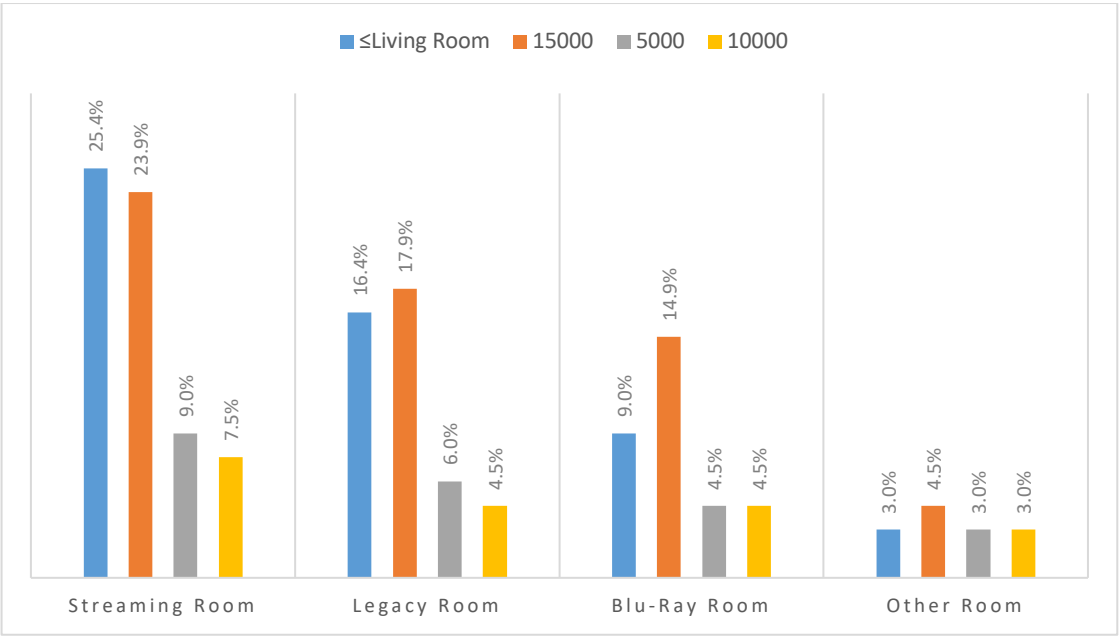

Figure 4-4 Room Size

4.3.2 Monitoring Practices and Room Correction

This section examines how mixers adapt their listening environment for non-theatrical content.

Layout Changes: Over 57% of participants (n=104) alter their monitor volume levels for TV/Streaming passes, and 35% change their speaker layout for example, switching from 7.1 to Stereo/5.1.

Room Correction: When asked about room calibration for television/streaming, 47.6% indicated using some form of correction. Of those, 40% utilise digital signal processing (DSP) solutions such as Trinnov or Dirac, while 28% rely on physical acoustic treatment (DIRAC, 2023; Trinnov, 2023).

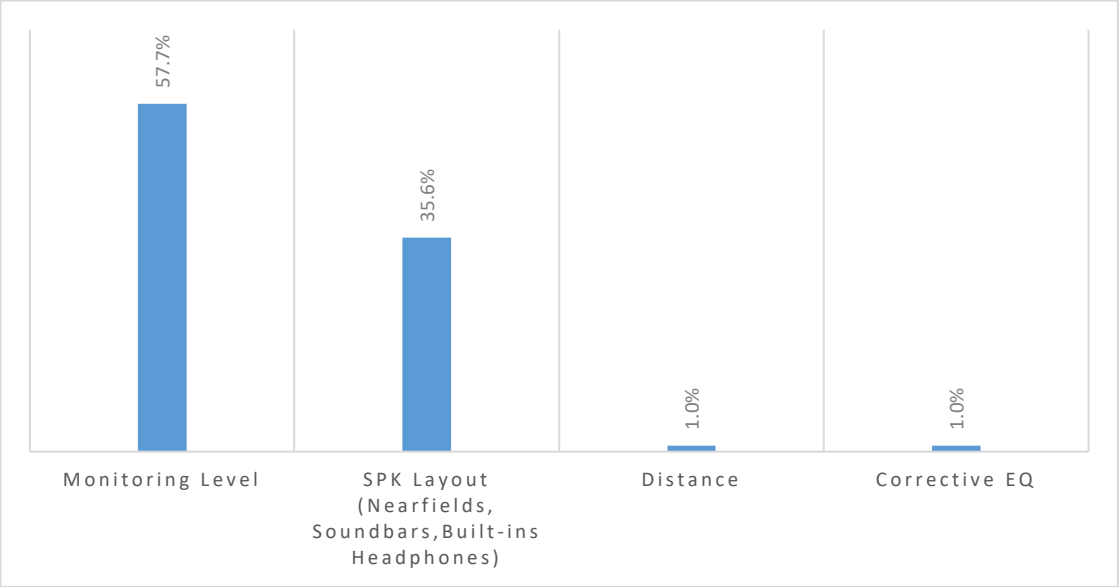

Figure 4-5 Speakers' layout

Reference Levels: Figure *4-6* shows the preferred monitoring levels for television/streaming mixes. The majority (65%) calibrate between 76–80 dB SPL, significantly lower than the cinema standard of 85 dB SPL.

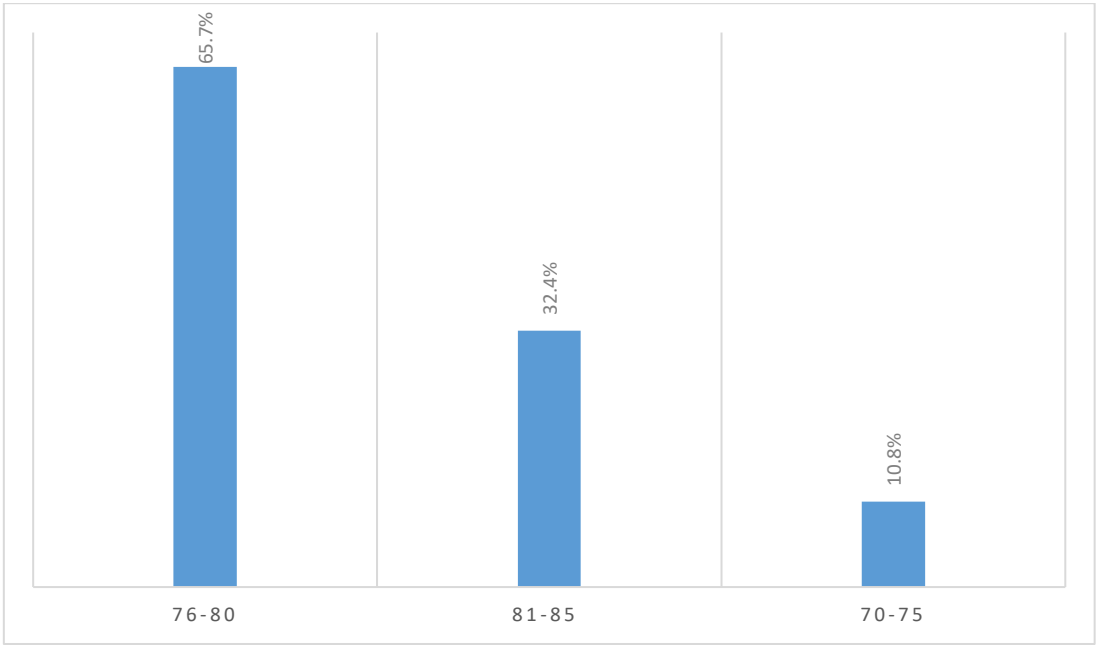

Figure 4-6 Monitor levels during mixing sessions

In conjunction with monitoring levels, the physical proximity of the mixer to the speakers influences perceptual decisions regarding the soundstage. Figure *3-7* outlines the mean physical distance established during these sessions.

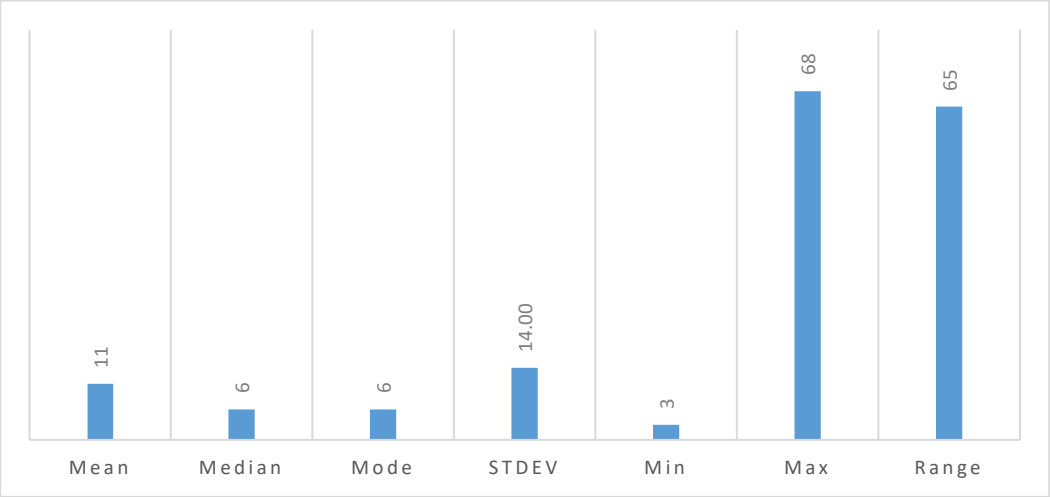

Figure 4-7 Mix position – distance from monitors

4.3.3 Mixing Approaches for Different Mediums and Stems

Participants were asked if they treat specific stems differently when mixing for Streaming/TV compared to Cinema (n=103).

- Dialogue: 63% indicated they approach the dialogue stem differently.
- Music: 54% reported a different approach for music.
- Effects: In contrast, 76% do not alter their approach for Foley, and 65% do not alter their approach for Spatial Effects.

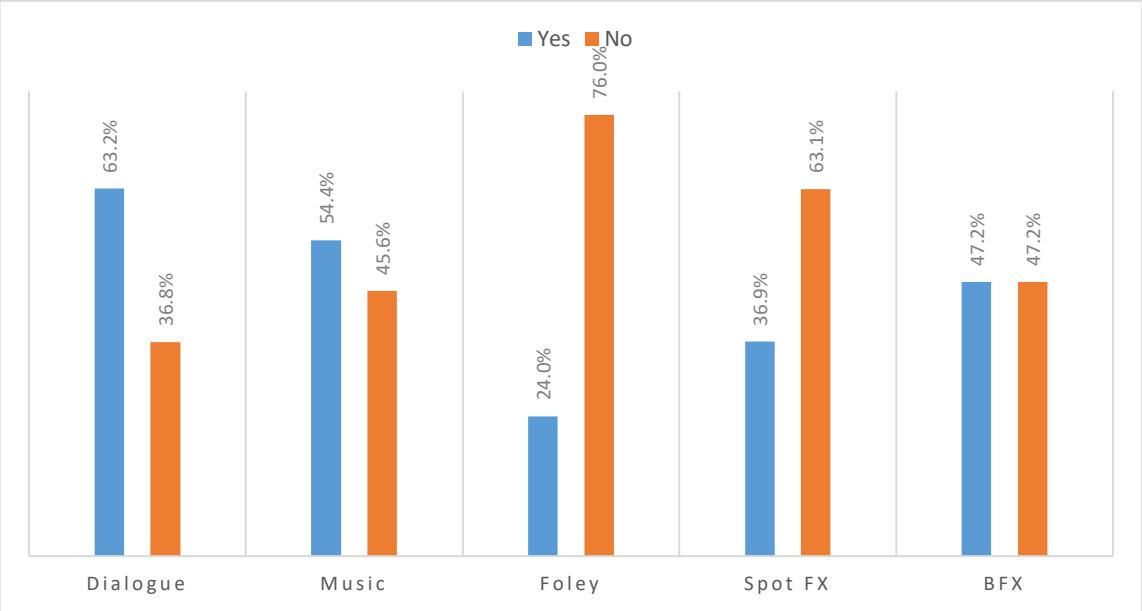

Figure 4-8 Stems approach (Mixers who change approach for non-cinema work)

Dynamic Range Processing: The most significant workflow change is in dynamics processing. As shown in Figure 4-9, 76% of participants reported applying

different dynamic range compression settings for TV/Streaming mixes compared to the theatrical version.

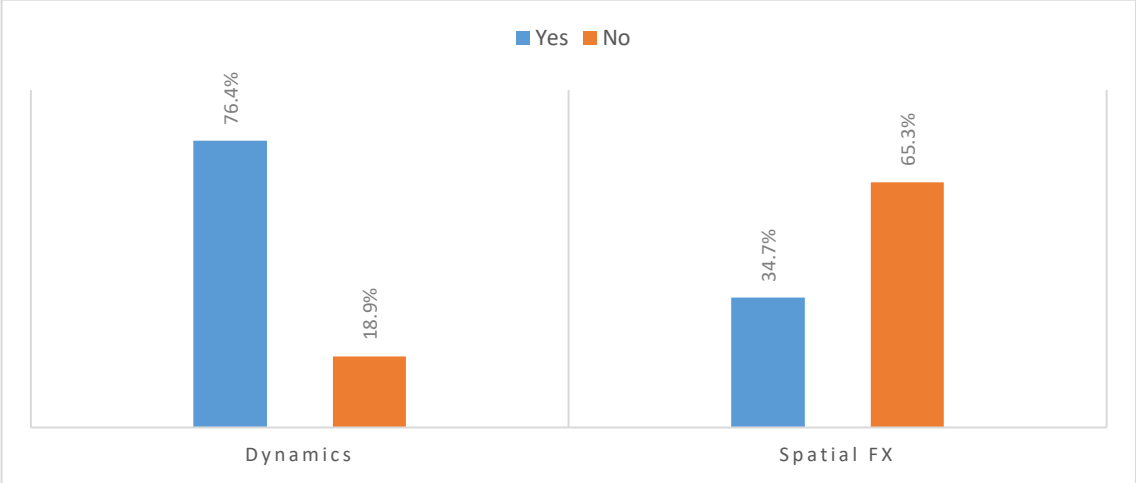

Figure 4-9 Stems approach (dynamics/spatial) Yes/No: Different approach to dynamics and spatial stems

4.3.4 Playback Devices, Entertainment Preferences, and Formats

- Consumer Checking: Over 80% of mixers (n=104) reported checking their mixes on consumer devices (e.g., soundbars, TVs) before final delivery.
- Screen Size Influence: 53% stated that the size of the mixing display affects their mix decisions, while 29.8% consider the target consumer screen size.

Personal Consumption: When asked about their own preferred format for entertainment (n=102), professional mixers favoured immersive formats:

- Dolby Atmos: 47% preferred.
- 5.1 Surround: 33% preferred.
- Stereo/Headphones: Less than 1% selected stereo or headphones as their preferred critical listening format.

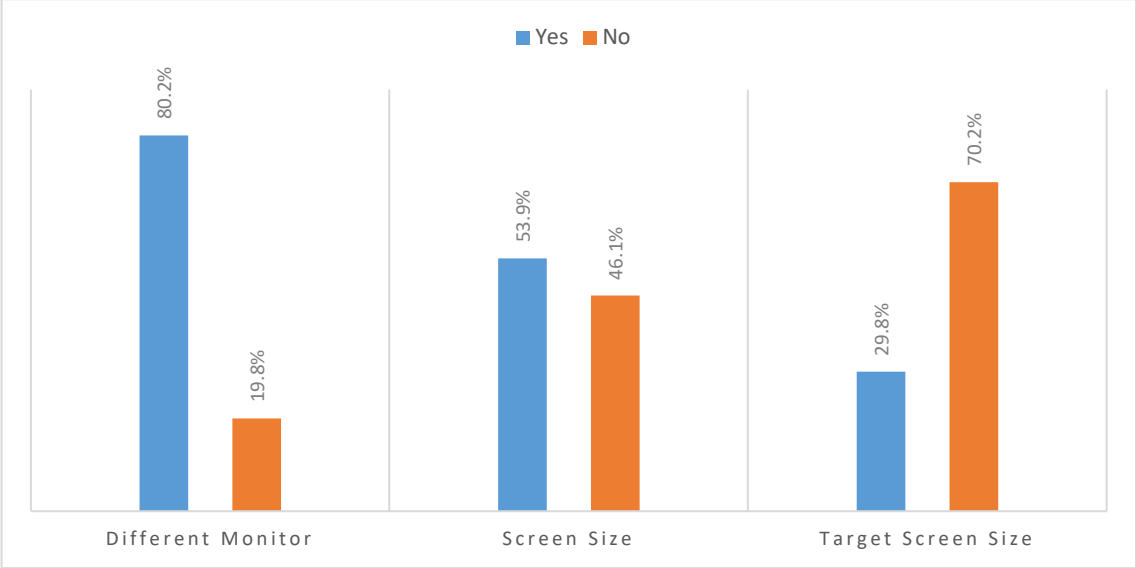

Figure 4-10 Yes/No: Influence of target screen, mixing environment, and consumer device checks

This awareness of consumer hardware limitations is further reflected in the professionals' own listening habits outside the studio. See Figure *4-11*.

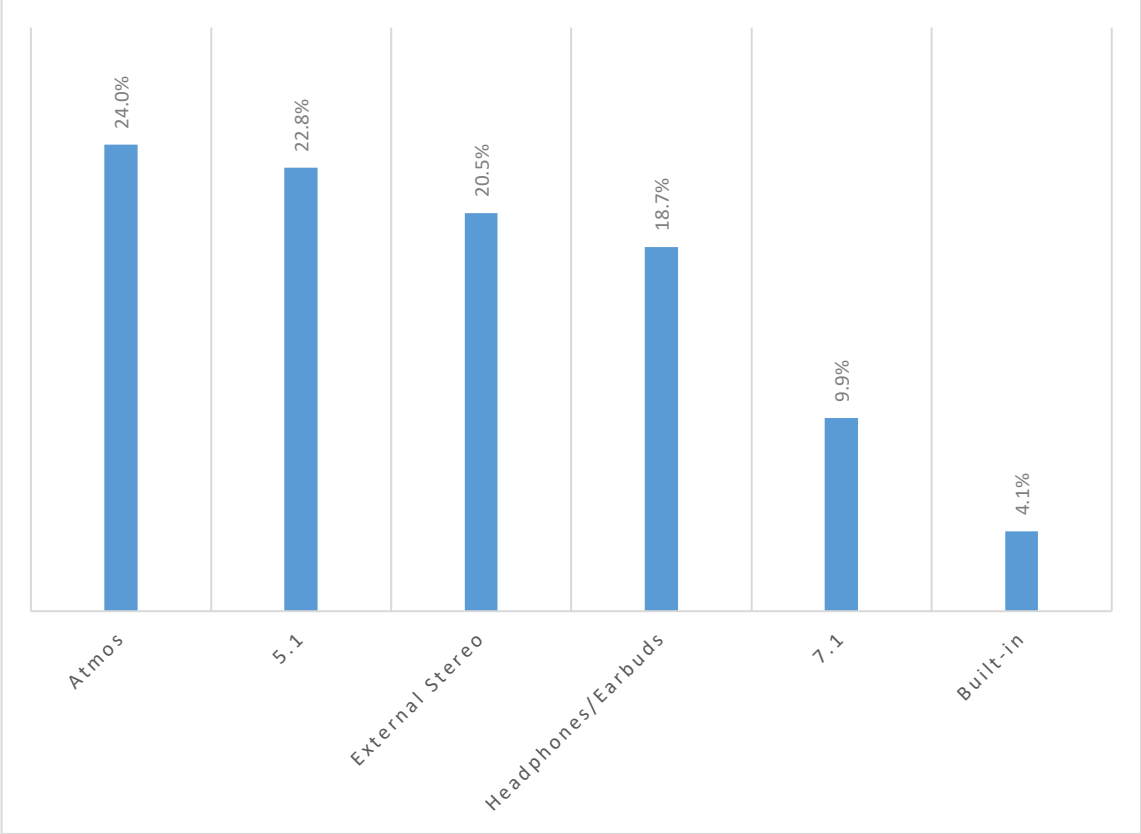

Figure 4-11 Playback format (entertainment)

When returning to the professional environment to build these soundtracks, the preference shifts to fully discrete, high-resolution formats. See Figure *4-12*.

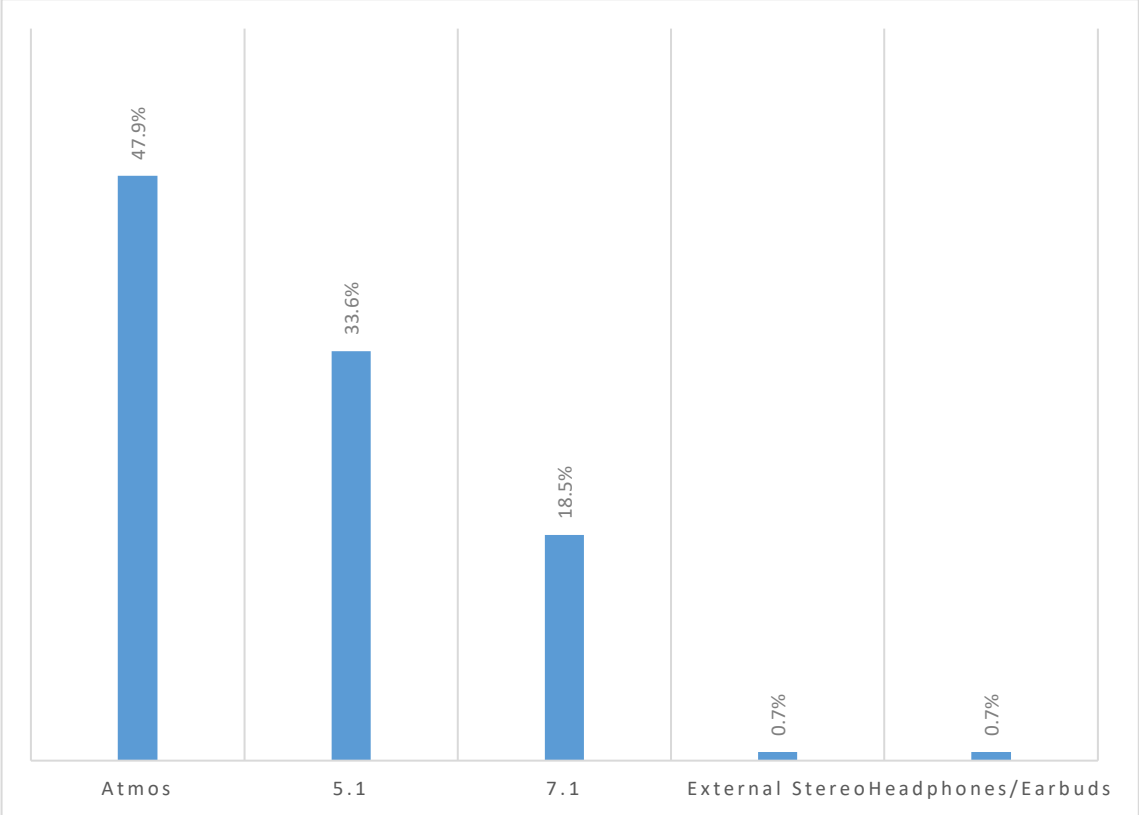

Figure 4-12 Preferred Mixing format

4.3.5 Qualitative Feedback: Workflow Improvements

Open-ended responses (n=85) regarding desired workflow improvements were analysed using Inductive Content Analysis. Responses were coded into themes such as schedule, standards, and translation as shown in Figure *4-13*.

1. Schedule Constraints (23.5%): The most frequent comment was that current schedules are insufficient to create separate, optimized mixes for every format.
2. Standards Enforcement (16.5%): Respondents indicated that current delivery standards are inconsistent or poorly enforced by streaming platforms.
3. Technology Adoption: Opinions on automated delivery technologies (e.g., Dolby Atmos automated downmixes) were split: 36% supported them as a solution for multiple deliverables, while 37.1% were opposed, citing a loss of creative control Figure 4-14.

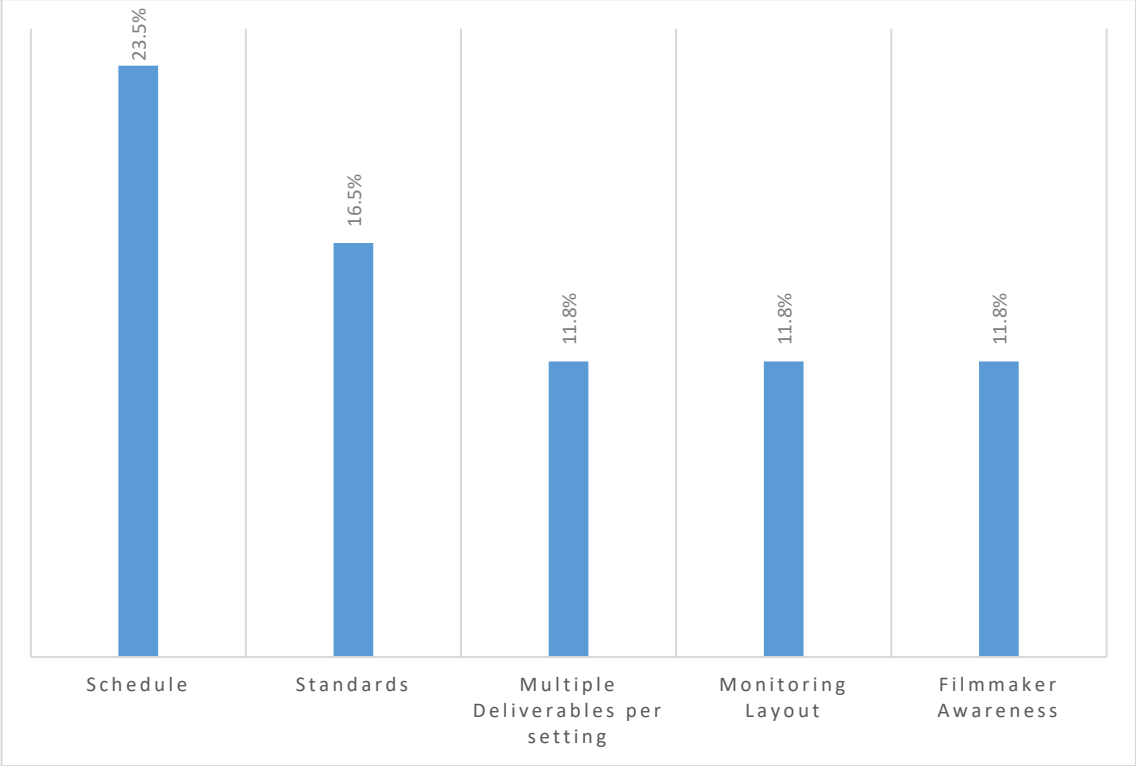

Figure 4-13 Improvements

To mitigate these scheduling and standards constraints, the industry is increasingly exploring automated software solutions and rendering algorithms. However, as Figure *4-14* illustrates, the adoption of these next-generation technologies remains polarized among practitioners.

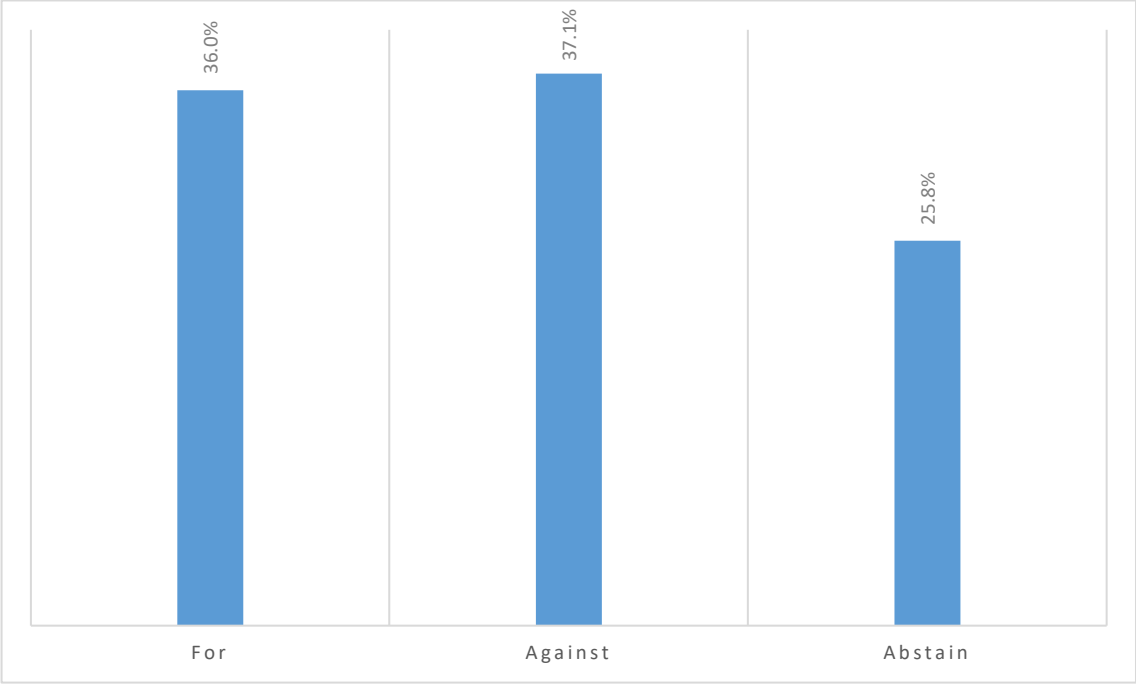

Figure 4-14 New generation technology for film mixing

4.4 Discussion

The survey results reveal a professional industry caught between the technical precision of the cinema and the highly variable reality of consumer playback contexts. While mixers possess the expertise to craft translations, they are constrained by time, budget, and the sheer volume of deliverables required. This section interprets these findings through the lens of the research gaps identified in Chapter 2.

4.4.1 The Translation Gap: Room Size and Monitoring

The results indicate a fundamental disconnect between creation and consumption environments.

- The Physics of Translation: With 31% of mixers working in large dubbing stages (>15,000 cubic feet), the acoustic translation to a consumer living room (<3,000 cubic feet) is physically compromised. As noted in the literature (Rumsey, 2012), mixes created in diffuse-field environments often lack clarity when played in the near-field conditions of mobile devices.
- The Living Room Pass: It is notable that 56% of mixers switch to smaller rooms for streaming passes. The widespread practice of checking mixes in smaller rooms suggests that translation across listening contexts remains an important professional concern.

4.4.2 The Stems Opportunity

One of the most significant findings for this thesis is the professional approach to stems.

- Independent Processing: The fact that 63% of mixers approach dialogue differently and 54% approach music differently for television/streaming indicates that stem-based processing is an established mental model for professionals. While not fully transitioned to an object-based paradigm, they clearly treat these core narrative elements as independent components.
- The Automation Gap: However, 76% of mixers reported simply applying global dynamic range compression to these stems. This brute force approach reduces dynamic range but does not solve the masking problem (for example, loud music burying quiet dialogue). The Adaptive Framework proposed in Chapter 5 addresses this by automating the level balance

between these stems based on the environment, rather than just compressing the peaks.

4.4.3 Constraints as a Driver for Automation

The qualitative feedback highlights that the barrier to better audio is not a lack of skill, but a lack of resources.

- The Schedule Bottleneck: 23.5% of respondents cited tight schedules as the primary reason for poor translation. They simply do not have the time to print a subway mix, a late-night mix, and a tablet Mix.
- The Solution: This finding is the strongest justification for the AMPLE framework. Since it is logistically impossible for a human mixer to author distinct mixes for every context, the only viable solution is to automate this personalization at the consumer endpoint.

4.5 Design Implications for the Framework

Based on these professional insights, the following design principles are established for Chapter 5:

- Respect the Stems: The framework must utilize the standard DME (dialogue, music, effects) structure that mixers are already comfortable with.
- Automate the Fader Ride: Since mixers cannot manually ride the faders for every consumer scenario, the system must algorithmically perform this function – boosting dialogue and ducking music for example – to mimic the professional's intent in a noisy environment.
- Preserve the Intent: By using the mixer's own stems rather than blind up-mixing, the framework maintains the creative integrity of the source material while adapting it for intelligibility.

The constraints identified here are derived from professional workflow practices and reflect the practical realities of contemporary film and streaming production. While Chapter 3 established audience-derived requirements centred on listening behaviour and user needs, the present chapter defines the operational boundaries within which any adaptive solution must function. Taken together, these complementary evidence streams provide the foundation for the Adaptive Mixing Framework (AMPLE), introduced in Chapter 5, which integrates user requirements with professional production constraints into a unified system design.

4.5.1 Limitations

Certain demographic and methodological limitations apply to this professional survey. While the sample size (N=104) captures a targeted cohort of experienced practitioners, it limits broader statistical generalisability beyond the professional contexts represented within the study. Respondent demographics also reflect established industry disparities, with participation heavily skewed toward male practitioners (94.2%) and geographically concentrated within North America and Europe (87.5% combined). Consequently, the findings primarily reflect the operational realities, production practices, and technical standards associated with post-production environments situated within these regions.

Additional limitations emerge from the purposive sampling strategy used to recruit guild-affiliated mixers and experienced industry practitioners. This approach may overrepresent workflows associated with higher-budget feature film and streaming productions when compared to independent, regional, or lower-budget broadcast environments. Accordingly, the findings should be interpreted as representative of specific sectors within contemporary professional post-production practice rather than the entirety of global mixing workflows.

Interpretation of the findings should also account for the reliance on questionnaire-based self-report methods. Responses concerning workflow constraints, translation practices, monitoring behaviour, and delivery requirements reflect participant perceptions and reported professional habits rather than independently verified production data. Although this approach provides valuable insight into practitioner experience and decision-making processes, variations in terminology, production workflow structure, and individual interpretation may influence the consistency of responses across participants.

The survey was designed to identify recurring workflow constraints, translation challenges, and practical production considerations associated with contemporary film and streaming audio delivery rather than establish universal production standards or definitive models of professional mixing behaviour. Findings from this chapter should therefore be interpreted primarily as requirements-oriented evidence used to inform the professional and operational constraints addressed within the proposed adaptive mixing framework.

4.6 Summary

The findings from this professional cohort provide direct justification for the design of the AMPLE Framework presented in Chapter 5 and inform the parameters of the mixing experiments in Chapter 6 and Chapter 7.

The survey indicates that while re-recording mixers are aware of translation challenges on mobile and domestic playback systems, budgetary and scheduling constraints limit their ability to produce device-specific or context-specific mixes. This constraint supports a user-centric approach, in which certain balancing decisions are deferred to playback rather than addressed exclusively during production. The framework therefore enables end-user control over selected stems, addressing intelligibility issues that cannot be resolved within existing professional workflows.

Dialogue intelligibility emerged as the most vulnerable element when soundtracks are reproduced on small speakers or in noisy environments. Professional respondents consistently identified dialogue as the primary stem requiring adjustment for non-cinematic playback. This finding underpins the prioritisation of dialogue separation within the framework and the subsequent experimental focus on independent stem control, allowing the narrative anchor to be adjusted without altering the overall mix balance.

More broadly, the results suggest that mixers already adopt different strategies for television and streaming delivery, particularly in relation to dialogue treatment and monitoring practices. However, these adjustments alone do not fully mitigate translation issues across diverse playback contexts. Combined with increasing production pressures, this reinforces the need for adaptive playback-side solutions that complement, rather than replace, professional mixing practice.

Chapter 5: The Adaptive Mixing Framework (AMPLE)

The preceding chapters identified a critical disconnect in the current film sound delivery ecosystem. Chapter 3 demonstrated that audiences frequently consume content on mobile and stationary devices in varying acoustic environments, often resulting in dissatisfaction with dialogue intelligibility and dynamic range. Simultaneously, Chapter 4 revealed that professional re-recording mixers, while aware of these translation issues, are constrained by legacy workflows, budgets, and schedules that prioritise theatrical standards over bespoke deliverables for every consumer context. This chapter therefore synthesises the audience-derived requirements identified in Chapter 3 with the professional production constraints established in Chapter 4, translating both into a unified adaptive playback architecture.

To bridge this gap, this chapter presents the adaptive mixing for a personalised listening experience (AMPLE) framework. This framework proposes a shift from the static, one-size-fits-all downmix model currently prevalent in the industry to a dynamic, user-centred rendering approach. The novelty of the AMPLE framework lies in its architectural integration of object-based audio, narrative prioritisation, and environmental context sensing, enabling adaptive rendering without altering established professional mixing workflows. By leveraging semantic metadata, the framework is designed to adapt the mix in real-time based on the listener's hardware capabilities, environmental context, and personal preferences.

This chapter defines the architecture of the proposed framework, detailing its inputs, processing logic, and output generation. This theoretical model serves as the basis for the experimental prototypes developed and evaluated in Chapters 6 and 7.

5.1 Rationale and Derivation

The development of the AMPLE framework is derived directly from the empirical findings of the audience and professional surveys.

5.1.1 The Audience Requirement

The data gathered in Chapter 3 indicates that listeners are forced into compensatory behaviours – frequently riding volume controls and relying on subtitles – to overcome environmental noise and dialogue masking. This active intervention suggests that a static mix often fails to maintain intelligibility in dynamic environments. Consequently, the framework must structurally separate dialogue from music and effects, treating them as distinct, adjustable objects to allow the user to recover the narrative without altering the global volume.

5.1.2 The Professional Constraint

Chapter 4 highlighted that while mixers create specific nearfield passes, it is logistically impossible to account for every permutation of playback hardware and acoustic environment (for example, a noisy bus commute versus a quiet bedroom). Since no production workflow can scale to address these infinite variables, the adaptation logic must inevitably shift from the production stage to the playback stage (rendering).

Therefore, AMPLE is designed not to replace the professional mixing process, but to operationalise the mixer's creative intent at the point of render, ensuring that the mix translates effectively regardless of the listening context.

5.2 Framework Architecture

The AMPLE framework is comprised of three core modules designed to function at the playback stage: the Input Module, the Adaptive Engine, and the Output Module. The architectural flow is illustrated in Figure 5-1.

5.2.1 Input Module

The Input Module ingests three distinct components: object-based audio stems specifically dialogue, music, and effects, metadata, which carries stem identification and detailed specifications, and environmental context data representing the listener's current acoustic environment. Functionally, the Input module produces a structured set of audio objects alongside a real-time environmental context parameter, which serves as the control signal for the subsequent processing stage.

5.2.2 Adaptive Engine

The adaptive engine serves as the decision-making logic of the framework. It adjusts the overall mix loudness and the individual stem loudness levels. These adjustments are calculated directly from the information gathered about the acoustic environment and the specific stem identifications provided by the Input module.

By monitoring these elements, if – for example – the acoustic environment masks the anchor element, dialogue, the engine applies loudness adjustments to the supportive stems music and effects to preserve intelligibility.

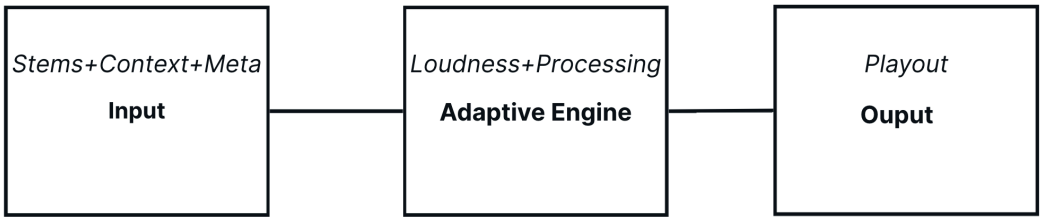

Figure 5-1 High-level architecture of the AMPLE Framework

5.2.3 Output Module

The output module executes the final audio rendering based on the instructions from the adaptive engine. Spectral correction, dynamic range compression, and

micro-level signal processing across the time and frequency domains are intentionally outside the scope of this framework. Instead, the system focuses exclusively on mix balance.

5.2.4 Example: The Commute Scenario

To illustrate the framework's operation during a noisy commute:

1. Input: The system receives the separated stems (dialogue, music, effects), the metadata carrying the stem IDs, and the acoustic context data detecting high background train noise.
2. Adaptive Engine: Based on the gathered acoustic environment information and the stem IDs, the engine calculates the required adjustments to maintain clarity.
3. Output: The engine adjusts the individual stem loudness – decreasing the music and effects while boosting the dialogue – ensuring intelligibility without simply raising the overall global loudness to unsafe levels.

5.3 The Role of Metadata

A critical innovation of the AMPLE framework is the operationalisation of metadata beyond simple spatial positioning. In current object-based workflows (for example, Dolby Atmos, ADM), metadata is primarily limited to technical rendering parameters such as azimuth, elevation, and divergence. The AMPLE framework proposes extending and operationalising this schema to include Narrative Importance tags, effectively adding a semantic layer that dictates how an object behaves under adverse listening conditions.

5.3.1 Semantic Priority Hierarchy

Rather than introducing a proprietary tagging system, the AMPLE framework leverages existing content classification metadata inherent in standardized object-based audio formats, such as the audio definition model (ADM) and Dolby Atmos (Dolby Atmos, 2015; EBU TECH 3364, 2014; Füg et al., 2016). Content creators already assign tags to audio objects during production. By parsing these standard tags, the adaptive engine maps the objects into a functional hierarchy of importance to dictate its suppression logic:

- Anchor Elements: Objects natively tagged as dialogue or speech are assigned the highest priority. These critical narrative elements must maintain intelligibility regardless of the acoustic environment.
- Focus Elements: Objects tagged as effects and music are assigned secondary priority. These supporting elements are attenuated by the engine only when the Anchor element is masked by environmental noise.

- Ambience Elements: Objects tagged as backgrounds or ambience are assigned the lowest priority. These contextual elements can be aggressively suppressed to preserve headroom and reduce spectral clutter during high-noise scenarios.

Dynamic vs. Static Assignment

Because this hierarchy relies on standard object metadata, content creators retain control over the adaptive logic. If a specific music cue serves a critical narrative function in a scene, the mixer can automate its tag or gain metadata within the ADM profile, which the adaptive engine will parse and protect accordingly. This ensures the machine acts strictly as an executor of the human mixer's intent.

5.4 Conceptual Positioning of the AMPLE Framework

To clarify the scope of this research, it is necessary to explicitly position the framework within the existing audio landscape. The AMPLE framework is not proposed as a new mixing format, a new loudness standard, or a replacement for established theatrical workflows. Instead, it is positioned as a playback-stage adaptation architecture that operates within existing production pipelines while addressing the translation gap identified in Chapters 3 and 4.

Current industry practice relies on static deliverables and downmixes strategies, even when object-based formats are employed. While object-based audio (OBA) provides technical flexibility, it is typically rendered according to predefined profiles that do not account for listener context, device variability, or user preference. As such, adaptation remains format-driven rather than context-driven.

The specific intellectual contribution of AMPLE lies in three integrated elements:

1. The relocation of adaptive control from production to playback. This acknowledges the finding from Chapter 4 that professionals cannot feasibly author bespoke mixes for every listening scenario, necessitating a systemic rather than manual solution.
2. The operationalisation of narrative-priority metadata as a control signal for real-time rendering decisions. This extends object-based paradigms beyond spatial placement into perceptual prioritisation.
3. The formal integration of environmental context, hardware capability, and user agency into a unified rendering logic, rather than treating these as separate accessibility or device-level features.

In this sense, AMPLE should be understood as an architectural framework for adaptive rendering rather than as a new aesthetic doctrine. It preserves creative

intent through metadata constraints while enabling context-sensitive personalisation at the endpoint.

5.5 Scope and Limitations

To ensure the contribution of this research is evaluated fairly, it is necessary to define the specific boundaries of the AMPLE framework. The framework is presented as an architectural and conceptual model for adaptive playback rendering, rather than as a complete commercial implementation. Its purpose is to demonstrate how object-based stems, narrative importance metadata, and contextual input signals may be integrated into a coherent adaptive system at the playback stage.

The framework assumes access to separated stems and structured metadata from the production workflow.

- Legacy Constraints: While such assets are standard in modern object-based formats for example, Dolby Atmos, ADM, this requirement limits immediate compatibility with legacy channel-based catalogues (e.g., 5.1/Stereo mixes) without additional preprocessing or source separation.
- Signal Processing: Environmental profiling is described at a functional level; detailed signal-processing optimisation, for example, specific DSP efficiency), privacy safeguards for microphone access, and device-level hardware integration fall beyond the scope of the present work.

The framework does not attempt to replace professional mixing practice or redefine creative authorship.

- It operates strictly within the existing stem structure authored by the mixer. Its aim is not to generate alternative mixes or remixes, but to preserve narrative clarity under variable listening conditions using bounded adaptive logic.
- The model is validated experimentally in controlled contexts in Chapters 6 and 7. Broader deployment scenarios, long-term user adaptation learning, and cross-platform standardisation remain areas for further investigation beyond this thesis.

In addition, the experimental validation associated with this framework was designed to evaluate behavioural response to discrete stem control rather than to provide a comprehensive usability or interface design assessment. While participants interacted directly with the prototype implementation, formal evaluation using established user-interface validation frameworks was not undertaken, as the research focus centred on perceptual prioritisation within the adaptive architecture. Statistical analyses were conducted to examine defined

behavioural relationships within the experimental configuration, and findings should therefore be interpreted within the scope of the specified hypotheses. The prototype was not benchmarked against existing commercial adaptive or streaming rendering systems, and conclusions relate specifically to the implementation developed for this research.

5.6 Summary

This chapter has articulated the AMPLE framework as a structured response to the empirical findings presented in Chapters 3 and 4. It translated audience requirements for improved intelligibility and contextual control, alongside professional constraints within existing production workflows, into a playback-stage architecture comprising defined input streams, adaptive processing logic, and context-aware rendering. By preserving stem separation and extending metadata to encode narrative priority, the framework reallocates adaptive control to the point of consumption while retaining creative influence at the production stage.

The following chapters evaluate specific components of this framework to test whether it can reconcile professional mixing practice with the diversity of contemporary audience listening contexts:

Chapter 6 evaluates the framework's reliance on separated stems and priority levels under controlled listening conditions. By isolating its functional behaviour, it tests how users adjust specific individual stem loudness (dialogue, music, effects) to maintain intelligibility.

Chapter 7 evaluates the framework's contextual input by introducing environmental variability. It assesses it under realistic listening constraints, testing the adjustments required by the adaptive engine when simulated acoustic noise is introduced to the environment.

Chapter 6: Prototype 1- Short Film

This chapter presents the first empirical evaluation of the AMPLE framework. Having defined the conceptual architecture and control logic in Chapter 5, the present chapter operationalises those principles in a controlled experimental setting. The purpose is not to test environmental robustness at this stage, but to examine whether independent stem control improves perceived balance and intelligibility under ideal listening conditions.

Prototype 1 therefore establishes a behavioural baseline. By isolating participants from the variability of consumer hardware and background noise, the study evaluates whether user-adjustable object separation produces measurable preference shifts when compared to a fixed mix. This controlled phase allows the core assumptions of the framework to be assessed without confounding variables.

The chapter outlines the experimental rationale, stimulus preparation, interface design, participant protocol, and evaluation measures. The findings provide the first stage of validation for the AMPLE approach. Chapter 7 then extends this investigation by introducing environmental noise and contextual variation, thereby testing the framework under more ecologically complex conditions.

6.1 Background

The preceding chapters established two interrelated issues: audience dissatisfaction with fixed mixes on personal devices and professional constraints that limit the feasibility of producing multiple context-specific deliverables. While Chapter 5 articulated a conceptual framework to address this gap through adaptive rendering and stem-level control, the assumptions underpinning that framework require empirical validation under controlled conditions before environmental complexity is introduced.

Prototype 1 is therefore designed to examine the behavioural premise of the framework in isolation. The experiment does not attempt to replicate real-world listening variability at this stage. Instead, it seeks to determine whether independent control over dialogue, music, and effects results in systematic preference shifts when compared to a fixed reference mix, provided that hardware quality and ambient interference are held constant. Establishing this baseline is essential in order to determine whether user agency at the stem level meaningfully alters perceived balance and intelligibility.

The controlled setting utilises calibrated reference monitoring to minimise acoustic variability. This allows the experiment to focus exclusively on the perceptual and behavioural dimensions of user-adjustable mixing. The intention is not to evaluate environmental robustness, which is addressed in Prototype 2, but to test whether the core architectural assumption of AMPLE, namely that decoupled stem control enhances perceptual alignment with listener preference, produces measurable outcomes.

Drawing directly from the requirements identified in Chapters 3 and 4, three design constraints informed the experimental interface. Independent stem separation preserves dialogue as a controllable anchor element distinct from

music and effects. Interaction is limited to level adjustment in order to isolate balance preference from processing variables such as equalisation or compression. Visual labelling is minimised to reduce cognitive bias, encouraging participants to rely on auditory judgement rather than semantic cues. Together, these constraints ensure that the experiment evaluates the fundamental behavioural mechanism proposed by the framework, rather than the influence of interface complexity or environmental confounds.

Accordingly, the experiment is structured to determine whether participant-driven adjustments produce consistent and measurable deviations from the fixed reference mix, thereby providing empirical support for the core behavioural assumption of the AMPLE framework.

6.2 Method

This experiment emulated a professional re-recording mixer's workflow, adapted for a non-expert audience via a custom touch interface. The objective was to determine how general audiences balance narrative elements when given direct control in a reference listening environment.

6.2.1 Participants

A total of 25 participants were recruited via academic networks, professional groups, and social media. The cohort represented a diverse general audience rather than a specialized audio engineering group.

- Demographics: Ages ranged from 18 to 70 (Mean = 43, Range = 42). Gender distribution was 64% male and 36% female.
- Hearing Health: 80% reported no hearing loss, 16% reported mild loss, and 4% reported moderate loss. All participants possessed normal or corrected-to-normal vision.
- Ethics: The study received full ethical approval from the University Research and Integrity Committee. Participants provided informed consent and received no monetary compensation. (See Appendix C for documentation).

6.2.2 Stimulus Material

The auditory stimulus was a 3-minute excerpt from the independently produced drama Samray (Samray, 2022).

- Content: The narrative depicts a tense confrontation involving a toy gun, addressing themes of racial injustice.
- Selection Criteria: This specific clip was selected for its high dynamic range and dense spectral content, relying on a complex interplay between

dialogue, Foley, and ambient backgrounds, making it an ideal candidate for testing intelligibility and balance preferences.

Figure 6-1 Film Screenshot 01:00:59:00

6.2.3 Listening Environment and Apparatus

The experiment was conducted in the University's Auralisation Suite (Prokofieva et al., 2014), a medium-sized critical listening room with a reverberation time (RT 60) of 215ms (0.5–2 kHz). See Figure *6-2*

Figure 6-2 Experimental apparatus ENUAuralisation suite

- Playback System: Audio was reproduced via a stereo pair of Genelec 8030a monitors positioned in an equilateral triangle relative to the listener (Genelec, 2026). The system was calibrated to 79dB(C) RMS to establish a standard baseline of reference.
- Audio Engine: A custom Max/MSP patch running on a MacBook Air M2 managed the audio playback via Ableton Live (Ableton, 2022; Apple Inc, 2024; Cycling74, 2022). Max/MSP was selected for its ability to handle low-latency Open Sound Control (OSC) data.

6.2.4 Interface Design

Designing an audio interface for a non-expert audience requires balancing professional functionality with accessibility. Alternative interface options typically found in consumer software – such as rotary knobs, multi-parameter equalizers, or semantically labelled sliders – were briefly considered. However, introducing complex visual representations or text-based semantic labels can induce cognitive overload or anchoring bias in untrained listeners, distracting from the primary task of critical auditory evaluation (Mycroft, 2017). Therefore, the interface was intentionally restricted to a simplified, unlabelled vertical fader topology. This design emulates the established physical workflow of professional mixing consoles while minimizing interaction friction for general users.

To achieve this, participants controlled the mix via an iPad Mini running a custom touch interface:

- Topology: The interface utilized five vertical faders (Range: +12dB to −∞) to emulate the established topology of professional mixing consoles, providing a standard physical reference for level adjustment.
- Unlabelled Fader Strategy: The five faders were assigned randomly, default interface colours (red: dialogue, grey: Foley, pink: music, yellow: SFX, green: backgrounds). Importantly, the faders were unlabelled on the interface. The colours carried no semantic meaning; they were chosen randomly to prevent visual bias (for example, boosting a fader simply because a text label reads dialogue). This unlabelled design forced participants to engage in active listening – identifying stems by manipulating the fader and observing the auditory result.

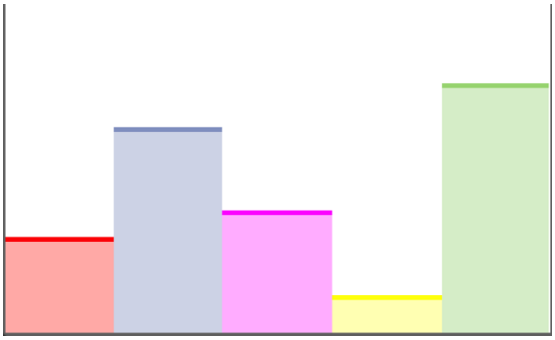

Figure 6-3 Experiment user interface

6.2.5 Procedure

The experiment followed a structured protocol:

1. Briefing: Participants read the information sheet and signed a digital consent form.
2. Training: A researcher demonstrated the interface. Participants performed a non-recorded 'dummy round' to familiarise themselves with fader sensitivity and stem separation.
3. The Task: Participants watched the film excerpt and were instructed to adjust the levels of the five stems to your personal listening preference. No target levels were provided and starting levels were randomised within ±3 dB to prevent anchoring bias.

4. Repetition: Participants could loop the film up to three times to refine their mix.
5. Exit Survey: Upon completion, participants completed a brief digital survey (Appendix C) designed to capture their demographic profile, listening habits, and task comprehension. Rather than relying on a standardized usability scale, the survey asked direct, targeted questions, including demographic inquiries (for example, what is your age?, what is your gender?), media consumption habits (for example, which streaming service do you prefer?). The questionnaire was formed by adapting the demographic and viewing habit constructs originally developed for the larger survey in Chapter 3.

6.2.6 Data Analysis Strategy

All fader movements were logged in real-time within Max/MSP. Data processing (fader gain values and survey responses) was conducted in Microsoft Excel.

- Quantitative Analysis: Descriptive statistics (Mean, Standard Deviation) were used to analyse fader positions. Bivariate correlation and regression analyses were performed to investigate relationships between participants' reported habits (for example, mobile usage) and their mixing decisions.
- Qualitative Analysis: Free-text comments regarding the interface were analysed using Inductive Content Analysis to identify recurring themes regarding usability and learnability.

6.3 Results

The following section details the analyses conducted and the results of the listening tests (N=25). The findings are presented in three stages: demographic validation, system usability, and the core analysis of participant mixing decisions.

6.3.1 Participant Demographics and Experimental Setup

The demographic profile of the experimental cohort broadly reflected the behavioural tendencies observed in the large-scale survey (Chapter 3).

- Viewing Preferences: Preferences were evenly distributed, with 52% indicating a preference for stationary devices and 48% favouring the cinema. This balance supports the suitability of the sample for evaluating user-adjustable mixing behaviour.

- Platform Usage: Netflix was the most frequently reported platform (>50%), while services like Discovery and Flex were selected by fewer than 5% (Figure *6-4*).
- Cross-Platform Behaviour: When asked whether they watch content across multiple platforms, responses were evenly divided (52% No, 48% Yes).

These measures suggest that the participant group reflects common contemporary viewing habits rather than a specialised cohort.

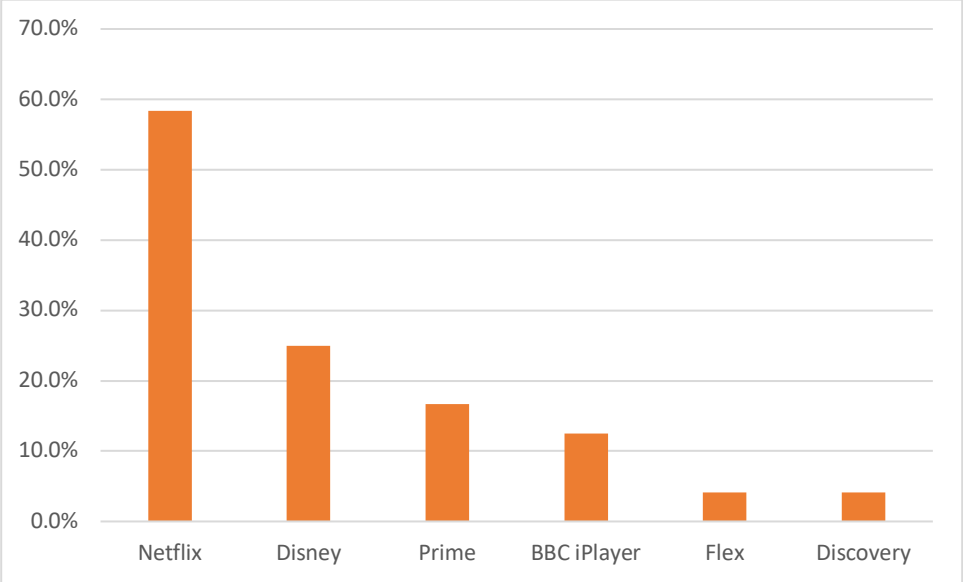

Figure 6-4 - Preferred Service

6.3.2 Audience Perception of Soundtrack Clarity

Participants were asked about the clarity of soundtracks on their current streaming services. While 50% rated current mixes as clear, a significant minority (36%) rated them as unclear or neither clear nor unclear Figure *6-5*, reaffirming the motivation for this study.

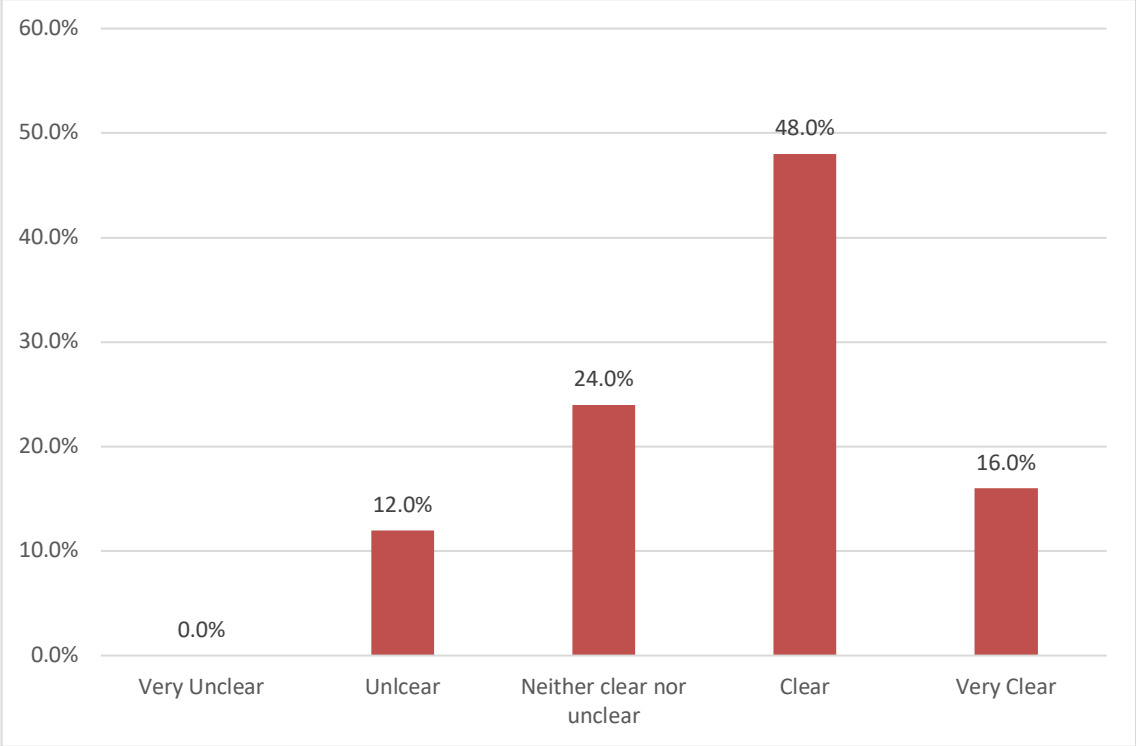

Figure 6-5 - Streaming Clarity

6.3.3 Task Comprehension

Because the objective of this study was to measure loudness balancing behaviour rather than commercial software design, standard interface evaluation frameworks, such as the System Usability Scale (SUS), were not employed. Instead, a targeted post-experiment survey (see Appendix C) asked direct questions regarding task execution, such as "How would you rate the system you just used in the experiment?" to ensure participants understood the mechanical assignment. 70% of respondents indicated the system was easy to use, while 20% noted they would use such a feature frequently (Figure *6-6*).

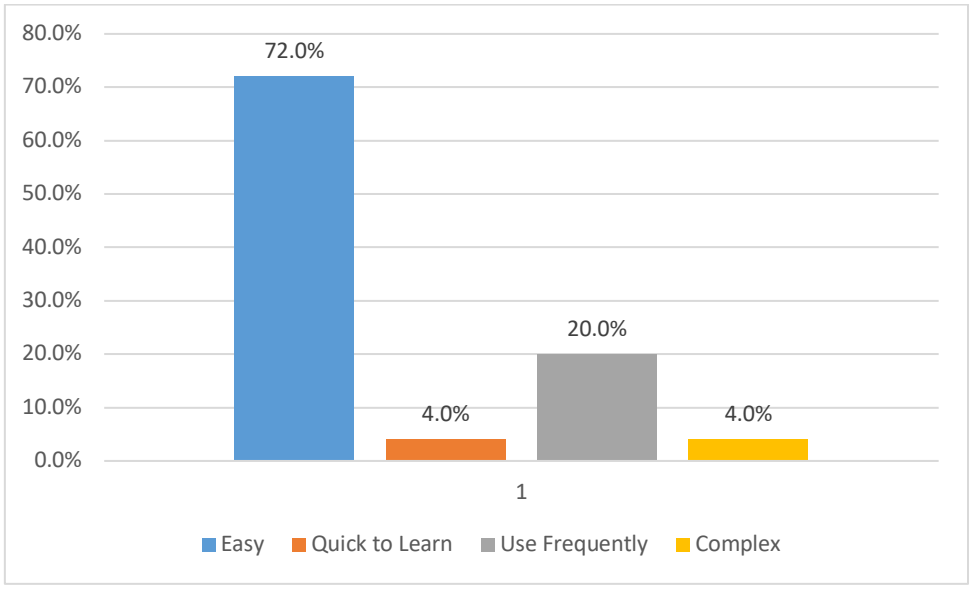

Figure 6-6 – System Interaction Rating

To contextualise these multiple-choice responses, participant free-text comments were analysed by assigning thematic tags to recurring statements and calculating their frequency – inductive content analysis (Elo & Kyngäs, 2008b). This allowed specific behavioural themes to emerge directly from the data (Figure 6-7)

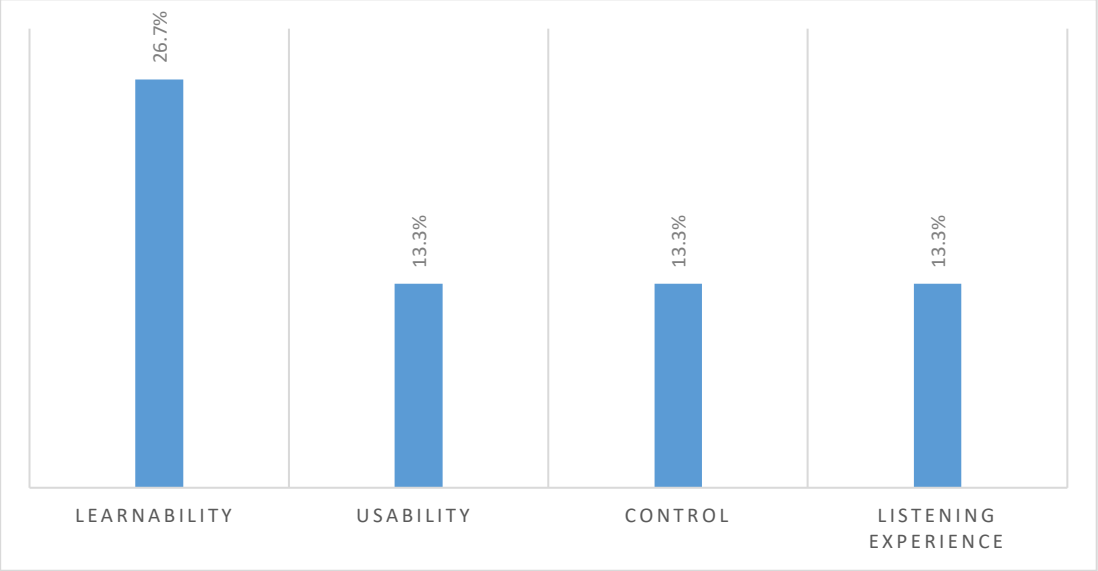

Figure 6-7 – Classification of participant comments

Table 6-1 – Attribute definitions - participants comments

Learnability	The capacity and the ability of a product to be learned
Usability	How easy a system is to use, and how well it allows users to accomplish their goals
Control	Ability to adjust loudness levels for each element
Listening Experience	Referring to self-contained event where a listener constructs meaning from what they hear when watching films.

6.3.4 Analysis of Participant Mixing Preferences

The core analysis focuses on the loudness levels set by participants for the five stem elements. The data reveals a distinct hierarchy of narrative importance.

Descriptive Statistics (Loudness) : The mean loudness (LUFS) and True Peak (TP) values for each stem are summarised below.

- Dialogue (The Anchor): Exhibited the tightest consistency among participants.
  - Mean: -18.4 LUFS (SD = 6.52).
  - Range: 28.2 dB.
- Music: Showed greater variability, often set significantly lower than dialogue.
  - Mean: -32.1 LUFS (SD = 9.97).

- Foley & Effects: Showed the highest variance, indicating these are subjective flavour elements rather than critical narrative anchors.
    - SFX Range: 130.0 LUFS (due to outliers).
    - Backgrounds Range: 126.9 LUFS.

Outlier Analysis:

Notable outliers were observed in the supportive stems. For example, Participant 7 completely silenced the backgrounds stem (−∞ dB), while Participants 1 and 11 significantly boosted sound effects. This extreme variability confirms that a one-size-fits-all mix cannot satisfy all users.

The data indicates that for dialogue, Foley, and music, there is a preferred range of levels that audiences tend to favour, however, the same cannot be concluded for sound effects and background elements. Furthermore, the observed variation suggests that a single metric cannot account for the preferences of every audience member.

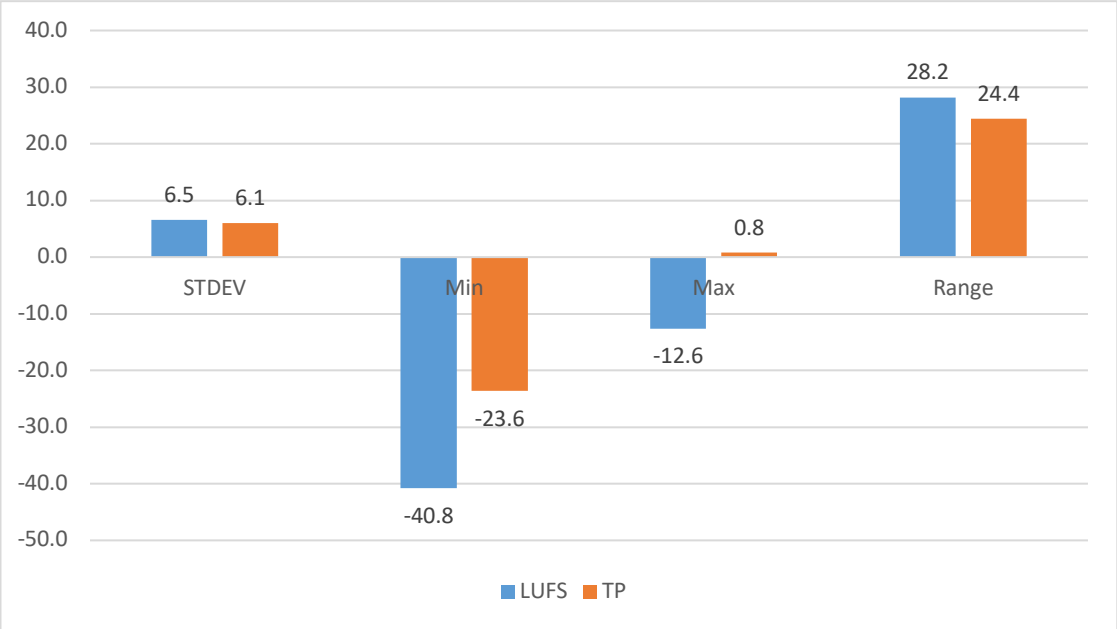

Figure 6-8 - Dialogue Levels - Prototype-1

While dialogue levels maintained a tight distribution, the functional layer of Foley introduced a slightly wider range of listener preference, reflecting its role as a supportive physical element (
Figure *6-9*).

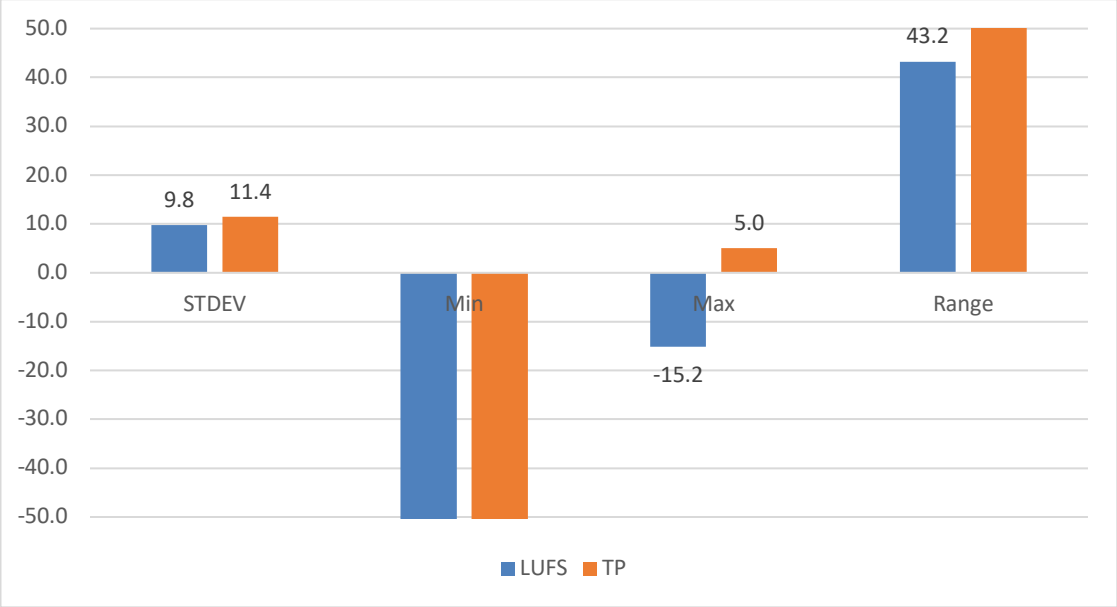

Figure 6-9 - Foley Levels - Prototype-1

Beyond the physical sounds, the musical stem exhibited greater variability, as participants actively balanced it against the dialogue track (
Figure *6-10*).

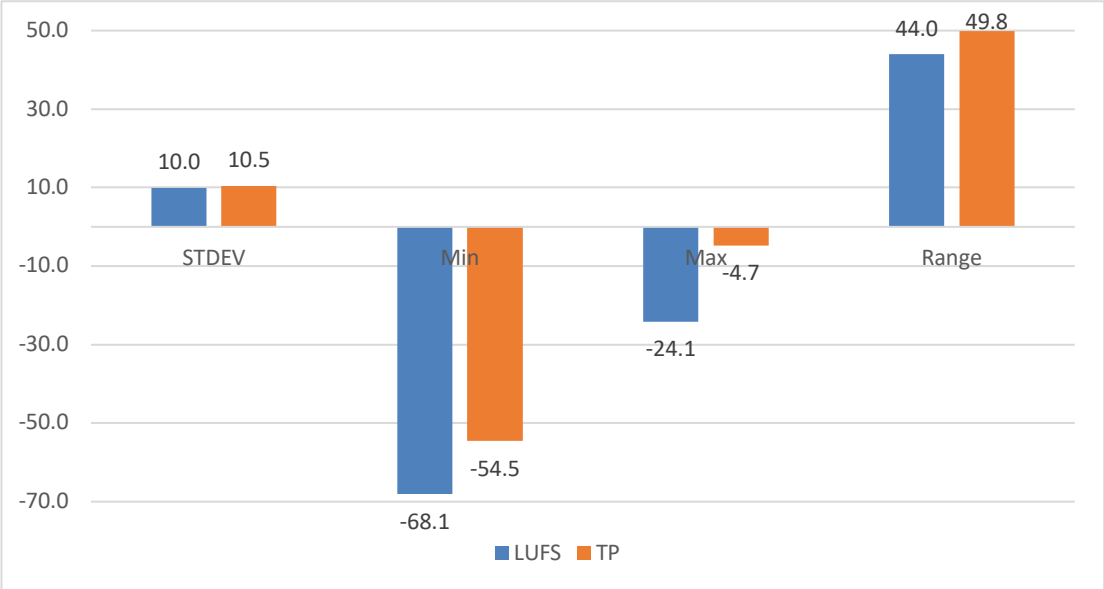

Figure 6-10 Music Level - Prototype-1

The most extreme variance was observed in the remaining supportive stems. Sound Effects demonstrated significant subjective spread (Figure 6-11).

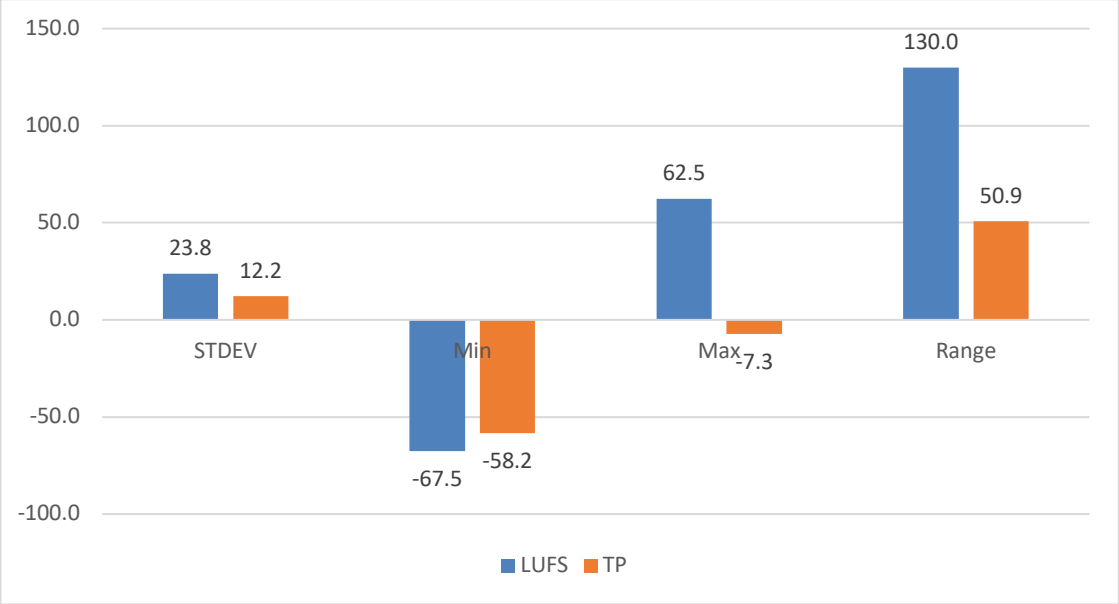

Figure 6-11 Sound Effects Level - Prototype-1

Similarly, backgrounds were subject to extreme adjustments (Figure *6-12*).

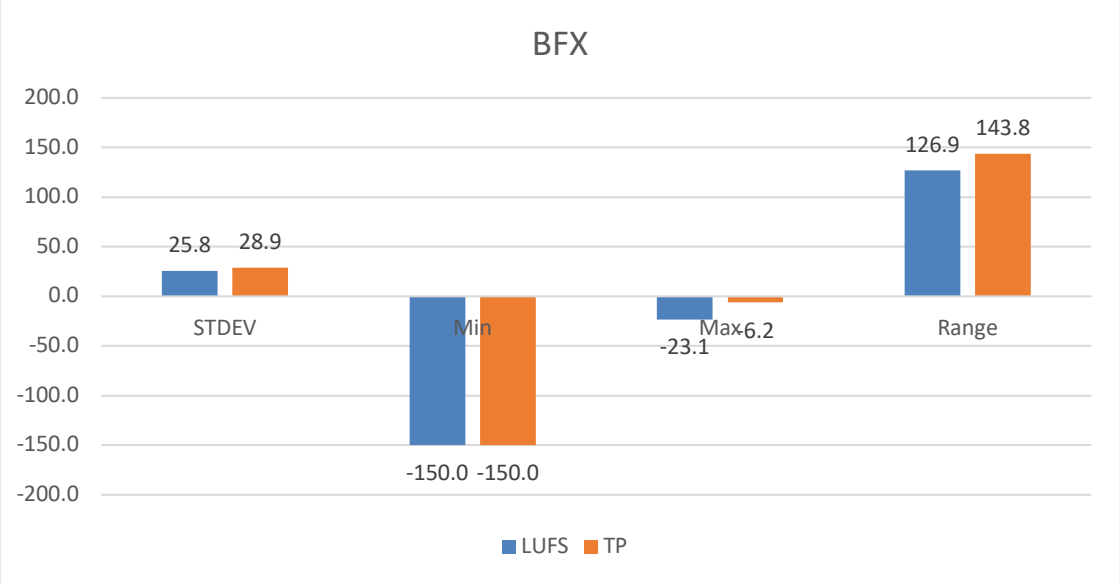

Figure 6-12 Background Effects Level - Prototype-1

6.3.5 Variability and Correlation of Sound Element Levels

A correlation matrix, and a boxplot, see Figure *6-13* and Figure *6-14* were generated to examine relationships between stems.

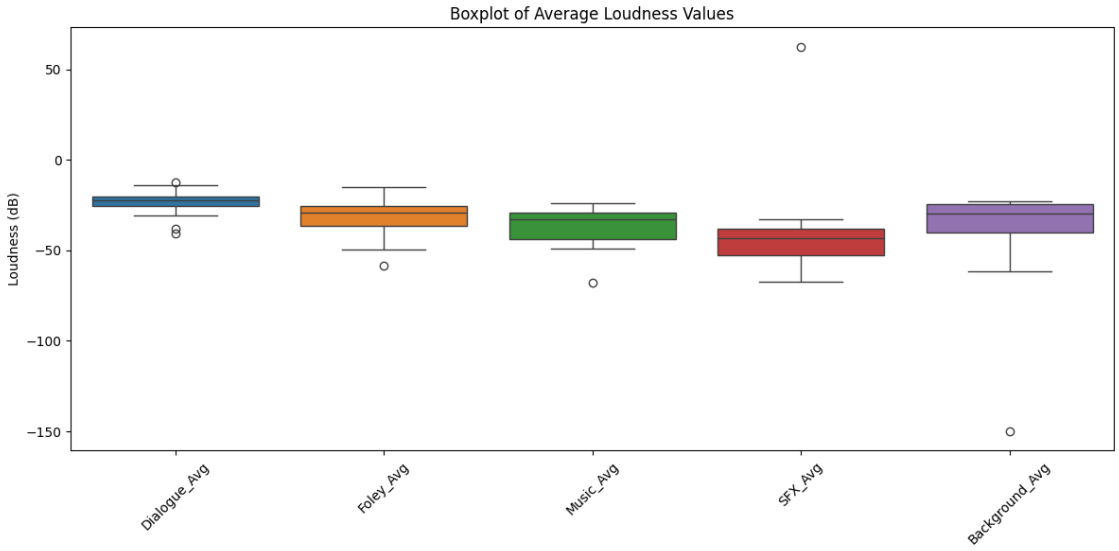

Figure 6-13 Average Loudness Sound Elements Boxplot

There was a weak correlation between dialogue levels and background levels, suggesting participants adjusted dialogue to a specific intelligibility target regardless of the background ambience level.

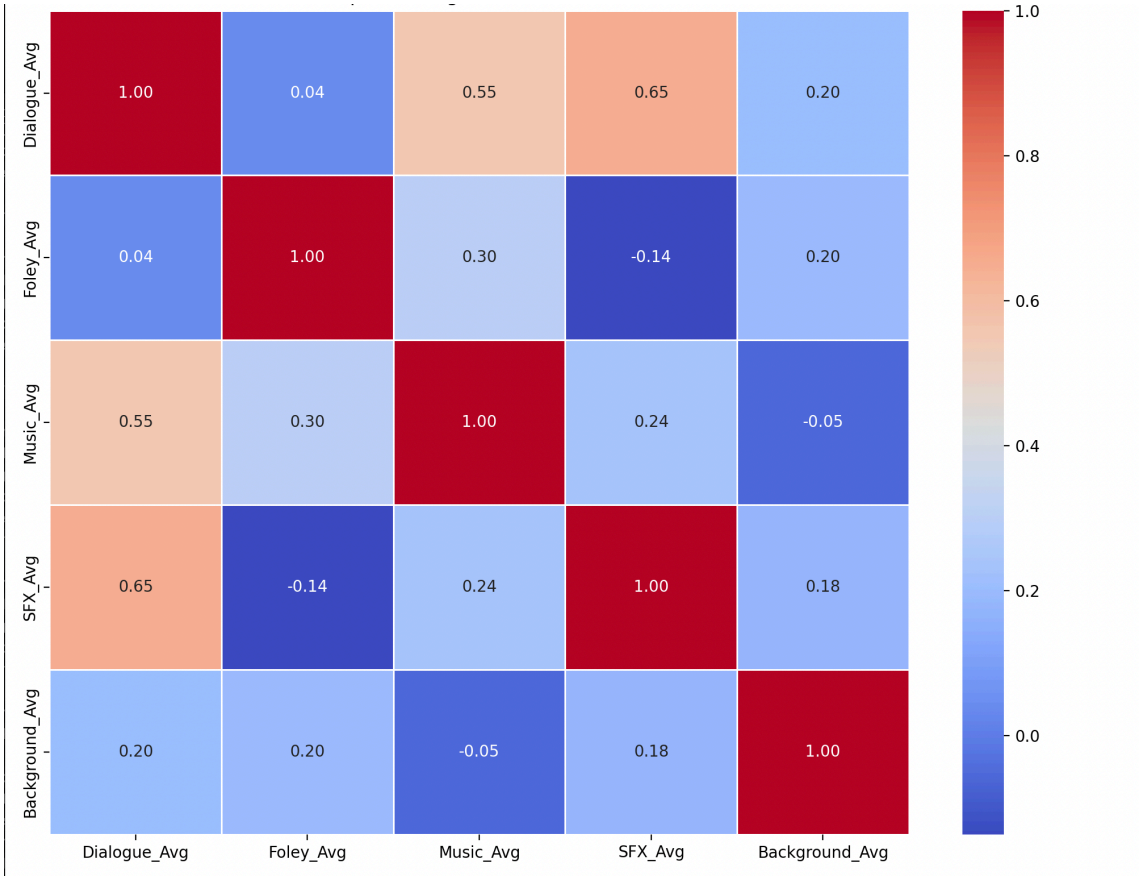

Figure 6-14 Average loudness between Sound Elements Matrix

6.3.6 Statistical Analysis of Mixing Decisions

To statistically verify the hierarchy of mix elements, non-parametric Wilcoxon Signed-Rank Tests were conducted comparing dialogue against all other stems (Woolson, 2007). This method was selected because the high subjective

variance in the supportive stems breaks the assumption of normal distribution required for parametric testing. The results reveal statistically significant differences ($p<.001$) in all cases:

- Dialogue vs. Foley: $W=45.0, p=.001$
- Dialogue vs. Music: $W=4.0, p<.001$
- Dialogue vs. SFX: $W=10.0, p<.001$
- Dialogue vs. Background: $W=25.0, p<.001$

Rather than proving the entirety of the framework's logic, these tests strictly demonstrate that within this controlled experiment, participants consistently prioritised dialogue as the dominant foreground element, setting it significantly louder than all supportive stems.

```
Performing Wilcoxon test: Dialogue vs Foley
Wilcoxon test statistic: 45.0
P-value: 0.0009118914604187012

Performing Wilcoxon test: Dialogue vs Music
Wilcoxon test statistic: 4.0
P-value: 4.172325134277344e-07

Performing Wilcoxon test: Dialogue vs SFX
Wilcoxon test statistic: 10.0
P-value: 2.562999725341797e-06

Performing Wilcoxon test: Dialogue vs Background
Wilcoxon test statistic: 25.0
P-value: 5.3882598876953125e-05

Summary of Wilcoxon Tests:
Dialogue vs Foley: Statistic = 45.0, P-Value = 0.0009118914604187012
Dialogue vs Music: Statistic = 4.0, P-Value = 4.172325134277344e-07
Dialogue vs SFX: Statistic = 10.0, P-Value = 2.562999725341797e-06
Dialogue vs Background: Statistic = 25.0, P-Value = 5.3882598876953125e-05
```

Figure 6-15 Wilcoxon Signed-rank Test

6.4 Discussion

The results from Prototype 1 provide initial empirical support for the key assumptions underpinning the AMPLE framework. By allowing a general audience to actively mix a cinematic sequence under controlled conditions, the experiment revealed a clear hierarchy of narrative importance that translates directly to the framework's adaptive logic.

6.4.1 The Primacy of Dialogue as the Anchor

The most significant finding is the statistical rigidity of dialogue levels compared to all other stems.

- Consistency: Despite the freedom to mix, participants exhibited a tight consensus on dialogue levels (Standard Deviation = 6.52), significantly lower than the variation seen in music or effects.

- Statistical Validation: The Wilcoxon signed-rank tests confirmed that dialogue is treated as a distinct, dominant category ($p<.001$ against all other stems).
- Implication for AMPLE: This confirms that dialogue acts as the anchor of the mix. Users do not view dialogue as a creative variable; they view it as a functional necessity. Therefore, the adaptive framework is justified in locking dialogue to an intelligibility target while manipulating other elements around it.

6.5 The Subjectivity of Immersion (The Chaos of Backgrounds)

In contrast to dialogue, the supportive stems, Foley, SFX, backgrounds, suggest extreme variability.

- The Background Split: The fact that one participant completely silenced backgrounds ($-\infty$ dB) while others boosted them highlights that immersion is highly subjective. For some users, background noise adds realism; for others, it is merely noise that competes with the narrative.
- Foley as a Functional Element: The weak correlation between Foley and other elements suggests it is perceived as a functional layer (footsteps, movement) rather than an emotional one, like music.
- Implication for AMPLE: This supports the need for a user-centric immersion slider. A static downmix cannot possibly satisfy both the user who wants cinema realism and the user who wants clean speech. The framework must allow the user to attenuate these supportive layers without breaking the narrative anchor.

6.5.1 Demographic Influences

While the sample size (N=25) limits broad generalisation, distinct trends emerged:

- Hearing Loss: Interestingly, mild hearing loss did not statistically alter dialogue levels. This suggests that for mild impairment, the issue is often masking signal-to-noise-ratio (SNR) rather than absolute volume. By lowering the music/effects, improving SNR, users with mild loss can achieve clarity without simply cranking the volume.

6.5.2 Usability and Control

The qualitative feedback strongly supports the concept of active user control over the listening experience. Participants explicitly valued the ability to correct the mix to their own preference. As one participant noted, "It was helpful to be able to change the sound dynamics of the film so simply" (P7), while another stated, "It was very exciting to be able to play with the settings 'live'."

Furthermore, the data suggests a desire for this control to translate to domestic environments. As one respondent highlighted: "I often find that audio mixes are poorly optimised for different audio systems. As somewhat of an audio enthusiast, I would like to have more adjustability in mixes on my home setup." This suggests that decentralized stem control is perceived as a highly desirable utility for consumer playback.

6.5.3 Validation of the Framework

Prototype 1 successfully demonstrated that audiences have the capability and the desire to rebalance audio for intelligibility. The data suggest that a single static mix is insufficient for a diverse audience.

6.6 Limitations

The experimental configuration in this chapter was designed to isolate listener behaviour under controlled reference listening conditions. While this approach strengthens internal validity by minimising environmental variability, it does not replicate the diversity of real-world acoustic contexts in which film sound is commonly consumed. The findings therefore reflect behavioural prioritisation under stable listening conditions rather than under environmental masking or mobility constraints, which are examined subsequently in Chapter 7.

The prototype interface was intentionally limited to discrete stem level control in order to evaluate perceptual hierarchy without introducing additional tonal, spatial, or interaction variables. The study did not implement iterative interface redesign or formal usability validation frameworks, as the research objective centred on behavioural response to independent stem manipulation rather than interface optimisation.

Statistical analyses were selected to test defined relationships between stem adjustments within the experimental configuration. The design did not incorporate comprehensive multivariate modelling across all possible interaction factors, and results should therefore be interpreted within the scope of the hypotheses examined.

Participant recruitment was purposive and not stratified across all demographics, hardware, or listening-experience variables. Consequently, the chapter does not claim universal representation of audience behaviour. Instead, it establishes baseline evidence for the dialogue-anchor hierarchy under controlled listening conditions, providing the comparative foundation for the contextual robustness testing presented in Chapter 7.

Finally, a minor interaction bias must be acknowledged regarding the experimental protocol. To ensure task comprehension, the researcher demonstrated the unlabelled interface to participants during the training phase. Crucially, this mechanical demonstration was conducted entirely without the experimental audio stimulus playing. While this step was strictly necessary to prevent mechanical user error and did not bias the participants' auditory judgements during the actual task, providing a visual demonstration of the system's functionality inherently precludes a strictly blind testing environment.

6.7 Summary

This chapter presented the first empirical evaluation of the AMPLE framework under controlled listening conditions. Participants were given independent control over five stems – dialogue, Foley, music, sound effects, and backgrounds – and were asked to adjust levels according to personal preference while viewing a short film excerpt. The objective was to examine whether users would demonstrate consistent balancing behaviour and whether a stable anchor element would emerge when control was decentralised.

6.7.1 Key Findings: Stability vs. Variability

The findings indicate that dialogue occupied a distinct and consistently prioritised position within the mix. Statistical comparisons confirmed significant differences between dialogue and every other stem, reinforcing its role as the primary narrative driver. In contrast, sound effects and backgrounds exhibited substantial variability across participants, confirming that these elements are treated more flexibly and are subject to individual interpretation regarding immersion.

Correlation patterns further clarified these relationships. In this cohort, dialogue showed meaningful alignment with sound effects and music, reflecting their narrative interplay, while Foley and backgrounds were adjusted with greater independence. Together, these results demonstrate that while audiences converge around dialogue prominence, they diverge considerably in how supporting elements are balanced.

These behavioural patterns have direct implications for the framework's architecture. Although some strong correlations (for example, dialogue/music) might theoretically justify automated stem coupling to simplify the interface, the observed variability in sound effects and backgrounds indicates that removing independent control would constrain legitimate user preference.

Accordingly, prototype 2 (Chapter 7) will retain discrete stem manipulation while introducing environmental complexity (simulated noise) to test the robustness of this control structure under adverse listening conditions.

Chapter 7: Prototype 2

Chapter 7 presents the second empirical evaluation of the AMPLE framework. Whereas Prototype 1 examined user-controlled stem balancing under controlled listening conditions, the present experiment introduces environmental masking in order to assess the framework's behavioural robustness under ecologically relevant constraints.

The transition from ideal to noisy conditions reflects a deliberate methodological progression. Prototype 1 established that dialogue functions as a stable anchor when listeners are granted independent control. Prototype 2 therefore investigates whether this anchoring behaviour intensifies, destabilises, or redistributes when spectral masking is introduced.

Environmental noise is not treated as a generic disturbance, but as a defined masking variable that simulates common listening contexts such as domestic background activity or transport environments. This allows the experiment to evaluate whether independent stem control meaningfully supports intelligibility when external interference is present.

The chapter outlines the experimental configuration, noise modelling approach, stimulus preparation, participant protocol, and evaluation measures. The findings provide a second stage of validation for the framework by examining whether adaptive control remains effective when listening conditions depart from the laboratory baseline.

7.1 Background

While prototype 1 (Chapter 6) established a baseline for user preference in ideal listening conditions, real-world consumption frequently occurs in environments where background noise fundamentally alters auditory perception. As identified in the audience survey (Chapter 3), mobile consumption is prevalent in high-noise settings, with 69% of participants reporting usage during transport (planes, trains, buses). In such environments, the dynamic range tolerance of listeners is significantly reduced due to auditory masking.

Therefore, prototype 2 is positioned as a stress-test of the adaptive framework. By introducing simulated background noise, this experiment aims to determine if the dialogue anchor behaviour observed in prototype 1 persists when the user is forced to compensate for environmental masking, or if their mixing strategy shifts to combat external distractions.

7.1.1 The Commute Scenario

To simulate a representative acoustic environment, the experiment utilises a commute noise floor. This scenario was selected for its ecological plausibility and characteristics; public transport noise contains significant energy that effectively masks dialogue fundamentals. By validating the framework under these conditions, we assess the system's strength in a high-masking environment that reflects user reality.

Conducting listening tests in actual public settings poses significant logistical challenges regarding the repeatability of the noise floor. To ensure internal

validity, this experiment utilises simulated background noise within a controlled environment. This approach allows for rigorous testing of how real-world distractions impact mixing choices while avoiding the unpredictable variables of field testing.

7.1.2 Experimental Controls

To isolate environment noise as the single independent variable, specific controls were implemented:

To manage the multichannel routing of the environmental noise, the audio engine was migrated from Ableton to Reaper . Consequently, the visual interface reflects Reaper's default aesthetic. However, the functional topology of the interface remained unchanged, maintaining the setup of five unlabelled vertical faders (Range: −∞ to +12 dB) utilized in prototype 1. Maintaining this functional consistency ensures that the resulting data reflects participants' response to acoustic environment context.

To reduce learning and familiarity effects across repeated listening conditions (plane, train, bus) participants mixed different excerpts from the stimulus for each acoustic environment. This prevented prior exposure to one mix from influencing subsequent mixing decisions through memory recall. All excerpts were selected from the same film to preserve consistency in narrative style and sonic characteristics across trials (Samray, 2022).

7.2 Method

This experiment employed a within-subjects repeated measures design to evaluate user mixing behaviour under adverse listening conditions. The experimental protocol was designed to replicate the prototype 1 process within the same physical Auralisation space (as in section 6.2.3), while introducing environmental masking as the primary constraint. To simulate an immersive acoustic environment, the masking noise was routed through the Auralisation suite's surround-sound array. By utilizing the upper, mid, and bottom speaker layers, the system generated a three-dimensional physical masking field around the listener.

7.2.1 Experimental Variables

To ensure analytical transparency, the study operationalised the following variables:

- Independent Variable (IV): Simulated environmental masking condition (plane, train, or bus).
- Dependent Variables (DVs): User-adjusted gain levels (dB), resulting Loudness (LUFS), and relationships between soundtrack elements.

7.2.2 Participants

To ensure independent validation and prevent practice effects, a completely new cohort of 25 participants was recruited. There was no overlap with the participants from the prototype 1 experiment. A total of 25 participants (N=25) were recruited via academic networks and professional mailing lists.

- Demographics: The cohort was aged between 21 and 67 (M=42.08, SD not specified). Gender distribution was 64% male and 32% female.
- Hearing Health: 80% reported no hearing loss, while 20% reported mild to moderate hearing loss.
- Ethics: Participation was voluntary with no monetary compensation. Full ethical approval was granted by the University Research and Integrity Committee (See Appendix D).

7.2.3 Technical Setup

The technical setup was designed to replicate the logic of prototype 1.

- Audio Engine: A MacBook Air M2 hosted the audio engine. The Digital Audio Workstation (DAW) was updated to Reaper (Figure 7-2) to better manage the multichannel noise routing and digital summing. While technical constraints required migrating the interface backend away from Max/MSP – resulting in minor visual changes to the fader colours – the functional topology of five unlabelled faders was strictly preserved to maintain experimental control (Apple Inc, 2022b; Reaper, 2025).
- Interface: Participants interacted via an Apple iPad running the interface. The interface retained the unlabelled fader topology from the previous experiment, range: −∞ to +12 dB (Apple Inc, 2022a). The exclusion of visual semantic labels (for example, dialogue, music) was maintained to prevent anchoring; participants were forced to rely entirely on their auditory perception and subjective preference to achieve intelligibility, rather than defaulting to preconceived visual hierarchies. See Figure 7-1
- Calibration: Playback was conducted in the University's Auralisation Suite. The system was calibrated to 79 dB(C) using pink noise to match the reference level of prototype 1 (Prokofieva et al., 2014).

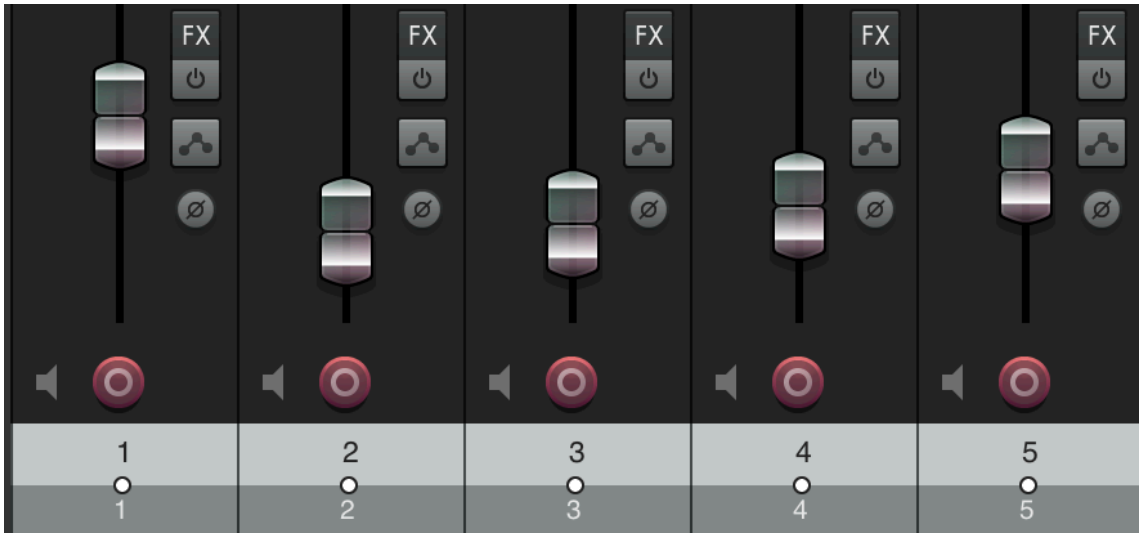

Figure 7-1 Experiment user interface

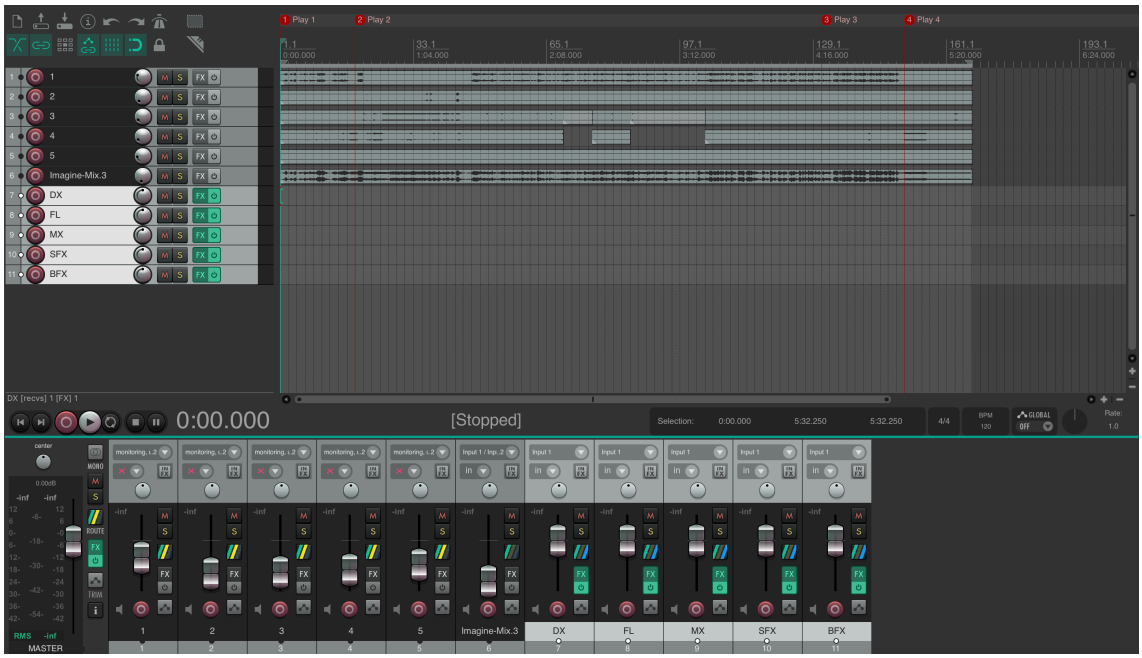

Figure 7-2 Experiment DAW session

7.2.4 Environmental Stimulus (The Masking Condition)

To mirror real-world commute conditions, a composite noise floor was generated based on published transport noise measurements (Basner & McGuire, 2018; Gershon et al., 2006; Neitzel et al., 2009). The noise profile included:

1. Subway/Underground: Intermittent high-amplitude rumble (≈80.4 dB A).
2. Bus Interior: Low-frequency engine idle (≈78 dB A).
3. Aircraft Cabin: Continuous broadband drone (≈78 dB A).

To accurately simulate the immersive acoustic environment of a commute, the masking noise was routed through the Auralisation suite's surround-sound array. By utilizing the upper, mid, and bottom speaker layers, the system generated a three-dimensional, physical acoustic masking field around the listener. The presentation sequence of the three contextual noise scenarios (plane, train, bus)

was randomised for each participant. This approach was implemented to mitigate order effects and prevent listener fatigue from skewing the data toward a specific environment

Participants could not attenuate this ambient noise; they were forced to mix the forward-facing stereo film content over the acoustic masking threshold, thereby simulating a highly realistic, real-world signal-to-noise ratio (SNR) challenge. The visual stimulus was a new three-minute excerpt from an independent drama addressing themes of community violence (Samray, 2022). This clip was selected to match the spectral density and production value of prototype 1 while altering the narrative content.

7.2.5 Procedure

The experimental protocol was standardized across all participants:

1. Familiarisation Phase: To minimise the startle response and optimise mechanical competence, a training phase was conducted. The researcher provided a mechanical demonstration of the UI, instructing participants on how to manipulate the faders, without providing any aesthetic guidance, target levels, or semantic identification of the stems. Participants then performed a non-recorded test round where they were exposed to an alternative high-level noise profile. They were instructed simply to mix the film to compensate for the simulated noisy environment. This ensured that subsequent data reflected genuine, unprompted listening preferences rather than an initial orientation reaction or interface confusion.
2. The Task: Participants watched the film excerpt while exposed to one of the simulated acoustic masking conditions. They were instructed to "adjust the levels of the five unlabelled stems to your personal listening preference".
    a. Looping: This task was repeated for each of the three acoustic contexts (plane, train, bus). For each specific environment, participants could loop the clip up to two times to finalize their mix.
    b. Initial fader positions were randomised (±3 dB) at the beginning of every condition. This prevented mechanical anchoring bias and forced active engagement with every stem.
3. Post-Test Survey: Immediately following the completion of all three masking conditions, participants completed a digital survey (Appendix D). This survey utilized the same adapted constructs as Prototype 1 to capture

demographic data, validate sample consistency, and gather qualitative feedback on their mixing strategies under acoustic stress.

7.2.6 Statistical Analysis Strategy

Data was exported to Microsoft Excel for processing and evaluation using Python's statsmodels and scipy.stats libraries (Microsoft Corporation, 2026; Python, 2026; SciPy, 2008). The analysis followed a multi-tiered approach:

- Linear Mixed Effects (LME) Modelling: To assess the predictive weight of the environmental conditions versus the narrative elements, an LME model was utilized. This approach was chosen over standard regression to mathematically account for the repeated-measures design by applying random intercepts for each participant (Gelman & Hill, 2006).
- Pairwise Comparisons (Wilcoxon): Non-parametric Wilcoxon Signed-Rank tests were selected to compare specific stem levels within environments. This method was used because the high subjective variance in loudness adjustments under severe masking violates the assumption of normal distribution necessary for parametric testing (Woolson, 2007).
- To control the false discovery rate across multiple pairwise comparisons without the extreme penalisation of a Bonferroni correction, a Benjamini-Hochberg (FDR-BH) correction was applied (Benjamini & Hochberg, 1995).
- Internal Validity: Participants were not informed of the specific hypothesis regarding dialogue dominance, ensuring no element was implicitly framed or prompted as more important than another prior to testing.

7.3 Results

The following section details the results of the listening tests conducted within the simulated acoustic environments. Unlike the data in Chapter 6 (prototype 1), this experiment introduced controlled noise masking to evaluate how participants manage their mixing strategies under acoustic stress.

The analysis focuses on three key areas:

1. Baseline Habits: Demographics and current streaming satisfaction.
2. Mixing Behaviour: Quantitative analysis of loudness levels (LUFS) applied under masking conditions.

3. Statistical Drivers: Linear Mixed Effects (LME) modelling and pairwise comparisons to evaluate the influence of environmental masking versus narrative priority. Participant Demographics and Baseline Habits

The sample cohort (N = 25) reported a strong preference for stationary home-listening (80%), with only 8% favouring the traditional cinema experience. Consistent with wider market trends, Netflix was identified as the dominant primary streaming platform (56%).

Streaming Experience and Perceived Clarity:

To establish a comparative baseline prior to the masking tests, participants rated the perceived clarity of soundtracks on their current commercial streaming services. Over 75% of the cohort considered current commercial mixes to be clear or very clear.

It is important to contextualise this finding against the larger audience survey detailed in Chapter 3 (N = 500), which revealed widespread dissatisfaction with television and streaming dialogue intelligibility. The positive baseline ratings from this smaller experimental group are secondary to the significant problem identification established in the wider quantitative study.

However, this baseline usefully suggests that the current cohort did not possess an inherent, pre-existing negative bias toward streaming audio prior to the experiment. Consequently, their mixing behaviours during the task can be attributed to the introduced environmental masking rather than an overarching frustration with commercial mix standards.

7.3.1 System Interaction Rating

As noted in Chapter 6, the unlabelled fader interface was previously demonstrated to be intuitive under ideal listening conditions. Consequently, usability metrics were briefly re-captured in this iteration strictly as an experimental manipulation check. The objective was to ensure that the introduction of severe acoustic masking did not induce cognitive overload or degrade the participants' mechanical competence. Quantitative feedback (Figure 7-3) suggests that the interface remained accessible, with 60% of participants still rating the system as either easy (48%) or quick to learn (12%) despite the environmental stress.

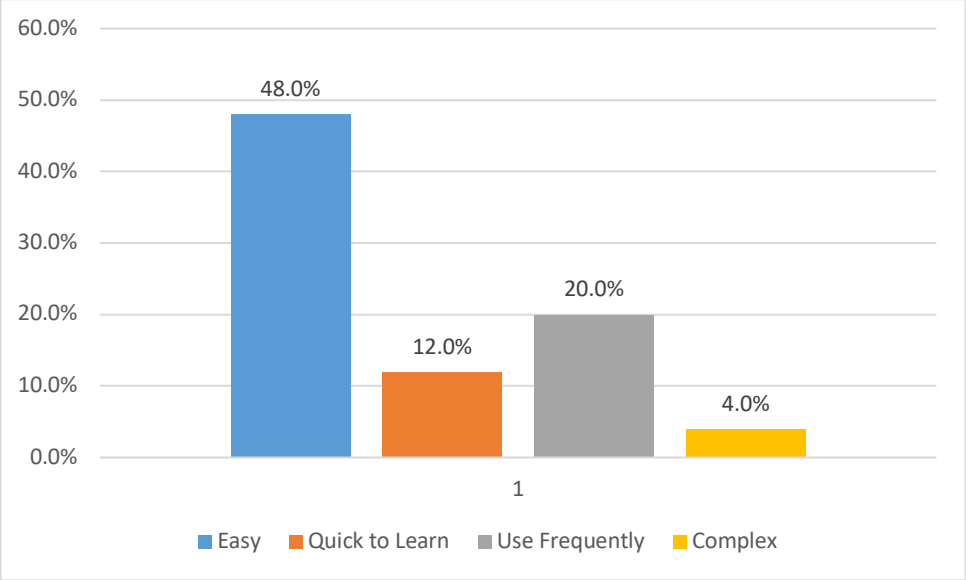

Figure 7-3 System interaction rating

To contextualise these multiple-choice responses, participant free-text comments were analysed by assigning thematic tags to recurring statements and calculating their frequency – inductive content analysis (Elo & Kyngäs, 2008b). This allowed specific behavioural themes to emerge directly from the data (Figure *7-4*)

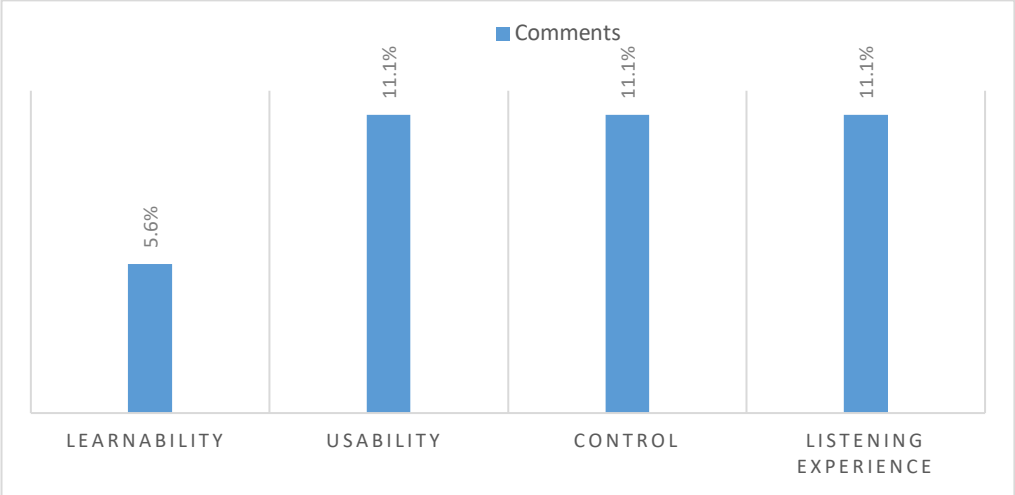

Figure 7-4 Comments

7.3.2 Loudness Analysis: The Dialogue Anchor

The following analysis details the behaviour of the primary narrative stem (dialogue) under acoustic stress. To examine the dispersion of dialogue levels, the data was analysed separately across the Plane, Train, and Bus simulations (Table 7-1 and Figure 7-5)

- Plane Condition: Average loudness exhibited a standard deviation of 15.1 LUFS, with values ranging between −65.2 and −8.4 LUFS (Range = 56.8). True peak (TP) values varied between −51.9 and 7.1 dBTP (SD = 15.9; Range = 59.0).

- Train Condition: Average loudness demonstrated a standard deviation of 15.1 LUFS, ranging from −61.0 to −8.4 LUFS (Range = 52.6). True peak values exhibited a slightly greater spread (SD = 16.6), extending from −45.9 to 25.6 dBTP (Range = 71.5).
- Bus Condition: Average loudness values ranged from −64.6 to −8.4 LUFS (SD = 15.7; Range = 56.2). True peak values spanned between −54.6 and 7.1 dBTP (SD = 16.7; Range = 61.7).

Across all environments, the dispersion of the dialogue stem remained consistent, with average level standard deviations tightly clustered between 15.1 and 15.7 LUFS.

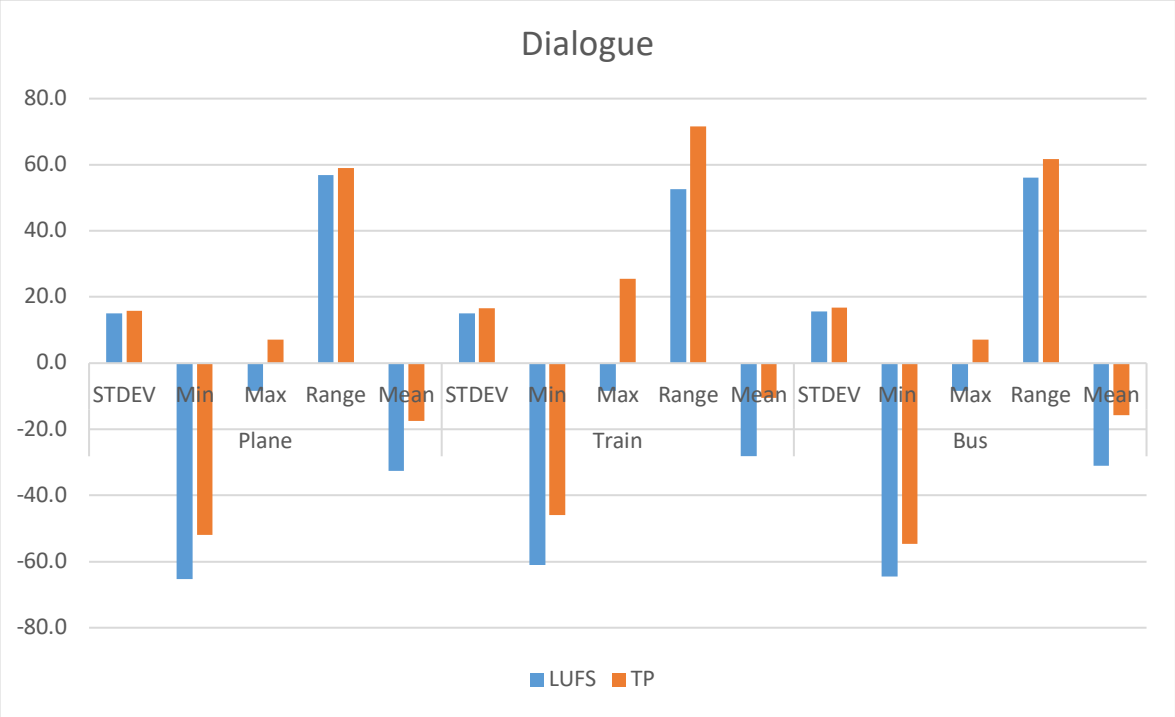

Figure 7-5 Dialogue Levels - Prototype-2

Table 7-1 Dialogue Levels Breakdown – Prototype -2

Environment	Metric	LUFS	TP
Plane	**STDEV**	15.1	15.9
	Min	-65.2	-51.9
	Max	-8.4	7.1
	Range	56.8	59.0
	Mean	-32.5	-17.4
Train	**STDEV**	15.1	16.6
	Min	-61.0	-45.9
	Max	-8.4	25.6
	Range	52.6	71.5

	Mean	-28.1	-10.4
Bus	**STDEV**	15.7	16.7
	Min	-64.6	-54.6
	Max	-8.4	7.1
	Range	56.2	61.7
	Mean	-31	-15.6

7.3.3 Loudness Analysis: Supportive Elements

Foley elements demonstrated a highly stable adjustment profile across all testing conditions, see Figure *7-6* and Table *7-2*. In the plane environment, average loudness (LUFS) had a standard deviation of 12.3 LUFS, with a mean of −38.6 LUFS and a range spanning from −62.9 to −15.2 LUFS. True Peak (TP) values for the plane condition had a standard deviation of 16.5 dBTP.

This distribution remained consistent in the train condition, where average loudness showed a standard deviation of 12.7 LUFS (Mean: −34.2 LUFS; Range: 44.8 LUFS), and TP values showed a standard deviation of 17.0 dBTP. In the bus environment, the dispersion metrics were comparable, showing an average loudness standard deviation of 12.8 LUFS (Mean: −33.0 LUFS; Range: 43.1 LUFS) and a TP standard deviation of 16.1 dBTP.

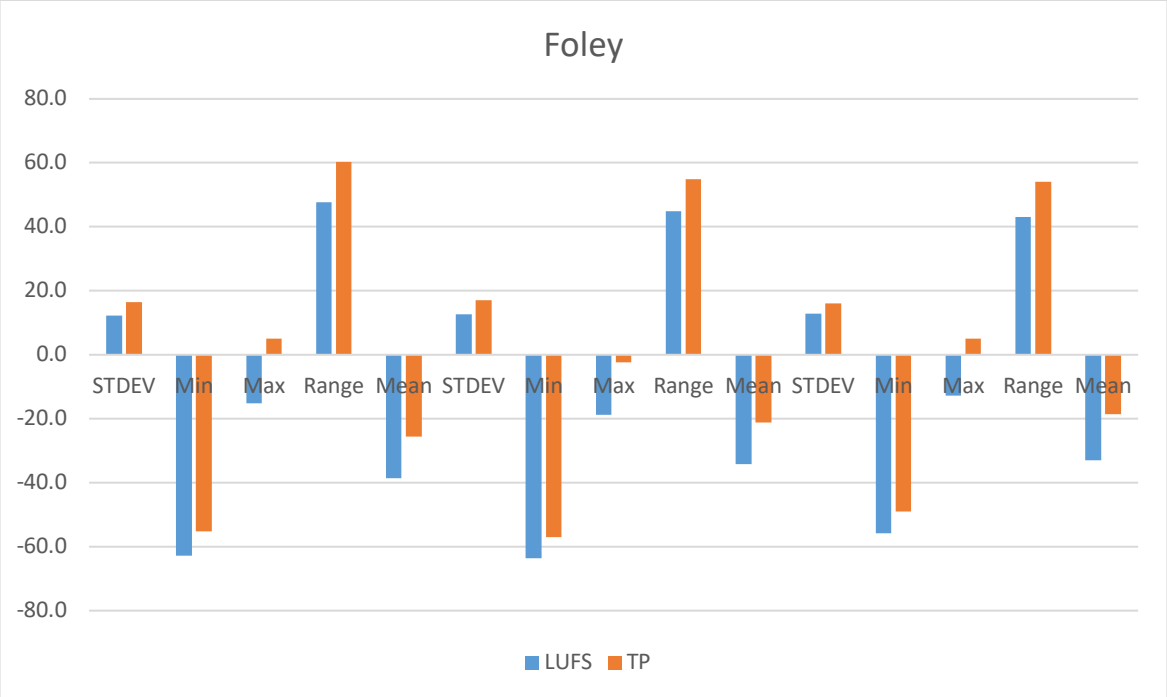

Figure 7-6 Foley Levels Aggregate- Prototype -2

Table 7-2 Foley Levels Breakdown Protoype-2

Environment	Metric	LUFS	TP
Plane	**STDEV**	12.3	16.5
	Min	-62.9	-55.2

	Max	-15.2	5.0
	Range	47.7	60.2
	Mean	-38.6	-25.5
Train	**STDEV**	12.7	17.0
	Min	-63.6	-57.1
	Max	-18.8	-2.3
	Range	44.8	54.8
	Mean	-34.2	-21.2
Bus	**STDEV**	-55.9	-49.0
	Min	-12.8	5.0
	Max	43.1	54.0
	Range	-33.0	-18.5
	Mean	-55.9	-49.0

Musical elements demonstrated a different adjustment profile (

Table *7-3*). In the plane and train conditions, average loudness (LUFS) maintained consistent standard deviations of 14.2 LUFS and 15.1 LUFS, respectively. Maximum mixed levels for these environments were −18.9 LUFS (Plane) and −17.3 LUFS (train).

In the bus condition, the standard deviation for average loudness remained comparable at 14.9 LUFS. However, the True Peak (TP) dispersion increased significantly. The TP standard deviation rose to 22.1 dBTP, and the overall TP range expanded to 92.2 dB (Min: −95.1 dBTP; Max: −2.9 dBTP). This indicates a high degree of variance in how transient musical peaks were managed under the bus environment.

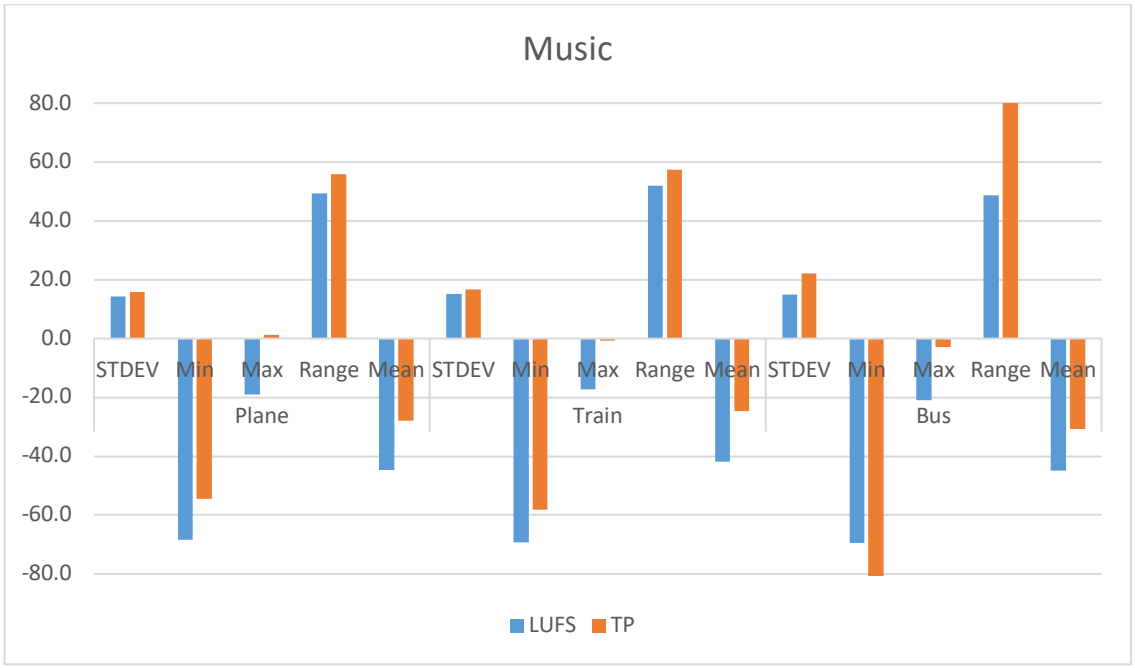

Figure 7-7 Music Levels- Aggregate - Prototype -2

Table 7-3 Music Levels Breakdown- Prototype-2

Environment	Metric	LUFS	TP
Plane	**STDEV**	14.2	15.9
	Min	-68.3	-54.5
	Max	-18.9	1.3
	Range	49.4	55.8
	Mean	-44.6	-28
Train	**STDEV**	15.1	16.7
	Min	-69.2	-58.2
	Max	-17.3	-0.8
	Range	51.9	57.4
	Mean	-41.9	-24.5
Bus	**STDEV**	14.9	22.1
	Min	-69.5	-95.1
	Max	-20.9	-2.9
	Range	48.6	92.2
	Mean	-44.7	-30.8

Sound effects (SFX) demonstrated distinct distribution profiles across the three testing environments (Figure *7-8* and Table *7-4*). In the plane condition, average loudness (LUFS) had a standard deviation of 13.3 LUFS, with a mean value of −42.5 LUFS and a range spanning from −63.8 to −20.8 LUFS. True Peak (TP) values in this environment showed a standard deviation of 16.2 dBTP.

In the train condition, dispersion metrics widened slightly, with the average loudness standard deviation rising to 16.6 LUFS (Mean: −36.3 LUFS; Range: 49.2 LUFS). The corresponding TP values for the train environment exhibited a standard deviation of 19.7 dBTP. For the bus condition, the average loudness distribution returned to a profile similar to the plane environment, yielding a standard deviation of 14.4 LUFS, a mean of −42.6 LUFS, and a range of 49.5 LUFS (Min: −68.7 LUFS; Max: −19.2 LUFS), with a TP standard deviation of 17.5 dBTP.

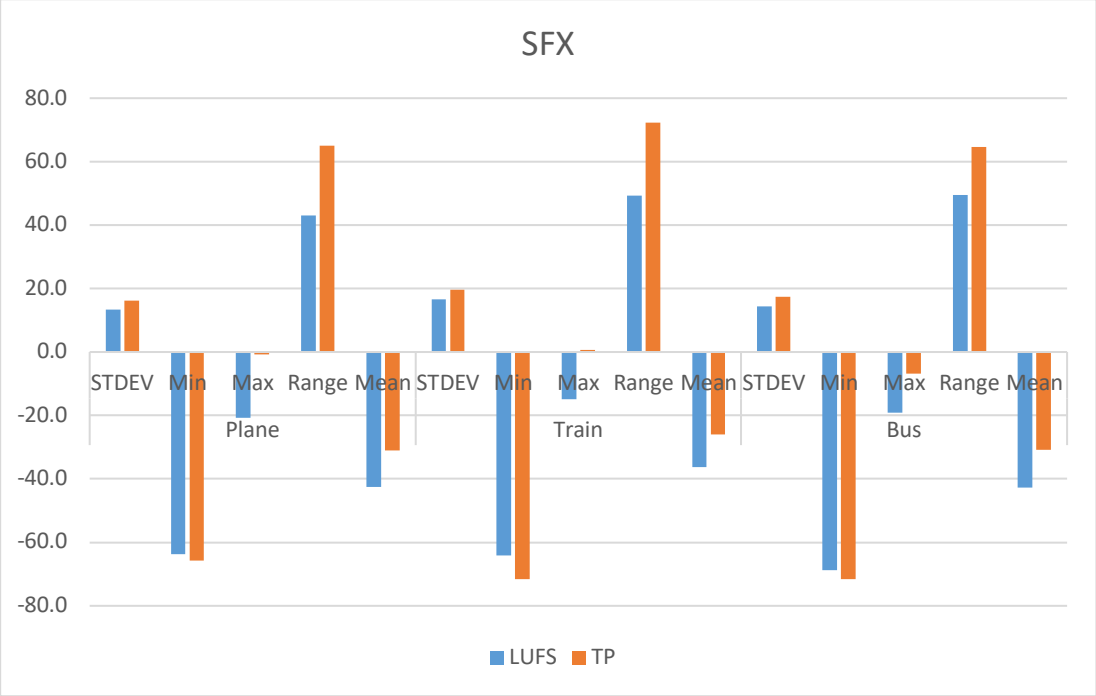

Figure 7-8 SFX Levels - Prototype 2

Table 7-4 SFX Levels Breakdown- Prototype-2

Environment	Metric	LUFS	TP
Plane	**STDEV**	13.3	16.2
	Min	-63.8	-65.8
	Max	-20.8	-0.8
	Range	43.0	65.0
	Mean	-42.5	-31
Train	**STDEV**	16.6	19.7
	Min	-64.1	-71.6
	Max	-14.9	0.7
	Range	49.2	72.3
	Mean	-36.3	-26
Bus	**STDEV**	14.4	17.5
	Min	-68.7	-71.6
	Max	-19.2	-6.9
	Range	49.5	64.7
	Mean	-42.6	-30.8

Background elements (BFX) demonstrated steady distribution profiles across the three testing environments (Table *7-5* and Figure *7-9*). In the plane condition, average loudness (LUFS) exhibited a standard deviation of 12.0 LUFS, with a mean value of −56.0 LUFS and a range spanning from −69.7 to −28.2 LUFS. True Peak (TP) values in this environment showed a standard deviation of 16.5 dBTP, with a mean of −45.1 dBTP and a range of 71.2 dB (Min: −80.8 dBTP; Max: −9.6 dBTP).

In the train condition, the metrics remained comparable, with the average loudness standard deviation measured at 13.3 LUFS (Mean: −51.9 LUFS; Range: 44.5 LUFS). The corresponding TP values for the train environment exhibited a standard deviation of 22.6 dBTP, a mean of −36.5 dBTP, and a range of 102.8 dB (Min: −75.3 dBTP; Max: 27.5 dBTP). For the bus condition, the distribution profile showed an average loudness standard deviation of 11.0 LUFS, a mean value of −51.1 LUFS, and an overall range of 38.6 LUFS (Min: −67.8 LUFS; Max: −29.2 LUFS), with a corresponding TP standard deviation of 17.5 dBTP and a mean of −40.8 dBTP.

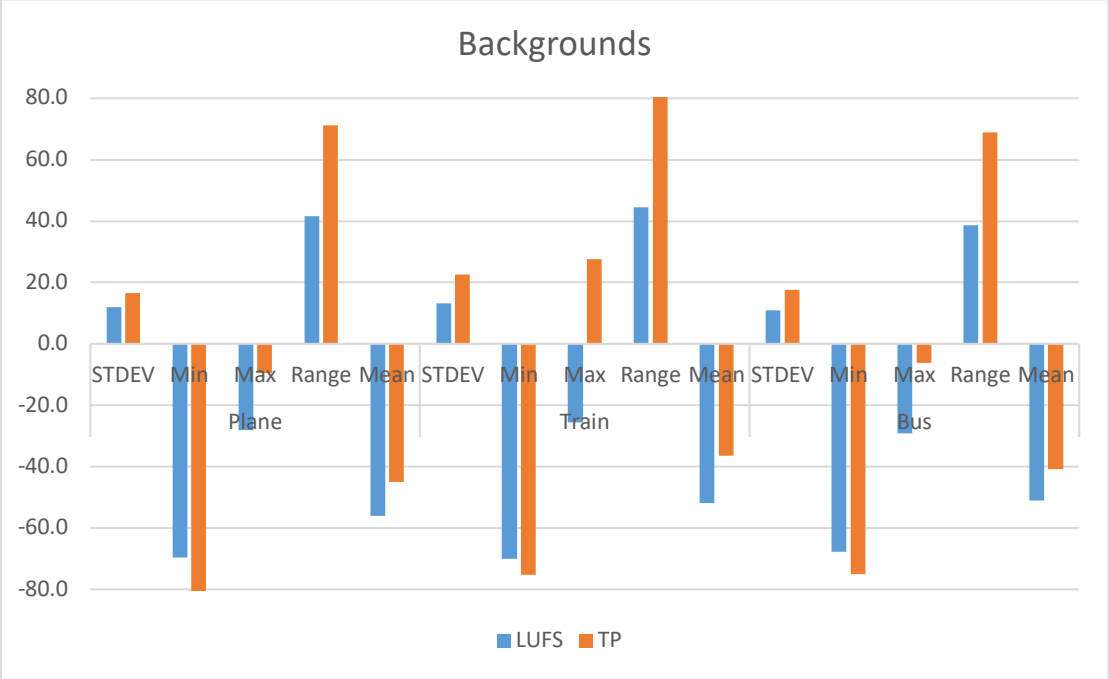

Figure 7-9 Background levels – Aggregate - Prototype

Table *7-5* shows the specific background attenuation across the transport types.

Table 7-5 BFX Levels Breakdown- Prototype-2

Environment	Metric	LUFS	TP
Plane	**STDEV**	12.0	16.5
	Min	-69.7	-80.8
	Max	-28.2	-9.6
	Range	41.5	71.2
	Mean	-56.0	-45.1
Train	**STDEV**	13.3	22.6
	Min	-70.0	-75.3
	Max	-25.5	27.5
	Range	44.5	102.8
	Mean	-51.9	-36.5
Bus	**STDEV**	11.0	17.5
	Min	-67.8	-75.1
	Max	-29.2	-6.2
	Range	38.6	68.9
	Mean	-51.1	-40.8

Figure *7-10* illustrates the overall distribution and variance of loudness levels for each of the five distinct sound elements.

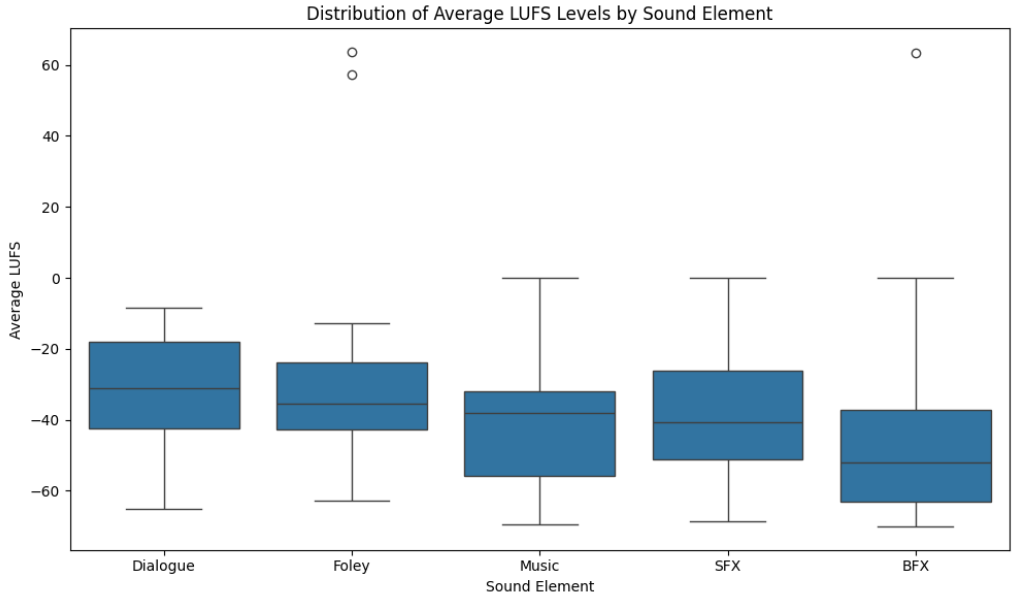

Figure 7-10 Loudness by Sound Element Prototype 2

7.3.4 Statistical Analysis of Mixing Drivers

A Linear Mixed Effects (LME) model was done using Python's statsmodels library (N=375 total observations across N=25 participant groups) (Gelman & Hill, 2006; Python, 2026; statsmodels, 2025). The model incorporated an interaction term

(environment × element) and applied treatment coding, establishing the plane environment – the least spectrally complex of the three – and the dialogue stem as the statistical reference baseline (Intercept = −32.58 LUFS). See Figure 7-11.

Main Effects: Relative to the dialogue baseline, statistically significant negative coefficients were observed for backgrounds (BFX) (β=−27.22 LUFS, p<0.001), sound effects (SFX) (β=−14.26 LUFS, p=0.021), and music (β=−12.12 LUFS, p=0.049). The main effect for Foley did not significantly differ from the baseline (β=−1.42 LUFS, p=0.817). No significant main effects were observed for the train environment (p=0.468) or the Bus environment (p=0.804).

Interaction Effects: Every interaction term within the model was statistically non-significant (all p>0.10). The individual interaction values were tracked at: train × foley (p=0.963), bus × Foley (p=0.953), train × music (p=0.490), bus × music (p=0.102), train × SFX (p=0.863), bus × SFX (p=0.494), train × BFX (p=0.708), and bus × BFX (p=0.319).

Non-parametric Wilcoxon Signed-Rank Tests (scipy.stats) with a Benjamini-Hochberg (FDR-BH) correction were conducted within each masking condition to compare dialogue against the remaining four supportive elements (Benjamini & Hochberg, 1995; Python, 2026; SciPy, 2008; Woolson, 2007). See Figure 7-12.

Plane Condition: Dialogue was mixed at statistically distinct levels compared to all four elements: BFX (corrected p<0.001, W=0.0), music (corrected p<0.001, W=4.5), SFX (corrected p<0.001, W=19.0), and Foley (corrected p=0.047, W=86.0).

Train Condition: Statistically significant differences were maintained between dialogue and music (corrected p<0.001, W=0.0), BFX (corrected p<0.001, W=23.0), and SFX (corrected p<0.001, W=39.0). The comparison between dialogue and Foley was non-significant (corrected p=0.081, W=87.5).

Bus Condition: Statistically significant differences remained between dialogue and music (corrected p<0.001, W=1.0), SFX (corrected p<0.001, W=3.0), and BFX (corrected p<0.001, W=3.0). The comparison between dialogue and Foley was non-significant (corrected p=0.252, W=119.0).

```
── Linear Mixed Effects Model Summary ──
                                Mixed Linear Model Regression Results
=============================================================================================================
Model:                                MixedLM                    Dependent Variable:                    LUFS
No. Observations:                     375                        Method:                                REML
No. Groups:                           25                         Scale:                                 475.5663
Min. group size:                      15                         Log-Likelihood:                        -1669.5395
Max. group size:                      15                         Converged:                             Yes
Mean group size:                      15.0
-------------------------------------------------------------------------------------------------------------
                                                                  Coef.  Std.Err.   z    P>|z|  [0.025  0.975]
-------------------------------------------------------------------------------------------------------------
Intercept                                                        -32.576    5.283 -6.166 0.000 -42.931 -22.221
C(Environment, Treatment("Plane"))[T.Train]                        4.476    6.168  0.726 0.468  -7.613  16.565
C(Environment, Treatment("Plane"))[T.Bus]                          1.528    6.168  0.248 0.804 -10.561  13.617
C(Element, Treatment("Dialogue"))[T.Foley]                        -1.424    6.168 -0.231 0.817 -13.513  10.665
C(Element, Treatment("Dialogue"))[T.Music]                       -12.116    6.168 -1.964 0.049 -24.205  -0.027
C(Element, Treatment("Dialogue"))[T.SFX]                         -14.260    6.168 -2.312 0.021 -26.349  -2.171
C(Element, Treatment("Dialogue"))[T.BFX]                         -27.216    6.168 -4.412 0.000 -39.305 -15.127
C(Environment, Treatment("Plane"))[T.Train]:C(Element, Treatment("Dialogue"))[T.Foley]   0.400    8.723  0.046 0.963 -16.697  17.497
C(Environment, Treatment("Plane"))[T.Bus]:C(Element, Treatment("Dialogue"))[T.Foley]    -0.512    8.723 -0.059 0.953 -17.609  16.585
C(Environment, Treatment("Plane"))[T.Train]:C(Element, Treatment("Dialogue"))[T.Music]  -6.028    8.723 -0.691 0.490 -23.125  11.069
C(Environment, Treatment("Plane"))[T.Bus]:C(Element, Treatment("Dialogue"))[T.Music]   -14.244    8.723 -1.633 0.102 -31.341   2.853
C(Environment, Treatment("Plane"))[T.Train]:C(Element, Treatment("Dialogue"))[T.SFX]     1.508    8.723  0.173 0.863 -15.589  18.605
C(Environment, Treatment("Plane"))[T.Bus]:C(Element, Treatment("Dialogue"))[T.SFX]      -5.972    8.723 -0.685 0.494 -23.069  11.125
C(Environment, Treatment("Plane"))[T.Train]:C(Element, Treatment("Dialogue"))[T.BFX]    -3.268    8.723 -0.375 0.708 -20.365  13.829
C(Environment, Treatment("Plane"))[T.Bus]:C(Element, Treatment("Dialogue"))[T.BFX]      -8.692    8.723 -0.996 0.319 -25.789   8.405
Group Var                                                        222.261    3.480
=============================================================================================================
```

Figure 7-11 LME Summary

```
── Wilcoxon Signed-Rank Results (FDR corrected) ──
Environment        Comparison  Statistic      p-value  p-corrected (FDR-BH)  Significant (α=0.05)
      Plane Dialogue vs Foley       86.0 3.954486e-02          4.745383e-02                  True
      Plane Dialogue vs Music        4.5 2.121759e-05          4.243517e-05                  True
      Plane   Dialogue vs SFX       19.0 1.127777e-04          1.933331e-04                  True
      Plane   Dialogue vs BFX        0.0 5.960464e-08          3.576279e-07                  True
      Train Dialogue vs Foley       87.5 7.411601e-02          8.085383e-02                 False
      Train Dialogue vs Music        0.0 5.960464e-08          3.576279e-07                  True
      Train   Dialogue vs SFX       39.0 4.296899e-04          5.729198e-04                  True
      Train   Dialogue vs BFX       23.0 1.742545e-04          2.613817e-04                  True
        Bus Dialogue vs Foley      119.0 2.521039e-01          2.521039e-01                 False
        Bus Dialogue vs Music        1.0 1.192093e-07          4.768372e-07                  True
        Bus   Dialogue vs SFX        3.0 2.980232e-07          7.152557e-07                  True
        Bus   Dialogue vs BFX        3.0 2.980232e-07          7.152557e-07                  True
```

Figure 7-12 Wilcoxon Signed Rank Prototype 2

Collectively, these analyses provide a picture of the variation and relationships in loudness choices made by participants across different sound elements and listening environments.

7.3.5 Qualitative Feedback: Reactive Mixing

Post-experiment free-text responses (n=14) were examined to provide subjective context for the quantitative mixing data. Tagging the responses revealed three distinct observational themes:

A subset of respondents (n=4) explicitly cited ambient noise or practical real-world scenarios as drivers for needing stem-based control. Participants noted that "whatever is happening in the room at the same time is a big factor," and highlighted that the interface "would be very useful in travelling and other scenarios and could assist if using non-noise-cancelling earphones." Others described using the control to soften uncomfortable noise or to correct common streaming imbalances where "the audio of the dialogue is normal/low, the background music and noises... are always high."

A similarly sized subset (n=4) provided constructive feedback regarding the physical mapping of the control surface. Participants expressed difficulty discerning the distinct functions of all five faders without visual aids, suggesting it would be helpful to "label the indicators on the device" and noting that "sound effects could be balanced with 3 sliders rather than 5."

Two participants (n=2) noted strong emotional reactions to the narrative content of the stimulus itself, indicating that the subject matter of the dialogue drew significant attentional focus during the task. These qualitative themes broadly aligned with the quantitative findings, particularly the prioritisation of intelligibility and the perceived value of listener control under masking conditions.

7.4 Discussion

The results from prototype 2 extend the findings of the previous chapter by demonstrating that the hierarchy of narrative importance observed in ideal conditions remains structurally consistent under acoustic stress. While prototype 1 established a baseline for user preference, prototype 2 shows how participants manage overall mix structures when the acoustic context changes, highlighting a consistent mixing strategy combined with localized tactical adjustments for transient peaks and functional narrative elements.

7.4.1 Sample Validation

The participant group (N=25) reflected a representative cross-section of the potential user base, with 20% reporting some degree of hearing loss and a balanced age distribution (21–67). Consistent with the survey data in Chapter 3, the majority (56%) preferred stationary viewing. However, the results indicate that even these stationary-preference users are capable of executing consistent mixing adaptations when simulated masking conditions are introduced.

7.4.2 The Elasticity of the Mix

Here, elasticity refers to variability in transient peak management while average loudness targets remain comparatively stable. A notable finding was the distinction between how participants managed average loudness (LUFS) and transient peaks (True Peak) under masking conditions.

- Despite the introduction of masking (plane, train, and bus noise), the standard deviations for average loudness (LUFS) across the supportive stems remained constricted and consistent. For example, Foley maintained a standard deviation between 12.3 and 12.8 LUFS across all three environments, while music remained stable between 14.2 and 15.1 LUFS. This indicates that users did not exhibit large or highly variable global level adjustments their target average levels are highly methodical.
- Where elasticity did manifest was in the management of True Peaks under specific acoustic stress. In the bus condition, the True Peak standard deviation for music spiked to 22.1 dBTP, with the overall range expanding to 92.2 dB. This suggests that while users kept the average loudness of the music stable, they struggled to consistently balance transient musical peaks against the bus environmental noise, leading to extreme variance at the limits of the mix.

7.4.3 Global Stability and Localized Convergence

The Linear Mixed Effects (LME) model directly challenges the assumption that distinct noise profiles force completely divergent mixing strategies.

- The interaction terms within the LME model (environment × element) were entirely non-significant (all $p>0.10$). This suggests that the relative loudness hierarchy between dialogue and the supporting stems remained structurally stable across the plane, train, and bus environments. The absence of significant environmental effects suggests that the presence of masking itself may be more influential than the specific transport context

in which that masking occurs. Participants therefore applied a similar prioritisation structure across the masking conditions evaluated in this study.

- While the global architecture is stable, the granular Wilcoxon pairwise comparisons revealed specific localized shifts. In the baseline plane condition, dialogue was kept statistically distinct from foley ($p=0.047$). However, in the train ($p=0.081$) and bus ($p=0.252$) conditions, this statistical gap disappeared. As acoustic stress increased, the distinction between dialogue and Foley became less pronounced. This suggests that participants prioritised Foley alongside dialogue when managing intelligibility under masking conditions.
- Because the global hierarchy is set across noise types, an adaptive system might not necessarily require highly complex, dynamically shifting suppression curves for average levels based on the specific noise profile (for example, plane vs. bus). Instead, it requires a stable, attenuation of immersive elements (music, backgrounds), combined with dynamic True Peak limiting to manage transients.

7.4.4 The Anchor Logic

The analysis suggests that narrative priority is a far stronger predictor of level than the environment itself.

- With dialogue serving as the anchor, purely immersive elements were consistently penalized. The LME main effects demonstrated that backgrounds ($\beta=-27.22$, $p<0.001$) and music ($\beta=-12.12$, $p=0.049$) were mixed significantly lower than dialogue.
- The results are consistent with the proposed anchor logic. Users apply a generalized prioritization structure regardless of the specific spectral profile of the masking noise and will consistently sacrifice environmental immersion to maintain that anchor.

7.4.5 User Interaction and Control

The post-experiment inductive content analysis (n=14) contextualises the above findings while highlighting areas for optimization.

- Environmental Drivers: A subset of participants explicitly cited the ambient noise floor as a driver for interaction, noting that "whatever is happening

in the room at the same time is a big factor" and acknowledging the interface's utility "if using non-noise-cancelling earphones." This provides validation that a subset of users actively desires masking compensation tools.

- Interface Ambiguity: Equally important, participant feedback highlighted cognitive load issues regarding the control surface itself. A few participants indicated difficulty distinguishing the five faders without visual labels, suggesting that an optimized framework might require a simpler, three-stem aggregation to reduce the burden on the user.

7.5 Limitations

The experimental design in this chapter was structured to test the robustness of the AMPLE framework under controlled environmental masking conditions. Although the acoustic profiles were derived from empirically informed simulations of public transport and aircraft cabin environments, participants remained within a laboratory setting. The study did not reproduce dynamic real-world variables such as physical movement, fluctuating crowd density, spontaneous device repositioning, or continuously shifting spectral noise characteristics. The findings should therefore be interpreted as controlled evaluations of masking influence rather than full ecological field validation.

Prototype 2 retained the interface architecture established in Chapter 6 to preserve methodological comparability and isolate the effect of environmental masking on stem prioritization. The study did not introduce alternative interaction paradigms, tonal equalization controls, or expanded interface elements, as the research objective was confined to evaluating whether the dialogue-anchor hierarchy remained stable under acoustic stress.

Statistical procedures were selected to evaluate defined within-condition and cross-condition behavioural patterns. The design did not incorporate exhaustive multivariate modelling across all demographics, hardware, or environmental permutations, nor was it structured as a longitudinal study of adaptation over time. As such, the findings should be interpreted strictly within the scope of the environmental masking scenarios examined here.

While the results demonstrate consistent prioritisation of dialogue and increased elasticity of supportive stems under masking, the chapter does not claim validation across all deployment ecosystems or streaming infrastructures. Broader platform integration, large-scale user modelling, and long-term adaptive learning mechanisms remain areas for future investigation beyond the boundaries of this thesis.

The experimental design was built to evaluate the framework under controlled environmental masking conditions. Although the acoustic profiles were empirically derived, participants remained seated within a static laboratory environment and were exposed primarily to sustained, steady-state noise.

Consequently, the framework's responsiveness to abrupt acoustic transitions, such as a listener moving from a quiet street onto a crowded train, remains untested. The absence of spontaneous device repositioning, fluctuating crowd density, and rapidly changing spectral conditions means that these findings represent a controlled evaluation of user adaptation rather than a full ecological stress test. Future real-world implementations will likely require temporal smoothing mechanisms to prevent abrupt level changes when ambient noise conditions shift rapidly.

7.6 Summary

This chapter concluded the evaluation of the AMPLE framework by examining audience mixing preferences under simulated acoustic stress. By introducing ecologically valid masking noise, the experiment tested the strength of the behavioural patterns identified in Chapter 6.
Taken together, the two experiments establish a directional behavioural hierarchy for film sound consumption:

Across both ideal (Chapter 6) and noisy (Chapter 7) conditions, dialogue maintained a statistically rigid position as the primary focus of the mix. This confirms the conceptual stability of the anchor assumption central to the AMPLE framework.

Prototype 2 revealed that while the overarching hierarchy between dialogue, music, and backgrounds remains globally stable across different noise environments, localized elasticity exists. Specifically, transient peaks exhibit high variance under specific masking condition, and tactical narrative elements – Foley – dynamically converge with dialogue as acoustic stress increases.

These findings provide the empirical mandate for the adaptive mixing framework proposed in Chapter 5. The data suggests that users treat intelligibility as a constant and immersion as a variable. The final chapter will evaluate the implications of these findings for the research questions and the field of adaptive audio.

Chapter 8: Discussion and Conclusion

This thesis set out to examine how film sound mixes translate across contemporary listening contexts and to determine whether a user-centred adaptive framework could address the persistent intelligibility and balance concerns identified in both audience and professional domains. The preceding chapters have moved systematically from problem identification to framework development and evaluation. This final chapter draws those strands together, articulating the central contributions of the work and situating them within current practice in film sound delivery.

8.1 The Problem Space: Systemic Translation Failure

The research began by establishing a disconnect between production standards and lived listening conditions. The large-scale audience survey (Chapter 3) substantiates that film, and streaming content is frequently consumed in acoustically variable environments, with dialogue intelligibility emerging as a recurrent concern. Professional insights (Chapter 4) suggested an acute awareness of these translation challenges yet also highlighted structural constraints within legacy workflows that limit the feasibility of producing bespoke mixes for every listening scenario. Together, these findings expressed a clearly bounded problem space: translation failure is not solely technical, nor solely behavioural, but systemic.

Together, the empirical findings from Chapters 6 and 7 provide supporting evidence suggesting that the behavioural assumptions underpinning AMPLE are stable across both controlled and environmentally masked listening conditions, positioning the framework as a viable response to the translation gap identified earlier in the thesis.

8.2 The Proposed Solution: The AMPLE Framework

The AMPLE framework was developed in response to this gap. Rather than proposing wholesale changes to creative practice, the framework reallocates certain aspects of control to the playback stage, aiming to preserve creative intent through structured metadata while enabling listener-level adjustment of core stems.

The two experimental prototypes then examined this premise empirically:

- Prototype 1 (Ideal Conditions): Under reference listening conditions, participants consistently prioritised dialogue as an anchor element while exhibiting flexibility in secondary stems
- Prototype 2 (Masking Conditions): Under simulated environmental noise, this anchoring behaviour persisted, though with greater individual variability in the supportive stems.

These findings provide behavioural evidence suggesting that participants adapted their mixing strategies in relation to context, supporting the conceptual assumptions of the framework.

8.3 Core Contributions

Taken together, the research makes three interlinked contributions:

1. It provides empirical documentation of audience mixing priorities under both ideal and adverse conditions.
2. It articulates a structured adaptive architecture grounded in those behavioural findings.
3. It suggests, through experimental evaluation, that user-focused stem control can function without undermining narrative coherence within the tested use cases.

Rather than arguing for the replacement of existing workflows, this work proposes a complementary layer that addresses environmental and demographic variability at the point of consumption.

The remainder of this chapter reflects on the theoretical implications of these findings, their practical relevance to industry workflows, the limitations of the present research, and directions for further investigation. In doing so, it situates the thesis within the evolving landscape of film sound production and delivery, where variability of context is increasingly the norm rather than the exception.

8.4 Summary of Key Findings

This research provides empirical evidence to address the central research questions posed in Chapter 1. By moving beyond theoretical critiques of industry standards, the thesis translates observed user behaviour into actionable architectural principles for potential future object-based audio delivery.

8.4.1 Addressing the Research Questions: Towards Context-Aware Audio Delivery

RQ1: How does the current delivered listening experience impact audiences across different platforms, and where do existing static delivery models fail to meet their needs?

The findings suggest that current delivery models do not consistently provide a satisfactory listening experience across platforms. Evidence from the requirements gathering reported in Chapter 3 indicates that more than 60% of participants using both mobile and stationary devices described frequent volume adjustment to maintain intelligibility. This pattern of behaviour indicates a mismatch between static delivery formats and the varied acoustic contexts in which end users consume media, supporting the interpretation that existing approaches are not reliably aligned with audience listening conditions.

RQ2: What are the audience preferences, and their listening behaviour when they watch films on mobile or stationary hardware?

Through controlled empirical testing (Chapters 6 and 7), this research suggests a stable perceptual hierarchy within the evaluated conditions. Participants

consistently prioritized dialogue as the primary anchor element. However, the research indicated a divergence in how secondary elements are treated. Purely immersive elements music, SFX, and backgrounds appear highly elastic; under environmental masking, listeners actively attenuated these stems to preserve narrative intelligibility.

Conversely, physical narrative sounds (Foley) were not sacrificed. As acoustic stress increased, users actively grouped Foley tightly with the dialogue anchor, demonstrating a clear behavioural demand for context-driven flexibility.

RQ3: What are the key components of an adaptive mixing framework that can effectively reflect audience preferences and listening behaviours?

Synthesising these behavioural findings with professional workflow constraints, the AMPLE framework (Chapter 5) proposes three structural pillars that future adaptive standards could incorporate:

1. Object-Based Stem Separation: To permit the independent control of narrative versus supportive audio layers.
2. Semantic Metadata: To explicitly tag and protect the anchor element (dialogue) from automated attenuation.
3. Context Awareness: To dynamically adjust the dynamic range of supportive elements relative to the local noise floor, aiming to preserve the signal-to-noise ratio without requiring manual user intervention.

8.4.2 Methodological Triangulation and Core Contributions

A key methodological strength of this thesis lies in the triangulation of large-scale audience surveys, professional workflow analyses, and controlled experiments. Bringing these parts highlights a point of convergence: the prioritization of dialogue as a narrative anchor is central to professional production workflows, and this hierarchy was consistently reflected by experimental participants adjusting their mixes during the listening tasks. The findings also indicate that existing production workflows often rely on a single fixed mix intended to serve multiple contexts, which appears less effective and more so when listening conditions become acoustically challenging.

Evidence from the empirical studies suggests that participants were willing to adjust elements of the soundscape, particularly secondary components, in order to maintain intelligibility under environmental masking. This behavioural pattern supports the interpretation that adaptive, user-responsive audio tools are not only technically achievable but also aligned with observed listening practices. Across the experimental prototypes, participants showed a consistent tendency to prioritise dialogue while allowing greater flexibility in supportive elements.

Across both experimental studies, narrative role emerged as a more consistent predictor of listener behaviour than environmental context. Participants consistently prioritised dialogue while treating supportive elements more flexibly,

regardless of the specific masking condition.Taken together, these findings support the formulation of a stable perceptual hierarchy within the tested cohort, which represents an original contribution of the thesis. Rather than framing audio post-production as a purely predictive exercise undertaken in the studio, the results suggest a shift towards adaptive delivery, where aspects of the mix can be resolved at the point of listening in response to local conditions.

8.4.3 Research Dissemination and Impact

Aspects of this research have been disseminated through peer-reviewed publications and conference presentations. These outputs have allowed for critical engagement with the academic and professional audio community, providing peer-reviewed support for the framework's approach and highlighting the relevance of the findings within the wider audio field.

Outputs from this research include:

- Shalabi, A., & McGregor, I. (2024). Capturing Film Mixers' Workflows. Journal of the Audio Engineering Society, 72(9), 603–612
- Gelby, A., & McGregor, I. (2023). *Capturing audience film sound preferences*. Audio Engineering Society.

8.5 Discussion of Findings in Relation to Literature

This research engages directly with the core challenges articulated in professional standards and academic literature, moving beyond descriptive comparisons to offer empirical, actionable insights for the field of audio delivery and distribution.

8.5.1 Methodological Triangulation

Existing literature in this domain tends to isolate single perspectives – focusing either on demographic audience studies (Atkin & Litman, 1986; BFI, 2017; Toole, 2015b), isolated workflow critiques (Rumsey, 2016b; Newell, 2014a), or theoretical standards. A key distinguishing feature of this thesis is its methodological triangulation. By integrating large-scale audience surveys, professional workflow insights, and controlled experiments, this research seeks to bridge the gap between creator intent and endpoint reality. This holistic approach provides measurable behavioural evidence of how participants compensated in practice, contributing to a research conversation that integrates theoretical policy with empirical response.

8.5.2 From Accessibility to Universal Context Responsiveness

Previous research driving adaptable audio has often isolated intelligibility challenges to specific demographic or audiological profiles (Lopez et al., 2018; Shirley et al., 2017; Ward, 2020). This thesis broadens that framing by suggesting that dynamic range compression and acoustic masking are also environmental challenges. By illustrating that participants required active mix alterations when subjected to simulated transport noise (Chapter 7), this research provides empirical evidence indicating that context-responsiveness may be a highly beneficial feature for wider audiences, rather than solely a niche accessibility tool.

8.5.3 Multi-Stem Adaptation vs. Single-Element Enhancement

Recent commercial developments have introduced basic dialogue enhancement tools to address widespread intelligibility complaints (Amazon, 2024; Apple Inc., 2025). However, the experimental findings of this thesis highlight potential limitations of these single-element solutions. The results suggest that simply boosting dialogue may be insufficient; participants actively and variably adjusted secondary elements (music, SFX, and backgrounds) to shape their acoustic environment.

These findings indicate that listeners may benefit from contextual multi-stem adaptation approaches, with participants adjusting secondary soundtrack elements in addition to overall level changes.

8.5.4 Justifying a Hybrid Adaptive Architecture

The consistent prominence of dialogue as the anchor element across both ideal and noisy experimental conditions aligns with previous works regarding narrative focus (Busselle & Bilandzic, 2009a; Jaeckle, 2013; Rushton, 2023). However, the high variance observed in the supportive stems questions the assumptions underlying the legacy one-size-fits-all delivery standards (BBC & DPP, 2020).

Ultimately, these findings illustrate predictable perceptual hierarchies – the dialogue anchor – alongside highly contextual secondary adjustments for immersion layers. This dual reality supports the rationale for the architecture proposed by the AMPLE framework.

8.6 Theoretical Contributions

This thesis operationalizes User-Centred Design (UCD) principles within the historically rigid domain of post-production audio infrastructure (Norman & Draper, 2017). Rather than treating the audience as passive receivers of a fixed playout, the UCD approach was explicitly integrated throughout the methodology: first by identifying user pain points through the audience survey (Chapter 3), and subsequently by granting users direct control over stem balancing during the experimental evaluations (Chapters 6 and 7). The contributions are defined across three conceptual shifts:

8.6.1 Reconfiguring Design Agency in Audio Delivery

Historically, film sound delivery operates on a model that grants the creator absolute agency over the final output. This research reframes adaptive audio not merely as an accessibility feature, but as a reallocation of design agency within the distribution pipeline. By allowing the listener's environment, hardware, and preferences to dictate the final balance of the stems, this thesis theoretically positions the user's playback device as an active, final stage of the mixing process.

8.6.2 Metadata as Adaptive Decision Logic

A conceptual advancement of this thesis is the repositioning of object-based audio metadata within streaming ecosystems. In legacy standards, metadata is predominantly classificatory – used for spatial rendering, archival organization, or binary personalization (for example, language track selection).

In contrast, the AMPLE framework reconceptualizes metadata as adaptive decision logic. Existing object-based systems primarily use metadata for rendering and classification. The AMPLE framework extends this role by treating metadata as a mechanism for adaptive decision-making based on listener context. By leveraging semantic tags alongside real-time environmental awareness, the framework transforms metadata from a passive descriptor into an active, real-time mixing agent. This enables the system to make context-sensitive decisions at the point of playout, adjusting the mix based on local conditions.

8.6.3 Non-Disruptive Innovation in Media Production

A prevailing assumption in media production research is that meaningful technological innovation inherently requires disrupting established production paradigms. This thesis offers an alternative perspective to that premise.

The novelty of this approach lies in illustrating how rich, user-aware adaptation can be integrated into existing professional supply chains. By proposing a shift of the adaptive processing burden entirely to the endpoint infrastructure, the framework indicates that personalization at scale could be operationally compatible with current post-production workflows. Rather than forcing re-recording mixers to abandon their creative practices or generate dozens of bespoke deliverables, this work provides a foundational theoretical model for future broadcast standards: achieving advanced endpoint personalization through delayed rendering, without disrupting the creative pipeline.

8.7 Original Contributions to Knowledge and Practice

This thesis presents, original contributions to the understanding of film sound perception and the application of object-based audio approaches within adaptive listening environments. The following sections outline these contributions by identifying aspects of the research that extend existing work, alongside their potential relevance to research and professional practice.

8.7.1 Empirical and Conceptual Contributions

This research provides behavioural evidence that informs understanding of how audiences engage with media under varied listening conditions.

While dialogue is widely regarded as a central component of film sound practice, the findings presented in this thesis suggest that dialogue functions as a stable perceptual anchor across both reference and environmentally masked listening conditions. Under increased acoustic masking, participants consistently prioritised narrative intelligibility often attenuating immersive soundtrack elements in order to preserve speech clarity.

The thesis further contributes to understanding audience listening behaviour by indicating differences in how supportive soundtrack stems are treated under acoustic stress. Findings suggest a conditional flexibility in listener preference, where atmospheric and musical elements were often attenuated in response to increased environmental masking in order to preserve intelligibility. Findings also suggest selective preservation of functional physical sounds such as Foley, rather than uniform attenuation across all non-dialogue elements. This pattern

indicates that listeners may apply more targeted adjustments to soundtrack components under masking conditions, rather than responding through uniform global changes. Collectively, these observations provide evidence that listener adaptation behaviour may involve more nuanced responses to environmental masking than assumed within static listening models.

8.7.2 Technical Contributions

The technical contribution of this work lies not in the creation of object-based metadata, but in providing the empirical parameterization required to make it functionally adaptive at the endpoint.

The primary technical contribution lies in extracting audience mixing data (LUFS boundaries, True Peak variances, and stem-level standard deviations) and utilizing those figures to operationalize existing object-based audio metadata. By modelling the observed behavioural patterns of the listener under acoustic stress – the dialogue anchor, the localized convergence of Foley, and the conditional attenuation of immersive stems – this research offers a logical framework for transitioning object-based audio from a static spatial tool into a dynamic, context-aware delivery system at the end point.

The thesis provides a tested technical framework (AMPLE) that integrates these empirical mixing rules with environmental awareness. It illustrates the technical feasibility of shifting the adaptive processing from the studio to the playback device. This suggests that endpoint personalisation could be achieved without requiring mixers to generate bespoke deliverables, thus protecting established post-production pipelines.

8.7.3 Methodological Contributions

The research design proposes a replicable approach for investigating audio delivery systems by breaking down traditional academic and industry silos.

Methodological Triangulation of the Audio Supply Chain: The most distinctive methodological contribution of this thesis is its triangulation of large-scale audience survey data, professional workflow analysis, and controlled testing. While previous studies have largely isolated these domains, this integrated approach provides a stronger framework for evaluating film sound holistically – from the creator's structural constraints to the end-user's reality.

Ecologically Valid Stress-Testing for Audio Frameworks: By moving beyond idealised critical listening rooms and introducing simulated, real-world acoustic masking (transport environments) as a controlled independent variable, the thesis sets a rigorous methodological precedent. This approach provides a blueprint for evaluating the ecological validity and robustness of future adaptive audio systems under the types of challenging acoustic conditions where static mixes typically fail.

8.8 Implications and Recommendations

The findings from this research offer empirically grounded implications for the future of film sound delivery. By indicating that listener preference is closely linked

to acoustic context, the data highlights the potential value of systems that can flexibly respond to both the user and their environment.

Furthermore, the results help reframe ongoing discussions around audio accessibility. Rather than viewing intelligibility purely as a niche audiological issue, the empirical data – specifically the consistent prioritisation of the dialogue anchor across both ideal (prototype 1) and masked (prototype 2) conditions – illustrates that masking is an environmental challenge affecting wider audiences. Forcing static, cinema-optimised mixes through diverse real-world devices can degrade the experience across a variety of listening contexts, suggesting that context-aware adaptation could serve as a broadly applicable feature, rather than solely an accessibility tool.

Based on these findings, the following recommendations are structured for the key stakeholders in the audio delivery pipeline:

8.8.1 Industry and Standards Bodies

The experimental findings indicate that listener behaviour is neither static nor uniform. In Prototype 1, participants consistently prioritised dialogue as the anchor element under controlled conditions. In Prototype 2, the introduction of simulated transport environments produced greater variability, particularly in the treatment of music, sound effects, and background layers. These results suggest that current one-size-fits-all delivery standards are misaligned with the observable diversity of listener behaviour across environments.

Rather than relying solely on incremental revisions to loudness targets or nearfield monitoring guidelines, standards bodies and platform providers could consider supporting optional adaptive rendering layers at the playback stage. This does not require replacing existing production workflows. Instead, it involves extending the use of object-based stems and narrative metadata already present in contemporary delivery formats, allowing rendering engines to respond to device capability and environmental context at the point of playout.

The evidence presented in this thesis suggests that such adaptation would not undermine creative intent. Dialogue was consistently preserved by listeners as the primary narrative driver, while secondary elements were adjusted flexibly. Supporting context-aware rendering therefore aligns with audience behaviour rather than contradicting it.

In this way, the research proposes a realistic pathway for standards evolution: one that maintains compatibility with established post-production pipelines while acknowledging that the listening environment is no longer fixed.

8.8.2 Dubbing Mixers and Professional Practice

For professional re-recording mixers, the implication of this research is not a displacement of creative authority, but rather a protection of it. By shifting the burden of environmental adaptation to the rendering stage at the endpoint, mixers could be relieved from the competing demands of creating a single compromise mix intended to survive both a critical listening room and a noisy commuter train. The framework is designed to protect their core narrative balance – the dialogue anchor, while allowing the supportive stems to flex according to the user's environment, with the goal of preserving the intelligibility of the mix against real-world acoustic degradation.

8.8.3 Academia

For academic researchers, this thesis contributes to a behavioural foundation that highlights specific, unresolved technical gaps. Having illustrated the potential benefits of adaptive mixing, future research could explore the automation of these processes. Specifically, academia might benefit from focusing on developing lightweight, automated environment-detection algorithms (for example, utilizing smart-device microphones to classify ambient noise profiles in real-time) and conducting longitudinal studies on the psychoacoustic impact of automated versus manual stem suppression at scale.

8.8.4 Implementation Pathway

Regarding implementation within the current supply chain, the optimal balance of device-side versus server-side adaptation remains an open technical challenge influenced by platform infrastructure, bandwidth, and user privacy considerations. Importantly, this thesis does not prescribe a single, rigid implementation pathway; rather, it helps to map the design space for future adaptive audio ecosystems. Pilot projects bridging content creators, technologists, and standards organizations will be crucial to refining these approaches and scaling them from theoretical frameworks into consumer realities.

8.9 Positionality and Researcher Perspective

As a professional operating within the audio industry, my background has naturally influenced both the direction and implementation of this research. Almost two decades of experience designing, integrating, and troubleshooting professional audio workflows for various media environments have provided practical insight into the systemic challenges faced by both practitioners and audiences. This standpoint informed the initial formulation of the research questions, the contextual framing of the translation gap, and the subsequent interpretation of the empirical data.

Being acutely aware of the impact of my own positionality, care was taken to ensure that the research design and analytical processes remained transparent. While my expertise inherently influenced certain methodological choices – such as the categorisation of the five discrete mix stems and the topology of the

experimental interface – I actively sought to decouple these decisions from my personal aesthetic preferences or ingrained professional habits.

To maintain methodological self-awareness, assumptions were actively tested and challenged throughout the research design. For example, rather than designing the prototype interface based solely on standard professional console layouts, the UI was iteratively stripped of semantic text labels, such as dialogue or SFX.

This was a deliberate reflexive choice to prevent my own industry assumptions from biasing the non-expert participants, prompting them instead to rely purely on psychoacoustic feedback. Similarly, when selecting the auditory stimuli, peer and supervisory feedback led to adjustments that broadened the range of material, with the aim of reflecting typical high-dynamic-range scenarios rather than niche critical-listening material. By grounding these choices in established literature and subjecting them to peer review, I aimed to mitigate the risk of reproducing unexamined industry norms.

Ultimately, my dual role as both practitioner and researcher is presented not as a definitive authority, but as a perspective that brings practical contextual realism to the study of audio workflows. This practitioner-researcher lens was intended to support the ecological validity of the prototypes, maintaining a commitment to the scientific method while offering insights of practical, actionable value to both the academic field and the wider professional audio community.

8.10 Limitations

While this research offers empirical evidence to support the AMPLE framework, it is important to acknowledge the boundaries of the study. The limitations can be categorised into methodological constraints, generalisability boundaries, and future system-level implementation challenges.

8.10.1 Methodological Constraints

A theoretical risk inherent to within-subjects experimental designs is listener fatigue and memory-based learning effects. While this was managed in the study design by utilising distinct narrative clips for each prototype, restricting the test stimuli to three-minute excerpts, and limiting the number of permitted playback loops, repeated critical listening tasks can nonetheless place a cognitive load on participants.

Furthermore, while the sample size (N=25 per prototype) was sufficient to observe primary psychoacoustic baselines and achieving statistical significance on the main variables (for example, dialogue vs. supportive stems), it limits the statistical power required to detect granular interaction effects. A larger sample size would be beneficial to more fully explore how specific demographic variables – such as varying degrees of age-related hearing loss (presbycusis) – interact with specific environmental masking profiles.

8.10.2 Generalisability and Ecological Validity

To support internal validity and isolate environmental noise as a variable, the experiments relied on simulated acoustic environments and calibrated playback

hardware (Genelec monitors at 79dB(C)). While this laboratory control was highly beneficial for rigorous data collection, it inherently bounds the ecological validity of the findings.

The tightly controlled playback context does not fully account for the substantial variability introduced by the wide range of consumer listening devices (e.g., mobile phone speakers, inexpensive earbuds, untreated domestic living rooms). These real-world acoustic variables introduce phase and frequency distortions that could influence perceptual outcomes and mixing behaviour in ways not captured within a critical listening room.

8.10.3 System-Level Boundaries and Implementation Challenges

Finally, it should be noted that this thesis outlines the behavioural requirements and logical architecture for adaptive delivery; it does not engineer a final, commercial software deployment. Moving this framework from an experimental prototype to a consumer-facing product introduces system-level challenges that remain beyond the scope of this study.

Implementing such a framework within existing linear production pipelines would require significant engineering refinement. Decisions regarding a mixed rendering architecture – whether processing occurs centrally at the server origin or locally at the edge (the user's device) for lower latency – would require careful Application Programming Interface (API) integration. Furthermore, an adaptive system that continuously monitors the user's environment to trigger noise compensation raises complex ethical considerations regarding continuous microphone access, secure data handling, and the safeguarding of user privacy.

8.11 Future Work

Building directly on the empirical foundations and limitations presented in this thesis, future research can be strategically categorised into immediate technical extensions of the AMPLE framework and longer-term applications across adjacent audio domains.

8.11.1 Immediate Extensions of the Framework

The findings from prototypes 1 and 2 highlight specific areas where the framework's mechanics can be refined to further reduce user friction:

- Automated Environment Detection and Responsive Mixing: In prototype 2, participants actively compensated for simulated transport noise by manually adjusting faders. While this suggested a desire for adaptation, manual mixing imposes a cognitive load. Future research could conduct controlled trials comparing manual user adjustment against fully automated, algorithm-driven adaptation (for example, using device microphones to detect ambient noise and automatically suppress the background stems). This would measure whether system-driven adaptation improves sustained engagement over manual control.

- User-Controlled Spectral Shaping: The interaction data from prototype 2 indicated that masking of the bus environment impacted the perception of music and Foley. Future experiments could expand the framework beyond simple gain (volume) control to include adaptive spectral balancing (EQ). Allowing users – or an automated system – to carve out specific frequencies dynamically could preserve the perceived loudness of supportive elements without masking the dialogue fundamental.

8.11.2 Long-Term Applications

As object-based audio delivery becomes more ubiquitous, the core logic of this thesis – prioritising the anchor and flexing the environment – can be applied to broader media contexts:

- Longitudinal Studies on User Adaptation: Extended, in-situ trials are required to observe how user preferences and interactions with context-aware mixing systems evolve over time outside of the laboratory. This would capture critical insights into learning curves, tweak fatigue, and sustained engagement in domestic settings.
- Live Events and Broadcast Accessibility: Future work should explore how real-time, metadata-driven mixing tools can be applied to live sports, concerts, or news broadcasts. Experiments could assess whether allowing users to dynamically prioritise commentary over crowd noise improves accessibility and satisfaction for individuals with diverse hearing profiles in real-time environments.
- Spatial and Immersive Media: Investigating the impact of context-aware mixing in 6 Degrees of Freedom (6DoF) environments, such as VR or AR applications. Research could measure how adaptive mixing responds not only to real-world background noise but to the user's spatial orientation and visual focus.

8.11.3 Concluding Remarks

This thesis illustrates the conceptual and technical viability of an adaptive, user-centred mixing framework under controlled experimental conditions. By foregrounding empirical audience behaviour and evaluating the impact of context-aware adaptation, the research provides a data-driven foundation for addressing the persistent translation gap in film sound delivery.

The empirical evidence indicates that while the prioritisation of dialogue remains stable, the perception of immersive, supportive audio is highly elastic and context

dependent. Consequently, the findings suggest that relying solely on static delivery standards and incremental loudness revisions is inadequate for modern consumption habits.

Ultimately, as media environments become increasingly unpredictable, this research proposes a pragmatic pathway forward. By intelligently leveraging object-based metadata to shift adaptive agency to the endpoint, the industry could accommodate the diverse acoustic realities of the audience. Such an approach aims to support the preservation of creative intent, narrative integrity, and speech intelligibility, regardless of where or how the listener chooses to engage with the content.

8.12 Summary and Conclusion

This thesis set out to investigate the persistent translation gap in film sound delivery, examining why professional mixes frequently fail to satisfy audiences across diverse playback contexts. By systematically moving from the documentation of widespread audience dissatisfaction and professional workflow constraints, through to the controlled empirical testing of listener behaviour, this research successfully developed and evaluated the AMPLE framework.

Rather than relying on theoretical advocacy, the research provides measurable, empirical evidence that shifting mixing agency to the playback endpoint could resolve the limitations of static legacy standards. The data suggest that audiences do not consume audio uniformly; however, their adaptive behaviour appears structurally predictable within the tested conditions. By providing evidence that listeners consistently lock dialogue as a stable narrative anchor, converge physical narrative cues – Foley – with that anchor under stress, and treat purely immersive stems as highly elastic variables, this thesis offers an empirically supported conceptual model for context-aware audio delivery.

As one potential implementation scenario, future streaming platforms could leverage this metadata-driven architecture to offer sophisticated, context-aware playback options. Rather than applying crude, destructive master-bus compression, an adaptive system could utilize device-level environmental data to resolve the stems in real-time – automatically attenuating intrusive background layers when commuter noise is detected or offering a true dialogue priority mode that protects the narrative anchor without degrading the dynamic range of the remaining mix. In doing so, personalization and creative intent can coexist at the point of playout.

Ultimately, this thesis resolves its central inquiry by indicating that the current delivered listening experience is systematically misaligned with modern consumption habits, and that an object-based, user-responsive framework has a strong potential to successfully bridge this gap. While the systemic engineering and integration of such technologies across the broadcast and postproduction supply chain remains an open challenge for the industry, this research empirically suggests that in an increasingly mobile and unpredictable media landscape, the future of film sound delivery is likely to become increasingly adaptive.

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Appendix A: Survey – Audience.

Page-1

Adaptive Mixing Study

Thank you for considering taking part. This study aims to investigate an adaptive mixing framework for a personalised listening experience for film. The project is being run by Ahmed Shalabi, Dr Iain McGregor and Dr Linda Shore from Edinburgh Napier University.

You will be asked to answer 45 short online questions, which should take approximately 30 minutes. Specifically, about the hardware you use to listen to film sound, and any modifications that you wish to make to the levels of the dialogue, music and sound effects.

All of your responses will be anonymised using the convention P01, P02…. Data will be stored for 10 years on a secure cloud service, and not shared with any other parties. The anonymised results will be presented at an audio conference, which has yet to be determined.

Identifying the hardware preferences and audience preferences with regards to film sound, may allow for a development of algorithm that will have an overall improvement and personalisation of film sound through correction factors tailored according to audience preferences and behaviour.

The results might also be of interest to those designing and mixing film sound, in respect of identifying the relative importance of different stems from a audience perspective, providing information about how each auditory element could be mixed, with minimum or no user intervention.

If you have any questions, please do not hesitate to contact Ahmed Shalabi on ahmed.shalabi@napier.ac.uk, Dr Iain McGregor on 0131 455 2788 or i.mcgregor@napier.ac.uk, Dr Linda Shore via l.shore@napier.ac.uk

If you have any concerns or wish to complain about the research please contact Professor Ben Paechter on 0131 455 2764 or via B.Paechter@napier.ac.uk.

Thank you for your time.

Continue later

Page-2

Adaptive Mixing Study

Edinburgh Napier University Research Consent Form

An adaptive mixing framework for a personalised listening experience

Edinburgh Napier University requires that all persons who participate in research studies give their written consent to do so. Please read the following and sign it if you agree with what it says.

1. I freely and voluntarily consent to be a participant in the research project on the topic of adaptive mixing framework for a personalised listening experience to be conducted by Ahmed Shalabi who is a student at Edinburgh Napier University, Dr Iain McGregor and Dr Linda Shore who are staff members at Edinburgh Napier University.

2. The broad goal of this research study is to explore audience' choice of reproduction hardware, levels of listening, exposure time and relative levels of sound stems (Dialogue, Music and Sound Effects). Specifically, I have been asked to complete an online questionnaire, which should take approximately 30 minutes to complete.

3. I have been told that my responses will be anonymised. My name will not be linked with the research materials, and I will not be identified or identifiable in any report subsequently produced by the researcher.

4. I also understand that if at any time during the survey I feel unable or unwilling to continue, I am free to leave. That is, my participation in this study is completely voluntary, and I may withdraw from it without negative consequences. However, after data has been anonymised or after publication of results it will not be possible for my data to be removed as it would be untraceable at this point.

5. In addition, should I not wish to answer any particular question or questions, I am free to decline.

6. I have been given the opportunity to ask questions regarding the survey and my questions have been answered to my satisfaction.

1. I have read and understand the above and consent to participate in this study. My agreement is not a waiver of any legal rights.

☐ Yes ☐ No

Continue later

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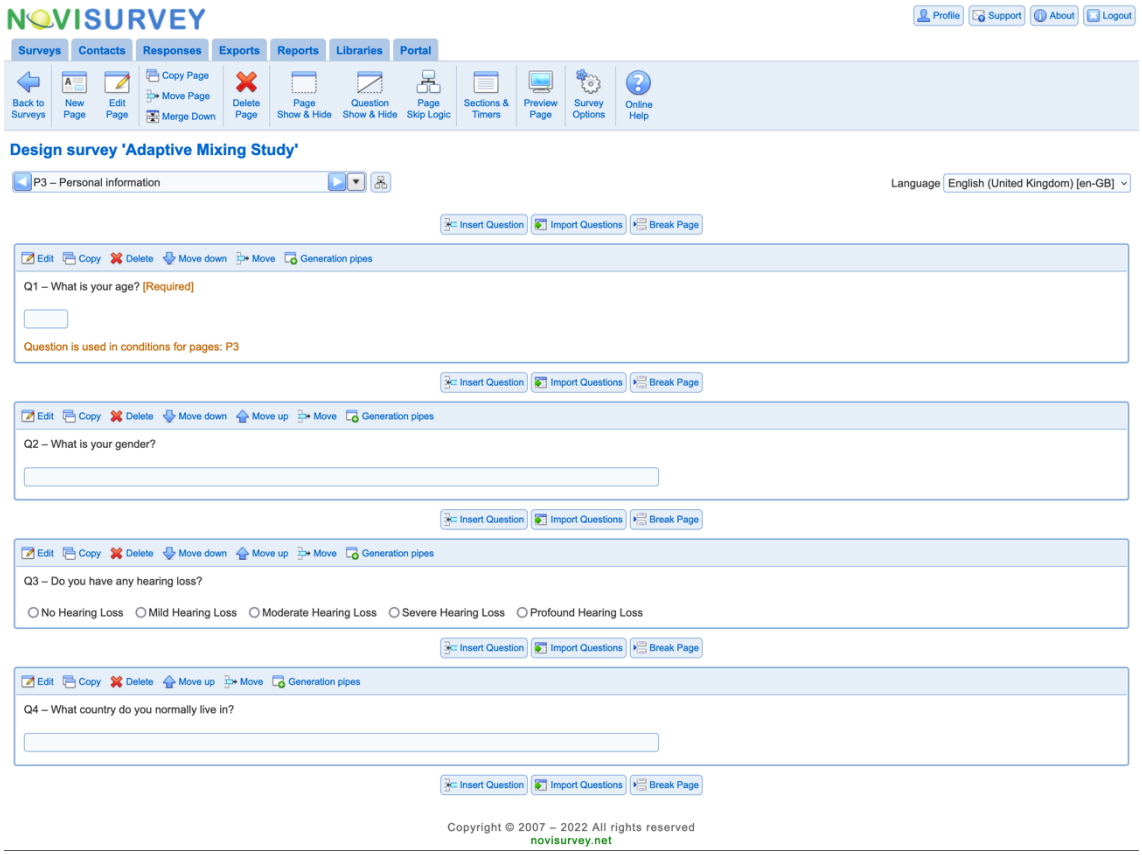
NOVISURVEY
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Contacts
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Reports
Libraries
Portal
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New Page
Edit Page
Copy Page
Move Page
Merge Down
Delete Page
Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P3 – Personal information
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move up
Move
Generation pipes
Q1 – What is your age? [Required]
Question is used in conditions for pages: P3
Q2 – What is your gender?
Q3 – Do you have any hearing loss?
No Hearing Loss
Mild Hearing Loss
Moderate Hearing Loss
Severe Hearing Loss
Profound Hearing Loss
Q4 – What country do you normally live in?
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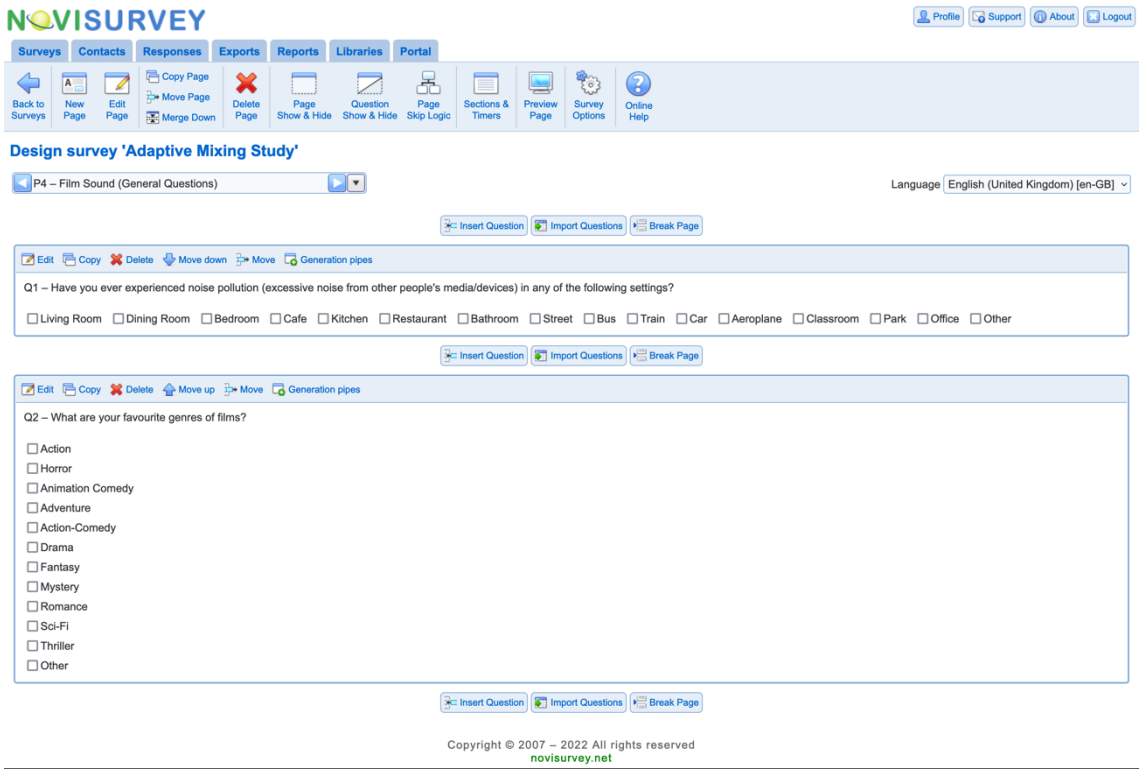
NOVISURVEY
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Surveys
Contacts
Responses
Exports
Reports
Libraries
Portal
Back to Surveys
New Page
Edit Page
Copy Page
Move Page
Merge Down
Delete Page
Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P4 – Film Sound (General Questions)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move up
Move
Generation pipes
Q1 – Have you ever experienced noise pollution (excessive noise from other people's media/devices) in any of the following settings?
Living Room
Dining Room
Bedroom
Cafe
Kitchen
Restaurant
Bathroom
Street
Bus
Train
Car
Aeroplane
Classroom
Park
Office
Other
Q2 – What are your favourite genres of films?
Action
Horror
Animation Comedy
Adventure
Action-Comedy
Drama
Fantasy
Mystery
Romance
Sci-Fi
Thriller
Other
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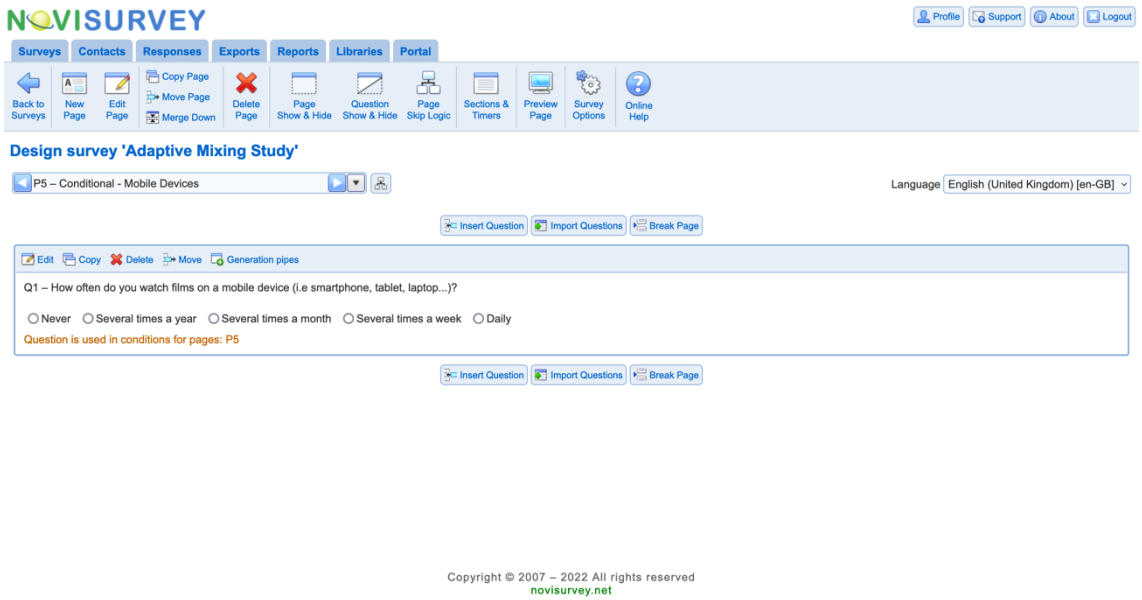
Page-5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Profile
Support
About
Logout
Surveys
Contacts
Responses
Exports
Reports
Libraries
Portal
Back to Surveys
New Page
Edit Page
Copy Page
Move Page
Merge Down
Delete Page
Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P5 – Conditional - Mobile Devices
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move
Generation pipes
Q1 – How often do you watch films on a mobile device (i.e smartphone, tablet, laptop...)?
Never
Several times a year
Several times a month
Several times a week
Daily
Question is used in conditions for pages: P5
Insert Question
Import Questions
Break Page
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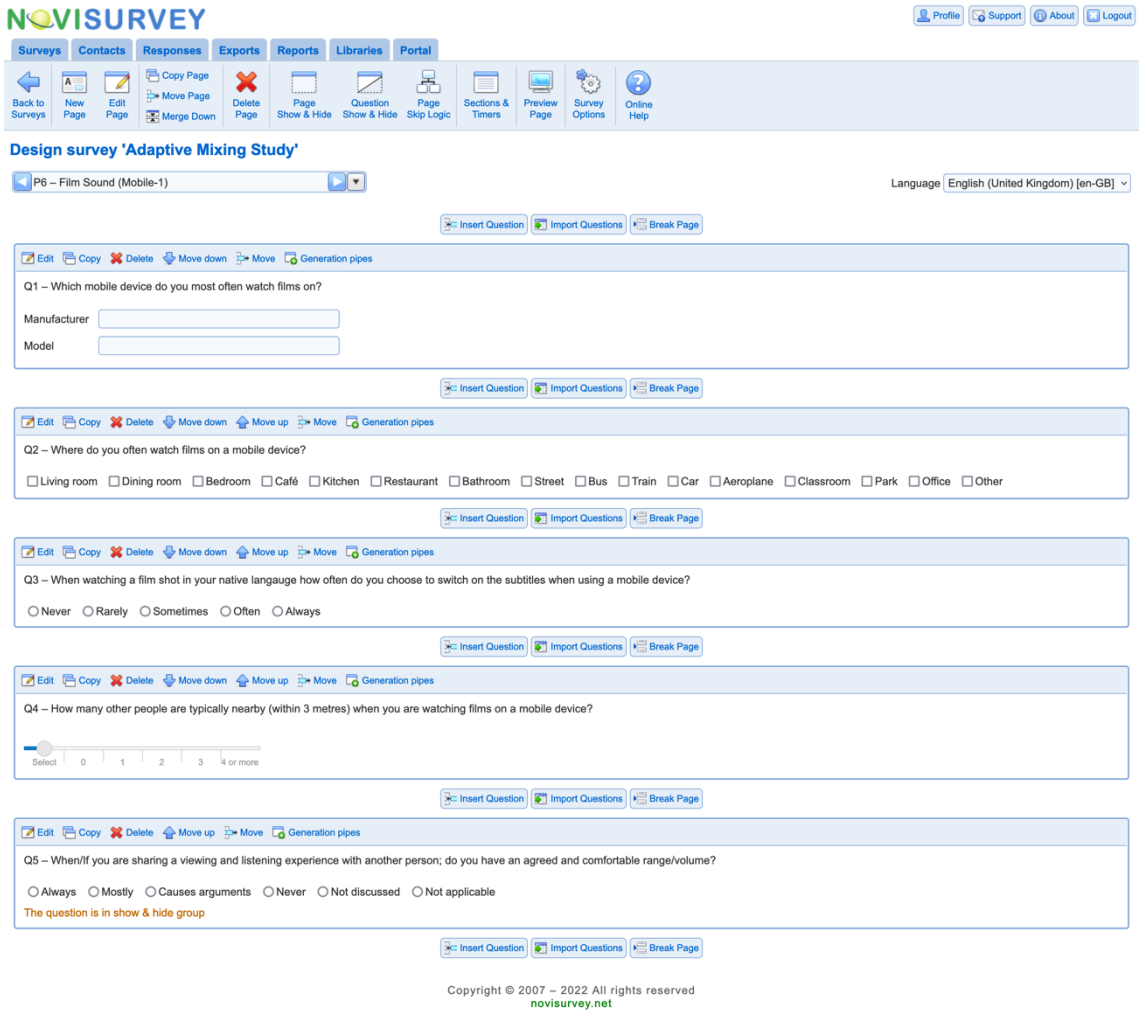
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New Page
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P6 – Film Sound (Mobile-1)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – Which mobile device do you most often watch films on?
Manufacturer
Model
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move up
Move
Generation pipes
Q2 – Where do you often watch films on a mobile device?
Living room
Dining room
Bedroom
Café
Kitchen
Restaurant
Bathroom
Street
Bus
Train
Car
Aeroplane
Classroom
Park
Office
Other
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move up
Move
Generation pipes
Q3 – When watching a film shot in your native langauge how often do you choose to switch on the subtitles when using a mobile device?
Never
Rarely
Sometimes
Often
Always
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move up
Move
Generation pipes
Q4 – How many other people are typically nearby (within 3 metres) when you are watching films on a mobile device?
Select
0
1
2
3
4 or more
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move up
Move
Generation pipes
Q5 – When/If you are sharing a viewing and listening experience with another person; do you have an agreed and comfortable range/volume?
Always
Mostly
Causes arguments
Never
Not discussed
Not applicable
The question is in show & hide group
Insert Question
Import Questions
Break Page
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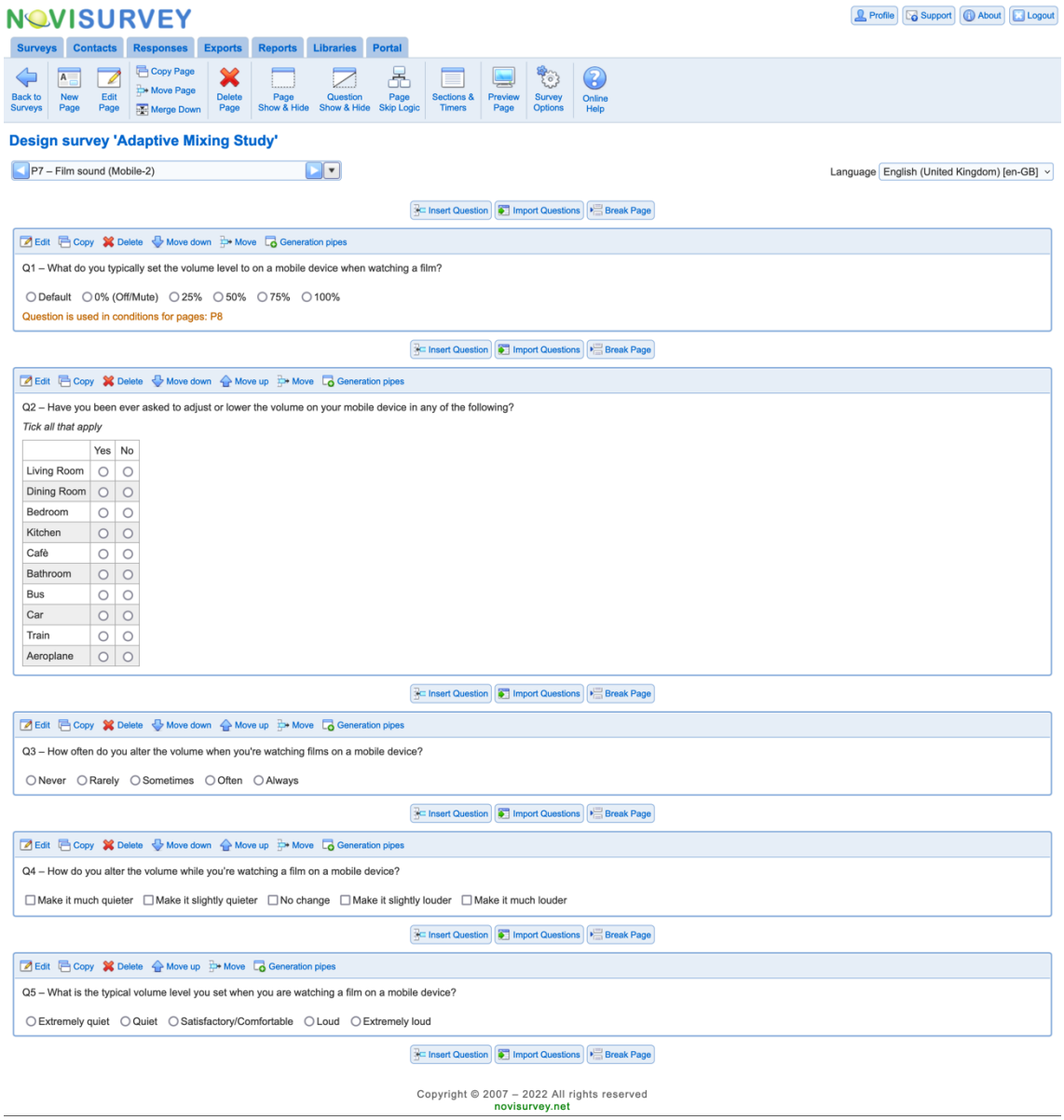
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P7 – Film sound (Mobile-2)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – What do you typically set the volume level to on a mobile device when watching a film?
Default
0% (Off/Mute)
25%
50%
75%
100%
Question is used in conditions for pages: P8
Move up
Q2 – Have you been ever asked to adjust or lower the volume on your mobile device in any of the following?
Tick all that apply
Yes
No
Living Room
Dining Room
Bedroom
Kitchen
Café
Bathroom
Bus
Car
Train
Aeroplane
Q3 – How often do you alter the volume when you're watching films on a mobile device?
Never
Rarely
Sometimes
Often
Always
Q4 – How do you alter the volume while you're watching a film on a mobile device?
Make it much quieter
Make it slightly quieter
No change
Make it slightly louder
Make it much louder
Q5 – What is the typical volume level you set when you are watching a film on a mobile device?
Extremely quiet
Quiet
Satisfactory/Comfortable
Loud
Extremely loud
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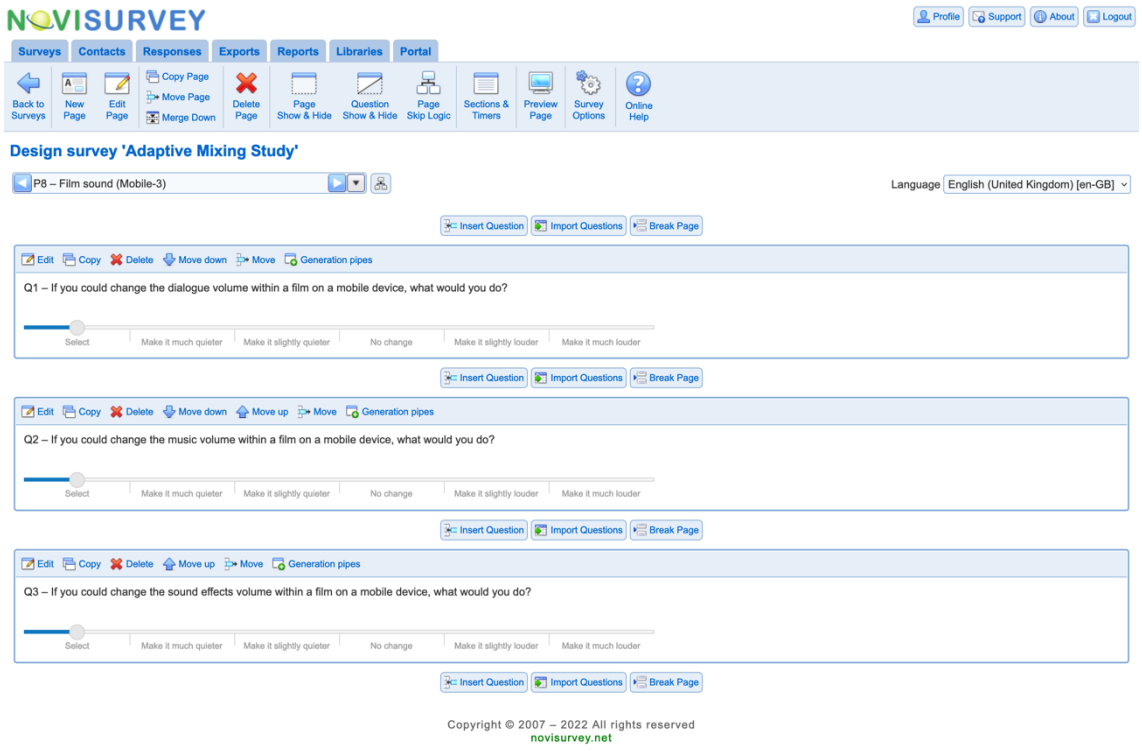
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P8 – Film sound (Mobile-3)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – If you could change the dialogue volume within a film on a mobile device, what would you do?
Select
Make it much quieter
Make it slightly quieter
No change
Make it slightly louder
Make it much louder
Move up
Q2 – If you could change the music volume within a film on a mobile device, what would you do?
Q3 – If you could change the sound effects volume within a film on a mobile device, what would you do?
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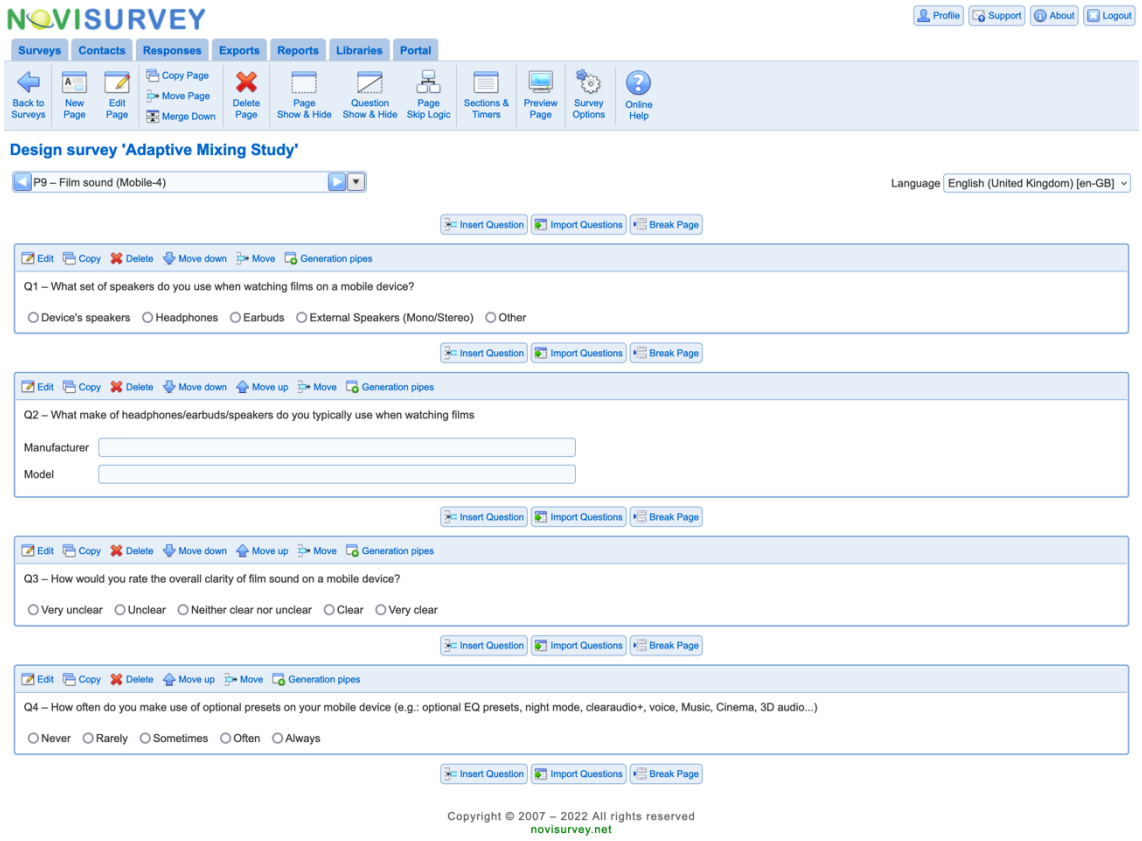
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P9 – Film sound (Mobile-4)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – What set of speakers do you use when watching films on a mobile device?
Device's speakers
Headphones
Earbuds
External Speakers (Mono/Stereo)
Other
Move up
Q2 – What make of headphones/earbuds/speakers do you typically use when watching films
Manufacturer
Model
Q3 – How would you rate the overall clarity of film sound on a mobile device?
Very unclear
Unclear
Neither clear nor unclear
Clear
Very clear
Q4 – How often do you make use of optional presets on your mobile device (e.g.: optional EQ presets, night mode, clearaudio+, voice, Music, Cinema, 3D audio...)
Never
Rarely
Sometimes
Often
Always
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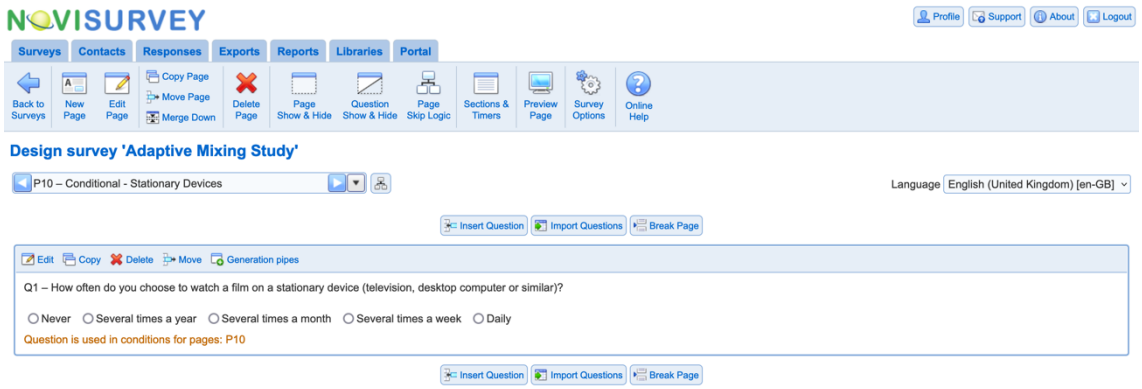

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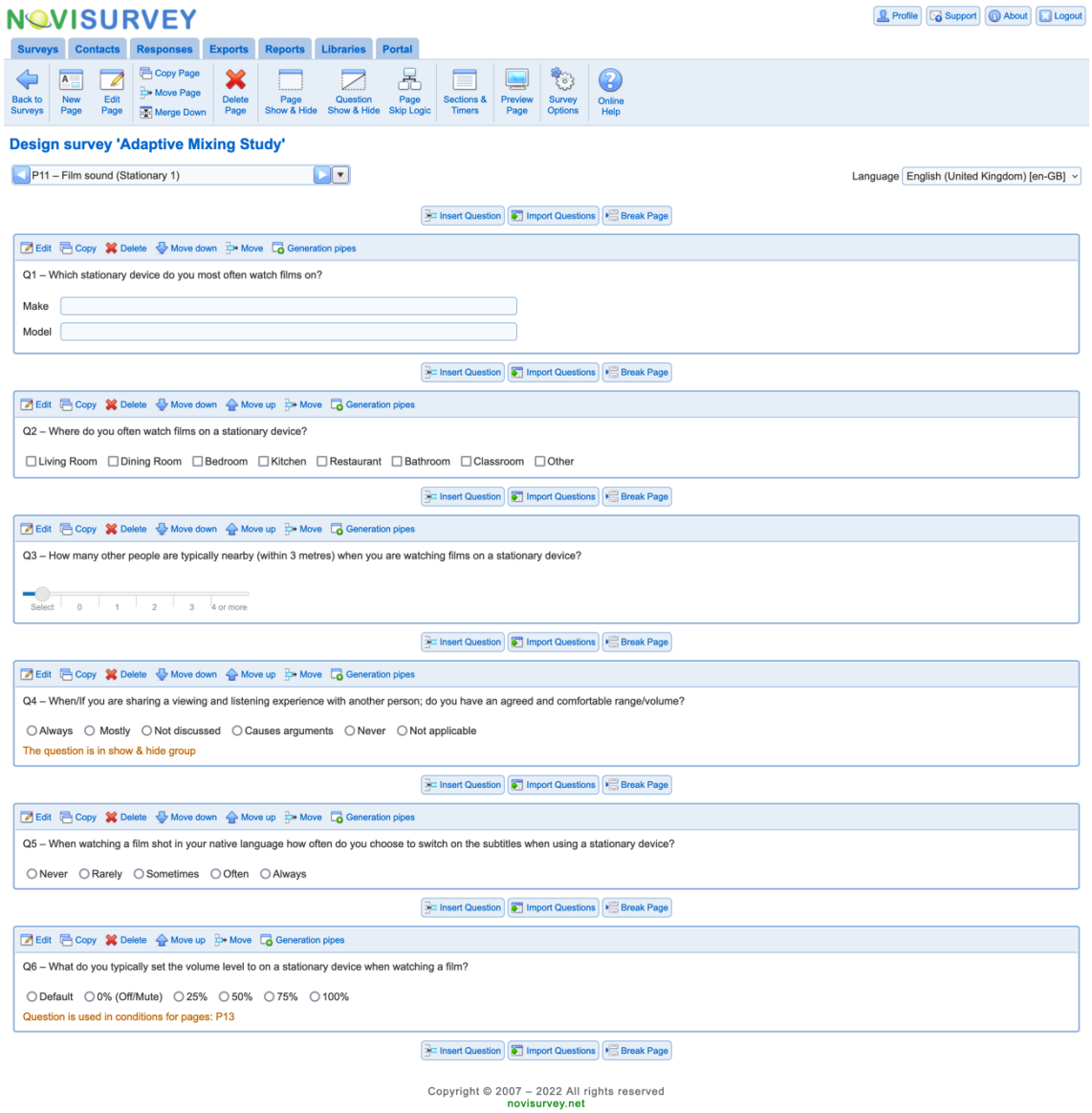
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P11 – Film sound (Stationary 1)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move up
Move
Generation pipes
Q1 – Which stationary device do you most often watch films on?
Make
Model
Q2 – Where do you often watch films on a stationary device?
Living Room
Dining Room
Bedroom
Kitchen
Restaurant
Bathroom
Classroom
Other
Q3 – How many other people are typically nearby (within 3 metres) when you are watching films on a stationary device?
Select
0
1
2
3
4 or more
Q4 – When/If you are sharing a viewing and listening experience with another person; do you have an agreed and comfortable range/volume?
Always
Mostly
Not discussed
Causes arguments
Never
Not applicable
The question is in show & hide group
Q5 – When watching a film shot in your native language how often do you choose to switch on the subtitles when using a stationary device?
Never
Rarely
Sometimes
Often
Always
Q6 – What do you typically set the volume level to on a stationary device when watching a film?
Default
0% (Off/Mute)
25%
50%
75%
100%
Question is used in conditions for pages: P13
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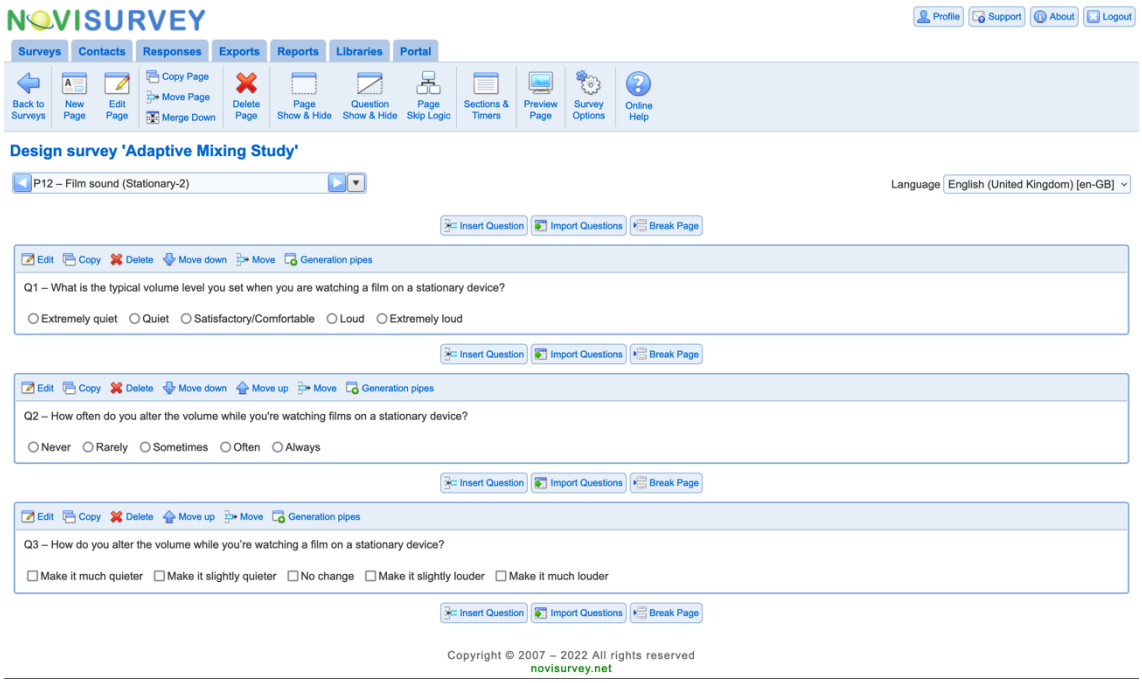

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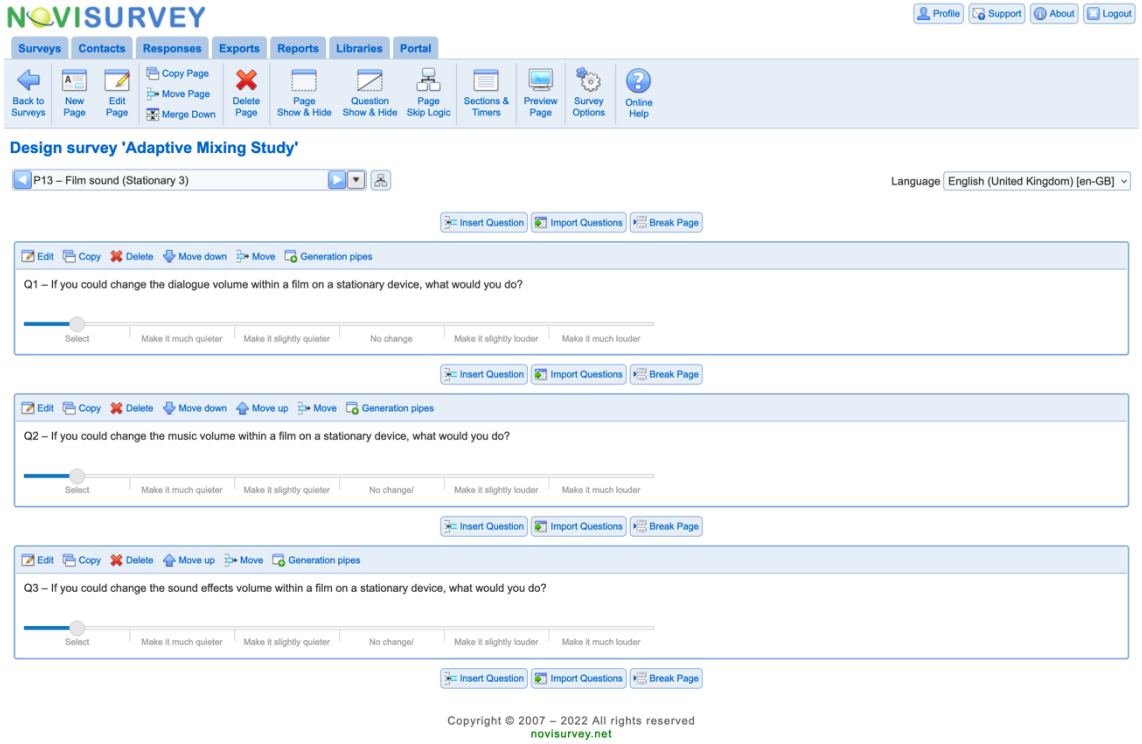

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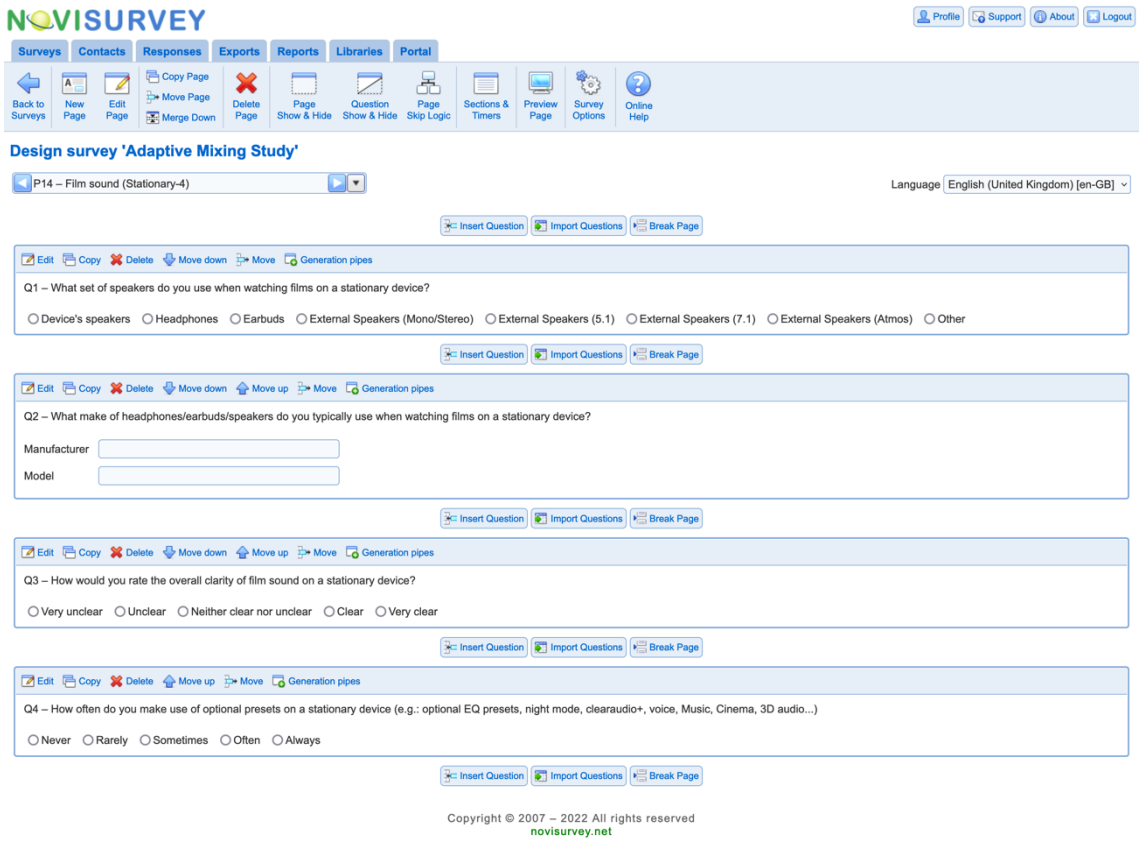
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P14 – Film sound (Stationary-4)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – What set of speakers do you use when watching films on a stationary device?
Device's speakers
Headphones
Earbuds
External Speakers (Mono/Stereo)
External Speakers (5.1)
External Speakers (7.1)
External Speakers (Atmos)
Other
Move up
Q2 – What make of headphones/earbuds/speakers do you typically use when watching films on a stationary device?
Manufacturer
Model
Q3 – How would you rate the overall clarity of film sound on a stationary device?
Very unclear
Unclear
Neither clear nor unclear
Clear
Very clear
Q4 – How often do you make use of optional presets on a stationary device (e.g.: optional EQ presets, night mode, clearaudio+, voice, Music, Cinema, 3D audio...)
Never
Rarely
Sometimes
Often
Always
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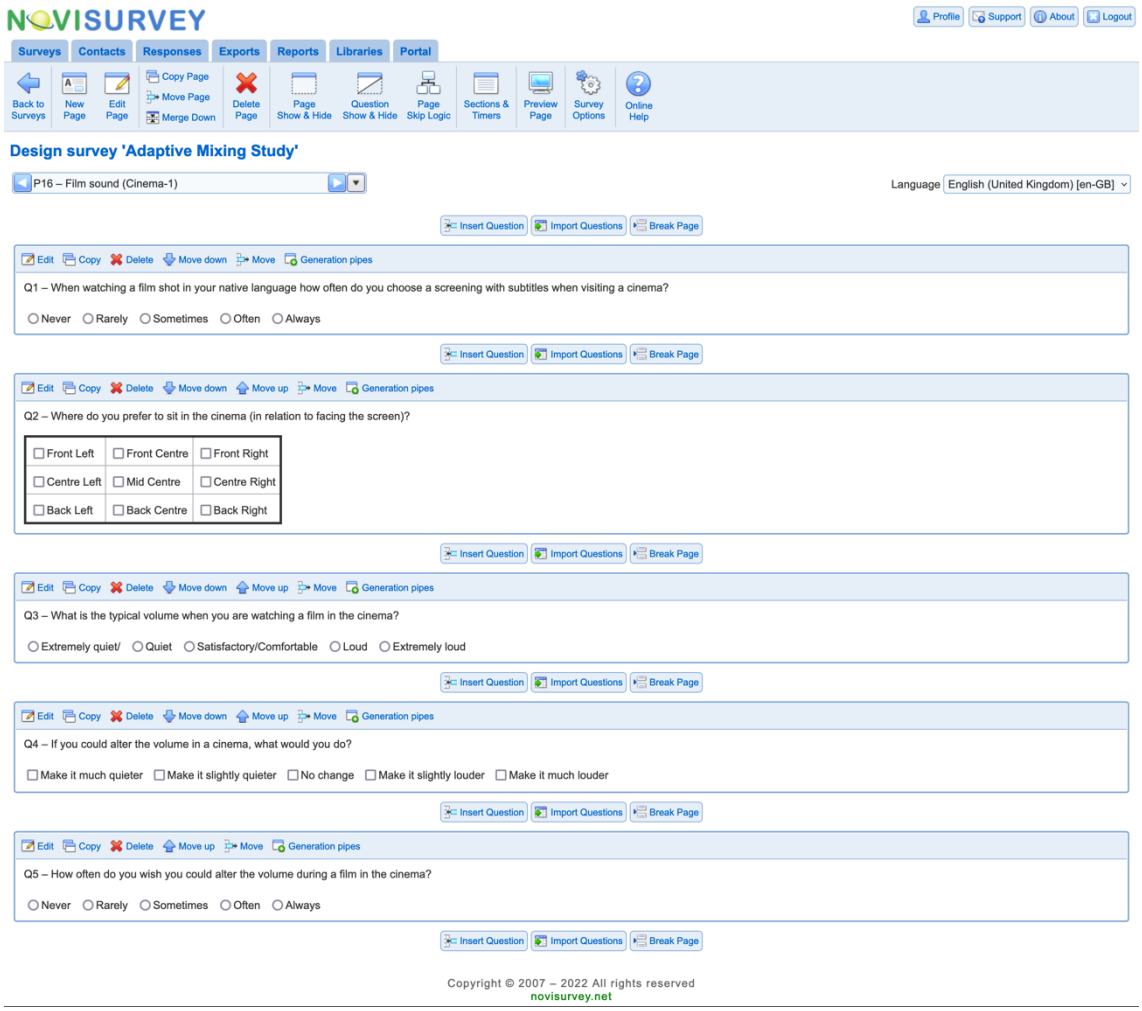
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P16 – Film sound (Cinema-1)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move up
Move
Generation pipes
Q1 – When watching a film shot in your native language how often do you choose a screening with subtitles when visiting a cinema?
Never
Rarely
Sometimes
Often
Always
Q2 – Where do you prefer to sit in the cinema (in relation to facing the screen)?
Front Left
Front Centre
Front Right
Centre Left
Mid Centre
Centre Right
Back Left
Back Centre
Back Right
Q3 – What is the typical volume when you are watching a film in the cinema?
Extremely quiet/
Quiet
Satisfactory/Comfortable
Loud
Extremely loud
Q4 – If you could alter the volume in a cinema, what would you do?
Make it much quieter
Make it slightly quieter
No change
Make it slightly louder
Make it much louder
Q5 – How often do you wish you could alter the volume during a film in the cinema?
Never
Rarely
Sometimes
Often
Always
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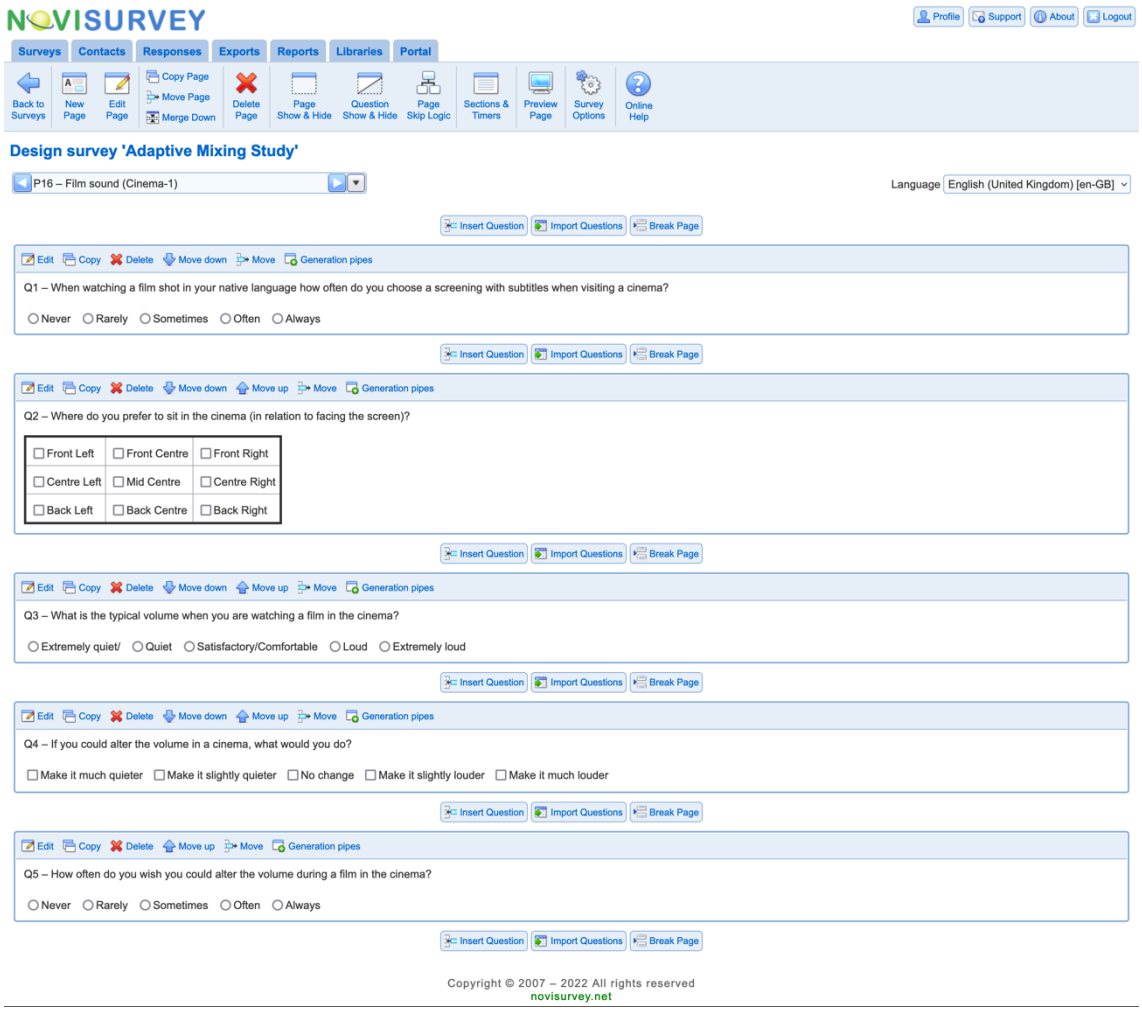
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P16 – Film sound (Cinema-1)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – When watching a film shot in your native language how often do you choose a screening with subtitles when visiting a cinema?
Never
Rarely
Sometimes
Often
Always
Move up
Q2 – Where do you prefer to sit in the cinema (in relation to facing the screen)?
Front Left
Front Centre
Front Right
Centre Left
Mid Centre
Centre Right
Back Left
Back Centre
Back Right
Q3 – What is the typical volume when you are watching a film in the cinema?
Extremely quiet/
Quiet
Satisfactory/Comfortable
Loud
Extremely loud
Q4 – If you could alter the volume in a cinema, what would you do?
Make it much quieter
Make it slightly quieter
No change
Make it slightly louder
Make it much louder
Q5 – How often do you wish you could alter the volume during a film in the cinema?
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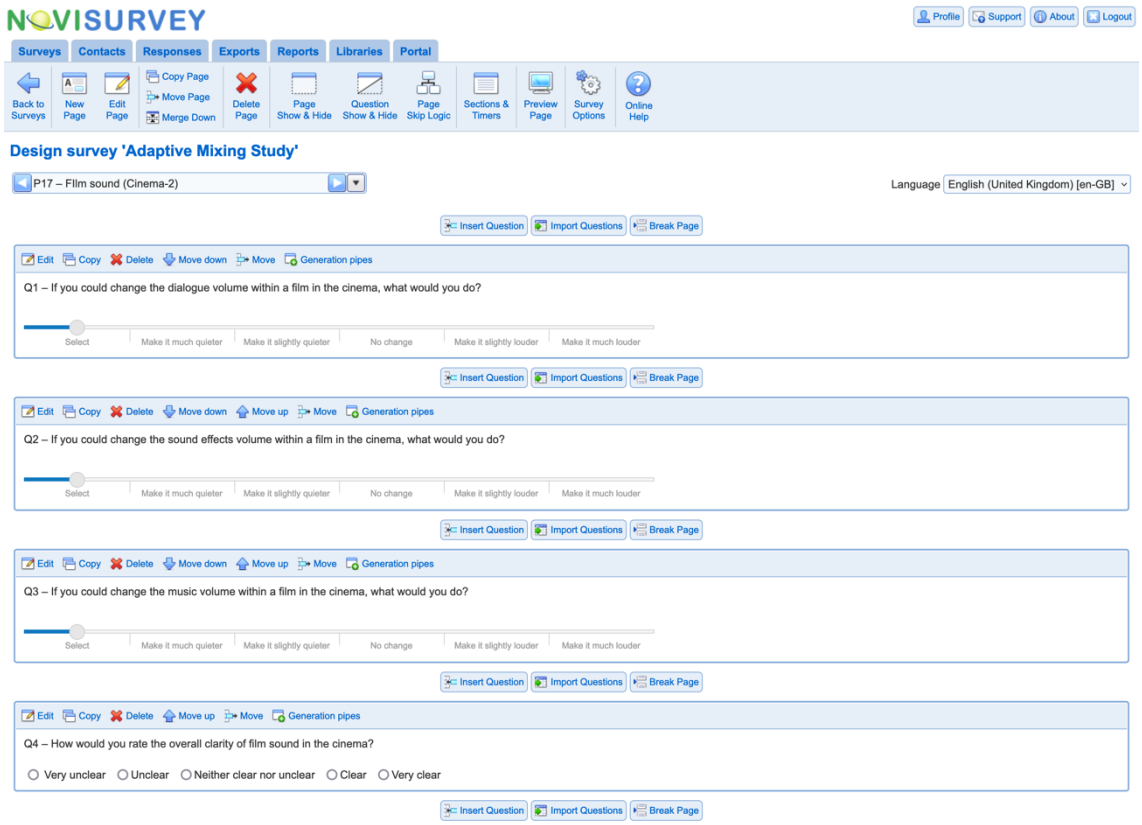

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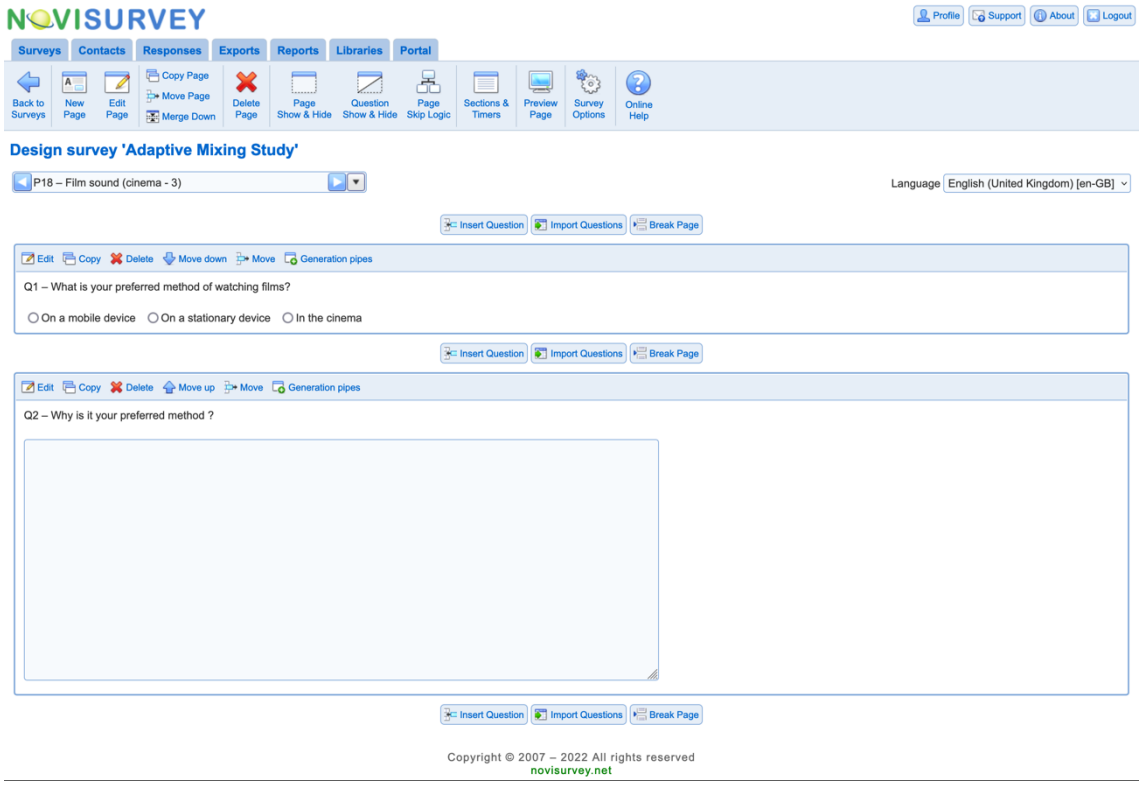

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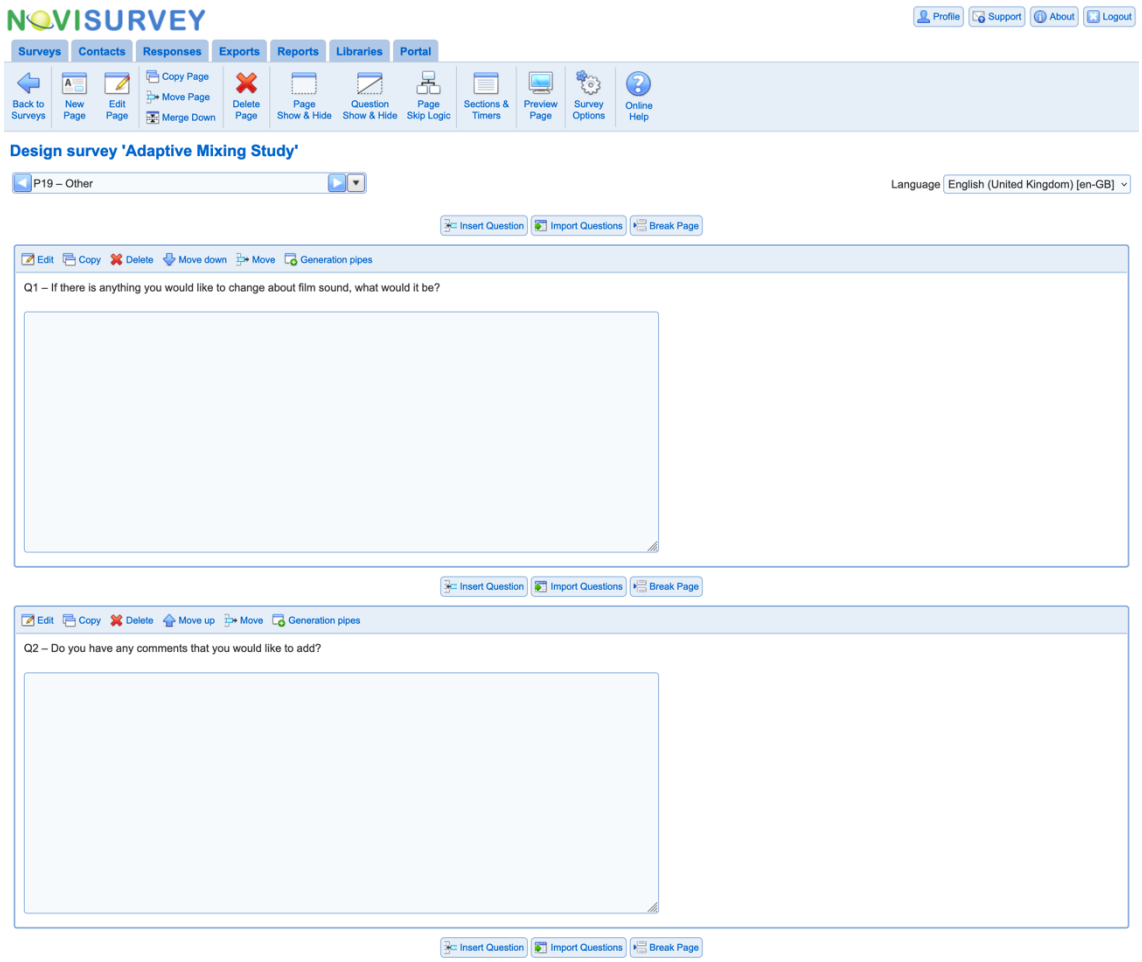
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P19 – Other
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – If there is anything you would like to change about film sound, what would it be?
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move up
Move
Generation pipes
Q2 – Do you have any comments that you would like to add?
Insert Question
Import Questions
Break Page
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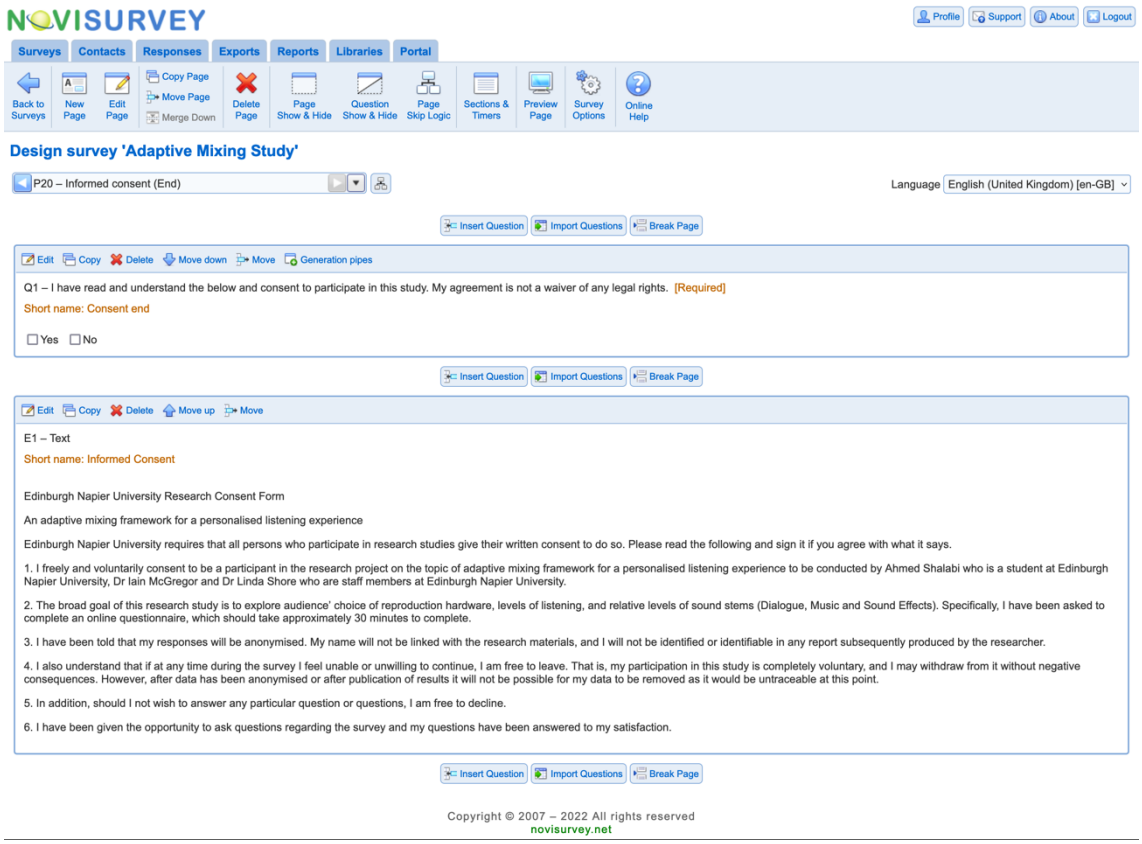
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Page Show & Hide
Question Show & Hide
Page Skip Logic
Sections & Timers
Preview Page
Survey Options
Online Help
Design survey 'Adaptive Mixing Study'
P20 – Informed consent (End)
Language English (United Kingdom) [en-GB]
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move down
Move
Generation pipes
Q1 – I have read and understand the below and consent to participate in this study. My agreement is not a waiver of any legal rights. [Required]
Short name: Consent end
Yes
No
Insert Question
Import Questions
Break Page
Edit
Copy
Delete
Move up
Move
E1 – Text
Short name: Informed Consent
Edinburgh Napier University Research Consent Form
An adaptive mixing framework for a personalised listening experience
Edinburgh Napier University requires that all persons who participate in research studies give their written consent to do so. Please read the following and sign it if you agree with what it says.
1. I freely and voluntarily consent to be a participant in the research project on the topic of adaptive mixing framework for a personalised listening experience to be conducted by Ahmed Shalabi who is a student at Edinburgh Napier University, Dr Iain McGregor and Dr Linda Shore who are staff members at Edinburgh Napier University.
2. The broad goal of this research study is to explore audience' choice of reproduction hardware, levels of listening, and relative levels of sound stems (Dialogue, Music and Sound Effects). Specifically, I have been asked to complete an online questionnaire, which should take approximately 30 minutes to complete.
3. I have been told that my responses will be anonymised. My name will not be linked with the research materials, and I will not be identified or identifiable in any report subsequently produced by the researcher.
4. I also understand that if at any time during the survey I feel unable or unwilling to continue, I am free to leave. That is, my participation in this study is completely voluntary, and I may withdraw from it without negative consequences. However, after data has been anonymised or after publication of results it will not be possible for my data to be removed as it would be untraceable at this point.
5. In addition, should I not wish to answer any particular question or questions, I am free to decline.
6. I have been given the opportunity to ask questions regarding the survey and my questions have been answered to my satisfaction.
Insert Question
Import Questions
Break Page
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Appendix B: Survey – Mixers

Page-1

Film sound study (Mixers)

Thank you for considering taking part. This study aims to investigate an adaptive mixing framework for a personalised listening experience for film. The project is being run by Ahmed Shalabi, Dr Iain McGregor, and Dr Linda Shore from Edinburgh Napier University.

You will be asked to answer a series of questions about your decision making and approach to mixing film sound, this should take approximately 45 minutes.

Specifically, about the monitoring environment, decisions you make to dialogue, music, and sound effects. You will also be asked to provide information about your age, gender, country of operation and whether you have any hearing loss, in addition to your preferred hardware, and make/model of speakers you use at the mixing stage.

All of your responses will be anonymised using the convention M01, M02…. The anonymous data will be stored for 10 years on the University's secure cloud service, and not shared with any other parties. The anonymised results will be written up for an audio journal, which is yet to be determined, and will form part of Ahmed Shalabi's PhD.

Results might improve mix translation across mobile and desktop devices and may allow for a development of an algorithm that will have an overall improvement and personalisation of film sound through correction factors tailored according to audience preferences and behaviour.

If you have any questions, please do not hesitate to contact Ahmed Shalabi on [redacted] or a.shalabi@napier.ac.uk, Dr Iain McGregor on 0131 455 2788 or via i.mcgregor@napier.ac.uk, Dr Linda Shore via l.shore@napier.ac.uk

If you have any concerns or wish to complain about the research, please contact the School of Computing Research Integrity Lead Dr Gemma Webster on 0131 455 2667 or via g.webster@napier.ac.uk.

Thank you for your time.

Ahmed Shalabi
PhD student
School of Computing
Edinburgh Napier University

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Film sound study (Mixers)

Edinburgh Napier University requires that all persons who participate in research studies give their written consent to do so. Please read the following and sign it if you agree with what it says.

1. **I freely and voluntarily consent to be a participant in the research project on the topic of adaptive mixing framework for a personalised listening experience to be conducted by Ahmed Shalabi who is a student at Edinburgh Napier University, Dr Iain McGregor and Dr Linda Shore who are staff members at Edinburgh Napier University.**
2. **The broad goal of this research study is to explore mixers' approaches to mixing content that requires delivery across multiple platforms. Specifically, I have been asked to take part in an online interview, which should take approximately 45 minutes to complete.**
3. **I have been told that my responses will be anonymised. My name will not be linked with the research materials, and I will not be identified or identifiable in any report subsequently produced by the researcher.**
4. **I also understand that if at any time during the interview I feel unable or unwilling to continue, I am free to leave. That is, my participation in this study is completely voluntary, and I may withdraw from it without negative consequences. However, after data has been anonymised or after publication of results it will not be possible for my data to be removed as it would be untraceable at this point.**
5. **In addition, should I not wish to answer any particular question or questions, I am free to decline.**
6. **I have been given the opportunity to ask questions regarding the interview and my questions have been answered to my satisfaction.**
7. **I have read and understand the above and consent to participate in this study. My signature is not a waiver of any legal rights. Furthermore, I understand that I will be able to keep a copy of the informed consent form for my records.**

1. **I have read and understand the above and consent to participate in this study. My agreement is not a waiver of any legal rights.**

   ☐ **Yes** ☐ **No**

Page-3

Film sound study (Mixers)

1. **What is your age?**

2. **What is your gender?**

3. **Do you have any hearing loss?**

   ○ **No Hearing Loss** ○ **Mild Hearing Loss** ○ **Moderate Hearing Loss** ○ **Severe Hearing Loss** ○ **Profound Hearing Loss**

4. **What country do you normally live in?**

5. **How many years have you been working in film sound?**

6. **How many years have you been mixing films professionally?**

Page-4

Film sound study (Mixers)

1. **Do you do diffferent mix passes such as a nearfield mix for home theatre to accommodate different formats?**

   ☐ **Yes** ☐ **No**

6. **If you work primarily for ONE mixing facility, what is the approximative size of the room (in cubic feet)?**

   ○ **< 1,499**
   ○ **1,500 < 4,999**
   ○ **5,000 < 9,999**
   ○ **10,000 < 19,999**
   ○ **Living room size or similar**
   ○ **Other**

Page-5

Film sound study (Mixers)

1. **Do you approach the mix differently if you know the film is going straight to streaming or TV?**

   ☐ **Yes** ☐ **No**

2. **When mixing films for streaming or TV (not trailers), what reference level do you set your monitoring to?**

3. **How far are you usually from the near-field monitors ?**

   Feet

4. **Do you do any acoustic correction or room optimisation before working on the near-field mix?**

   ☐ **Yes**

   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

Page-6

1. **Do you approach the dialogue stem differently in streaming and TV mixes? (if Yes, how)**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

2. **Do you approach the music stem differently in streaming and TV mixes? (if Yes, how)**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

3. **Do you approach Foley stem differently in streaming and TV mixes? (if Yes, how)**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

4. **Do you approach the spot effects differently in streaming and TV mixes? (if Yes, how)**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

5. **Do you approach the ambient/background sounds differently in streaming and TV mixes? (if Yes, how)**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

6. **Do you approach dynamics differently (macro and micro) in streaming and TV mixes? (if Yes, how)**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

7. **Do you approach spatial effects differently in streaming and TV mixes? (if Yes, how)**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

8. **Do you playback the mix on on other devices (e.g, TV with built in speakers or soundbar)?**

   ☐ **Yes**
   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

Page-7

1. **Does the screen size at the mixing stage affect the way you mix? If so, how?**

   ☐ **Yes**

   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

2. **Does the target screen size ever factor into the way you mix? If so, how?**

   ☐ **Yes**

   ☐ **No**

   Please give as much detail as possible!

   0 / 4000

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1. **What's your favorite playback format for entertainment purposes ?**

   ☐ **Built in device speakers** ☐ **External stereo speakers** ☐ **Over Ear Headphones** ☐ **In Ear Headphones** ☐ **On-Ear Headphones.** ☐ **Closed-Back Headphones** ☐ **Open-Back Headphone** ☐ **Earbuds** ☐ **Bluetooth Headphones** ☐ **Noise-Cancelling Headphones** ☐ **5.1** ☐ **7.1** ☐ **Atmos** ☐ **Other**

   Please give details on why it's your preferred option

   0 / 4000

2. **What is your preferred format to mix on (Stereo, 5.1, 7.1, Atmos/which config) and why?**

   0 / 4000

3. **What would you like to be done differently at the mixing stage to ensure optimum translation?**

   0 / 4000

4. **What are your thoughts on other object-based technologies like MPEG-H and the flexibility for audiences to enhance dialogue?**

   0 / 4000

Page-9

Film sound study (Mixers)

1. Do you have any comments that you would like to add?

0 / 4000

Film sound study (Mixers)

Edinburgh Napier University requires that all persons who participate in research studies give their written consent to do so. Please read the following and sign it if you agree with what it says.

1. **I freely and voluntarily consent to be a participant in the research project on the topic of adaptive mixing framework for a personalised listening experience to be conducted by Ahmed Shalabi who is a student at Edinburgh Napier University, Dr Iain McGregor and Dr Linda Shore who are staff members at Edinburgh Napier University.**
2. **The broad goal of this research study is to explore mixers' approaches to mixing content that requires delivery across multiple platforms. Specifically, I have been asked to take part in an online interview, which should take approximately 45 minutes to complete.**
3. **I have been told that my responses will be anonymised. My name will not be linked with the research materials, and I will not be identified or identifiable in any report subsequently produced by the researcher.**
4. **I also understand that if at any time during the interview I feel unable or unwilling to continue, I am free to leave. That is, my participation in this study is completely voluntary, and I may withdraw from it without negative consequences. However, after data has been anonymised or after publication of results it will not be possible for my data to be removed as it would be untraceable at this point.**
5. **In addition, should I not wish to answer any particular question or questions, I am free to decline.**
6. **I have been given the opportunity to ask questions regarding the interview and my questions have been answered to my satisfaction.**
7. **I have read and understand the above and consent to participate in this study. My signature is not a waiver of any legal rights. Furthermore, I understand that I will be able to keep a copy of the informed consent form for my records.**

1. **I have read and understand the below and consent to participate in this study. My agreement is not a waiver of any legal rights.**

   ☐ **Yes** ☐ **No**

Appendix C: Prototype 1

Prototype Survey

Page 1 of 2

Thank you for considering taking part. This study aims to investigate an adaptive mixing framework for a personalised listening experience for film. The project is being run by Ahmed Shalabi, Dr Iain McGregor and Dr Linda Shore from Edinburgh Napier University.

You will be asked to answer 45 short online questions, which should take approximately 30 minutes. Specifically, about the hardware you use to listen to film sound, and any modifications that you wish to make to the levels of the dialogue, music and sound effects.

All of your responses will be anonymised using the convention P01, P02…. Data will be stored for 10 years on a secure cloud service, and not shared with any other parties. The anonymised results will be presented at an audio conference, which has yet to be determined.

Identifying the hardware preferences and audience preferences with regards to film sound, may allow for a development of algorithm that will have an overall improvement and personalisation of film sound through correction factors tailored according to audience preferences and behaviour.

The results might also be of interest to those designing and mixing film sound, in respect of identifying the relative importance of different stems from a audience perspective, providing information about how each auditory element could be mixed, with minimum or no user intervention.

If you have any questions, please do not hesitate to contact Ahmed Shalabi on ahmed.shalabi@napier.ac.uk, Dr Iain McGregor on 0131 455 2788 or i.mcgregor@napier.ac.uk, Dr Linda Shore via l.shore@napier.ac.uk

If you have any concerns or wish to complain about the research please contact Professor Ben Paechter on 0131 455 2764 or via B.Paechter@napier.ac.uk.

Thank you for your time.

Edinburgh Napier University Research Consent Form

An adaptive mixing framework for a personalised listening experience

Edinburgh Napier University requires that all persons who participate in research studies give their written consent to do so. Please read the following and sign it if you agree with what it says.

1. I freely and voluntarily consent to be a participant in the research project on the topic of adaptive mixing framework for a personalised listening experience to be conducted by Ahmed Shalabi who is a student at Edinburgh Napier University, Dr Iain McGregor and Dr Linda Shore who are staff members at Edinburgh Napier University.

2. The broad goal of this research study is to explore audience' choice of reproduction hardware, levels of listening, exposure time and relative levels of sound stems (Dialogue, Music and Sound Effects). Specifically, I have been asked to complete an online questionnaire, which should take approximately 30 minutes to complete.

3. I have been told that my responses will be anonymised. My name will not be linked with the research materials, and I will not be identified or identifiable in any report subsequently produced by the researcher.

4. I also understand that if at any time during the survey I feel unable or unwilling to continue, I am free to leave. That is, my participation in this study is completely voluntary, and I may withdraw from it without negative consequences. However, after data has been anonymised or after publication of results it will not be possible for my data to be removed as it would be untraceable at this point.

5. In addition, should I not wish to answer any particular question or questions, I am free to decline.

6. I have been given the opportunity to ask questions regarding the survey and my questions have been answered to my satisfaction.

1. I have read and understand the above and consent to participate in this study. My agreement is not a waiver of any legal rights.

☐ Yes ☐ No

2. What is your age?

3. What is your gender?

4. Do you have any hearing loss?

○ No Hearing Loss ○ Mild Hearing Loss ○ Moderate Hearing Loss ○ Severe Hearing Loss ○ Profound Hearing Loss

1. How would you rate the experiment instructions?

○ Very unclear ○ Unclear ○ Neither clear nor unclear ○ Clear ○ Very clear

2. How would you rate the system you just used in the experiment?

○ I would like to use this system frequently
○ I found the system unnecessarily complex
○ I thought the system was easy to use
○ I would need technical person to be able to use this system.
○ System functions were well integrated
○ There was too much inconsistency in this system
○ It's quick to learn
○ I felt very confident using the system.
○ I needed to learn a lot of things before I could get going with this system.

3. What is your preferred method of watching films?

○ On a mobile device ○ On a stationary device ○ In the cinema

4. Do you subscribe to a streaming service? If so, which is the one you use often ?

5. How would you rate the overall clarity of film sound on your streaming service?

○ Very unclear ○ Unclear ○ Neither clear nor unclear ○ Clear ○ Very clear

6. Do you ever watch the same film across different platforms?

☐ Yes ☐ No

7. Do you have any comments that you would like to add?

0 / 4000

Appendix D: Prototype 2

Prototype Survey

Page 1 of 2

Thank you for considering taking part. This study aims to investigate an adaptive mixing framework for a personalised listening experience for film. The project is being run by Ahmed Shalabi, Dr Iain McGregor and Dr Linda Shore from Edinburgh Napier University.

You will be asked to answer 45 short online questions, which should take approximately 30 minutes. Specifically, about the hardware you use to listen to film sound, and any modifications that you wish to make to the levels of the dialogue, music and sound effects.

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Identifying the hardware preferences and audience preferences with regards to film sound, may allow for a development of algorithm that will have an overall improvement and personalisation of film sound through correction factors tailored according to audience preferences and behaviour.

The results might also be of interest to those designing and mixing film sound, in respect of identifying the relative importance of different stems from a audience perspective, providing information about how each auditory element could be mixed, with minimum or no user intervention.

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Thank you for your time.

1. How would you rate the experiment instructions?

○ Very unclear ○ Unclear ○ Neither clear nor unclear ○ Clear ○ Very clear

2. How would you rate the system you just used in the experiment?

○ I would like to use this system frequently
○ I found the system unnecessarily complex
○ I thought the system was easy to use
○ I would need technical person to be able to use this system.
○ System functions were well integrated
○ There was too much inconsistency in this system
○ It's quick to learn
○ I felt very confident using the system.
○ I needed to learn a lot of things before I could get going with this system.

3. What is your preferred method of watching films?

○ On a mobile device ○ On a stationary device ○ In the cinema

4. Do you subscribe to a streaming service? If so, which is the one you use often ?

5. How would you rate the overall clarity of film sound on your streaming service?

○ Very unclear ○ Unclear ○ Neither clear nor unclear ○ Clear ○ Very clear

6. Do you ever watch the same film across different platforms?

☐ Yes ☐ No

7. Do you have any comments that you would like to add?

0 / 4000

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An adaptive mixing framework for a personalised listening experience

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6. I have been given the opportunity to ask questions regarding the survey and my questions have been answered to my satisfaction.

1. I have read and understand the above and consent to participate in this study. My agreement is not a waiver of any legal rights.

☐ Yes ☐ No

2. What is your age?

3. What is your gender?

4. Do you have any hearing loss?

○ No Hearing Loss ○ Mild Hearing Loss ○ Moderate Hearing Loss ○ Severe Hearing Loss ○ Profound Hearing Loss